

This International Student Edition is for use outside of the U.S.

FOURTH EDITION

TECHNICAL WRITING FOR ENGINEERS & SCIENTISTS



**Mc
Graw
Hill**

Leo Finkelstein Jr. • Jeanine Elise Aune • Leslie A. Potter

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TECHNICAL WRITING FOR ENGINEERS & SCIENTISTS



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Hill

Leo Finkelstein Jr. • Jeanine Elise Aune • Leslie A. Potter

Technical Writing for Engineers & Scientists

Fourth Edition

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TECHNICAL WRITING FOR ENGINEERS & SCIENTISTS

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Dedication

In memory of Dr. Leo Finkelstein, Jr. We are honored to continue his legacy.

This has been one of my favorite projects of all time due, in no small part, to the support of and collaboration with some of my favorite people. It is dedicated to Siggi for unwavering support, to my children for just being, to my pets and pony for the cuddles and chances to recharge my soul, to my mom for being my cheerleader, to my siblings for their wit and humor, and to the legendary Grandma Glenna for showing me the way to teaching and about not letting one's standards slip. And last but not least, to Leslie, for her creativity, enthusiasm, and for keeping my nose to the grindstone.

—Jenny

I have always loved language: it was difficult for me to choose between a career in engineering and a career in English. I am grateful for the opportunity to combine both interests in this book and couldn't have asked for a better writing partner—thank you, Jenny! I dedicate it to John, for all that he has done these past 30 years to make it possible for me to tackle this project; as well as to our three boys for their never-ending good humor, generosity of spirit, and willing assistance; to my parents for giving me the best of each of themselves, including my mom's amazing editing skills and my dad's spot-on advice; to my brother who supports me always; and to my in-laws whom I love dearly. Shout-out to my dogs, too, who keep me grounded.

—Leslie

Purpose

The purpose of this book is to succinctly explain the content and structure of concepts and genres common to communication in engineering and science disciplines. Much like Dr. Finkelstein did in the first three editions, we aim to avoid “the sterile, encyclopedic treatment of writing concepts” that exists in many textbooks about writing. While such textbooks can be helpful for writing instructors who want to cover all of the ins and outs of technical writing theory, concepts, strategies, and genres in writing classes, such comprehensive textbooks might not be the most useful for instructors looking to incorporate writing assignments into their already-packed classes, or for students looking for the nitty-gritty details about what they need to do to get the writing project done in their engineering and science classes.

Approach

Our approach to revising this textbook was based on our combined 50+ years of teaching experience. We have endeavored to bring our approach to teaching to *Technical Writing for Engineers and Scientists*, 4th edition.

Theoretical Foundation

Technical communication is most effective when it considers audience, purpose, and context. Audiences can be categorized in many ways, but one of the most utilitarian methods is to think of them as decision-makers, advisors, and implementors. For example, when you are writing an abstract or summary, you are typically writing for decision-makers and/or advisors. When you are researching and writing a feasibility or recommendation report, you are writing for decision-makers and advisors. When you are writing descriptions or instructions, you are most likely writing for

implementors. We have considered these three categories of audience as we have revised the content in this book.

In addition to audience, technical communication is most effective when it considers purpose and context. If we have been hired by the CEO and founder of a major pasta producer to write a report on the feasibility of moving a pasta factory to the upper Midwest, and we write a description of a pasta factory and its components, we will have singularly failed in understanding our purpose. The CEO is already familiar with a pasta factory; they need an evaluation of a solution based on a set of criteria, like proximity to rail lines for shipping raw ingredients. As technical writers, we must anticipate how our communication will be used and in what context. For example, electronic instructions for an executive in their corner office have an entirely different context than those required by a worker standing underneath a molten iron transfer line in a foundry.

Restructured for Easier Understanding

Over the years, we have learned that our students do best when they can see a finished example before we get into the details—much like assembly instructions and recipes, it helps the implementor to look at a picture of the final product before they begin crafting it. Therefore, we restructured the textbook's genre chapter content to provide a definition first, followed by an overview, a basic general outline, a complete example, that example broken down into the logical moves that the writer needs to make, and if relevant, additional examples illustrating the range of that technical writing genre. The intent is that students see the overall document to “get a feel” for it before they examine the breakdown of the components in that example document.

We have also incorporated references to other chapters along with small excerpts of copied-and-pasted text throughout most of the book. We did this to facilitate the use of individual chapters rather than expecting a student to read and remember the entire book. Bonus: repetition is how we move knowledge into our long-term memory.

Analogies

Over our years of teaching, we have developed the tendency to explain new concepts and ideas to students using analogies (lightbulb moments!). In this

revision, we also tried to connect technical writing concepts and genres to a framework with which (we hope!) students are familiar. While not everyone loves dogs, and some may be allergic or even avoid canines for religious reasons, we anticipate that most everyone will at least be familiar with the concept of the dog, its multiple variations and roles in society, and purposes behind those variations. We hope that our readers find the analogies helpful, if somewhat wacky, in learning about technical writing concepts and genres.

Although we do assume that most everyone is familiar with dogs Page v in general, we do not assume that everyone knows all of the various dog species by name, and here we took our own advice from [Chapter 4](#) on Technical Definitions:

In some situations, you might need to sacrifice desired precision in your definition to achieve the required level of communication.

To ensure that our readers clearly recognize our references to specific dog breeds and their connection to genres, we have treated dog breed names as proper nouns despite generally accepted capitalization guidelines for dog breeds. For example, in most writing situations, “border collie” would not be capitalized, nor would the “pinscher” in “Doberman pinscher.” However, we have capitalized all words in each dog breed name so that readers will know we are talking about Border Collies and Doberman Pinschers as dog genres, analogous to writing genres.

Embraced Our Inner Goofy

Finally, we have tried to incorporate the same Goofy (dog pun intended) sense of humor that we try to share in our courses to make writing as fun and interesting for our students as we can. It was Finkelstein’s light-heartedness and willingness to poke fun at himself that initially attracted us to his textbook, and we are more than happy to continue the tradition.

Disclaimers

As part of our attempts to be light-hearted, we have used numerous fictitious names in examples throughout the book. Any similarity to actual humans, towns, or organizations is completely coincidental.

Also, just as many sources for students writing technical reports have moved online in the past decade, so have they migrated online for authors writing textbooks. We have cited numerous websites throughout the text, including access dates, but understand that these addresses might change over time. Our intent was to provide enough information for readers to be able to search the topics successfully even if the websites change.

Organization

We organized the fourth edition around three major sections: the first one discusses fundamental material, the second describes how to write the most common technical documents, and the third provides useful information that, frankly, does not fit neatly in the first two sections.

Section I: Fundamentals

[Chapters 1](#) through [6](#) deal with basic considerations, including the component skills you will need to produce effective technical writing. Expanding on the successful approach used in the first three editions, this section includes

- **[Chapter 1: Introduction](#)** explains what technical writing is and the basic concepts needed in technical writing.
- **[Chapter 2: Ethical Considerations](#)** focuses on ethics in technical writing and includes a general discussion of ethical considerations for technical writers.
- **[Chapter 3: Note-taking](#)** provides both the “why” and the “how” of taking notes, including techniques and legal/ethical considerations.
- **[Chapter 4: Technical Definitions](#)** explains the “nuts and bolts” of writing effective technical definitions because the ability to define is one of the primary skills needed for most technical writing.
- **[Chapter 5: Description of a Mechanism](#)** explains another primary skill needed for technical writing, which is being able to describe mechanisms precisely, accurately, and at a level the audience can understand.

- **[Chapter 6: Description of a Process](#)** explains how to describe processes, that is, third-person descriptions of events that do not directly involve the reader.

Section II: Technical Documents

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- **[Chapter 7: Instructions and Manuals](#)** explains how to write instructions, or second-person descriptions for human involvement that provide specific directions so that the reader can perform a task or series of tasks.
- **[Chapter 8: Proposals](#)** explains the three necessary things that all proposals must include, provides multiple examples of informal proposals, and parses the “why” and “how” for each section.
- **[Chapter 9: Progress Reports](#)** builds on the proposals in [Chapter 8](#), explaining the necessary elements of a progress report and showing how to construct an effective progress report.
- **[Chapter 10: Feasibility and Recommendation Reports](#)** explains how to develop objective documents that identify and evaluate solutions to problems and explains the difference between a feasibility and a recommendation report.
- **[Chapter 11: Laboratory and Project Reports](#)** explains the difference between and the purpose of laboratory and project reports, as well as their various elements.
- **[Chapter 12: Research Reports](#)** explains the focused, objective nature of research reports, the general structure, and the wide variety of content based on audience need.
- **[Chapter 13: A3 Reports](#)** shares the history of the document, templates and examples, and an understanding of the usefulness in business.
- **[Chapter 14: Abstracts and Summaries](#)** explains the purpose of abstracts and summaries and provides examples of four different kinds of summations with a focus on both academic and business situations.

Section III: Other Useful Stuff

- **[Chapter 15: Style and Mechanics](#)** highlights common issues with basic building blocks, including how style is related to mechanics (spelling, punctuation, and grammar).
- **[Chapter 16: Documentation](#)** shows examples of how to cite many different kinds of online, print, and in-person sources.
- **[Chapter 17: Visuals](#)** includes basic guidelines for when to use what kinds of visuals, as well as tips for constructing them as accurately and effectively as possible.
- **[Chapter 18: Presentations](#)** provides in-person and online presentation tips, including an example set of slides for an update presentation.
- **[Chapter 19: Business Communication](#)** provides some general guidelines, outlines, and examples for business communication.
- **[Chapter 20: Communication with Future Employers](#)** details six specific kinds of job-seeking communication, including both written (e.g., resumes) and oral (e.g., interviews).
- **[Chapter 21: Team Writing](#)** addresses important considerations for accomplishing a collectively written document, both as a student and as a professional.

We believe that our revisions will be useful to students and instructors who choose to use this book. However, we acknowledge (and appreciate!) that language and communication are always evolving and what is considered acceptable today might be adapted by tomorrow. We have done our best to capture generally accepted formulations and long-lived rules.

Being an effective technical writer continues to increase in importance. In fact, we have heard from some employers that they would rather hire good writers with average technical skills (because they can teach the technical skills themselves), than hire someone with high technical skills who communicates poorly. We would be delighted if our textbook could help students gain the communication skills that employers want.

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 Department of English
 Iowa State University

Leslie A. Potter, Teaching Professor
Department of Industrial and Manufacturing Systems Engineering
Iowa State University

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About the Authors

Leo Finkelstein, Jr., received a bachelor's degree from the University of North Carolina at Chapel Hill in 1968; a master's from the University of Tennessee at Knoxville in 1969; and a Ph.D. from Rensselaer Polytechnic Institute at Troy, New York, in 1978. He was Lecturer and Director of Technical Communication for the College of Engineering and Computer Science, Wright State University, Dayton, Ohio. He directed the technical writing program at the U.S. Air Force Academy while also serving as adjunct faculty for the University of Colorado at Colorado Springs. He wrote, produced, and directed technical films in Southern California and commanded a combat-documentation, photographic unit in Southeast Asia during the Vietnam War, flying combat missions as an aerial photographer. In addition, his military service included experience in both space and logistics systems. He held FCC commercial and amateur radio licenses, had a black belt in tae kwon do, and was an avid user of all types of gadgets.



Jeanine Elise Aune

Jeanine (Jenny) Elise Aune is a Teaching Professor and the Director of ISUComm Advanced Communication (AdvComm) program at Iowa State University (ISU). She has an M.A. and Ph.D. from the University of Wisconsin-Madison and has been teaching writing in the Department of English at Iowa State University since 1999. She was the Coordinator of ISU's Learning Community (LC) English links for 11 years. Her responsibilities included helping linked discipline faculty communicate their expectations for students' communication skills to English instructors, and helping English instructors re-design writing classes to help students

develop those skills. The number of LC-linked English sections more than doubled during her tenure. She has been directing or co-directing the Advanced Communication program since 2011. She has worked with stakeholders across campus to build standardized curricula for the program's four courses—business communication, proposal and report writing, biological communication, and technical communication—in both face-to-face and online mediums. These four courses are taught by ~40 instructors in ~200 sections and enroll ~4,800 students every academic year. Most recently, the online business communication course earned QM certification¹ at 97%, and the online technical communication course earned QM certification at 98%.



Leslie A. Potter

Leslie A. Potter is a Teaching Professor in the Industrial and Manufacturing Systems Engineering (IMSE) department at Iowa State University (ISU). She has a B.S. in industrial engineering from ISU and an M.S. in industrial engineering with an emphasis in manufacturing from The Pennsylvania State University. She worked as an engineer and supervisor for John Deere for seven years before joining IMSE at ISU, where she has taught undergraduate courses across the curriculum for the past 20+ years, ranging from freshman problem-solving and programming to capstone design. As part of her research at ISU, she co-developed a professional communications course within the industrial engineering curriculum. From those efforts, she has developed and incorporated substantial writing and speaking curricula that are used by many of her peers. She was the co-founder in 2013 as well as the co-chair for the IMSE Undergraduate Research program for seven years, supporting hundreds of students with writing and presenting their research. She regularly requires writing and

presentation assignments in her engineering courses, and has collaborated with faculty in the Iowa State University Department of English since 2007.

¹<https://www.qualitymatters.org/reviews-certifications>

Email to a Technical Writer

Technical writers must consider their audience, purpose, and situational context. The following is an example of an email that the authors of this textbook could send to potential users of this textbook.

From: JennyandLeslie@Finkelstein
Sent: Monday, June 14, 2021 1:32 PM
To: Student
Subject: Email to a Technical Writer

Dear Student,

Your decision to study a technical field is most admirable, and we applaud your fortitude and motivation in dedicating your time and effort to learning a subject that has the power to change the world. However, your knowledge and expertise can only be understood by others if you can communicate your ideas, thoughts, findings, recommendations, and preferences with both technical and non-technical audiences. It is there, at the point of communicating ideas as non-abstractly as possible, that we can support you. We hope that after reading a few pages of our not-typically-super-serious textbook, having a conversation or two with others about it, and allowing yourself to enjoy the relatively formulaic processes of technical writing, you will come to appreciate the power of audience, purpose, and context. With this note, we offer two thoughts.

First: Writing in the real world is very different from the writing you have likely done to date.

You will need to change your mindset. No longer will you write for an audience of one. No longer will you write to a teacher to show them how much you know or have learned. We must now ask you to put aside the unintended mindset you might have of, “I will write what my instructor wants.” To become an effective technical writer, you must anticipate who your audience will be, recognizing that you might have multiple audience types, and why you will write for each of them. Do you wish to inform, persuade, or simply create goodwill? Ask yourself: what outcome do I need? And how can I make it happen? Then write with this at the front of your mind.

Second: While technical writing is a serious endeavor, one (and by that we include you) should not take oneself too seriously.

Accomplishment is one of life’s greatest rewards. Perhaps you have already been successful in calculus, soccer, robotics, speech, origami, research, baking, dance, or gaming. You sit in our classroom—we know how amazing you are! And we know that you can be an accomplished technical writer, too. As with your other successes, it requires only a willingness to improve through practice and reflection, but this is ever-so-much-more enjoyable with a quick laugh and permission to enjoy the sometimes painfully iterative improvement process. Somewhat related,

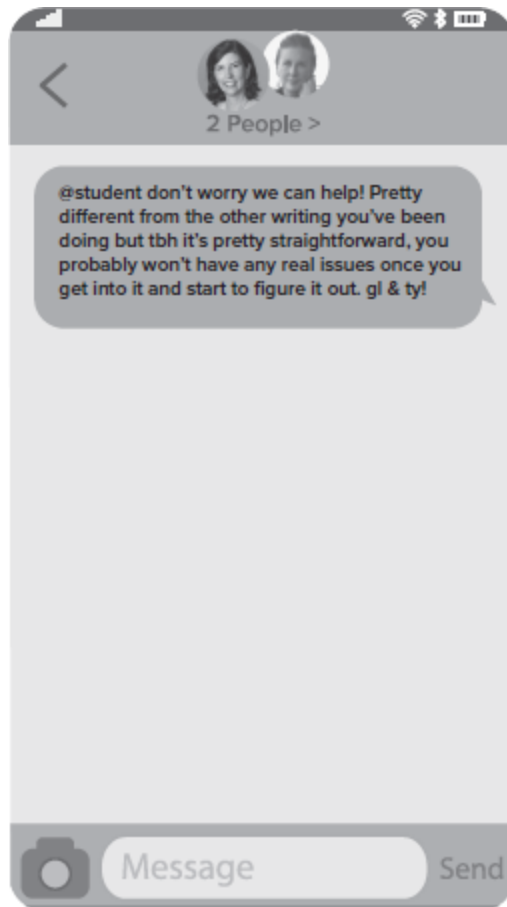
we thank you for indulging our, shall we say “quirky,” sense of humor (some will say “bad”—we happily accept that). In short, we say lighten up! Enjoy the ride. And the write!

We thank you for the confidence you will place in our ability to communicate about communication, knowing full well that the initial decision was not your own, but your instructor’s.

Yours very truly,
Jenny and Leslie

IM to a Technical Writer

The same information can be presented in different ways. For example, the authors of this textbook could also choose to send an IM to potential users of this textbook. The basic idea is the same, but much detail has been removed and the language is much more informal.



Acknowledgements

Acknowledging those who have helped us with the 4th edition of *Technical Writing for Engineers and Scientists* is a daunting task. While our names are listed as authors, they are only listed there due to the opportunity provided to us by McGraw Hill and the support of our families, friends, and colleagues.

Thank you to McGraw Hill for accepting our proposal for revising the textbook and giving us the opportunity to teach on a broader scale than our classrooms allow. Thank you to Shannon O'Donnell for the preliminary conversation a couple of years back about the potential for a revision of the textbook and supporting our authorship, Theresa Collins for getting us started on the process, and Erin Kamm for guiding us through to the end. In addition, thank you to Beth Bettcher for supporting our endeavor; Beth Cray for helping gain permissions; and Maria McGreal, Dheeraj Kumar, and team for getting our image plans to paper. There are so many elements, tiny details, and untold hours of work that go into creating a published textbook from a text document; hopefully, we have not missed anyone involved in that monumental task.

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CHAPTER ONE

Introduction



Technical writing is a fundamental skill for virtually everyone working in science and engineering—and that includes a broader range of people than just scientists and engineers. Most science and engineering activities require technical communication, whether in written, electronic, or some other form. Research, development, finance, manufacturing, and a host of technical commercial services rely on precise communication to relay complex information to a wide range of audiences for many purposes. Technical writing is the means by which such communication is produced.

What Is Technical Writing?

To define what technical writing is, it might be useful to first clarify what it is not. Technical writing is not what one does out in the meadow under an elm tree; that is creative writing (unless, of course, you are a meadow habitats specialist!). Creative writing, while challenging, is an artistic activity. It flourishes in a world of nuance, interpretation, and subtlety. Technical writing, however, is the opposite of artistic. Technical writing is writing with precision and eliminating the possibility of (mis)interpretation. Technical writing is not what most people commonly do for fun or relaxation.

Imagine, if you will, a learned, sophisticated person sipping fine wine, eating imported cheese, and watching the sunset. This person sounds like a creative writer who, with paper and pen in hand, might reflect on the nature of time as the last rays of sunlight slowly disappear over the horizon. Our creative writer might even develop a definition of **time** that goes something like this:

Time is a river flowing from nowhere¹ through which everything and everyone move forward to meet their fate.

A creative, sensitive person sees an inspiring sunset, is moved to words, and writes the “river from nowhere” definition. Obviously, this approach is metaphorical. But what if someone inspired by that magnificent sunset were to write the following definition of **time**?

Time is a convention of measurement based on the microwave spectral line emitted by cesium atoms with an atomic weight of 133 and an integral frequency of 9,192,631,770 hertz.

Perhaps not! It would certainly take a unique individual to be emotionally moved by a sunset and then come up with microwave spectral lines and cesium 133. The second definition of time is technical. It is designed to be objective, direct, and precise. Consequently, it lacks the emotional impact of the first definition because, as technical writing, it avoids the use of rich metaphors and figures of speech, substituting instead precise, empirical facts.

The difference between the two definitions shows the fundamental distinction between technical writing and creative writing (and all the other kinds of writing that fall in between). Technical writing is precise, objective, direct, and clearly defined.

Reducing Abstraction

In linguistic terms, technical writing is writing that displays a relatively low level of abstraction. To clarify the concept of abstraction, consider the funnel of abstraction in [Figure 1.1](#). Here, various levels of abstraction are being used to refer to the specific merle pigmentation of your friend’s new puppy, Sparky-the-Merle. Imagine that you are highly allergic and entirely unfamiliar with anything dog-related, but your friend is extremely excited about their puppy being a merle. They explain that merle is a **dog genetic mutation**. However, that does little to help you understand their excitement —“dog genetic mutation” is a high-level abstraction and could be anything from all kinds of dog coat variations to unmatched eye color to hip dysplasia (as seen at the top of the funnel in [Figure 1.1](#)).

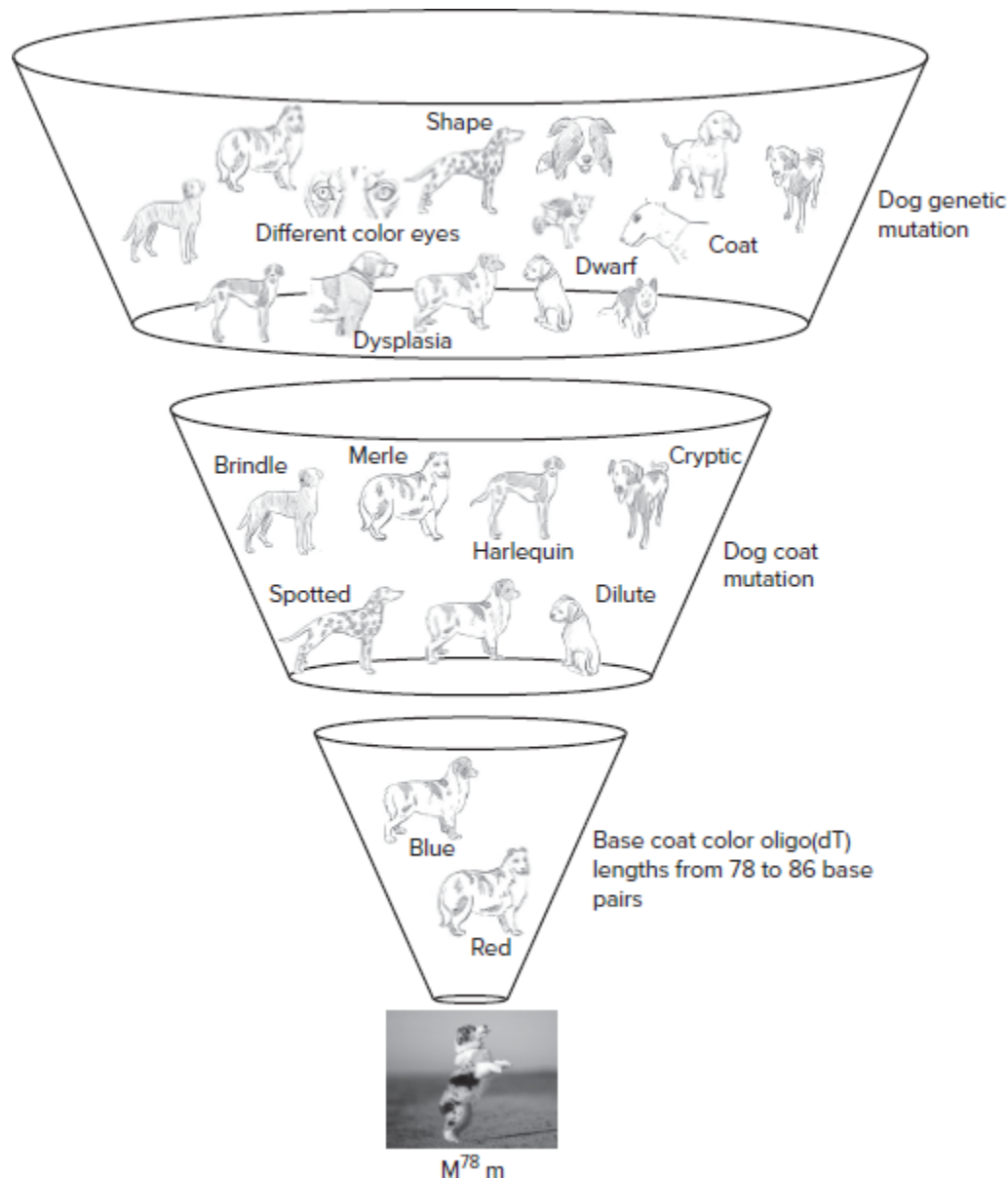


Figure 1.1

A funnel of abstraction. The high level of abstraction at the top of the funnel allows the audience to consider too many concepts or items. Abstraction is reduced through the use of non-abstract language. At the bottom of the funnel is the lowest level of abstraction, where only one concept or item is described by the language.

otsphoto/Shutterstock

When you ask for more clarity, your friend explains that **merle** is Page 3 a term used to describe a dog's splotchy coat, a type of **dog coat mutation**, which could mean any type of coat coloring with irregularly shaped spots. This description allows you to envision what merle means,

and could include a dog with a coat that fits in many different categories of merle, such as cryptic, dilute, harlequin, or, in this case, standard. As we filter the level of abstraction, by adding some objective precision in our words, the reference becomes increasingly precise.

Imagine how lucky the happenstance that you and your friend are both geneticists, and you both quickly get to an even more filtered and precise description. Your friend refers to the pigmentation of a standard merle dog, which is described as those with **base coat color oligo(dT) lengths from 78 to 86 base pairs**. You can specifically describe the genetics of a dog's coat type by using words alone.

The lowest level of abstraction would be Sparky's actual DNA. However, we normally do not paste genetic material into technical documents; so to be precise, we must substitute something else, such as a photograph of his coat and a standard description of his genetics, which in this case is M⁷⁸m. That is why a photograph of Sparky the merle-coated dog has been placed at the narrow end of the abstraction funnel—because that is the most concrete way available to refer to it in a document.

Let's look at another example. Imagine that you need an explanation of a 33K, one-watt carbon resistor and you read that it's an **electrical device**. Not helpful. "An **electrical device**" could be anything that is electrical, from a fluorescent light, to a computer keyboard, to your phone's SIM card (see [Figure 1.2](#)). Of course, mixed in with all those electrical devices is the electrical device also referred to as a 33-kilohm, 1-watt carbon resistor, but it would be difficult to hone in on it with all the other noise of **electrical devices**.

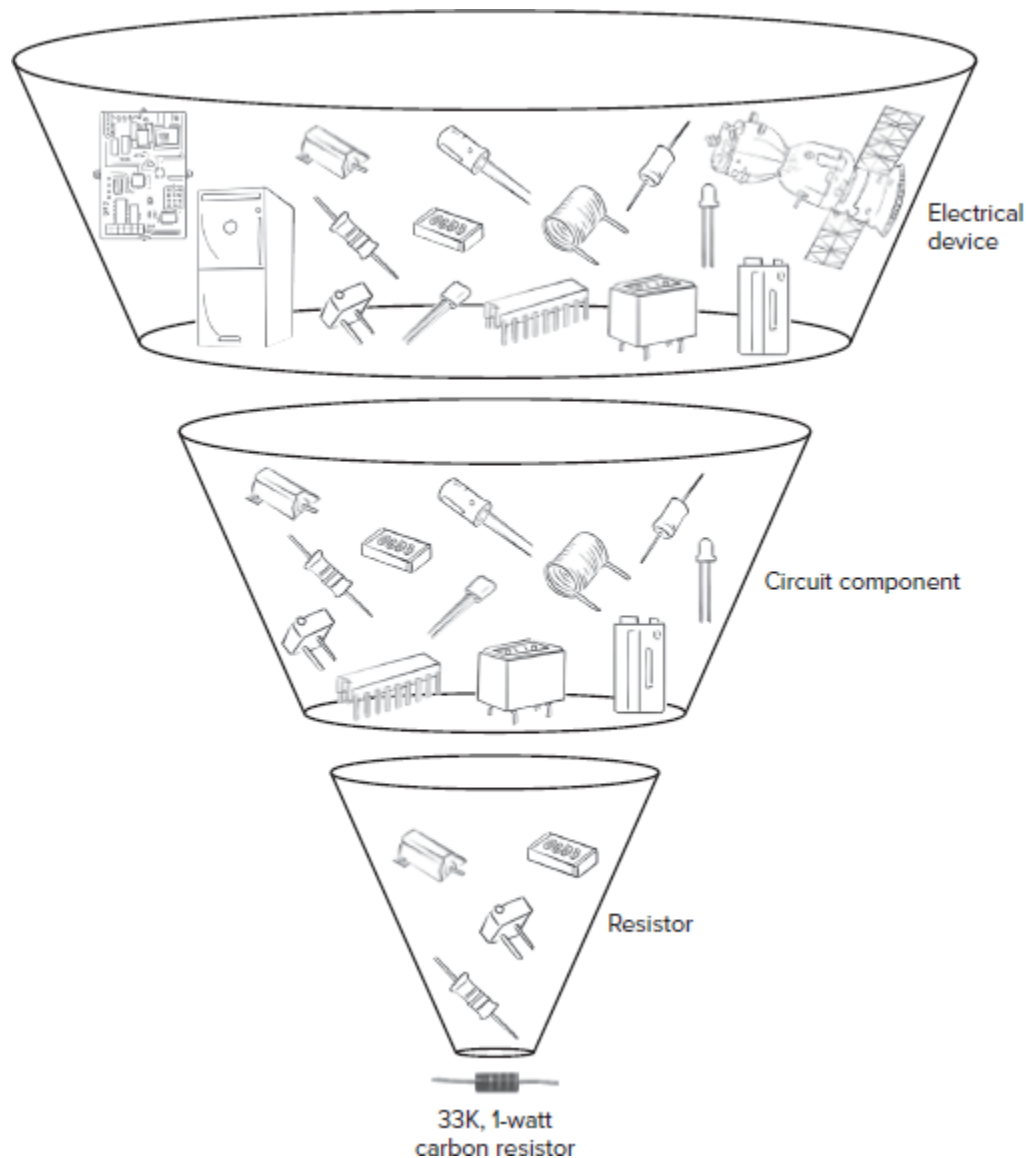


Figure 1.2

A funnel of abstraction. The high level of abstraction at the top of the funnel allows the audience to consider too many concepts or items. Abstraction is reduced through the use of non-abstract language. At the bottom of the funnel is the lowest level of abstraction, where only one concept or item is described by the language.

You read on that it is a **circuit component**. That's certainly more precise, but it could mean any electrical device in the circuit, such as a capacitor, inductor, diode, or, in this case, a resistor. As the explanation becomes more precise, it includes the word **resistor**, which means any circuit component that impedes the flow of current.

The next most concrete way of describing the resistor by using words alone is to precisely label it—as we do when we give our newborn children distinctive names and government taxpayer identification numbers. In this case, the resistor is labeled a **33-kilohm 1-watt carbon resistor**.

As in the case of Sparky, the lowest level of abstraction would be the resistor itself, which we cannot paste into a technical document. We substitute the resistor with a photograph, which is why a photograph of the resistor has been placed at the bottom of the funnel. This precision, of course, assumes that the audience knows what a resistor looks like. If not, the photograph can be pretty abstract, and for that matter, so can the actual resistor.

In fact, when we write technical documents for our audience, we need to carefully consider our audience's familiarity with our topic and carefully calibrate the level of abstraction to the lowest level that will help them understand given their current knowledge of the topic. Technical writing is **not** about how well the writer understands the topic; technical writing is about **how well the writer can explain the topic to someone else**.

As one moves down the funnel of abstraction, the symbols become more precise and less vague. In effect, this reduced abstraction gives the reader less freedom to interpret meaning as they want and more like the writer intended. In creative writing, this lack of flexibility is probably not good; in technical writing, such lack of flexibility is good. The goal of technical writing is to eliminate abstraction. Simply speaking, successful technical writing restricts the reader's freedom of interpretation so that only one meaning can be concluded—the meaning intended by the writer.

What sets technical writing apart, then, is its precision. How it achieves this precision is, in fact, the art and craft of technical writing—an activity that involves definition and description; data and analysis; photographs, diagrams, and charts; and often specialized language. The goal of technical writing, then, is **not** to be creative, or interesting, or to employ rich imagery or powerful metaphors. The goal of technical writing, first and foremost, is to communicate complex information clearly and precisely for the audience and the purpose at hand. Clarity and precision are the overriding goals for any technical writer, and understanding the audience, purpose, and context is the primary consideration for achieving those goals.

Audience, Purpose, and Context

The measure of how well a technical writer has written something depends on three things:

1. how well the **reader** understands, precisely, the writer's intended meaning,
2. how well that understanding fulfills the intended **purpose**, and
3. how well the communication fits into the broader situational **context**.

Consequently, technical writing must be geared directly to its Page 5 audience, purpose, and context. Remember, there is always a specific requirement for technical writing: a scientist must write a grant proposal, a programmer must document software prior to distribution, a lab supervisor must justify new equipment with a feasibility report, and an industrial engineer must convince a business manager to change work practices for the long-term health of employees.

Relating audience, purpose, and context requires that the writer consider the potential reader's knowledge, skill level, and specialization. The writer must anticipate fully responding to the needs of the reader in terms of the requirements of the situation. Let us take a closer look at each of the three separate yet interrelated components.

Audience

Technical writing is only effective if the **audience** understands the communication. To ensure that our writing is understood by the audience, we must contemplate and often even research our audience. Some things to consider include the following questions.

- How much does my audience know about the topic?
- How much background information about the topic might my audience need to understand my communication?
- How motivated is my audience to read my communication?

- What level of detail does my audience need to understand or to accept my communication?
- What are my audience's primary, secondary, and tertiary concerns?
- What concepts in my message would be better and more quickly understood with supporting visuals?
- How much documentation will my audience want and/or require?
- What format does my audience expect?
- How much time will my audience have to read my communication?
- Where will my audience use my communication and how does that affect how I should structure and format my communication?
- Will my audience be resistant to my communication? (the idea, message, topic, etc.)
- Will my audience be resistant to the communication coming from me personally?

Consider the industrial engineer who must convince a business manager to change work practices for the long-term health of employees. How will they answer these questions about the audience? For example, if there have been employees missing work because of carpal tunnel injuries, the business manager might be very receptive to a plan for reducing absenteeism. But if the same manager is struggling with immediate warranty costs caused by quality issues in an assembly area, long-term concerns like carpal tunnel may seem less urgent in the moment.

Purpose

Technical writing must fit with the **purpose** of communication, which is why you are communicating in the first place. There are several things to consider, starting with what we want our audience to do after they have read our communication. Do we want them to

- know more facts?
- better understand how something works?
- know the status of a large project?

- provide money for a project?
- assemble something?
- accept a recommendation?
- change their position on an issue?
- make a decision?
- some combination of things?

Once we have determined what we want or need our audience to do after they have read our communication, we must determine how to get our desired result.

- Do I need to inform? If so, to what level of comprehension?
- Do I need to persuade? That is, do I need to change someone's position on a topic, their belief about something, or their behavior or actions? If so, is there information that I should emphasize or highlight?
- Do I need to build goodwill? Sometimes this is a necessary requirement by itself, but other times we incorporate this into our communication which informs and/or persuades.

Again, consider the industrial engineer who must convince a business manager to change work practices for the long-term health of employees. How will they answer these questions about purpose? For example, they might need to talk with machine operators to get more information, but that means overcoming a possible mistrust of management. Once the industrial engineer has the necessary facts, they must think about how to request money from their manager. Is goodwill already established? Can they move to considering how to persuade?

Context

In addition to relating to a specific audience and for a specific purpose, technical writing must fit within the broader situational context. Unfortunately, much of the broader context is outside our control as communicators, yet we must account for it nonetheless. Context includes things like history, language, geography, politics, culture, economics—

basically anything in the world that affects how our communication is received by our audience. It could be something as specific as personality: maybe someone in management doesn't really like you, and you don't know why, but you know that they immediately respond negatively to anything you suggest. It could be something like company culture: perhaps you tend to write informally and use contractions, slang, and emoticons for emphasis ☺, but the company culture is formal. It could be something historical, like a chemical accident that killed an employee over 20 years ago—it is not a topic that is actively discussed, but it lingers deep in peoples' minds. Communication does not exist in a vacuum.

Let us go back to the industrial engineer. What contextual factors might affect the engineer's ability to convince their business manager to change work practices? Perhaps a recent OSHA law was passed that mandates regulations be implemented by a certain date or a company will be fined. Maybe a competitor in the industry had a lost-time accident occur and suddenly other companies take notice. What if there is a tight job market and employees care about safety and use it to decide where to go to work?

In technical communication, the audience and purpose are almost always well defined in advance—often by the supervisor, customer, or teacher. This is because technical writing is usually commissioned by someone else for a specific purpose and audience. Normally, technical communication aims to share objective information with an interested, educated audience. Technical communication is simply communication on technical subjects that shares information in a precise way.

However, the broader contextual situation, or context, is usually outside a technical writer's control. Remember our industrial engineer who wants to convince a business manager to change work practices for the long-term health of employees? Well, the engineer did their due diligence on researching the frequency and severity of injuries, the biomechanics of how the injuries happen, and the audience's concerns. They wrote a precise, meticulous, eloquent feasibility report. However, behind closed doors, upper management had discussed the closing down of that assembly line in the next three to four years. Despite the impressive quality of the engineer's report, it was shelved due to a factor outside the engineer's control. The engineer didn't have all of the necessary context needed to write an informed proposal.

Genre Decisions

Technical writing must work within the parameters of genre expectations in addition to relating specifically to the audience, purpose, and context. Genres are categories of things that have a similar subject matter, form, and style. You might not have heard of the term **genre** before, but it's a concept that you likely use every day.

- You are going for a run, so you listen to high-energy rock songs
- You are going for a three-hour drive, so you play podcasts on scientific discoveries
- You have had a bad day at work, so you kick back with an action movie

We all choose categories such as these, called **genres**, because we know what they are (generally speaking) going to be about, how they will be organized, and what they will sound, read, and feel like.

When we think about categories, or genres, that are understood the same way by people from all over the world, we can draw an analogy to dogs. Dogs, just like communication, have been around for thousands of years, and both have evolved with the right breed/tool identified for specific needs and jobs. When you need a mouser, you look for a terrier. When you need to herd livestock, you get a herding dog. When you need to retrieve water fowl, you choose a sporting dog. From the Chihuahua to the Collie to the Great Dane, you recognize all of them as dogs, but you also know that they do different things and have different purposes (see [Figure 1.3](#)). The same is true in communication. When you need to convey information to a certain kind of audience in a particular way, you choose the appropriate formulation. When you need to get funding for a project, you write a proposal. When you need to explain how to put that new basketball hoop together, you write instructions. When you need to get an internship or a job, you write a resumé. Each of these technical writing genres has a specific purpose or function.



Figure 1.3

Much like different dog breeds have specialized purposes (to hunt, to herd, to protect, etc.), communication genres have specialized purposes (to inform, to persuade, to build goodwill, etc.).

belchonock/123RF

However, once you've chosen which technical writing genre is most appropriate for your audience, purpose, and context, you still have many decisions to make. You must appropriately adjust the amount of content, level of detail, structure, format, tone, style, and length of each element of that genre in your particular document. Consider the farmer who needs a herding dog—what other factors do they need to consider? Well, to start, what type of animal will the dog be herding? Under what climate conditions? Will it also be a family dog? How experienced is the farmer with training dogs? How protective should the dog be? Does it matter how loud the dog is? How much it sheds? How many years does the dog typically work? There may be other factors to consider, too, but the farmer shouldn't randomly choose one of the approximately 70 breeds in the herding group. The farmer should carefully consider as many contextual factors as they can and make strategic determinations. Similarly, as a technical writer, to communicate effectively, you will also need to make decisions about elements of the genre you have chosen.

Characteristics of Technical Communication

So, finally, to answer the question posed at the beginning of this chapter—What is Technical Writing?—technical writing can be characterized as follows:

- **Technical communication is non-abstract and precise.** The goal of technical communication is to eliminate abstraction and communicate complex information clearly and precisely to the audience at hand.
- **Technical communication considers audience, purpose, and context.** Technical communication must not only relate specifically to the audience who will be using the communication, but also relate specifically to the purpose and context at hand.
- **Technical writing deals with technical information.** Technical writing is designed to deal with technical subjects.
- **Technical writing relies heavily on visuals.** As shown in the abstraction funnel of [Figures 1.1](#) and [1.2](#), a photograph or diagram may be the least abstract, and therefore the most precise, way to communicate something to an audience. Visuals—whether they are equations, photographs, tables, graphs, or drawings—are powerful tools for presenting a large amount of information effectively and efficiently. However, they almost always require interpretation or explanation in the text of the report, especially when the audience may not be familiar with their content.
- **Technical writing uses numerical data to precisely describe quantity and direction.** In many cases, mathematical equations and formulas, experimental results, and financial analyses provide the real substance of a technical report.
- **Technical writing is accurate and well documented.** Generalized, unsupported assertions have no place in technical writing. Conclusions, recommendations, and judgments are always based on clearly presented evidence or established expertise, and technical writing is always technically correct.
- **Technical writing is stylistically and mechanically correct.** This kind of correctness is far more than simply a matter of personal and corporate pride. Style and mechanics (spelling, punctuation, and grammar) often go to the heart of the author's credibility. In other words, right or wrong, fair or unfair, if you write shoddily, the reader may well perceive you as careless and your organization as a collection of people who do not care about details. Whether or not this perception is true is irrelevant. Where technical writing is concerned, perception is often more important than reality.

Notes

- [1.](#) The idea of a “river from nowhere” came from the Steve Winwood song “The Finer Things,” which includes a reference to time as a “river rolling into nowhere.”

==

CHAPTER TWO

Ethical Considerations



*Be the person your dog thinks you are.*¹

Why is ethics the focus of a lead chapter in this book? It is here because of increasing emphasis being placed on ethics in technical writing, as well as in engineering and the sciences as a whole. Technology and the communication of technical ideas represent powerful tools for promoting good or evil in society—and ethics is an important backdrop for all the material provided in the chapters that follow.

Ethical considerations are not separate from the work that engineers and scientists do, but are entwined in it. For example, the American Chemical Society has a standing Ethics Committee; one of their goals is to “increase the members’ recognition of ethics in their careers and its impact on society.”² In another example, many undergraduate engineering programs are accredited through ABET, which requires program achievement of “an ability to recognize ethical and professional situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.”³ Boston University published “The Ethics of Scientific Research—A Guidebook for Course Development” (Stern and Elliott), which includes the objectives: “a. Keep good notebooks” and “d. Record and report your work accurately” 1997.⁴ Companies strive to be named on the list of “World’s Most Ethical Companies”⁵ and publish their own reports about what ethics means to them (*Ethics in Scientific Research—An Examination of Ethical Principles and Emerging Topics*), Rand Corporation 2019.⁶ Ethical considerations are an inescapable part of our broader human context.

Humanity has been struggling for a long time over exactly what constitutes good and evil, and ethical and unethical behavior. It is unlikely that this chapter in a technical writing book will provide the definitive answer that will end this timeless struggle. The hope, however, is that this chapter will provide an awareness of the ethical dimensions of technical writing and some sensitivity to the real issues that exist within these dimensions. Additionally, this chapter will discuss several key areas of ethics in technical writing, including ethical writing, academic and professional dishonesty, copyright infringement, image alteration, and personal online presence.

What Is *Ethics* in Technical Writing?

The first thing we must develop is a working definition of ethics in technical writing. What exactly is ethics for a technical writer? For the purposes of this book, the answer is relatively simple. Collectively, **ethics** is a set of rules and standards for using communication skills and resources with the intention of doing good. Ethical behavior for technical writers, then, involves their moral duty and obligation to apply the power of technical communications to the purpose of doing worthy things. But what do **good** and **worthy** mean?

You probably already know what you think they mean, but not everyone would agree with you. The global technical writing community is composed of many disparate cultures. To some extent, concepts of good and bad, and ethical and unethical, are culturally relative. However, that does not mean we can just ignore the issue or take the viewpoint that our ethics are better than others' ethics. We must approach the matter realistically in the context of the diverse world in which we function. That is what this chapter attempts to do.

Begin by considering the following situation:

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Three technical writers write an irreverent, humorous technical writing book using, as its examples, reports on fictitious technologies based on real theory. Their laboratory report chapter deals with a fictional, high-power transmitting tube called the *SuperXLTube*, their research report chapter describes a quantum computing processor known as the *QuantumCPU*, and their process description chapter includes a figure skating jump called the *QuadFINKEL*.

For the sake of argument, assume that the book is a big hit, the royalties pour in, and the authors deposit the checks and live the good life.

Now consider this situation:

Three technical writers write a professional proposal from a fake corporation. They invite individual investment of funds in developing and marketing the fake company's fictitious product line of high-technology devices. They cite fabricated test data and technical research information demonstrating the efficacy of these products, and they develop altered photographs of the product line. In fact, they develop a professional, full-color brochure complete with tables of data, charts, diagrams, and photographs—and promise huge returns on investment. They mail this brochure to thousands of nursing home residents across the country. When many of these people send the technical writers their life savings, the authors deposit the checks and live the good life.

In both situations, the technical writers have used their knowledge and resources to develop published materials around fictitious technologies. In both situations, they have embellished the capabilities of these technologies with professionally developed photographs, diagrams, charts, and data. In both situations, they make money from the published product.

In ethical terms, how do these situations differ? That is an important question; the difference gets at what makes the behavior of technical writers ethical or unethical. Before we focus on this difference, however, we need to look briefly at the kinds of ethical constructs that are traditionally used in technical writing and the implications of these constructs for technical writers:

- **Technical writers must be accurate in their work.** Either technical writers must be precisely correct at all times, or they are unethical.
- **Technical writers must be honest in their work.** Technical writers who write untruths are unethical.
- **Technical writers must always honor their obligations.** Technical writers who do not produce the documents and other materials they are responsible for producing within the agreed-upon time frame are unethical.
- **Technical writers must not substitute speculation for fact.** Technical writers who do not clearly separate opinion from accepted truth are unethical.
- **Technical writers must not hide truth with ambiguity.** Technical writers who either omit, deemphasize, or misrepresent facts that would

be contrary to the theses of their reports are unethical.

- **Technical writers must not use the ideas of others without giving proper credit.** Technical writers who fail to document the sources of all nonoriginal ideas, except for common knowledge, are unethical.
- **Technical writers must not violate copyright laws.** Technical writers who fail to document the use of copyrighted materials when used with permission or under “fair use” are unethical. Additionally, technical writers who use any copyrighted materials without permission when these uses are not covered by fair use are unethical whether the materials are documented or not.
- **Technical writers must not lie with statistics.** Technical writers who manipulate data or graphical representations of data, use inappropriate or improper statistical tests, or employ loaded statistical samples are unethical.
- **Technical writers must not inject personal bias into their reports.** Technical writers who are less than objective in everything they write are unethical.

What a list! These rules are clearly well intentioned, and they Page 11
provide generally useful guidelines for writers. But there is a real problem with rules like these: they miss the mark where ethics are concerned. For example, being accurate and precise in technical writing is not the real ethical issue here. Intending to **be** good and to **do** good are the real issues. Of course, it is true that in technical writing “being good and doing good” usually mean being accurate and precise, but not always.

In the first situation, as the authors in question, we obviously do not believe that being accurate and precise is necessary to achieving our pedagogical goals for this book. In this case, we believe that just the opposite is true. The freedom to invent obviously fictitious technical writing examples supports the book’s educational objectives. We are being inaccurate, but we are not being deceptive. For example, in the process description chapter, an exercise deals with the QuadFINKEl figure skating jump, an athletic achievement so technically demanding that anyone landing it always receives the gold medal. Do you suppose anyone really believes that a QuadFINKEl exists? In fact, most readers probably have

serious doubts as to whether the authors can even stand up on figure skates—and these doubts would be well-founded.

We would be deceptive, and therefore unethical, only if we asked that you, the reader, truly believe these examples. Our clear intent is not to mislead; rather, it is to use the fictitious and humorous examples to help students learn—and we believe learning is still considered a “good” thing to do. However, we wouldn’t do well with that list of rules, would we? These rules do not get at ethics; rather, they get at behavior that is often correlated with ethics.

In the second example, the rogues trying to rip off nursing home residents are obviously not the kind of people you would want managing your money. They clearly intend to use their technical communications skills and resources to deceive. To make matters worse, they are also targeting a vulnerable group of people for this scam—the elderly, who are least able to afford this kind of attack. So unlike in the first situation, the tricksters are falsifying information not to do good, but to do bad. That is why the technical writers in the second example are absolutely unethical (not to mention being guilty of violating numerous local, state, and federal laws).

[Figure 2.1](#) presents a model for ethics in technical writing. Clearly, it does not represent the big breakthrough in the philosophy of ethics for which humanity has been waiting so long. But it does help make the point and clarify the issue. The rules may contribute to ethics and our understanding of what is involved in ethics, but they are not what constitute ethics.



Figure 2.1

Ethics model for technical writing.

Finally, ethics in technical writing, to some extent, must be Page 12 relative to societal values. If, on the one hand, you use technical communications to deceive your reader with the goal of doing bad things (defining bad by the standards of your society and perhaps by the standards of civilization as a whole), you are unethical. If, on the other hand, you use technical communications with the intention of doing good things (defining good in the same manner), you are ethical.

Academic and Professional Dishonesty and Copyright Infringement

When you submit work in a class—whether it be a paper, test, quiz, homework assignment, or something else—you are saying that the work you put into that submission was your own and that you followed the instructor’s requirements. Using someone else’s work without acknowledging the source constitutes **academic dishonesty** in a school setting and **professional dishonesty** in industry. Academic and professional dishonesty encompasses multiple types of stealing, although the types might have different labels:

- **Using dishonestly obtained information.** “Dishonestly obtained” could be anything from wholesale copying of someone else’s work (from a friend, a student sitting in the chair next to you, an online study site, a co-worker, or a competitor); buying work from a student, a writer contracted to write for you, or from an online source; working with others on an individual project; or even sneaking a disallowed peak at your own notes during an exam. If you use these methods to complete your own work, you are being dishonest.
- **Helping others get dishonestly obtained information.** Even if you are not the student or employee actually submitting the work, you can still be academically or professionally dishonest. This type of dishonesty includes giving or selling your own work—whether it be homework, exam questions, data, or papers—and sharing course materials developed by your instructors with an online study site. If you are helping others to be dishonest, you are being dishonest.
- **Failing to participate.** If you fail to participate in an assigned group project and make the other group members do all of the work, you are being dishonest.
- **Making up data and evidence.** If you fabricate data instead of conducting an experiment and recording the results, or fabricate evidence instead of researching a topic, you are being dishonest.
- **Gaining academic and professional advantage dishonestly.** If you destroy the work of others so they cannot complete the assignment (or complete it on time) or if you try to offer money, gifts, or services to others in order to gain an academic or professional advantage for yourself, you are being dishonest.
- **Stealing an idea or an expression of that idea (plagiarism).** If you steal a person’s idea, or their expression of an idea—whether in words, images, sounds, photographs, etc.—and then represent it as your own, you are a plagiarist. (For stealing the expression of the idea, you could also be guilty of a copyright violation.)

All of the acts listed above aim to deceive others into thinking the work submitted by the student or professional is original work.

The problem of academic and professional dishonesty has increased dramatically because of the internet and technology. Many years ago,

people who wanted to get information dishonestly or copy the works of others for an assignment had to go to the library and look through a limited collection of materials likely known to their teachers or supervisors. Students or professionals might even have had to use a copy machine (sometimes involving many dollars' worth of quarters) or endure the drudgery of transcribing the material manually. They would have to physically share papers or gain physical access to data or exams (perhaps even breaking into an office or file cabinet of an instructor or coworker!). When someone copied another's work, the act itself involved time and effort, and included the ever-present risk of detection. In many cases, plagiarism actually cost more in time and money than it was worth.

Those days are over. Now the internet and its multiple search engines have extraordinary search capabilities that provide people with quick access to literally billions of sources and documents. With a press of a touch pad, the drag of a cursor, and a few simple key strokes, any image, graph, or line of text can be copied and pasted into a plagiarist's document in privacy. For some, copying material from the internet may seem less wrong than copying pages from a book. Some see information available on the internet as public property that is theirs for the taking—not unlike old furniture a neighbor has put out at the curb on trash collection day. The ephemeral nature of files on the internet, the lack of material tangibility, the speed and impunity with which this material can be located and reproduced, and the ease with which it can be used in private are all factors that contribute to the perception that somehow stealing a file from the internet is less wrong than copying from a hardcover journal or book.

The risk of detection for the copy-and-paste type of plagiarism is Page 13 increasing as academic institutions employ the use of plagiarism detection software. Such software compares submissions against previously submitted work and content available on the internet. It produces a report showing which portions, if any, of the submissions are copied and the sources from where that material was copied. This does not mean the end of academic dishonesty and plagiarism, however. Plagiarists and cheaters can now hire essay writing services to write custom work that students can pretend is their own and submit for classes. The increase in essay writing service sites for students is so plentiful that there are now reviews to help students choose which service to hire.² It shouldn't need to be said, but if

you hire one of these sites to complete work for you, you are a plagiarist. By definition, this makes you dishonest.

For some struggling with the ethics of plagiarism, the fact that “everyone” seems to be doing it somehow makes doing it more acceptable. That is a logical fallacy, more specifically a bandwagon fallacy,⁸ and just because it seems like “everyone” is doing an unethical thing, does not magically transform an unethical action into an ethical action. It is still wrong. Fortunately, most students and professionals are not ethically challenged.

Are there implications beyond the classroom? Yes. More and more employers, institutions, and external agencies are requesting student files prior to offering full-time employment because they only want to hire ethical employees. For example, law schools and government or military agencies will often request that students sign a waiver or release of information to ask for information on a student’s disciplinary record at their university, even if that information wasn’t provided on an official transcript.

Academic and professional dishonesty is unethical and has serious consequences, but it is not illegal. Copyright infringement **is** illegal. **Copyright** is a type of intellectual property protection that provides federal and international protection for the expression of ideas, but not for the ideas themselves. When someone disregards, or infringes on, a creator’s copyrights, that someone is guilty of **copyright infringement**, an act that has legal consequences. Copyright laws protect “original works of authorship”⁹ by giving their creators a “set of rights to protect their work” and “to protect the creator from OTHERS who might be doing the same without permission, attribution, or payment”¹⁰:

- The right to reproduce the work in copies
- The right to create derivatives of the work
- The right to perform the work publicly and digitally
- The right to display the work publicly
- The right to the work digitally
- The right to let others exercise these rights

Before electronic publishing and the internet, the expression of ideas basically took the form of images printed on a page, whether those images

were text, photographs, or illustrations. In other words, the ideas were encoded into traditional media and were protected in that form. In many cases, it was impossible to separate an idea from its expression (as with, say, a photograph)—and publishers exercised substantial control over what the expression looked like, what form it took, and where and how it was distributed.

In effect, the limitations of the medium provided reasonably good protection for the ideas it carried. You could copy or duplicate a printed page, an illustration, or a photograph—but you had to have an original or good-quality copy to do so. After one or two generations of “analog” duplication, the copy would become so degraded that it would not be worth having. And, of course, copying usually was not free. These inherent limitations helped to reduce the widespread pirating of copyrighted material that is much more common today.

Now when an electronic document is placed on a web server, it is Page 14 likely accessible to anyone anywhere in the world. Viewing it requires that it be downloaded and copied to the viewing computer’s memory—and probably, as well, to the browser’s cache on the hard drive of that computer. Anyone viewing a web document has no need to make a local copy because, given the nature of the technology and the way web browsers work, that copy already exists. One needs only to specifically name and save it as a file to permanently retain an exact duplicate of the original. The hypertext, graphics, and sound files are just as good, quality-wise, as those on the original server from which they were taken.

Consequently, copyrighted material on the web is often available to anyone who wants it, and copies can spread rapidly with little or no degradation and virtually no control. It is imperative that you consider any technical report you place on a web server to be public material in a global information infrastructure; expect that people will look at, copy, and use it freely. Of course, you could encrypt the documents, password-protect them, or even place the materials on a secure server. But these inherent limitations to access may undermine the purpose of having the materials online in the first place. Moreover, no protection scheme is totally hackproof.

Other possibilities for protecting electronic documents exist, including digital watermarks and various licensing schemes. But when all is said and done, the internet is still a global community—and the rules and values you embrace may not apply everywhere. For copyright protections to be

effective, they must be international and enforced for all users who have access to the information. That may be where we are headed, but we are not there yet.¹¹ Consider carefully what you put online and why you are putting it there, before you actually do so.

What does all of this mean for you as a user of copyrighted material? It means that you cannot share movies or songs—or even large portions of them—without the copyright owner’s permission. Copyright is the reason that fans of the breakthrough TV show *Northern Exposure* cannot view it on a streaming platform; “the show’s creators didn’t want to compromise by replacing the music used at the time with generic music (like you’ll hear in streaming versions of *Alias* or *Beverly Hills 90210*, for example),” but using the original music means paying fees to the songs’ copyright holders and “no one wants to pay artists for rights to use their songs in the streaming version.”¹² Copyright means that you cannot scan, duplicate, copy, and then use an image (photograph, drawing, etc.) without the copyright holder’s permission or without citing it, and copyright is why you can’t use a corporate logo in your own works without the company’s permission. Copyright is why you cannot make copies of or share games, apps, or software. Copyright is also the reason that instructors cannot save you money by just posting copies of course reading materials, whether original electronic publications or scans of original print publication, for you to access for free.

Copyright laws protect the creator’s rights and the consequences of infringing on those rights can be steep. The original Napster, a peer-to-peer file sharing network, allowed people to share MP3 files, meaning that one person would purchase a song file and then share it with a large number of people, who could download copyrighted music for free. Recording artists and their labels were upset at the free distribution of their work by Napster, who was earning money from the platform that allowed the free distribution. Napster was taken to court by several recording companies and found guilty of contributing to copyright infringement.¹³ Napster “was forced to shut down the site after making a public apology and paying \$26m in damages.”¹⁴ Music sampling, which makes great music, is subject to copyright, and musicians who sample without permission can be found guilty of copyright infringement.

Another intellectual property protection that you will likely encounter someday, if you have not already, is a **nondisclosure agreement**. A

nondisclosure agreement is a legally binding contract that identifies confidential information that the parties involved in the agreement will share with each other, but which they may not share with others. Everyday examples of unilateral nondisclosure agreements for which you do not need to sign a contract but you still participate in are professor–college student, doctor–patient, priest–penitent, banker–client, and attorney–client. Unilateral means that while two parties are involved in the agreement, one person discloses the information (college student, patient, penitent, client) and the other party maintains confidentiality (professor, doctor, priest, banker, attorney). There are also mutual agreements in which all parties are expected to maintain confidentiality about the information identified in the agreement. If you intern or work for a company that wants to protect its trade secrets, you may very well be asked to sign a nondisclosure agreement. If you are asked to sign a nondisclosure agreement, make sure that you fully understand why you’re being asked to sign one, the details of the agreement (what you are agreeing to), and the consequences of breaking the agreement. Have your questions or concerns addressed before you sign it.

The bottom line is that in technical writing, as in all intellectual Page 15 pursuits, stealing ideas or the expression of ideas from others is not ethical and is absolutely unacceptable. It does not matter that “everyone” seems to be doing it, or that online resources make locating and using such material a trivial task. If you use the ideas or the expression of ideas (words, images, sounds, photographs, etc.) of others without acknowledging the source, you are stealing from the author, you are stealing from those students who are doing their own work, and you are stealing from yourself by turning what could be a valuable educational experience into an exercise in thievery. Also, if you commit plagiarism, especially in the professional world, you might find yourself on the wrong side of copyright, patent, trade secret, or trademark protections. That could subject you and your employer to severe criminal and civil sanctions, which, at the very least, would not be a career enhancement.

Learn to do it right the first time while in school. Document all sources with citations or notes at the point in the paper where the materials are used, and include a complete list of sources, usually at the end of the paper. See [Chapter 16](#) for a more complete discussion of documentation and documentation requirements.

Image Alteration and Ethics

Without a doubt, digital image processing lends itself to abuse and misuse. The relative ease, availability, and effectiveness of hardware and software, combined with the inherent credibility of photographic images, provide a powerful tool for misrepresenting reality. As previously discussed, modern technology provides almost anyone with the power to steal or plagiarize intellectual property in digital form. Additionally, this same technology provides technical writers with the means to modify material with the intent of misleading the reader. Concerning the ethics of image alteration, the distinction between the intent to do good or to deceive applies.

This is especially true in the case of altered photographs. Technical writing always must be accurate and correct, and serious ethical concerns exist regarding accuracy and correctness when images are edited. [Chapter 17](#) provides a discussion of image alteration and makes the point that modifying images is ethical or even desirable in technical writing when these modifications serve to clarify or complement—but not when they are used to misrepresent or deceive. If the image alteration has fundamentally changed the information it represents for the purpose at hand, that fact must be acknowledged. To avoid any ethical questions when you are using such altered images, always indicate to your reader that the images are altered. The easiest way is to acknowledge in the image’s title and caption that it is, in fact, an edited image and then to briefly describe the nature and purpose of the alteration.

Social Media Presence

Using communication skills and resources with the intention of doing good extends to social media presence. Students and employees are de facto representatives of their schools and places of work, and their public communications reflect on their school or employer. This is why news of employees or students behaving rudely or offensively is of concern to organizations, institutions, and businesses.

At the writing of this book, employment in the United States Page 16
(with the exception of the state of Montana) is “at will,” meaning

that employers can—except for firing someone because of their gender, race, religion, or other protected status—terminate employees without reason. There are some exceptions. For example, most states have a policy exception that prevents employers from retaliating against an employee who follows public policy or refuses to do something that would go against public policy. There are also implied contract exceptions, implied-in-law exceptions, and statutory exceptions.¹⁵ Personal communication outside of the workplace can have an effect at work. For example,

- **Personal TikTok.** An oncology nurse bragged that, in times of COVID-19, they don't wear a mask unless they're at work.¹⁶ They were put on administrative leave.
- **Personal Tweet.** A communication director lost their job after making a racist tweet before boarding a plane to join their family on vacation.¹⁷
- **Personal Facebook post.** A daycare worker lost her job after complaining about how much she hates her job and being around kids.¹⁸
- **Personal Tweet on corporate account.** A contractor for Chrysler lost their job after posting an explicit tweet from the official Chrysler account.¹⁹ Their firm also lost their account with Chrysler.
- **Personal Instagram post.** A marketing executive made a celebratory post about a new client—against company policy and the executive's contract—that only a small number of followers would see (~200).²⁰ They lost their job.

Things that you say online can cost you at work, and companies are starting to have policies to clarify what is and is not acceptable on social media in order to protect their organizations' brands.

It isn't just after you have gotten a job that you should be careful about what and how you communicate on social media. Employers check applicants' social media accounts, and employers use online search engines to learn more about the people applying for jobs at their organizations.²¹ Be proactive and build a professional and ethical online persona. If you aren't

already, be cognizant of how you use and conduct yourself on social media. Some things to avoid posting on social media include:

- Complaints about your job, supervisor, employer, customers, clients, classes, instructors, school
- Offensive content (racist, sexist, ageist, etc.)
- Profanity
- Illegal or sexual content
- Partying
- Information that reveals or announces things prematurely or inappropriately
- Information that isn't yours to announce

In summary, ethical considerations are entwined in the work that you do as a representative of your discipline and organization. When you communicate, your moral duty and obligation is to apply the power of technical communication to the purpose of doing good and worthy things. Be good and do good.

Plagiarism Exercise

Situation 1

You are enrolled in a technical writing course and have been assigned the task of writing a process description on the operation of a four-stroke, internal combustion engine. A close friend wrote a similar description for another technical writing course at a different university last year and earned an A. They offer you their paper as a gift, along with the original word computer file. They tell you to change the name, print it again, and turn it in. Since the original author has given you the paper and you now own it, is it plagiarism for you to do as they ask? After all, they argue, you are not stealing anything—the paper is now yours!

Situation 2

What if you actually wrote the paper that your close friend turned in under their name a year ago and for which they received an A? Now you are in a technical writing class at another university and have been given the same assignment. You still have the computer file for their paper on your hard drive. Since you were the original author, is it plagiarism if you change their name to yours and turn in the paper for credit? If so, do you even need their permission?

Image Alteration Exercise

The author and family wanted all three sons to be pictured together in front of a campus landmark on graduation day so that they could complete the family's scrapbook of a crazy year of COVID-19. The family members were the only people in the area and had been quarantined together for two weeks.

For the following two hypothetical situations, is the use of the altered image ethical? Why or why not? Explain your reasoning.

Situation 1

The author thought the image might make a great example of how people should deport themselves on campus to comply with the university's guidelines for keeping students, faculty, and staff safe and healthy. The author shows [Figure 2.2\(a\)](#) to students in the department's orientation class as an example of how students used to stand before COVID-19 and how quarantined groups can stand outdoors, and [Figure 2.2\(b\)](#) as an example of how students should stand post-COVID-19 (or until the vaccines effectively prevent the spread of COVID-19).



(a)



(b)

Figure 2.2(a–b)

2.2(a) is the original photo taken by the author. 2.2(b) is an altered version of the original photograph.

Leslie Potter

Situation 2

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The author is meeting with potential students and parents about their department and the university. As part of the discussion, the parents express concern about their student's safety on campus and the university's COVID-19 guidelines, so the author shows them [Figure 2.2\(b\)](#) as evidence that all students (and their families) are following COVID-19 guidelines.

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CHAPTER THREE

Note-taking



Imagine that your job is to understand and manipulate hundreds of information sources efficiently and effectively. The information includes subcontractors, their locations, products, logistics of getting products to the main production facility, schedules, contract limitations, impending weather constraints (blizzard or hurricane, anyone?), and more. Could that ever happen? Absolutely! Is it intimidating? It does not have to be, and it will not be with a good understanding of how to herd the information.

What Is Note-taking?

Note-taking is the act of recording information—whether writing, typing, voice-recording, or something else—from a variety of sources into a single location, usually a text or document. In general, everyone understands the idea of what it means to take notes because everyone has done it . . . maybe in a class, at the doctor’s office, or when your parents are reminding you of all the things you need to do before you move to college. Sometimes we rely on mental note-taking. In short-term or short-list situations, this can work well; we are likely to remember and act as needed. Oftentimes in our technical jobs, though, we need to document our notes. There are several reasons for this.

As individuals, we

1. must be able to remember a lot of information efficiently and
2. might need to access the information over long periods of time.

As part of a team, we

1. want to contribute to the institutional knowledge base and
2. must be sure that our actions are ethical and legal, including both the notes we take and do not take.

To be effective and efficient note-takers, we must understand why and have a strategic plan for taking them. Let us return to the opening scenario of this chapter, the one where we are trying to keep track of and herd hundreds of sources of information. We take our cue from Border Collies—naturals at quickly processing and sorting information. Perhaps you have seen them amaze in agility like *Pink* at the Westminster Dog Show.¹ Or maybe you have seen Border Collies in herding competitions like *Skid* in the Scottish National Herding trials.²

Let us check in with Skid, whose job is to herd sheep . . . lots of them! On any given day, his job is to literally move hundreds of sheep from Point A to Points B and C efficiently and safely. This move requires absorbing copious amounts of information, including understanding what Skid's farmer wants, what each sheep is doing and will be doing, the terrain, ambient conditions, the physical constraints of fences and gates, and a whole lot of other sensory data. For most, this results in sensory overload . . . game over! But for Border Collies, this is all in a day's work. They are successful at it because they can take in all this information, pare it down quickly to key points, and make good decisions based on their mental notes. Their farmer aids this process by reducing very complex and intricate commands into notes as well. When Skid's farmer is thinking, "Skid, I want you to round up half of those sheep and move them from the pasture through the north gate and into the corral, and put the other half in the south paddock" but says, "Skid, come-bye, walk, in here" and gestures or whistles, Skid understands instantly and processes the information with a quick assessment . . . in other words, Skid takes good notes (see [Figure 3.1](#)).

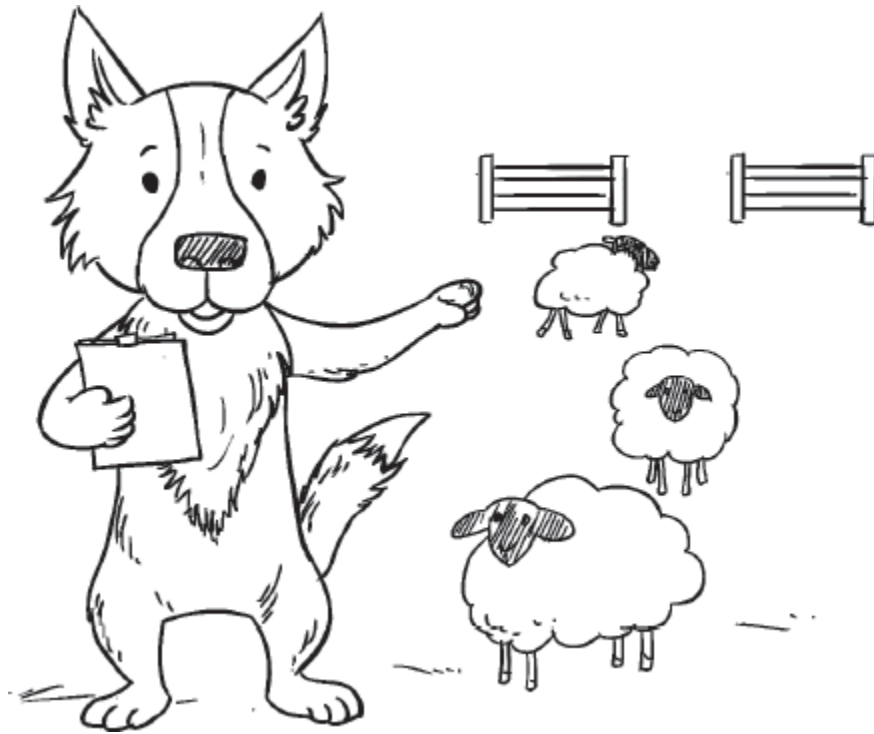


Figure 3.1

Usually people take notes for themselves, with the purposes of learning, documentation, and future reference. In these cases, you are your own audience, and you will know your purpose and the context of the situation. Sometimes, however, we take notes for others or potentially for others to reference. In these cases, consider what your audience needs and why.

Border Collies are born with a herding instinct. They want to do Page 20 their jobs. They know WHY—because when they do their job well, it comes with treats and praise. But they still must be taught what to do and how to do it. People are much the same. While we all know how to write words on paper or type on a computer, we benefit from training in what to do and how to do it. People also benefit when they understand WHY as well.

A quick search of the internet for “good note-taking” returns approximately three billion hits in less than one second. There are several different standard processes for good note-taking, including but not limited to

1. Outline
2. Mapping

3. Sketch-noting

4. Cornell

You are probably familiar with outlining—this works well if you know ahead of time what the agenda will be. It is harder to use this system in real time if you are trying to process new information without any prior knowledge of what will be covered. It is also a little overwhelming to go back and review these notes later as there is not any intuitive summarization beyond the notes themselves.

Mapping involves connecting ideas together. You can draw a main idea “cloud” and connect nodes of ideas to it. You can also connect cloud to cloud. If you are a visual person, this might be a good system. Another good system for visual notes is sketch-noting (see [Figure 3.2](#)). One drawback to sketch-noting is that it is difficult to be thorough in real time.

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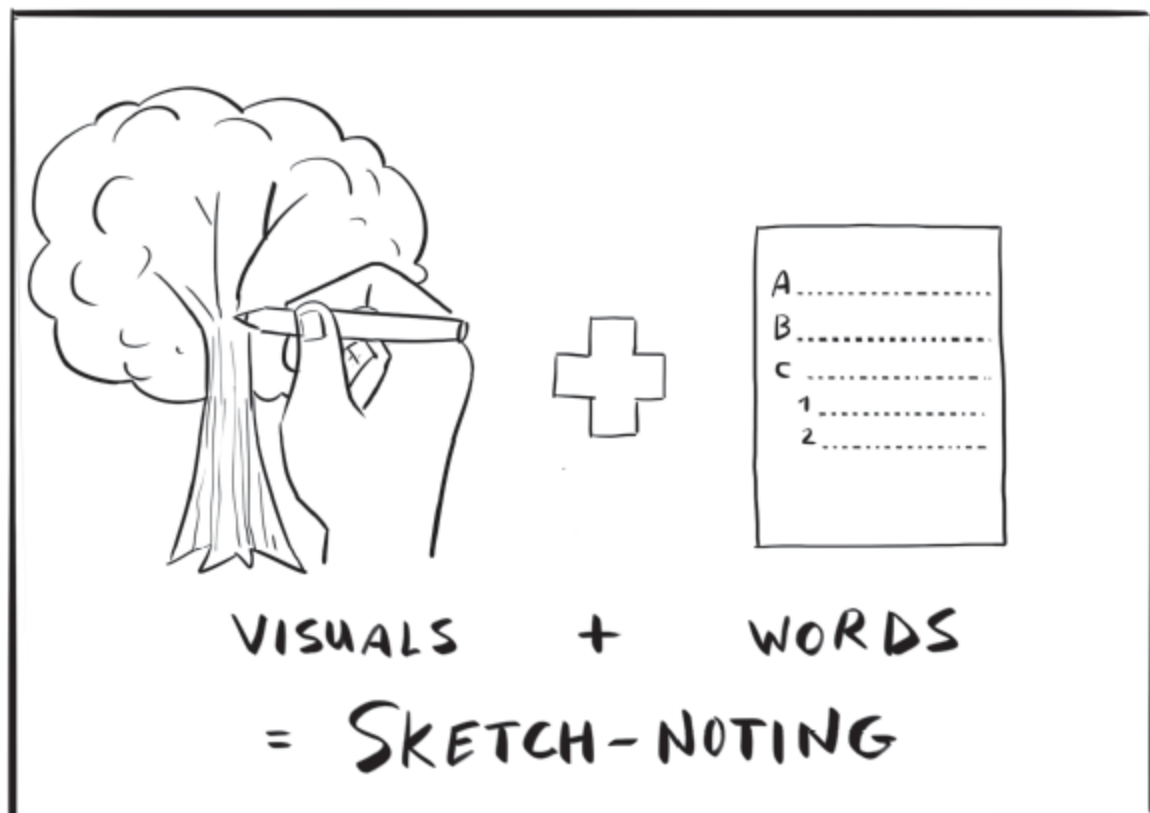


Figure 3.2

Sketch-noting is the process of using both visuals and text to conceptualize ideas, connect those ideas, and add details.

The Cornell method of note-taking is a great system for taking notes in real time, to which you can add visual cues later. The essence of the Cornell system is to divide your document into four sections as seen in [Figure 3.3](#). A title at the top provides the summary topic for the entire set of notes, while the summary section at the bottom provides a summary for the page. The right side of the document is for notes taken in real time, and the left side of the page is for key ideas. Sketches of information or key ideas can be included if the visuals help you quickly reference information.³

●	Name	Date
	Subject	Page #
	Key Ideas	Notes
●	Summary	

Figure 3.3

Cornell note-taking is a process that facilitates organized and easily referenced notes.

With any note-taking system, it is important to capture citation information as you go. It is painfully inefficient and often ineffective to try to remember where you found information days or even hours later. You should document source information including websites, access dates, authors, titles, page numbers, and anything else that could be required by the style guide you are using.

Our focus in this chapter is to remind you of some of the “how” options, but also to have you understand the WHY you want to be good at taking notes. There are three very critical reasons, including the utility of the

exercise, the ethical implications, and the potential legal ramifications of note-taking.

Utility

The science is clear; taking notes helps you retain information more effectively and efficiently.⁴ Note-taking forces your brain to think about things, and the result is that you are more likely to store the information in long-term memory. More complete notes are also better for learning than brief notes, so there is a balancing act between too much and not enough.

That said, one thing you will never see a Border Collie do is take notes about sheep on a computer, and it turns out that if your purpose for taking notes is to learn, you should follow their lead. There are many sources and studies which concur: using a computer to take notes is not as effective for learning as writing on paper.^{5,6}

Ethical Implications

Recognition of the importance of ethics is pervasive in science and engineering fields, and ethics includes taking notes. The National Society of Professional Engineers (NSPE), for example, has numerous case studies involving the ethics of note-taking in specific situations. If an engineer has information in their notes, is it ethical to report them or unethical to report them? You might not be surprised to read that “it depends,” and that it is your responsibility to know what “it depends” means for your discipline, field, position, and the situation. The important takeaway is that note-taking matters, and as a professional, you should investigate the ethics of your professional note-taking.

One example of a real NSPE case involving the ethics of note-taking is titled “Duty to Report Unrelated Information Observed During Rendering of Services.” It involved an engineer who discovered information pertinent to a preexisting bridge defect that might have contributed to a death. The engineer documented what they saw in their notes, but was then asked by a public agency to not include that finding in a final report. While the case

requires extensive background context and has multiple findings, among the findings, the NSPE Board determined that the engineer did the appropriate thing by documenting the information for possible future reference as appropriate.⁷

Another case from NSPE, titled “Employment–Employee/Employer Files,” states that

NSPE Code Section III.9.d. makes it clear that an engineer’s designs, data, records, and notes referring exclusively to an employer’s work are the employer’s property and not the property of the engineer.⁸

Notes are an important part of learning, recalling, documenting, and justifying information in technical jobs. So maybe you are now thinking, “Okay, my job is my job . . . but that doesn’t affect me personally, right?” Wrong. Potentially affecting us on a very personal level, studies show that jurors who take notes during trials perform more appropriately for the cases for which they serve. In one study called “Effects of Notetaking on Verdicts and Evidence Processing in a Civil Trial,” some interesting observations include that the more competent jurors took notes during the trial, and that jurors who took notes had better recall of information than those who did not take notes.⁹ If and when you serve the public as a juror, do you want to do the best job possible?

Legal Implications

If the ethics of note-taking are not enough incentive, consider the potential legal ramifications. You or your company can be held legally responsible if you know something that is detrimental to the safety or welfare of others and do not share it with the appropriate parties, or if you knew something and didn’t record it in an attempt to not have a written record. According to Christopher Proskey, patent attorney and industrial engineer,¹⁰ “notes are often important documents in litigation, as they represent the closest thing you can get to what a witness was thinking at the time.”

Every litigation begins with what is called the “discovery process,” when each party tries to determine what the other party knows about the subject of the litigation. One step in the discovery process is sending the other side “Requests for Production of Documents” (RPDs). RPDs demand the other side to produce all documents that are relevant to whatever topic is

specified in the RPD. Each party that receives an RPD has a legal obligation to provide the requested documents. Often notes are relevant to an RPD and therefore the notes must be produced to the other side—irrespective of whether the notes in question are good or bad for the witness or the side they are on. If the requested party fails to provide relevant documents in their possession, they are breaking the law (which means they can go to jail).

Notes produced in response to an RPD often provide the other side with valuable insight and information. These notes can then be used to test witnesses' opinions or recollection during depositions or at trial, and they can be used as evidence in motions or at trial. Many witnesses have been placed in uncomfortable positions on cross-examination when they say something very different from their notes taken previously. For these reasons, notes have a substantial impact on the outcome of many cases.

We see similar concerns again with the NSPE regarding the failure to share information. In Case 95-5, called “Failure to Include Information in Engineering Report,” an engineer chose not to report all of the technical information available and relevant to the situation. The findings reiterate that the NSPE Code of Ethics requires that engineers “shall include all relevant and pertinent information in such reports, statements or testimony.”¹¹ Mr. Proskey continues:

Note-taking also has particular importance in patent law. For many years, the U.S. was under the “First to Invent” rule, which meant that the first person to “invent” something was awarded the patent for it. This resulted in many litigations where Party A filed a patent first, but Party B challenged them saying they invented it first, and therefore Party B should own the patent instead of Party A. This fight often came down to a battle of the notes. Essentially, whoever could prove, using their notes, that they invented it first, was awarded the patent.

The law changed with the Leahy-Smith America Invents Act of 2012, which changed the law to the “First to File” rule. This new law means that the first party to file a patent is awarded the patent. However, this change did not make notes any less important in patent law. Following the Leahy-Smith America Invents Act, inventor notes can be very useful in litigation as a defense against a claim for patent infringement.

For example, if Party A files a patent and accuses Party B of infringing their patent, Party B can try to invalidate Party A's patent by proving Party B was using it first. If Party B can prove that Party B was using the subject of Party A's patent before Party A filed the patent, Party B can claim Party A's patent is invalid for being in public use or available to the public before Party A applied for the patent. Inventor notes are often critical in this proof.

Digital Footprints

While most of this chapter is about the notes that **you** will take, we would be remiss if we did not address digital footprints, which are the “notes” that

others take **about** you.¹² Whether you know it or not, every click of a mouse and every keystroke are being saved for posterity when you are working on the internet. If you save to a cloud-based server, your files are out there forever, regardless of “deleting” them. If you (responsibly!) back up your work to a remote back-up service, it is there forever. If you post to social media sites (e.g., LinkedIn, Snapchat, Instagram, YouTube, Facebook, Twitter, TikTok, etc.), your digital presence continues to expand and, theoretically, can be found and used to define you. “But I deleted the file! But I clicked on the wrong link! But I didn’t mean to post it publicly!” None of those matter. You must remember that your digital footprint is forever and could potentially be accessed by anyone. Consider that carefully before you start typing.

That said, there might be situations where you will appropriately comment within a digital platform or file, but then inappropriately share that information. What happens then? The NSPE addresses an example of that as well. In Case 09-11, titled “Discovering Embedded Comments in Electronic Documents Damaging to Adversary,” they note that the ethics and legalities of some situations overlap, and that in such cases, the legal ethics must be addressed first.¹³

In summary, note-taking is a critically useful tool for how we do our jobs and live our lives, but we want to make good, informed decisions. Knowing what system works best for a given situation, understanding what to document and when, and being cognizant of the permanence of our notes are all critical to being successful.

Note-taking Checklist

- ☐ Should we take notes?
- ☐ What method of notes works best for us and the situation at hand?
- ☐ Will these notes be the property of my company or are they personal?

- If using the Cornell system, do I have templates pre-drawn?
- If using the outline system, do we know any guiding information ahead of time?
- How will we store/maintain/disseminate our notes?

Exercise

1. Create a template for Cornell note-taking, go back to the start of this chapter, and take notes. Can you summarize?
2. Watch the video on Sketch-noting and create notes for it:
https://www.youtube.com/watch?v=pZgMpjjgCRA&feature=emb_logo.
3. When you take notes in a class, who is the audience (for the notes)? What is the purpose?
4. When you take notes while meeting with a client, who is the audience (for the notes)? What is the purpose?

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CHAPTER FOUR

Technical Definition



There are many well-understood powerhouses in this world: the hard-working, no-questions-asked, get-the-job-done kinds of creatures and things. They are not fancy, and they do not subscribe to needing recognition, but they are dependable and solid. Think of your toolbox—perhaps your trusty hammer comes to mind. After a snowfall, you grab that same old shovel you have had for ten years. In your kitchen, your can opener never fails. Likewise, the working dogs include many breeds that are just like this: solid, simple, and straightforward. The Doberman Pinscher’s job is to announce change. None do it better, quickly and easily convincing all that they mean business (see [Figure 4.1](#)). The Samoyed is built to pull sleds in -60° F weather. They manage this easily with their beautiful fur coats. The Newfoundland is literally born to swim, with webbed feet that allow them to haul in fishing nets and specialize in water rescues. Working dogs get the job done and do not ask for appreciation. In technical writing, there are some working dogs. The first of these, called the technical definition, helps people understand the meaning of words: not a fancy job, but a very important one.



Figure 4.1

The working dogs of technical writing, represented here by the Doberman Pinscher, include technical definitions, descriptions of mechanisms, and descriptions of processes.

Virtually any kind of technical writing includes one or more technical definitions, whether it is a formal technical report or an informal presentation. Technical writers must be able to effectively define terms for their audience, no matter their audience's level of knowledge. In fact, much of a technical writer's job will be to write for non-expert audiences with little to no knowledge or experience with the subject.

What Is a Technical Definition?

In technical writing, **definition** is the process by which one assigns a precise meaning to a term. If the goal of technical writing is to reduce abstraction and avoid (mis)interpretation, the goal of writing a technical description should be to define a term so well that there is only one way to understand what the term means. To define a term, it must be placed into a classification and then differentiated from other terms in that same classification. Technical definitions are relatively easy to write, except for some pitfalls that will be addressed later. The format for a technical definition is straightforward and works like this:

Term = *Classification* + Differentiation

For example, if a writer were to define the stall condition that an airplane experiences when it loses lift, they could start with the term **stall**, add a classification—*flight condition*—and then differentiate it from all other flight conditions, in this case, by a stall's unique characteristics. The definition might read something like this:

A **stall** is a *flight condition* in which the lift produced becomes less than the weight of the airplane, and the airplane stops flying.

That seems simple enough; but what happens when a term, such as **stall**, has multiple definitions in many contexts? For example, in a sports context, **stall** can mean a tactic to drain time on the clock and in an agricultural context, a **stall** is a compartment in a stable or barn for a domesticated animal (think horse or goat). In such cases, it might be necessary to add a qualifier in front of the definition statement to supply the necessary context. The qualifier is important when the general context for a definition needs to be established up front. If the context is known or is obvious, a qualifier is unnecessary. For example, in an aeronautics study on aircraft wing design, the context of **stall** is obvious. It is clear that **stall** in this case has to do more with the loss of lift than with, say, a defensive maneuver employed by a basketball team to slow down the pace of the game or where a horse stays overnight.

When context is needed, the format for the definition is

(Qualifier +) **Term** = *Classification* + Differentiation

Note: The parentheses and boldface have been added here and earlier for clarity. Visually emphasizing the term using boldface or italics, however, is a good practice when writing any definition.

Look at [Figure 4.2](#), which provides three definitions of the same term in different contexts. Notice how the term **stall** has three totally different meanings depending on the context, and how each definition begins with a qualifier that makes the context clear from the start.

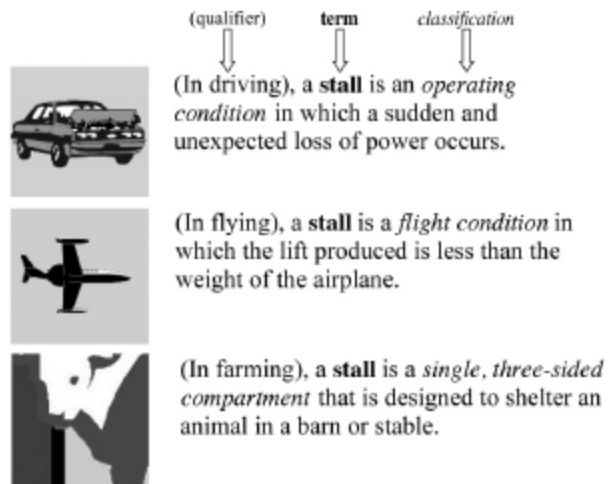


Figure 4.2

The **term** stall has multiple meanings, three of which are defined here using the format for technical definitions: (Qualifier +) Term = *Classification* + Differentiation.

In the first example, **stall** refers to a car that has suddenly stopped running. This condition typically occurs in the middle of heavy traffic, in bad weather, or when you are desperately trying to make a flight for which you have nonrefundable tickets.

In the second example, **stall** refers to what happens when an airplane does not go fast enough to stay in the air. Pilots routinely stall their airplanes right above the runway when landing them, in which case stalling is good. Sometimes, however, they stall them inadvertently in situations when they really want their planes to keep flying, in which case stalling is bad.

The third example of **stall** relates to farms, barns, stables, and livestock. The authors have had their fair share of experiences with stables and barns—though, hopefully, they have grown past some early childhood observations that they “were raised in a barn.”

Classifications and Classes

Often, the most difficult part of writing a technical definition lies in determining the proper classification for the term. The class should be a general category in which the term fits, but it cannot be too general. For

example, consider that 33-kilohm, 1-watt carbon resistor from the abstraction funnel in [Chapter 1](#). In the following sentence, the term is defined by using the very generic classification of *device*.

The **33-kilohm, 1-watt carbon resistor** is a *device* that impedes the flow of electric current.

The problem with this classification is that *device* could mean all kinds of different things, most of which have nothing to do with circuit components; consequently, its inclusion does not really help specify the meaning of the term. By changing *device* to *circuit component*, however, the meaning can be narrowed considerably for the reader, even before the classification is differentiated.

One trick for classifying a term is to build an abstraction funnel and use the filter one abstraction level above the term. In the abstraction funnel shown in [Figure 4.3](#), moving up a filter level from the term *33-kilohm, 1-watt carbon resistor* to the term *resistor* is not an option because the classification would be derived from the original term. That would yield the following circular definition:

The **33-kilohm, 1-watt carbon resistor** is a *resistor* that impedes the flow of electric current.

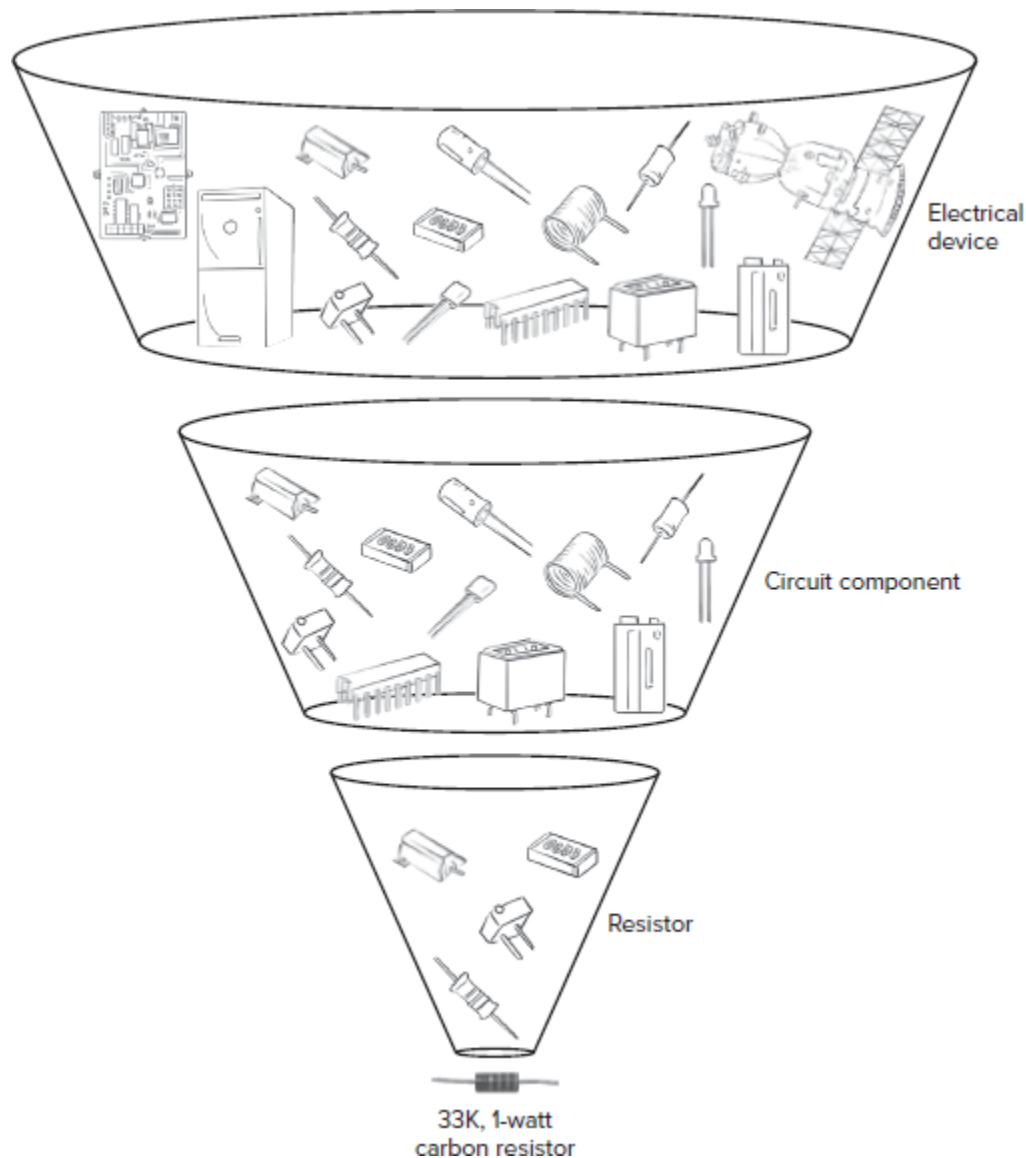


Figure 4.3

A funnel of abstraction. The high-level of abstraction at the top of the funnel allows the audience to consider too many concepts or items. Abstraction is reduced through the use of non-abstract language. At the bottom of the funnel is the lowest level of abstraction, where only one concept or item is described by the language.

Because the 33-kilohm, 1-watt resistor is obviously a resistor, this classification does not help define the term. In fact, it contributes nothing other than useless circularity. If possible, try to find a classification that is not derived from the term. In this case, the easiest solution is to just move up the abstraction funnel one more filter. *Circuit component* helps to define the term and has no circularity with the term.

The next step in defining the term is to differentiate it from all the other members of its class. Differentiation involves narrowing the meaning of the term to just one possibility within the class. Clearly, it would be easier to narrow the class of “circuit components” to a particular resistor than it would be to narrow the class of “device” to a particular resistor. There are all kinds of devices in the world and relatively few circuit components.

The class “circuit components,” however, still contains many possibilities: capacitors, diodes, switches, potentiometers, inductors, transistors, and IC chips, to name a few. In this case, a good approach is to focus on the function of a resistor, which is to impede the flow of electric current, and to use that function to differentiate the class. Doing so yields the following definition:

The **33-kilohm, 1-watt resistor** is a *circuit component* that impedes the flow of electric current.

Avoiding Imprecision

In writing technical definitions, it is easy to do something that is “not good.” For example, when we are defining the common computer term **hard drive**, it is probably best to add a qualifier unless the context of computing is obvious. (There are other kinds of hard drives, such as one through the Mojave Desert in July without air conditioning.)

So how about this definition?

In computing, a **hard drive** is an *input/output device* for the nonvolatile storage and retrieval of data.

Think about this one for a minute. If the reader does not know what a **hard drive** is, what is the likelihood that the reader will know what an *input/output device* is, much less the meaning of nonvolatile storage and retrieval?

How about this definition of a water softener’s resin?

In water softening, **zeolite** is an *exchange resin* that gains sodium ions while releasing calcium and magnesium ions.

This definition is fine as long as you are sure your audience will know what *ions* are, not to mention *exchange resin*, *calcium*, and *sodium*. An audience familiar with water softeners and basic chemistry certainly would, but what about a less technical audience—perhaps a nontechnical manager who controls your funding? For a less technical audience, we might rewrite this definition as follows:

In water softening, **zeolite resin** is a *special material* that releases sodium into the water while absorbing calcium and magnesium from the water.

Notice that there is no mention of ions or ionic exchange. A nontechnical audience would not know (or probably even care) about such things as ionic charge and exchange.

Extensions

You must always consider the reader's knowledge and skill level when defining terms and concepts. However, at times you may have no choice but to violate the principle to achieve your goal. Sometimes a simple term or concept that the audience will understand is just not available. In fact, sometimes the only thing to do is to define the term as best as you can, then add extensions to your definition to clarify your meaning.

Extensions can take many forms. In the following examples of the most common types of extensions, the original definition is included in brackets followed by the extension.

Further Definition

Use further definition when you need to define terms you used in your original definition.

[In water softening, **zeolite** is an exchange resin that releases sodium ions while gaining calcium and magnesium ions.] Zeolite is a collection of small polystyrene beads that forms a resin carrying a negative charge.

Comparison and Contrast

Use comparison and contrast when you need to show differences or similarities.

[In water softening, **zeolite** is an exchange resin that releases sodium ions while gaining calcium and magnesium ions.] Zeolite collects calcium and magnesium ions in the same way a feather broom collects dust.

The comparison or contrast can be with a concept your audience is familiar with, which can make understanding a new concept easier.

Classification

Use classification when you need to organize information into categories.

[In water softening, **zeolite** is an exchange resin that releases sodium ions while gaining calcium and magnesium ions.] Zeolite is used for ion-exchange “home” water softening, while lime and soda ash are used in precipitating municipal water softeners.

Much like comparison and contrast, classification can help your audience by connecting to concepts with which they are familiar.

Cause and Effect

Use cause and effect when you need to demonstrate why something happens or when you need to trace results.

[In water softening, **zeolite** is an exchange resin that releases sodium ions while gaining calcium and magnesium ions.] Zeolite carries a negative charge and attracts the positively charged calcium and magnesium ions in water.

Process

Use process when you need to list the steps of a procedure.

[In water softening, **zeolite** is an exchange resin that releases sodium ions while gaining calcium and magnesium ions.] First, the zeolite is charged with sodium ions from a salt brine solution in the regeneration step. Once charged, the zeolite absorbs calcium and magnesium ions from passing hard water while releasing sodium ions into the resulting soft water.

Exemplification

Use exemplification when you need to give real or analogous examples.

[In water softening, **zeolite** is an exchange resin that releases sodium ions while gaining calcium and magnesium ions.] Examples of the zeolite group include analcimes, chabazites, and gismondines.

Etymology

Use etymology to show the linguistic genesis of the term.

[In water softening, **zeolite** is an exchange resin that releases sodium ions while gaining calcium and magnesium ions.] The term **zeolite** comes from the Greek words *zeon*, meaning “to boil,” and *lithos*, meaning “stone.” It was first coined by the Swedish mineralogist Alex Fredrick Cronstedt in the 18th century.

Required Imprecision

In some situations, you might need to sacrifice desired precision in your definition to achieve the required level of communication. At times, for example, it is foolish to attempt to achieve expert-level precision with a non-expert audience.

Consider the following two definitions of a **black hole**—the Page 33 astrophysical phenomenon that exists in space.

Definition 1

In astrophysics, a **black hole** is a set of events from which it is not possible to escape to a large distance. A black hole gets its name from its boundary, called an **event horizon**, which is formed by the paths in space-time of rays of light that just fail to get away, hovering instead forever on the edge and, consequently, moving on paths parallel to or away from one another.

Definition 2

In astrophysics, a **black hole** is a collapsed neutron star whose gravity is so great that even light cannot escape. Although fusion reactions within this collapsed star still may emit brilliant rays of light, when it is viewed from the outside, the black hole appears to be a totally dark void in space.

Definition 1 functions at the expert level. For theoretical physicists, it provides a precise and accurate description of a black hole and thus is appropriate for their needs. But for the less knowledgeable, reading it represents a mind-twisting experience that, in many cases, can leave readers more confused about the term than they were before they read it.

Definition 2 functions at the level of the average reader. It is not nearly as precise or accurate as the first definition, but it communicates the basic gist of what constitutes a black hole in space.

The problem is that to be precisely correct and absolutely accurate, you would have to function at a level where you cannot effectively communicate with the average reader. If you were writing for the average reader, you would have to make a decision here: either be absolutely correct and communicate less effectively, or be less than absolutely correct and communicate more effectively. Since your goal is effective communication, your decision is obvious.

A Word about Defining Specifications and Standards

Defining specifications and standards is a specialized activity. Such documents can take many different forms, depending on what areas of engineering and science are involved and whether the documents are designed to meet commercial, industrial, or government standards. Writing these kinds of definitions is far more complex than simply defining terms.

Specification documents precisely state particulars, including requirements, designs, implementations, and testing. In engineering and science, specifications normally involve goods and services being developed under some type of contractual obligation. The specification precisely defines the quality of work and performance standards required by the contract. In addition, as will be shown in [Chapter 5](#), specifications can describe mechanisms in terms of their physical and functional attributes.

Specifications also form the basis of technical standards. Standards are accepted or established methods, measures, or designs for accomplishing specific tasks. Standards exist for everything from data transfer protocols and cable connectors to air conditioning coolants and drinking water.

Definition Checklist

- ☐ Have I fully analyzed the purpose of my project, and do I understand the skill and knowledge level of the audience?

- Have I defined the term by first classifying it in a way that adds
- precision and understanding for my audience and that serves my purpose?
 - Have I differentiated this classification to distinguish this term from other members of its class?
 - Have I determined whether the context is clear and, if not, whether it is critical to the definition? If the context is unclear and critical, have I used a qualifier before the term?
 - Have I avoided defining a term using the same term?
 - Have I avoided using terms that themselves need to be defined? If not, have I explained these terms with extensions?
 - Have I chosen extensions to my definitions that are appropriate for my audience and purpose?
 - Have I compromised my fundamental purpose (communicating with the reader) by including inappropriate or irrelevant information or precision?

Exercise

Read each of the following definitions and try to determine the context, audience level, accuracy, and purpose for which they were written.

- A **resistor** is a small electronic part that reduces the amount of electricity flowing through a circuit.
- A **resistor** is a circuit component that converts electrical energy to thermal energy and, in the process, determines the current produced by a given difference of potential.
- **Resonance** is a systemic condition in which small amplitudes of a periodic agent produce large amplitudes of oscillation or vibration.

- **Resonance** is a natural means of amplification that makes a musician's horn sound louder.
- **Ionization** is the electrostatic process by which a neutral atom or molecule loses or gains electrons, thereby acquiring a net charge.
- **Ionization** is the phenomenon that creates lightning in thunderstorms.
- **Ergonomics** is the field of study by which we make machines easier to use.
- **Ergonomics** is the systematic consideration of physical, psychological, and social characteristics of human beings in the design of tools and equipment, the workplace, and the job itself.

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CHAPTER FIVE

Description of a Mechanism



Much like technical writing typically includes one or more technical definitions, almost every kind of technical writing includes a technical description of either a mechanism ([Chapter 5](#)) or a process ([Chapter 6](#)). Whether it is a formal technical report, a proposal, a refereed article, or an informal presentation, you may need to include a technical description. There is no standard length for technical descriptions because they are what the audience needs them to be. Technical descriptions can be a paragraph within a proposal, a couple of paragraphs in a technical report, or even a stand-alone document of several pages.

Technology involves physical devices called **mechanisms**. Being able to describe mechanisms precisely and accurately, and at a level and in a way that the reader needs and can understand, is perhaps the most essential component skill of technical writing. This skill is particularly important for those who produce documents involving specifications and instructions.

Just like technical definitions, when it comes to descriptions of mechanisms, we have another working dog. While at first glance you might not fully appreciate them, they are no-nonsense, elegant, strong, and powerful (see [Figure 5.1](#)). Their grace and beauty show through when your audience fully understands the technical mechanism you are describing with minimal to no effort.



Figure 5.1

The Samoyed is another working dog whose ability to get the job done is sometimes underappreciated. Note that the dogs are pulling a sled and wearing harnesses, which are mechanisms that could be described.

What Is a Description of a Mechanism?

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Descriptions of mechanisms are precise portrayals of material devices with two or more parts that function together to do something. Mechanisms can range in complexity from circuit components and mechanical fasteners, to supercomputers and space shuttles, to a bike that parents must assemble for a birthday.

Know Your Audience

If you are describing a mechanism (like a bicycle kickstand) for a patent application, this has a very different audience from the bike distributor reading a sales brochure or the frazzled parent after an 8-year-old's birthday party reading a set of assembly instructions. See [Chapter 3](#) for a teaser on legal implications associated with patent law, but know that you will want to consult a patent attorney if your audience is at that level of need.

In [Figure 5.1](#), the Samoyed dogs are pulling a sled, which is how the breed has earned their keep in Siberia for centuries. The sled is a mechanism. So are the harnesses that the dogs wear. How would you explain the technical requirements for either of these to an audience that is considering making or using them? As you might notice in [Outline 5.1](#), the primary focus in writing a mechanism description is on the physical characteristics or attributes of a device and its parts. Mechanism description documents are built around precise descriptions of size, shape, color, finish, texture, and material, and they can be written for a wide range of audience skills. Such documents also normally include figures, diagrams, or photographs that help to clarify the worded description. Some descriptions (e.g., components for space systems) might require additional attributes such as weight or mass.

Outline 5.1 Description of a Mechanism

Introduction

- Define the mechanism with a technical definition (see [Chapter 4](#)), and add extensions to discuss any theory or principles necessary for the reader to understand what you are saying. Always make sure you add only what the reader needs for the purpose at hand. If the reader does not need any theory or operating principles, do not provide any. Consider any relevant context for the situation as well.
- Describe the mechanism's overall function or purpose.
- Describe the mechanism's overall appearance in terms of its relevant attributes—e.g., shape, color, material, finish, texture, mass/weight, and/or size.
- List the mechanism's parts in the order in which you plan to describe them.

Discussion

- **Part 1**

- Define the first part with a technical definition, adding extensions as needed to deal with theory or operating principles.
- Describe the part's overall function or purpose.
- Describe the part's shape, color, material, finish, texture, mass/weight, and/or size (as well as any other physical attributes appropriate for the mechanism and its function), using precise measures and descriptors. Also, be sure to use figures, diagrams, and photographs as necessary.
- Transition from this part to the next part.
- **Parts 2–n**
 - For each remaining part, repeat the pattern of defining, describing, and transitioning established for Part 1.

Conclusion

- Briefly summarize the mechanism's function and relist the parts described.
- Give a sense of finality to the paper.

A variation of the traditional mechanism description extends the Page 37 treatment from just physical attributes to include a more expansive discussion of function and theory of operation. The organization of this type of mechanism description is still based on the parts, but the discussion of function and theory may be extended.

A more specialized mechanism description, the functional mechanism description, describes a mechanism with key physical attributes related to operating parameters. This type of description, which often includes specific operational standards, can be used to provide specifications for the mechanism. Technical specifications are normally written for an expert audience that actually needs to design, maintain, use, or evaluate the mechanism. For example, if you were the design engineer for the skis on a dog sled, you would need to include the material specifications (what kind of steel, how hard must it be, what must the surface finish be, how should it

be formed, etc.). Your description should be fully understandable by the engineers who must make the product.

Moving from an Outline to a Complete Description

Outline 5.1 provides a model for a general mechanism description. To show how this model works, we will use it to describe a relatively simple mechanism: the 33-kilohm, 1-watt carbon resistor discussed in [Chapters 1](#) and [4](#). The resistor is a relatively simple mechanism; most mechanisms are much more complex. Assume, for the purpose of illustration, that this description is for the average technical reader who requires only a general description of the resistor.

Example 5.1 General Mechanism Description

Description of a 33-Kilohm, 1-Watt Carbon Resistor

Introduction

The **33-kilohm, 1-watt carbon resistor** is a circuit component that impedes the flow of electric current. The resistor impedes the flow of current by converting a portion of the electrical energy flowing through it into thermal energy, or heat. This particular resistor can safely convert 1 watt of electrical energy into heat.

The 33-kilohm, 1-watt carbon resistor looks like a small cylinder with wire leads extending from each end. The casing's surface is composed of smooth, brown plastic with a shiny finish. Four equally spaced color bands (three orange, one gold) circumscribe the cylinder, starting at one end.

The resistor consists of the following parts: the carbon element, the wire leads, the casing, and the color bands (see [Figure 5.2](#)).

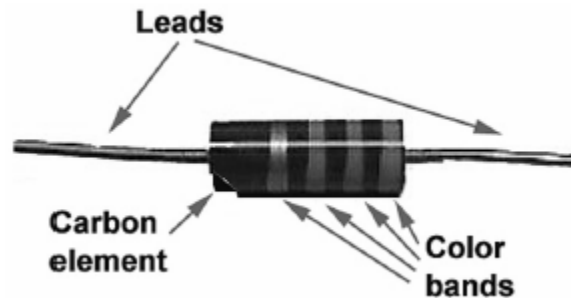


Figure 5.2

Parts of the resistor.

Discussion

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For the following discussion, refer to [Figure 5.3](#) for a notional X-ray view of the resistor's parts, along with its physical dimensions.

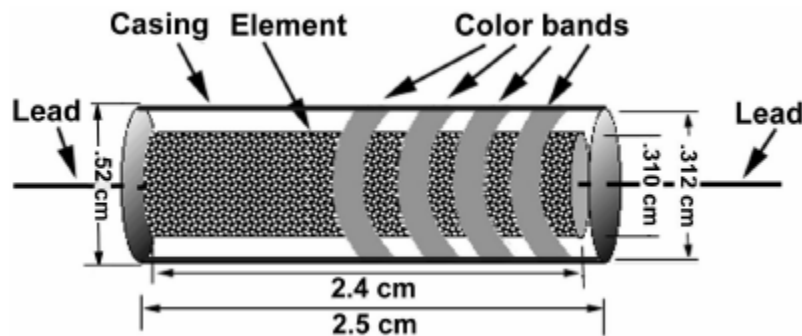


Figure 5.3

X-ray view of resistor.

Carbon Element

The **carbon element** is the capsule of resistive material that converts electrical energy into heat. The carbon element serves as the primary active component of the resistor by providing the necessary 33,000 ohms of resistance. The element functions by blocking, to some degree, the

flow of free electrons passing through it. The energy released by these blocked free electrons is then dissipated in the form of heat.

The carbon element is cylindrically shaped and is 2.4 centimeters long with a diameter of 0.31 centimeters. It is composed of finely ground carbon particles mixed with a ceramic binding compound. The element is gray with a dull, matte finish. The carbon element is electrically connected to the leads.

Leads

The **leads** are two conductive wires connected to opposite ends of the carbon element. The leads have two functions. First, they provide electrical connectivity from the carbon element to the circuit; and second, they provide a mechanical means of mounting and supporting the resistor in the circuit environment. The leads, which have a dull, silver color and smooth texture, are composed of tinned copper wire. Each lead is 4 centimeters long and is 20 gauge in thickness. The leads connect to the carbon element through the ends of the casing.

Casing

The **casing** is a cylindrical enclosure that surrounds the carbon element. The function of the casing is twofold. First, it physically protects and electrically insulates the carbon element from the outside environment. Second, it provides the heat-exchanging medium needed to dissipate the thermal energy generated by the carbon element. The casing is a brown, plastic cylinder that is 2.5 centimeters long, with a 0.312-centimeter inside diameter and a 0.52-centimeter outside diameter. It snugly fits over the carbon element. The outside of the casing is circumscribed by four color bands.

Color Bands

The **color bands** are visual indicators that describe the resistance and tolerance of the resistor. Using the standard color code for commercial, four-band resistors and starting with the band at the edge of the cylinder, the first three bands represent the value of the resistor in ohms. The fourth band indicates the tolerance or accuracy of the resistor. Each color band is 0.1 centimeters wide and is circumscribed on the outside of the

casing and parallel to the edge of the casing. Each color band is smooth and shiny.

The first color band, which starts flush with one end of the casing, is orange and represents a value of 3. The second color band, which starts 0.1 centimeters away from the inside edge of the first color band, is orange and also represents a value of 3. The third color band, which starts 0.1 centimeters away from the inside edge of the second color band, is orange and represents a multiplier of 1000. The fourth color band, which starts 0.1 centimeters away from the inside edge of the third color band, is gold and represents a tolerance of 5 percent.

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Conclusion

The 33-kilohm, 1-watt carbon resistor is a circuit component that impedes the flow of electric current through the use of a carbon element. The resistor is made up of four parts: the carbon element, which impedes the flow of current by converting a portion of the electrical energy applied to heat; the wire leads, which electrically connect the element to the circuit and support the resistor mechanically; the casing, which encloses and insulates the element and dissipates heat from it; and the color bands, which indicate the resistance and tolerance of the device. Together, these parts form one of the most commonly used circuit components in electronic systems today.

Dissecting [Example 5.1](#)

Now that you have read an example of a description of a mechanism, let us take a closer look at each of its elements and what they do.

Introduction Section

Technical Definition

Following the outline, first introduce the mechanism with a technical definition and extensions that describe its overall function and purpose (go back to [Chapter 4](#) if you need or want a refresher):

The **33-kilohm, 1-watt carbon resistor** is a circuit component that impedes the flow of electric current.

Add Extension

Next, use extensions to discuss any theory or operating principles necessary for the reader to understand the description. Be careful here; think about what you are doing, for whom you are doing it, and why you are doing it. Do not lose sight of the reader's knowledge and skill level or forget the purpose of the report. The goal is not to show how smart you are, but rather, to communicate the information. So, when you are discussing theory in a mechanism description, a good rule is to do what is necessary, but only what is necessary. For example, consider this theoretical discussion:

The resistor impedes the movement of free electrons, thereby generating a thermal response depending on temperature, cross section, and length of the resistive element. The resulting resistance is measured in ohms; a resistor is defined as having 1 ohm of resistance when an applied electromotive force of 1 volt causes a current of 1 ampere to flow. This resistor has 33,000 ohms, meaning that when 1 volt is applied, a current of 0.00003 amperes would flow. In addition, the square of the current flowing in amperes, times the resistance in ohms, determines the power dissipated in watts. This particular resistor can safely and continuously dissipate 1 watt of electrical energy as heat.

This extension might be fine for readers with more specific technical purposes, but not for the average person who needs only a general description—the audience specified earlier for this example. Always remember your audience and purpose. The goal of this type of mechanism description is not to provide a detailed description of how a resistor works or to show how to measure its effect in a circuit. In addition, if this theoretical discussion were to be used for the intended audience, several additional concepts (such as current, voltage, and power) would have to be defined and discussed.

Make Sure the Extension Is Appropriate for Audience and Purpose

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Given the reader's skill level and purpose, the following two sentences might provide a simplified theoretical extension that is more appropriate:

The resistor impedes the flow of current by converting a portion of the electrical energy flowing through it into thermal energy, or heat. This particular resistor can safely convert 1 watt of electrical energy into heat.

Notice that this description does not get into Ohm's law. It also does not explain that if we placed 1 million volts across this resistor, it would draw 30.3 amperes of current, resulting in the dissipation of more than 30 million watts of power, thereby generating an inferno that would destroy our houses and probably the entire neighborhood. This information is also not accurate (the resistor would vaporize well short of those power levels); and if it were accurate, it is still not relevant to describing the mechanism. It might be relevant to describing the operation of the mechanism (or, if accurate, in answering the civil suits brought by the neighbors), but that is not the purpose here.

Describe the Overall Function or Purpose

The next step in the introduction is to describe, in general terms, the mechanism's overall appearance—its shape, color, material, finish, texture, weight/mass, and/or size. In this case, one could describe the device as follows:

The **33-kilohm, 1-watt carbon resistor** looks like a small cylinder with wire leads extending from each end. The cylinder's surface is composed of smooth, brown plastic with a shiny finish. Four equally spaced color bands (three orange, one gold) circumscribe the cylinder, starting at one end.

List the Mechanism's Parts in the Order You Will Describe Them

Finally, to complete the introduction, list the mechanism's parts in the order in which they will be described. This listing is an "organizational blueprint" for the remainder of the mechanism description. Consequently, the decision regarding the order in which to list the mechanism's parts is not trivial. It effectively determines the structure for the rest of the mechanism description. There are two ways to order the parts:

- **Spatially.** Using a spatial organization, you can move from left to right, or top to bottom, or inside out, or outside in. Using an inside-out spatial approach, the final sentence of the introduction might read this way:

The resistor consists of the following parts: the carbon element, the wire leads, the casing, and the color bands.

- **Functionally.** Using a functional approach, you can order the parts in terms of how the parts function with one another. Functionally, for example, you could start with the leads, which connect the circuit to the carbon element, which is protected by the casing, and around which the color bands are painted to indicate resistance and tolerance values. Using this approach, you might write the final sentence of the introduction as follows:

The resistor consists of the following parts: two wire leads, the carbon element, the casing, and the color bands.

Both approaches are fine as long as they are logical and make sense to the reader. In any case, once the parts have been listed, move on to the discussion section, where detailed descriptions of the parts are provided. Note: If in writing the body of the paper you decide to change the order in which the parts are discussed, be sure to also change the order in which the parts are listed in the introduction.

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Discussion Section¹

The discussion section of a mechanism description precisely describes, in necessary detail, each part of the mechanism. It always follows the organizational pattern established by the listing of parts at the end of the introduction. For example, if we are using inside-out spatial ordering, the discussion section will have four subsections: the carbon element, the leads, the casing, and the color bands.

The discussion section is laid out in a relatively simple manner in the order of the listing. Create a subsection for each part and then describe the first part in the first subsection, and so forth.

Carbon Element

First, define the part with a technical definition:

The **carbon element** is the capsule of resistive material that converts electrical energy into heat.

Second, provide an extension that describes the part's function and gives any needed theory:

The **carbon element** serves as the primary active component of the resistor and provides the necessary 33,000 ohms of resistance. The element functions by blocking, to some degree, the flow of free electrons passing through it. The energy released by these blocked free electrons is then dissipated in the form of heat.

Third, describe the part's shape, color, material, finish, texture, weight/mass and/or size, using precise measures and descriptors. Be sure to include necessary visuals such as diagrams and photographs. Two visuals were included in this example; for now, concentrate on describing the part's physical attributes:

The carbon element is cylindrically shaped and is 2.4 centimeters long with a diameter of 0.31 centimeters. It is composed of finely ground carbon particles mixed with a ceramic binding compound. The element is gray with a dull, matte finish.

Fourth, transition to the next part by showing how this part relates to it:

The carbon element is electrically connected to the leads.

Now handle the rest of the parts in similar fashion.

Leads

Define the part:

The **leads** are two conductive wires connected to opposite ends of the carbon element.

Describe the function of the leads and provide any needed theory:

The leads have two functions. First, they provide electrical connectivity from the carbon element to the circuit; and second, they provide a mechanical means of mounting and supporting the resistor in the circuit environment.

Describe the part in detail:

The leads, which have a dull, silver color and smooth texture, are composed of tinned copper wire. Each lead is 4 centimeters long and is 20 gauge in thickness.

Transition to the next part:

The leads connect to the carbon element through the ends of the casing.

Casing

Define the part:

The **casing** is a cylindrical enclosure that surrounds the carbon element.

Describe its function and provide any needed theory:

The function of the casing is twofold. First, it physically protects and insulates the carbon element from the outside environment. Second, it provides the heat-exchanging medium needed to dissipate the thermal energy generated by the carbon element.

Describe the part in detail:

The casing is a brown, plastic cylinder that is 2.5 centimeters long, with a 0.312-centimeter inside diameter and a 0.52-centimeter outside diameter. It snugly fits over the carbon element.

And transition to the next part:

The outside of the casing is circumscribed by four color bands.

Color Bands

Define the part:

The **color bands** are visual indicators that describe the resistance and tolerance of the resistor.

Describe the function of the bands and provide the needed theory:

Using the standard color code for commercial, four-band resistors, and starting with the band at the edge of the cylinder, the first three bands represent the value of the resistor in ohms. The fourth band indicates the tolerance or accuracy of the resistor.

Describe the part in detail:

Each color band is 0.1 centimeters wide and is circumscribed on the outside of the casing and parallel to the edge of the casing. Each color band is smooth and shiny. The first color band, which starts flush with one end of the casing, is orange and represents a value of 3. The second color band, which starts 0.1 centimeters away from the inside edge of the first color band, is orange and also represents a value of 3. The third color band, which starts 0.1 centimeters away from the inside edge of the second color band, is orange and represents a multiplier of 1000. The fourth color band, which starts 0.1 centimeters away from the inside edge of the third color band, is gold and represents a tolerance of 5 percent.

Now that all the parts have been described, it is time to add a conclusion.

Conclusion Section

The final section of the mechanism description serves two purposes: it summarizes the description of the mechanism, and it provides a sense of

finality to the document.

Summarize

First, briefly summarize the mechanism's function and relist its parts:

The 33-kilohm, 1-watt carbon resistor is a circuit component that impedes the flow of electric current through the use of a carbon element. The resistor is made up of four parts: the carbon element, which impedes the flow of current by converting a portion of the electrical energy applied to heat; the wire leads, which electrically connect the element to the circuit and support the resistor mechanically; the casing, which encloses and insulates the element and dissipates heat from it; and the color bands, which indicate the resistance and tolerance of the device.

Give Sense of Finality

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Now give a sense of finality to the paper. Include a sentence that by tone and content indicates to the reader that the mechanism description is complete:

Together, these parts form one of the most commonly used circuit components in electronic systems today.

This final sentence tells the reader not to look for anything else because the document is ending; it is a courtesy to the reader.

Visuals and Mechanism Descriptions

When you are designing visuals such as diagrams, schematics, and photographs in support of a mechanism description, be sure to include only information that directly relates to the mechanism description and is specifically keyed to the description provided in the text. Do not include visuals that do not match the mechanism description in subject matter or terminology. Also, avoid visuals that include too much or too little complexity for the level of discussion in the document. See [Chapter 17](#).

- Visuals should relate specifically to the text discussion, and they should be referred to in the text before actually being used. The reader should never encounter a visual and wonder why it is there or what its purpose is. Always tell your reader specifically when to look at a

visual, and try to do so *before* the point in the document where the visual actually appears.

- Each visual should have an assigned sequence number and name, such as “Figure 5.2 Parts of the resistor.” These labels reference precisely the visuals that are included. It is a good idea to use compound numbers—to show both the section or chapter number and the sequence number. For example, in the label “Figure 5.2 Parts of the resistor,” the 5 indicates that the visual occurs in the fifth chapter of this book, and the 2 refers to the sequence in which the figure occurs within the chapter. Numbering visuals in this way allows you to add or delete visuals in one section without having to renumber all the visuals in subsequent sections. Also, if possible, run separate sets of sequence numbers for each type of visual. For example, figures should have their own set, as should equations and also tables.
- Each visual must be included for a specific purpose. Do not add visuals to make the page look “pretty” or to “add visual interest.”

[Examples 5.2](#) and [5.3](#) provide mechanism descriptions complete with visuals, including photographs, equations, and diagrams, all of which enhance the description. Besides noting how the sections and their components of each description follow [Outline 5.1](#), also pay particular attention to how the visuals follow the guidelines previously described.

Mechanism Description with Expanded Functional Theory

Normally, mechanism descriptions focus on physical attributes with only a brief treatment of the mechanism’s process of operation; however, depending on the audience and purpose, you might need to expand a mechanism description to include more information about how the mechanism and its parts work. The following mechanism description of a common wire nail includes an expanded theoretical discussion of how the nail operates. It is designed to explain the nail’s theory of operation to an average technical audience.

Description of a Common Wire Nail

Introduction

In construction, a **common wire nail** is a sharp, cylindrical fastener used to join two or more pieces of wood or other fibrous substances (called target materials). Wire nails use the force of friction to hold target materials together. Typically, a wire nail is driven into target materials with a hammer. The nail is composed of a disc-shaped head at one end of a shaft and a sharp point at the other end. See [Figures 5.4](#) and [5.5](#).

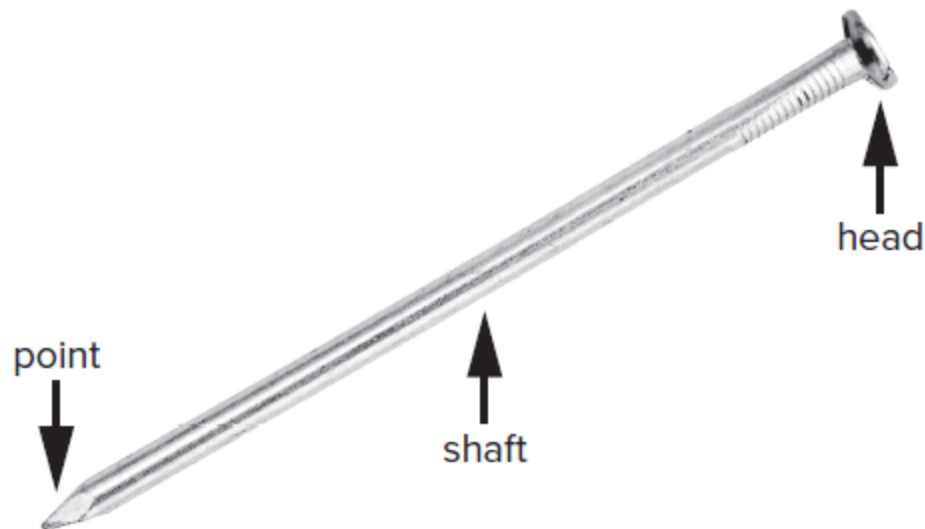


Figure 5.4

Common wire nail.

Lipskiy/Shutterstock

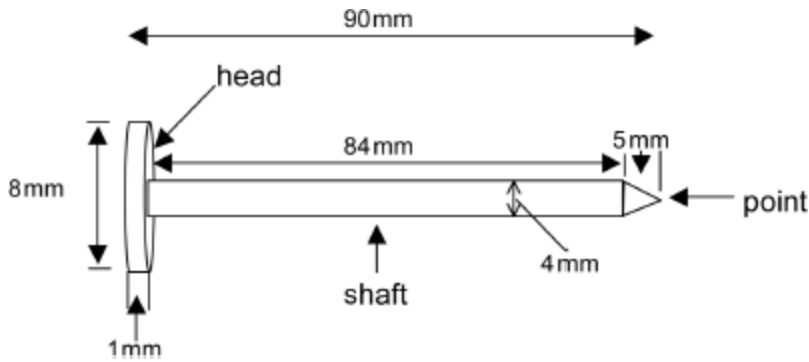


Figure 5.5

Diagram of a nail.

Overall, the nail is 90 mm long and is composed in its entirety of hardened steel. The nail is metallic gray in color with a smooth texture and shiny appearance.

Functional Overview

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The nail's head acts as an impact surface that receives a driving force from a hammer. This force is then transmitted through the shaft to the point. The point magnifies the pressure, creating an entry pathway into the target material, as well as a channel through which the shaft can move. Once driven into the target material, friction between the surface of the shaft and the surrounding material holds the nail in place. The nail consists of three functional parts: the **head**, **shaft**, and **point**. For a functional overview of the nail's operation, see [Figure 5.6](#).

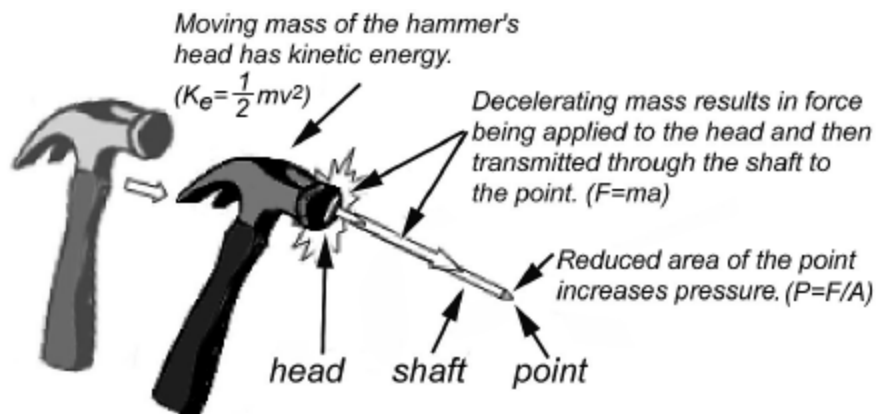


Figure 5.6

Nail's operation.

Discussion

Head

The nail's **head** is a hardened steel disc that acts as an impact surface for a hammer. The hammer receives kinetic energy (see Equation 5.1) as a result of its mass being placed in motion.

$$K_e = \frac{1}{2}mv^2 \quad \text{(Equation 5.1)}$$

where:

K_e is kinetic energy of the hammer in motion

m is mass of the hammer

v is velocity of the hammer.

The nail's head is 8 mm in diameter and 1 mm thick. It is metallic gray in color with a smooth texture and shiny appearance. When the accelerating mass of the hammer's head strikes the head of the nail, the kinetic energy of the moving hammer's head is converted into force (see Equation 5.2) that acts on the nail's head.

$$F = ma \quad \text{(Equation 5.2)}$$

where:

F is force

m is the mass of the hammer's head

a is the deceleration, or change of velocity (Δv), of the hammer's head upon striking the nail's head.

Extending from the center of the other flat side of the disc (the side opposite the striking surface) and at a right angle to the plane of the disc is the shaft.

Shaft

The **shaft** is a hardened steel rod that provides a friction surface with the target material it enters. It also transmits to the point the force received by the head. The friction between the shaft's surface and the surrounding material holds the nail in place. The shaft is 84 mm long and 4 mm in diameter. It is composed of hardened steel and has a metallic gray color with a smooth texture and shiny appearance. The shaft transmits the force acting on the nail's head to the nail's point, which facilitates the nail's entry into, and movement through, the material.

Point

The **point** is a tapered, 4-faceted wedge fashioned into the end of the steel rod comprising the shaft (see [Figure 5.7](#)). It is located on the end of the shaft opposite the end connected to the head and extends 5 mm beyond the length of the shaft. The point tapers over its 5 mm length, extending from the end of the shaft and converging on a single, geometric point. The nail's point is composed of hardened steel that is metallic gray in color. Each facet is flat and has a smooth texture and shiny appearance.

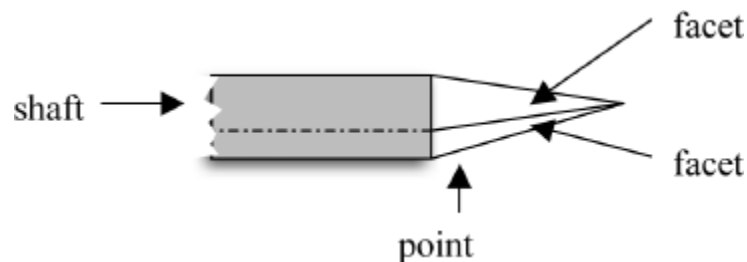


Figure 5.7

Detail of the nail's point.

The point's function is to increase the pressure resulting from the force (see Equation 5.3) transmitted from the head and through the shaft.

$$P = F/A \quad \text{(Equation 5.3)}$$

where:

P is pressure

F is force

A is the cross-sectional area of the end of the point.

The nail's point, by reducing the cross-sectional area, increases the pressure applied to the target material. The point pushes into the material, separating the material's fibers and providing an entry channel for the shaft. Once the shaft is inside the channel, the friction between the outer surface of the shaft and the channel's surrounding material holds the nail in place.

Conclusion

A **common wire nail** is a sharp, cylindrical fastener used to join together pieces of wood or other fibrous materials using friction. It consists of a head, which receives force from a hammer; a shaft, which transmits the force to the point and provides the needed friction surface; and the point, which magnifies pressure and creates a channel for the nail to enter. The common wire nail is one of the most ubiquitous fasteners used in construction today.

Specifications Using a Functional Mechanism Description

Functional mechanism descriptions used in specifications focus primarily on those physical attributes that relate directly to the device's operating parameters and capabilities. These descriptions provide precise technical information for a mechanism, including standards for its use. Functional descriptions, which are used frequently to provide specifications, normally include the following:

- a general description of the mechanism,
- visuals related to the device's structure and function, and
- a listing of physical attributes that directly affect or describe the mechanism's operational profile, capabilities, and limitations.

The following example is a codified form of a functional mechanism description called a **specification sheet**. This example provides a one-page functional mechanism description for the SuperXLTube transmitting tube. It is designed as a quick reference for an expert audience; as such, it assumes substantial knowledge about high-power transmitting tubes on the part of the reader.

Example 5.3 Codified Form of a Functional Mechanism Description

SuperXLTube Specification Sheet

General Description

The SuperXLTube high-power, ceramic tetrode is a class C transmitting tube for broadcast service at frequencies up to 150 megahertz. The tube has a thoriated-tungsten mesh filament mounted on water-vapor-cooled supports. The outer casing is composed of steel and ceramic and is circumscribed with heat radiation fins around the base of the anode dome structure extending up from the plate (P) electrode. Maximum anode dissipation is 1000 kilowatts in continuous service with high-level amplitude modulation. Water-vapor cooling ports extend from the base. Flange electrodes are used for plate, screen grid, and ground; coaxial terminals provide connectivity for the control grid and RF filament (See [Figure 5.8](#)).

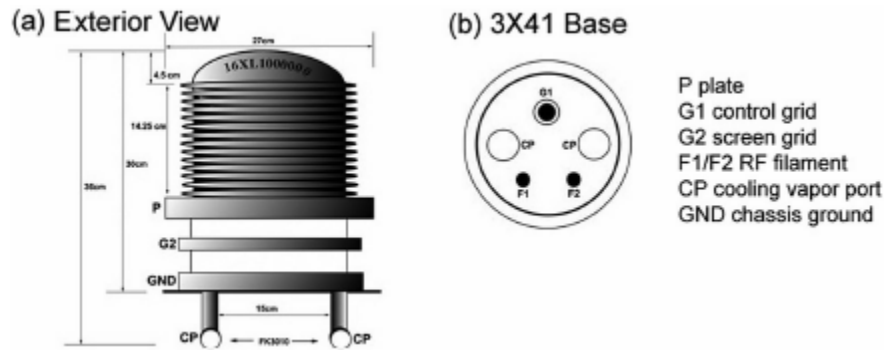


Figure 5.8

SuperXLTube Composite.

Specifications

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Maximum useful frequency = 300 megahertz

Frequency for ratings (max.) = 200 megahertz

Amplification factor (G1–G2) = 6.2

Plate dissipation (max.) = 1,400,000 watts

Plate voltage (typ.) = 22 kv

Plate current (typ.) = 60 amperes

G2 dissipation (max.) = 10,000 watts

G1 dissipation (max.) = 2000 watts

Heat dissipation (max.) = 800,000 watts

Cooling = water vapor

Cooling port connector = FK-3010

Core temperature (max.) = 2008C

Weight = 98 kilograms

Capacitance (cathode GND) in = 1.1 millifarad

Capacitance (grid GND) out = 175 picofarad

Capacitance (grid GND) in = 490 picofarad

Capacitance (feed-through) = 6 picofarad

Capacitance (cathode GND) out = 185 picofarad

Capacitance (feed-through) = 0.8 picofarad

Filament = thoriated-tungsten mesh

Filament power = 16.2 volts @ 600 amperes

Base = 3X41 coaxial

Filament RF connector = FK-3000

Grid RF connector = FK-3010

Mechanism Description Checklist

- ☐ Do I understand who my audience is for this description?
- ☐ Do I know how this description will be used?
- ☐ Have I considered all of the relevant context for the situation?
- ☐ Have I defined the mechanism, and have I extended this definition with any theory or principles necessary for my reader's understanding?
- ☐ Have I included theory or principles that are neither above nor below the level of my reader?
- ☐ Have I described the mechanism's overall function or purpose?
- ☐ Have I described the overall appearance of the mechanism in terms of its shape, color, material, finish, texture, weight/mass and/or size?
- ☐ Have I listed the main parts in the introduction in the order that they will be described?
- ☐ Have I defined each of these parts?
- ☐ Have I described each part's function or purpose?

- ❑ Have I discussed the needed theory or operating principles for each part?
- ❑ Have I precisely described each part's physical structure?
- ❑ Have I provided transitions from one part to the next?
- ❑ Have I concluded by summarizing the mechanism's function and parts?
- ❑ Have I given a sense of finality to the description?
- ❑ Have I only provided visuals that relate to my text?
- ❑ Have I referred to the visuals in my text before they are used by my audience?
- ❑ Have I given each visual an assigned sequence number and name?
- ❑ Have I run separate sets of sequence numbers for different types of visuals?
- ❑ Does each visual have a specific purpose?

Exercise

Apply the checklist just given to the following mechanism description of the FinkelBOAT 688 Attack Submarine.

- What problems exist? Are any parts missing? Do the definitions work? Are any of them circular? Do some require additional definitions or extensions?
- Can you determine the intended audience and the purpose of the description? Pay particular attention not only to the size and

complexity of this mechanism, but also to the purpose, presentation, and quality aspects of the description.

- What about the visuals? Is the level of complexity appropriate for this description? Are the visuals properly marked? Do they appear in the correct place? What purpose, if any, do they serve? How well integrated are the visuals into the text's description?
- What sections provide precise technical writing, and what sections feature abstract, imprecise concepts?

Mechanism Description of a FinkelBOAT 688 Attack Submarine

Introduction

The **FinkelBOAT 688 submarine** is a nuclear-powered, attack, undersea watercraft characterized by high-speed, ultra-quiet operation, high cost, and an awesome array of conventional and special weapons including unicorn dust. The submarine, which is used primarily for peaceful undersea research, has a top submerged speed of 44 knots and carries both UGM-109 cruise missiles and Mk-48 torpedoes (see [Figure 5.10](#)). The FinkelBOAT 688 consists of a main tube, two hemispheric caps, a propeller, and a fairwater.

Discussion

Main Tube

The **main tube** is a claustrophobic cylinder of pure YM-31 high-tensile titanium that provides the body of the boat. The main tube is 75 feet long, 6 inches thick, and has an inside diameter of 33 feet. The main tube is covered with an anechoic/decoupling glitter coating designed to defeat active sonars and contain internal noise. At each end of the main tube are hemispheric caps. See [Figure 5.9](#).

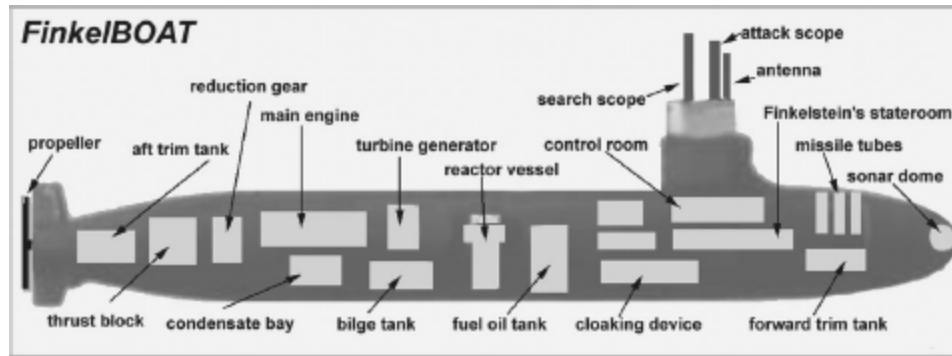


Figure 5.9

FinkelBOAT.

Hemispheric Caps

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The **hemispheric caps** are cones of pure YM-31 high-tensile titanium that enclose both the stern and bow ends of the main tube, thereby completing the substructure of the boat. The caps are tapered and welded to the main tube's barrel sections. The boat's propeller is located directly behind the stern hemispheric cap.

Propeller

The **propeller** is a seven-blade propeller that provides high thrust with little or no cavitation. The propeller, which is made of a special bronze alloy, is 30 feet in diameter and can silently move 500 acre-feet of water per second at flank speed. On the top of the substructure, and 55 feet forward toward the bow hemispheric cap, is the fairwater.

Fairwater

The **fairwater** is the raised, enclosed observation post of the submarine that supports periscopes and communications antennas ([Figure 5.10](#)) and that also provides a modest bridge area. Often called a sail or conning tower, the fairwater, which is 25 feet high, 15 feet long, and 10 feet wide, is also composed of pure YM-31 high-tensile titanium.

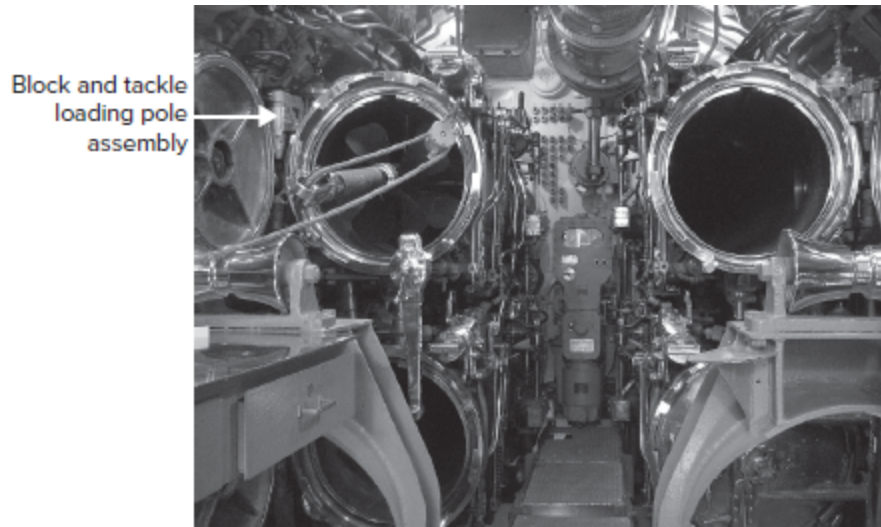


Figure 5.10¹

The after-torpedo room. Note the use of a block and tackle and loading pole to position torpedoes in the tube.

Leo Finkelstein



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Figure 5.11²

Fairwater of nearby submarine seen through the FinkelBOAT's attack periscope.

Leo Finkelstein

Conclusion

As [Figure 5.9](#) shows, the FinkelBOAT has many parts that work together to make a submarine.

Notes

- [1.](#) Photo taken aboard the U.S.S. *Bowfin* at the U.S.S. Bowfin Submarine Museum and Park, Pearl Harbor, Hawaii, August 2005.
- [2.](#) Photo of the U.S.S. *Bowfin* taken through a nearby display periscope at the U.S.S. Bowfin Submarine Museum and Park, Pearl Harbor, Hawaii, August 2005.

¹By “Discussion,” we mean “body of the document.”

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CHAPTER SIX

Description of a Process



Process descriptions are similar to mechanism descriptions in organizational structure, but they differ in content. Whereas mechanism descriptions detail the physical attributes of a mechanism, process descriptions describe the steps of a process. You might be thinking you would rather poke your eyes out with a stick than have to write such a (dry? boring?) document, but once again, we are talking about one of the fundamental “working dogs” of technical writing.

Processes are important in technical writing because much of what must be communicated involves unfolding events or actions. In fact, process descriptions are key elements of **many** technical reports. Process descriptions are typically objective, third-person portrayals of events that do not directly involve the reader, such as explaining how a space shuttle operates, how an airplane flies, how a piano makes sound, or the biomechanics of a golf swing. In the case of the Samoyed dog sledding team (see [Figure 6.1](#)), instead of talking about the description of the *sled*, we would perhaps describe the process for *dog sledding*: How many dogs? What is the order and function of each dog? What is the best order for harnessing dogs based on ability and personality? How much weight can they pull? How many hours can they run? What temperatures and wind chills can they handle? When do we yell, “Mush!”? All of these questions and more might be addressed in a process description. The question you might have is *why*? What is the purpose of a process description?



Figure 6.1

A process description for how to run a sled dog team would be very helpful for a rookie Iditarod competitor.

What Is a Process Description?

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Process descriptions are precise portrayals of events occurring over time that lead to a desired outcome. Process descriptions describe either the steps in the operation of a mechanism or the steps of a conceptual process. Unlike mechanism descriptions, a process description of an operational mechanism focuses less on the physical attributes of the mechanism, and more on a mechanism's function and how the parts work together. For example, a description of a car's four-cycle, internal combustion engine in operation would not focus on the parts (such as the pistons, rods, cylinders, valves, and spark plugs); rather, the description would focus on the steps of the engine's operation (such as intake stroke, compression stroke, ignition stroke, and exhaust stroke).

Know Your Audience

If you are describing a process for a patent application, this has a very different audience from the typical audiences that we are addressing

in this textbook. See [Chapter 3](#) for a teaser on legal implications associated with patent law, but know that you will want to consult a patent attorney if your audience is at that level of need.

Third vs. Second Person

Many process descriptions are written in third person: “1. The piston moves down and draws in a mixture of air and fuel. 2. The piston moves up, compressing the air-fuel mixture.” Third person does not tell the reader what to do, but explains the process as an observer.

However, some process descriptions are written in second person as directives: “1. Do this. 2. Do that.” These steps **imply** a directive to the reader, i.e., “You do this. You do that.” An example of an explicitly second person process description is seen in [Outline 6.1](#). These descriptions share similarities with instructions (see [Chapter 7](#)).

As with mechanisms in operation, conceptual process descriptions discuss the steps of a process, but these steps do not involve a physical mechanism. For example, a process description for operations research might include a decision tree; however, this tree is not the kind you have to plant, water, and fertilize. In fact, it is not a tangible thing at all. It exists only conceptually in the imagination. The same is true for a conceptual system. In a laboratory, there might be a specific protocol for obtaining approval to conduct research which involves human subjects. The system’s process description describes how multiple entities, including the research team members, the institutional research board, the grantor, and others, must follow a specific and sequential set of steps to get approval before data are collected. Getting approval is the desired outcome, and by clearly defining the process description, the likelihood of future projects gaining approval increases. Whether a process involves something physical is irrelevant. The organization of a process description is basically the same, whether it describes a mechanism in operation or a conceptual process.

Outline of a Process Description

Notice that the two outlines for process descriptions are similar. [Outline 6.1](#) shows a model for describing the operation of a mechanism, such as an air conditioning system that includes compression, condensation, expansion, and evaporation as its steps. [Outline 6.2](#) provides a similar pattern for describing a conceptual process, such as a sort algorithm that includes the iterative steps of evaluation, identification, and insertion.

Outline 6.1 Process Description of a Mechanism in Operation

Introduction

- Define the mechanism with a technical definition (see [Chapter 4](#)), and add extensions to discuss any theory or principles necessary for the reader to understand what you are saying. Make sure you include only what the reader needs for your purpose.
- Describe the purpose, function, and operation of the mechanism.
- List the major steps of the mechanism's operation in the order that they will be discussed.

Discussion

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- **Step 1**
 - Define the step with a logical definition and discuss it in reasonable depth.
 - Describe the equipment, material, or concepts involved in this step.
 - Describe what happens during this step.
 - Show the relationship between this step and the next step with a transition statement.
- **Steps 2–n**
 - For each remaining step, repeat the pattern established for Step 1. The final step discussed may not have a transition, unless it cycles back to a previous step.

Conclusion

- Briefly summarize the mechanism's function and the major steps of its operation.
- Give a sense of finality to the paper.

Outline 6.2 Description of a Conceptual Process

Introduction

- Define the process with a technical definition (see [Chapter 4](#)), and add extensions to discuss any theory or principles necessary for the reader to understand what you are saying. Make sure you include only what the reader needs for your purpose.
- Describe the purpose and function of the process.
- List the major steps of the process in the order that they will be described.

Discussion

- **Step 1**
 - Define the step with a logical definition, and discuss it in reasonable depth.
 - Describe the purpose and function of this step.
 - Describe what happens during this step.
 - Show the relationship between this step and the next step with a transition statement.
- **Steps 2–n**
 - For each remaining step, repeat the pattern established for Step 1. The final step discussed may not have a transition, unless it cycles back to a previous step.

Conclusion

- Briefly summarize the function and the major steps of the process.
- Give a sense of finality to the paper.

This chapter has three descriptions that illustrate the use of these models to describe processes. The first example develops a process description of an air conditioner's operation. The second example describes the purely conceptual process of a sort algorithm, which does not involve a material mechanism. The third example describes the process of softening water.

To provide an example using [Outline 6.1](#), this chapter will describe the process of the operation of a mechanism: in this case, a notional air conditioning system. The audience for this description is a general, technical reader who does not have specific expertise in air conditioning, fluid mechanics, or thermodynamics.

Example 6.1 Description of a Mechanism in Operation

Description of the Operation of an Air Conditioner

Introduction

An **air conditioner** is a mechanical device used to refrigerate a controlled environment. The air conditioner accomplishes this refrigeration by transferring heat within the environment to an area outside the environment. The air conditioner transfers heat by using a fluid refrigerant. This refrigerant is pumped through both the controlled environment and the outside area. At the same time, the refrigerant is cycled at strategic points between liquid and vaporous states. This change in state provides the means for transferring thermal energy.

The air conditioner's operation is centered on four major components: the compressor, condenser coil, expansion valve, and evaporator coil. The operation of the air conditioner relates directly to these parts and includes the following steps: compression, condensation, expansion, and evaporation. See [Figure 6.2](#).

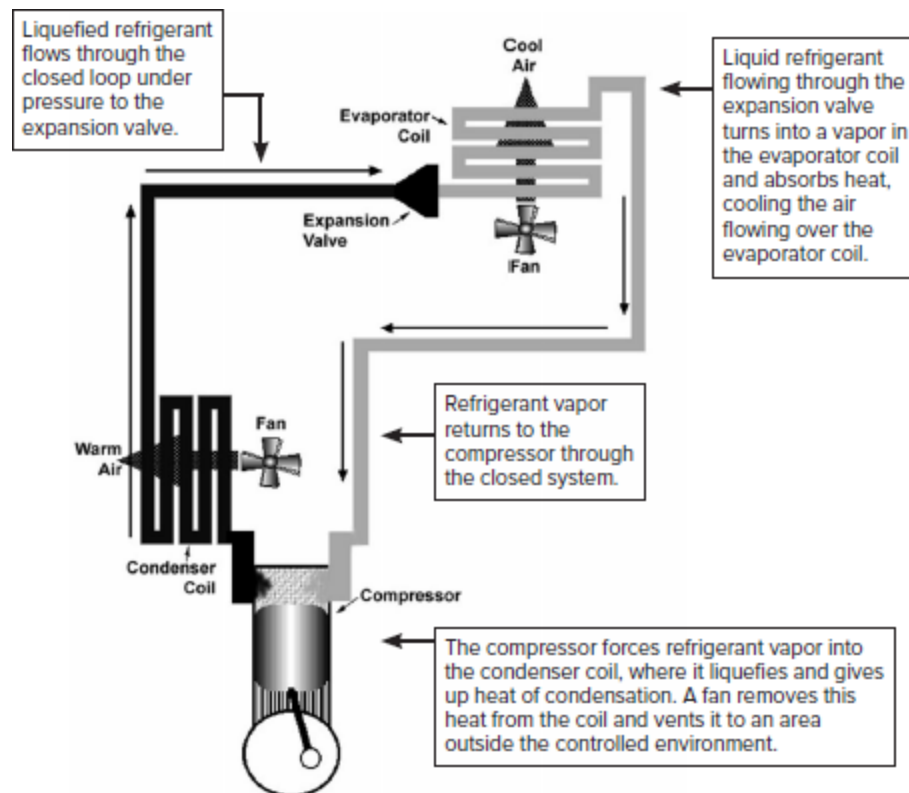


Figure 6.2

Air conditioning process.

Discussion

Compression

Compression is a fluid-dynamics process in which a given volume of refrigerant vapor is forced to occupy a smaller volume of space. Compression occurs when the compressor forces hot refrigerant vapor under pressure into the compression chamber. The chamber is composed of a cylinder/valve arrangement. The piston draws refrigerant into the cylinder through an intake valve; the intake valve closes; and the piston

pushes up into the cylinder, compressing the refrigerant vapor. The vapor then exits through an exhaust valve and enters the condenser coil, where the condensation step occurs.

Condensation

Condensation is a fluid-dynamics process in which the hot refrigerant vapors, pumped from the compressor to the condenser coil, cool and change into a liquid. As this change occurs, the heat of condensation is released from the refrigerant in the condenser coil, heating the coil. This heat, in turn, is drawn from the coil by a fan, which passes relatively cooler air across the coil, picking up the heat and venting it outside. The refrigerant liquid then flows through a closed loop to the expansion valve, where rapid expansion occurs.

Expansion

Expansion is a fluid-dynamics process in which the condensed liquid refrigerant, under relatively high pressure from the compressor, is forced through an expansion valve into an area of substantially lower pressure. The expansion valve acts as a nozzle, constricting and then accelerating the liquid refrigerant until it passes through a threshold where the constriction is removed and rapid expansion occurs. At this point, the expanding refrigerant enters the evaporator coil, where it will cool and change state again.

Evaporation

Evaporation is a fluid-dynamics process in which the rapidly expanding refrigerant liquid changes into a vapor. As the liquid enters the evaporator coil, it also enters an area of substantially lower pressure. As a result, it vaporizes and, in the process, absorbs heat. Air in the controlled environment is circulated across this coil, which, in turn, absorbs heat from the air. The fan distributes the resulting cooler air throughout the controlled environment. The refrigerant vapor in the evaporator coil is then drawn back through the closed loop to the compressor, at which point the entire cycle repeats.

Conclusion

The operation of an air conditioner involves four steps. First, the compressor pumps refrigerant under pressure into the condenser coil. Here it is liquefied, giving up heat that is removed by a fan circulating air over the condenser coil. The liquid refrigerant moves through a closed loop, through an expansion valve, and into the lower-pressure evaporator coil. Here, the refrigerant changes into a vapor, absorbing heat from air passing over the evaporator coil. This air then cools the controlled environment, while the refrigerant is drawn back into the compressor. At this point the cycle is complete, and the process repeats.

Dissecting [Example 6.1](#)

Now that you have read an example of a process description, let us take a closer look at each of its components and what they do.

Introduction

Begin by introducing the mechanism with a logical definition (see [Chapter 4](#) for a refresher); then add extensions to provide any needed theory or principles:

An **air conditioner** is a mechanical device used to refrigerate a controlled environment. The air conditioner accomplishes this refrigeration by transferring heat within the environment to an area outside the environment.

Next describe the function and operation of the mechanism:

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The air conditioner transfers heat by using a fluid refrigerant. This refrigerant is pumped through both the controlled environment and the outside area. At the same time, the refrigerant is cycled at strategic points between liquid and vaporous states. This change in state provides the means for transferring thermal energy.

You may want to list the primary operational parts of the mechanism, especially if they relate directly to the process steps. In any case, be sure to list the major steps of the mechanism's operation in the order in which they will be described. Choosing this order may not be easy. Ideally, the steps will follow a logical timeline from the start to the finish of the process;

however, in some cases, such as transactional, iterative, and branching processes, that may not be possible. In such cases, you must describe the process steps in the way that is clearest for your reader and the purpose at hand.

The air conditioner's operation is centered on four major components: the compressor, condenser coil, expansion valve, and evaporator coil. The operation of the air conditioner relates directly to these parts and includes the following steps: compression, condensation, expansion, and evaporation.

Discussion

In the discussion section, treat each step as a separate subsection. Define each step of the process; then extend these definitions as necessary to address the equipment, material, or concepts involved. Be sure to tell your reader clearly what happens in each step and provide linking transitions between the steps.

Compression

Discuss **compression** in reasonable depth for the audience and purpose at hand. First define compression, and then deal with the equipment and concepts involved. Describe what happens during compression, and then show the relationship with the next step, condensation:

Compression is a fluid-dynamics process in which a given volume of refrigerant vapor is forced to occupy a smaller volume of space. Compression occurs when the compressor forces hot refrigerant vapor under pressure into the compression chamber. The chamber is composed of a cylinder/valve arrangement. The piston draws refrigerant into the cylinder through an intake valve; the intake valve closes; and the piston pushes up into the cylinder, compressing the refrigerant vapor. The vapor then exits through an exhaust valve and enters the condenser coil, where the condensation step occurs.

Condensation

Describe **condensation** in much the same way as you dealt with compression. Start with a technical definition of condensation, and then extend this definition as necessary to describe the parts of the mechanism involved and the concepts and theory the reader may need. Next describe what happens in this step of the process, and transition to the third step, expansion:

Condensation is a fluid-dynamics process in which the hot refrigerant vapors, pumped from the compressor to the condenser coil, cool and change to a liquid. As this change occurs, the heat of condensation is released from the refrigerant in the condenser coil, heating the coil. This heat, in turn, is drawn from the coil by a fan, which passes relatively cooler air across the coil, picking up the heat and venting it outside. The refrigerant liquid then flows through a closed loop to the expansion valve, where rapid expansion occurs.

Expansion

Treat this third step in the same manner as the first two. First define **expansion**, and then extend the definition to describe the parts of the mechanism involved and to provide needed theory and concepts. Explain what happens during this step, and provide a transition to the fourth step, evaporation:

Expansion is a fluid-dynamics process in which the condensed liquid refrigerant, under relatively high pressure from the compressor, is forced through an expansion valve into an area of substantially lower pressure. The expansion valve acts as a nozzle, constricting and then accelerating the liquid refrigerant until it passes through a threshold where the constriction is removed and rapid expansion occurs. At this point, the expanding refrigerant enters the evaporator coil, where it will cool and change state again.

Evaporation

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Finally, define the last step, **evaporation**; extend the definition; and describe what happens, as in earlier steps. However, because this process is cyclical and repetitive, the transition from this last step should link back to the first step.

Evaporation is a fluid-dynamics process in which the rapidly expanding refrigerant liquid changes to a vapor. As the liquid enters the evaporator coil, it also enters an area of substantially lower pressure. As a result, it vaporizes and, in the process, absorbs heat. Air in the controlled environment is circulated across this coil, which, in turn, absorbs heat from the air. The fan distributes the resulting cooler air throughout the controlled environment. The refrigerant vapor in the evaporator coil is then drawn back through the closed loop to the compressor, at which point the entire cycle repeats.

Conclusion

The last part of the process description summarizes the process and provides a sense of finality as a courtesy to the reader:

The operation of an air conditioner involves four steps. First, the compressor pumps refrigerant under pressure into the condenser coil. Here it is liquefied, giving up heat that is removed by a fan circulating air over the condenser coil. The liquid refrigerant moves through a closed loop, through an expansion valve, and into the lower pressure evaporator coil. Here

the refrigerant changes into a vapor, absorbing heat from air passing over the evaporator coil. This air then cools the controlled environment, while the refrigerant is drawn back into the compressor. At this point the cycle is complete, and the process repeats.

Visuals and Process Descriptions

When you are designing visuals such as diagrams, schematics, and photographs in support of a process description, be sure to include only information that directly relates to the process description and is specifically keyed to the description provided in the text. Do not include visuals that do not match the process description in subject matter or terminology. Also, avoid visuals that include too much or too little complexity for the level of discussion in the document. See [Chapter 17](#).

- Visuals should relate specifically to the text discussion, and they should be referred to in the text before actually being used. The reader should never encounter a visual and wonder why it is there or what its purpose is. Always tell your reader specifically when to look at a visual, and try to do so *before* the point in the document where the visual actually appears.
- Each visual should have an assigned sequence number and name, such as “Figure 6.2 Air conditioning process.” These labels reference precisely the visuals that are included. It is a good idea to use compound numbers—to show both the section or chapter number and the sequence number. For example, in the label “Figure 6.2 Air conditioning process,” the **6** indicates that the visual occurs in the sixth chapter of this book, and the **2** refers to the sequence in which the figure occurs within the chapter. Numbering visuals in this way allows you to add or delete visuals in one section without having to renumber all the visuals in subsequent sections. Also, if possible, run separate sets of sequence numbers for each type of visual. For example, figures should have their own set, as should equations and also tables.
- Each visual must be included for a specific purpose. Do not add visuals to make the page look “pretty” or to “add visual interest.”

There is one important additional guideline for providing visuals for process descriptions: process visuals usually need to show something happening and, as such, tend not to be static representations of something seen in a “slice of time.” Often you can show movement through space and time with a technique as simple as a well-placed arrow. At other times you may need a coherent series of visuals to demonstrate your point.

The following is a brief, notional example of an iterative, [Page 60](#) conceptual process. Computer programmers use this particular process, which is roughly analogous to a selection sort algorithm, to put lists of unsorted data into either ascending or descending order. In reality, the actual implementation of a selection sort algorithm in modern, higher-level languages is far more complex and efficient than what is represented here.

This example, which integrates two visuals into the description, is written for a general, technical audience that does not have specific expertise in computer programming and information systems. For illustration, this example process for sorting data in descending order uses the following list of six numbers: 6, 32, 8, 19, 3, 20.

Example 6.2 Description of a Conceptual Process

Process Description of Selection Sort

Introduction

Selection sort is an iterative process in which a list of unsorted data is placed in either ascending or descending order. Selection sort gets its name from the way in which the data are sorted. During each iteration, the smallest value from a list of unsorted numbers is selected and then inserted in the front of the list. As the process continues, the sorted area in the front of the list gets larger, while the remaining unsorted area gets smaller. At some point, when all the numbers from the unsorted list have

been selected and inserted, the list is sorted, and the process is complete. The selection sort process involves the following iterative steps: evaluation, identification, and insertion. See [Figure 6.3](#) for a flowchart of the process.

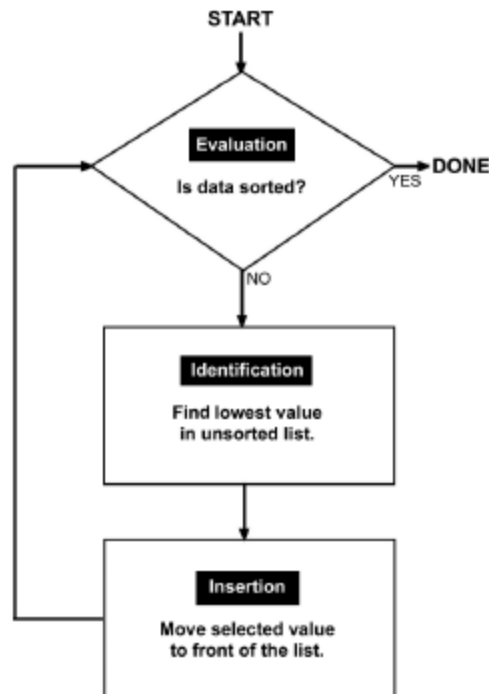


Figure 6.3

Selection sort flowchart.

Discussion

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The following example sorts the list in descending order.

Evaluation

Evaluation is the scanning process in which the list of numbers is tested for proper order. This step normally involves a pass through the list where a series of comparisons of adjacent numbers determines whether the list is ordered. If the list is ordered, the process is complete and ends. If the list is not ordered, then the lowest value is identified.

Identification

Identification is the scanning process by which the lowest value in the list is located. Normally, the lowest value is identified by a series of comparisons and swaps. Once identified, this value can be inserted.

Insertion

Insertion is a relocation process in which the lowest value in the unsorted area of the list is placed or inserted in the front of the list. On the first iteration, this value starts the ordered or sorted area of the list, while the remaining numbers constitute the unordered or unsorted area of the list. Once the lowest value has been inserted, the process returns to the evaluation step, where the iteration continues until no unordered numbers remain (see [Figure 6.4](#) for an example).

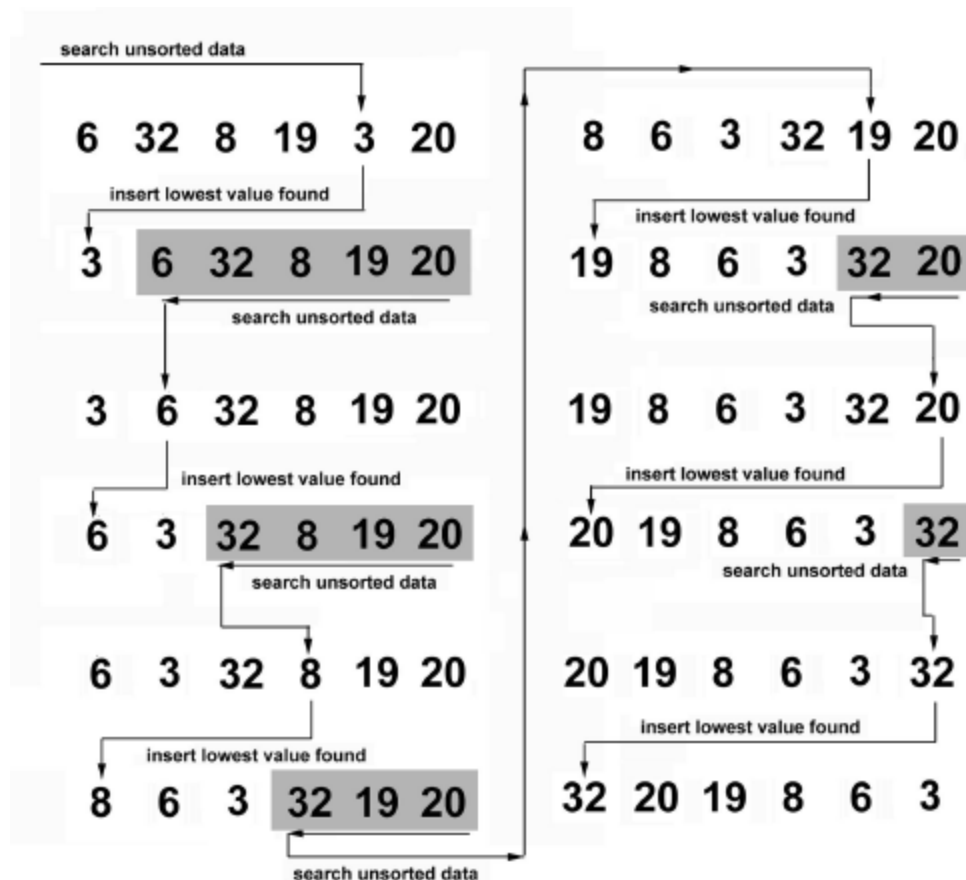


Figure 6.4

Selection sort.

Selection sort is an iterative process in which a list of unsorted data is placed in either ascending or descending order. During each iteration, the list is evaluated to see if it is, in fact, ordered. If not, the smallest value from a list of unsorted numbers is selected and then moved to the front of the list. As the process continues, the sorted area in the front of the list gets larger, while the remaining unsorted area gets smaller. At some point, when all the numbers from the unsorted list have been moved, the list is sorted and the process is complete. Insertion sort is one of the many algorithms used by computer programmers to sort lists of data.

The content and detail of process descriptions will vary significantly depending on the purpose and audience. The following brief example describes the water softening process frequently used by home water softeners. Notice that the emphasis here is on the ion exchange process by which the water softener works and not on the water softener itself. The topic is effectively treated as a conceptual chemical process, even though a mechanism (the water softener) is involved. In other words, the focus is on ion exchange in describing the process, while the parts of the actual water softener that accomplish this ion exchange (resin tank, brine tank, timer/controller, etc.) are largely ignored. The resin tank and brine tank are mentioned in passing as needed, but the timing mechanism that controls the process is not mentioned at all. Also notice that no diagrams are provided—only the chemical equations that describe the ion exchange process.

Example 6.3 Description of a Physical Process

Process Description of Water Softening

Introduction

Water softening is a chemical process by which certain minerals contained in hard water are displaced by sodium. Hard water is “hard” primarily because of the dissolved calcium it contains. The calcium in solution degrades the effectiveness of soaps and detergents by preventing soap from dissolving completely, forming a coagulated soap curd instead. Not only does incomplete dissolution result in the need for more soap, the resulting curd and available calcite (hard form of calcium) also cling to the materials being washed and can damage plumbing systems. The water softener contains two tanks: an exchange resin tank and a brine tank. Both tanks are used in the softening process, which includes **exchanging** and **regeneration**.

Discussion

Exchanging

Exchanging is the process where calcium ions (Ca) in the water displace sodium ions (Na) attached to the resin. The resin is composed of polystyrene beads of zeolite, which carry a negative charge. Attached to the resin are sodium ions, which carry a positive charge. Calcium in water carries a stronger positive charge than sodium. As hard water passes through the resin (R), the calcium that the hard water contains in solution clings to the beads, displacing the sodium ions from the resin into the water. The result is water containing sodium chloride (NaCl) and a resin rich in calcium chloride (CaCl₂) (Equation 6.1).



Regeneration

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Regeneration is the process by which calcium ions are flushed from the resin and replaced with sodium ions. First, the resin is backwashed with water to remove any dirt and debris. Next, the resin is recharged with a strong sodium chloride brine solution flowing through it. The sodium

ions cling to the resin, displacing the calcium ions collected earlier in the **exchanging** step (Equation 6.2).



The resin is then flushed of excess brine, and the brine tank is refilled. The water softener is now ready for the process to repeat.

Conclusion

Water softening is the process of removing calcium from hard water to improve the function of soaps and detergents, and to prevent the negative effects of hard water curd and calcite deposits. The process exchanges sodium ions for calcium ions. The resulting soft water is free of the calcium that compromises its use for cleaning and causes damage to plumbing systems.

Process Description Checklist

- ☐ Have I defined the mechanism or the process, and have I extended this definition with any theory or principles necessary for the reader's understanding? Have I included theory or principles that either are not needed or are above or below the level of the reader?
- ☐ Have I described the process's overall function or purpose?
- ☐ Have I described the overall function or operation involved in the process?
- ☐ Have I listed the main steps of the process in the introduction in the order that they are discussed?

- ❑ Have I defined each of these steps?
- ❑ Have I described each step's function or purpose?
- ❑ Have I discussed the needed theory or operating principles for each step?
- ❑ Have I precisely described what happens in each step?
- ❑ Have I provided transitions from one step to the next step?
- ❑ For iterative processes, have I clearly shown when branching occurs and when the process is complete?
- ❑ Have I concluded by summarizing the process's purpose and function?
- ❑ Have I given a sense of finality to the description?
- ❑ Have I only provided visuals that relate to my text?
- ❑ Do the visuals show something happening (dynamic, not static) whether as an individual visual or a series?
- ❑ Have I referred to the visuals in my text before they are used by my audience?
- ❑ Have I given each visual an assigned sequence number and name?
- ❑ Have I run separate sets of sequence numbers for each type of visual?
- ❑ Does each visual have a specific purpose?

Exercise

Apply the checklist just given to the following process description of the QuadFINKEl figure skating jump. In particular, look at the following:

- Is the subject a process? If so, what areas of this process description are not technical writing? Do any areas deal with abstract, imprecise value judgments and concepts? Also, does this description involve a mechanism in operation, and if so, what is the mechanism?
- For what audience was this description written? Is there a consistent audience? What areas of expertise would be required for the reader to understand the various parts? At what level of expertise is this paper written?
- How is the paper organized? Does the introduction properly introduce the process? Are the steps well defined? Are they adequately described? Does each step transition well to the next step?
- Look at the figures. One exists for each step of the process. Are these visuals useful? Are they properly marked? Do they have adequate labeling? Do they all match the text discussion? Are they placed effectively in the document? Are they well integrated into the text's description?
- Look at the photograph. Does it add anything of value to the description? Is it ethical to use it? If not, why not?

Process Description of a QuadFINKEl

Introduction

In figure skating, the **QuadFINKEl** is a competitive jump that combines a Lutz glide, Axel vault, high-speed spin, and Salchow landing. The QuadFINKEl is regarded by the skating world as the most difficult maneuver in the history of the sport. The QuadFINKEl, which requires a skater of great power, skill, and courage, has six steps: the glide, the turn, the vault, the spin, the stall, and the landing (see [Figure 6.5](#)).

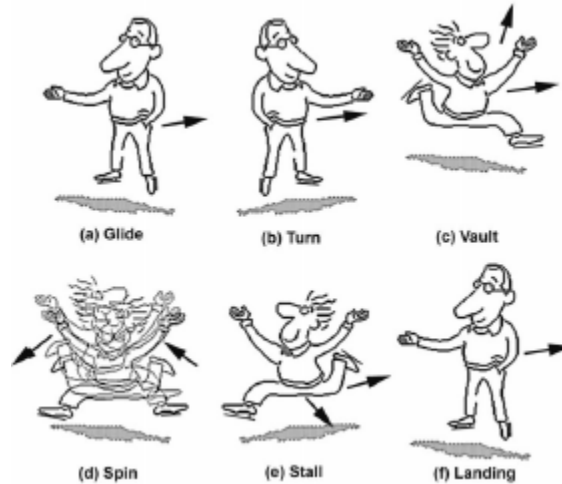


Figure 6.5

QuadFINKEL.



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Figure 6.6

Landing the QuadFINKEL. A sports photographer's camera captures Finkelstein a few moments after he lands the QuadFINKEL on a frozen lake in East Greenbush, New York. Notice the dissipating low-pressure vortex trailing him, as well as the remaining boundary layer turbulence along the length of his pant legs.

Leo Finkelstein

Discussion

The Glide

The **glide** is an acceleration maneuver by which the skater establishes high-speed movement on the right back outside edge. As the glide speed approaches 55 miles per hour, the left hand is inserted into the left pocket to reduce aerodynamic drag, while the right hand is extended to provide rudder control; see [Figure 6.5\(a\)](#). When backward velocity reaches 60 miles per hour, the skater is ready to turn.

The Turn

The **turn** is a transition maneuver in which the skater's body rotates 180° counterclockwise while maintaining the original vector. Additionally, the skater shifts from the right back outside edge to the left front outside edge and spreads both arms in preparation for the vault; see [Figure 6.5\(b\)](#).

The Vault

The **vault** is a propulsion maneuver in which the skater leaps off the right front outside edge, accomplishing an Axel-entry blade jump. Once airborne, the skater extends both hands up and away from the body to provide rotational stability and artistic elegance; see [Figure 6.5\(c\)](#). This action prepares the skater for the spin.

The Spin

The **spin** is a rotational maneuver in which the airborne skater rapidly moves their extended arms and hands to create both substantial artistic impression and a low-pressure vortex with resulting counterclockwise torque; see [Figure 6.5\(d\)](#). Generally, spin speeds exceeding 2000 revolutions per minute are possible using this technique. The skater then turns their arms to simulate a Bernoulli surface, which provides sufficient aerodynamic lift to sustain autogyration flight. (Note: In the southern hemisphere the rotation would be clockwise, so the QuadFINKEL is not possible south of the equator.) Once the rotational torque is established and the skater has made three revolutions, they are ready for the stall.

The Stall

The **stall** is an aerodynamic maneuver in which the skater uses their outstretched hands as spoilers to slow rotation to the point where lift is no longer sufficient to sustain flight; see [Figure 6.5\(e\)](#). This action is

taken after three revolutions, so that a stall condition occurs at precisely four revolutions. The stall leads immediately to landing.

The Landing

The **landing** is the final jump maneuver by which the skater touches down on the right back outside edge; see Figure 6.5(f). The skater's left hand is reinserted in the pocket while the outstretched right hand is used again for rudder control. The left edge must not touch the ice until the skater's speed has been reduced to zero. At this point, the QuadFINDEL jump is complete.

Conclusion

The QuadFINDEL is the most difficult of all competitive figure skating jumps. To do a QuadFINDEL, the skater does an Axel vault off the left front outside edge after a Lutz acceleration glide on the right back outside edge. The jump uses sophisticated aerodynamic forces to sustain a high-speed spin, followed by a Salchow landing on the right back outside edge. Because of its hypnotic artistic impression and demanding technical qualities, it has been said that "One who achieves the QuadFINDEL always lands on the gold."

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CHAPTER SEVEN

Instructions and Manuals



You’ve read instructions before: “Do this. Do that. Do this next.” And so it continues until you have accomplished the job or (yikes!) given up. Unclear instructions are incredibly frustrating, so to help achieve clarity, they are often written in the second person with an implied **you**. “[You] do this. [You] do that.” The brevity increases their readability, though when on the receiving end of instructions, you might interpret them as demanding.

Explaining how to do something can be a challenging task. If you do not believe it, try showing a young child how to tie their shoes or explaining to an elderly neighbor how to use the new smart phone they just received for their birthday. Even more challenging, consider how difficult it is to give instructions across species—a well-trained guide dog does just that for its visually impaired owner. The guide dog must very clearly, efficiently, and safely convey what the owner must do. The dog takes its owner through a process: “Stop here. Wait. [It is safe so] walk” (see [Figure 7.1](#)).



Figure 7.1

A guide dog gives clear, efficient, and safe instructions.

Both instructions and manuals describe processes so that readers can accomplish the steps required to successfully complete a task. In technical writing, instructions deal with narrowly defined topics where the goal is to explain how to do something and not much more. Broader requirements demand complex descriptions in the form of manuals. Both instructions and manuals have the same objective, but manuals contain more than just instructions for carrying out a simple task. Fundamentally, however, they both do the same thing and work the same way.

What Are Instructions?

Instructions are process descriptions for human involvement. Instructions are second-person descriptions for human involvement that provide specific

directions so that the reader can perform or accomplish the process. When you give instructions, you are describing not only the steps of a process, but also how to accomplish these steps. That means you are accepting the additional responsibilities of describing the process accurately and in a way that your readers can follow safely and effectively. It is the difference between describing to a medical student what happens when an inflamed appendix is removed and describing to a surgical resident, hovering over a patient with scalpel in hand, how to remove an inflamed appendix. To appreciate the difference, just imagine that you are the patient!

Instructions can be written as formal or informal, and can be presented as hardcopy or electronic. Formality is **not** determined by length. It is determined by criticality. You might be tasked with writing very formal instructions that must also be as concise as possible; for example, if you are a process engineer for a tractor assembly line, one of your responsibilities might be to write assembly instructions for the line operators. These instructions could be provided in hardcopy format, but increasingly they are provided online, accessible via monitors at workstations. If an operator is supposed to build tractor model X1L, they click on the instructions for X1L and the standard work required to complete the work at that station opens. These instructions must be very concise and very visual. They must provide clear, efficient, and safe sequential steps. Often, a picture really is worth a thousand words—the operator can understand what to do much faster and better by looking at how an o-ring is seated in a groove rather than by reading a long explanation or even just a concise set of words (see [Figure 7.2](#)). An operator who has been building model X1L for a long time will probably not use the instructions once they become proficient; however, when they are first training for the job, they will need excellent instructions. Likewise, if they miss work because of vacation or because they are ill, or if they must attend a required meeting, a replacement operator will also require good, clear instructions. Often, showing an audience the final product first—in other words, what they are trying to accomplish—helps them better understand the sequential steps that are then shown and/or described. These instructions, regardless of the amount of text, are formal standards and must be followed to ensure operator safety and product quality.

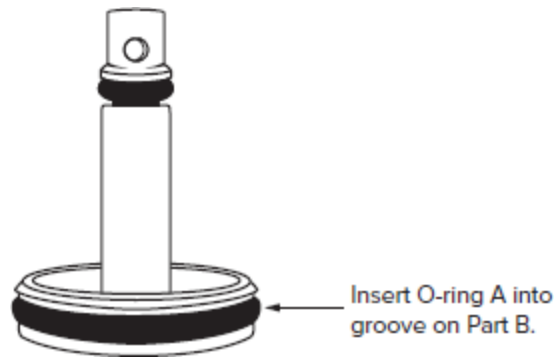


Figure 7.2

Example of formal instruction that is visually-based and works for both hardcopy and electronic instructions.

Presenting instructions electronically vs. hardcopy requires conscious decisions about font types and sizes, colors, and the many other visualization choices discussed in [Chapter 17](#). A color picture that prints nicely on paper might not show up as well on a screen in less-than-ideal (e.g., poorly lit, dusty, hazy, etc.) environments. You must also consider software and hardware differences; will your electronic files be viewed on a large laptop or a phone application? Will you use an HTML interface or PDF files? Consider your audience and the environment in which they will be reading your instructions when you write them. An audience reading electronic instructions from the comfort of their corner office rather than from underneath a molten iron transfer line in a foundry can handle a very different interface. The more challenging the environment, the more concise and visual the instructions must be, regardless of how “expert” your audience is.

It is also important to note that your audience might not read Page 69 English as their first language. You might write product instructions for international, non-English-speaking customers, or perhaps process instructions for ESL (English as a Second Language) employees in your own organization. If you are writing instructions for how to use a product that will be sold internationally, you could take a chance on your two years of high school language translation abilities, but a better choice would be to employ someone fully bilingual with demonstrated excellence in both languages. If you are writing instructions for ESL operators, again, the same is true—have someone well-versed in the second language confirm that the instructions read correctly. There are many publicized instances of

unintended translation errors, ranging from funny (“Eating carpet strictly prohibited”) to downright dangerous (the wrong body part is surgically altered).¹ Finally, it is also not uncommon to write instructions for people who **speak** English but **do not read** English. The general rule of thumb is to assume an unknown audience has an 8th-grade-level education, but there are still people who are unable to read. If the likelihood of your audience includes people with reading challenges, more pictures and fewer words is the best choice.

Visual Instructions

IKEA is well-known for their visually focused, low-to-no word instructions; if this is an option, you can avoid the need to translate significant amounts of text. You can see their product instructions as PDFs on their website.²

Outline 7.1 Instructions

Introduction

- *Definition* Define the overall process.
- *Overview* Describe its purpose and provide an overview.
- *Theory* Explain any needed theory or principles.
- *Steps* List the steps.

Discussion

- *Definition* For each step listed above: Define the step.
- *Overview* Describe what happens in this step.
- *Background* Provide needed context specific to this step.

• <i>Dangers/cautions</i>	Note any dangers and cautions the reader should be aware of.
• <i>Equipment</i>	List the equipment and tools required for this step.
• <i>Directions</i>	Provide specific directions for executing this step.
• <i>Result</i>	Describe the result that should occur.
• <i>Transition</i>	Move coherently from this step to the next step.
Conclusion	
• <i>Summary</i>	Briefly summarize the steps of the process.
• <i>Information</i>	Tell the reader where to find any needed, additional information.

There are entire textbooks, courses, and professions that specialize in human computer interface, graphic design, and ESL. We recommend that you consult an expert from these areas when it is prudent. This chapter addresses the most writing-intensive examples of instructions and manuals. As [Outline 7.1](#) shows, providing technical instructions is actually straightforward if you understand the process and the specific needs and skill level of your audience. Note, however, that this outline models a complete, formal instruction set. Informal instructions, especially for an expert audience, may well omit unnecessary elements such as theory or equipment.

Always remember that the reader's skill level is of paramount concern, since the goal of instructions is to show the reader how to actually do something. How you approach instructions is determined substantially

by the task at hand and the skill level of the audience. This first example illustrates how to use [Outline 7.1](#) by developing instructions designed to show a nontechnical audience how to set up an ergonomically correct computer workstation. The second example provides instructions to a modestly experienced equestrian for wrapping a horse's leg.

Example 7.1 Instructions for a Non-Expert

Instructions for the Layperson

The following example develops a set of general instructions geared to the lay reader for setting up a workstation.³ Notice how the theoretical discussions are limited and the language is more appropriate for a nontechnical audience. Also note that the actual execution steps are sequentially numbered, not only to clarify the sequence but also to set off the execution instructions from the rest of the document.

A layperson is someone who is not an expert in a given field.

Instructions for Setting Up a Computer Workstation

Disclaimer

The following instructions for setting up a computer workstation summarize several ergonomic guidelines for the average person with a typical use profile. However, not all guidelines are included, and no single workstation arrangement can possibly accommodate everyone's

ergonomic requirements. In addition, workstations designed for specialized, nonstandard functions may require unique approaches developed by a qualified ergonomics engineer.

Introduction

Setting up a computer workstation is an ergonomics-based process designed to enhance the user's health, comfort, and productivity. Generally speaking, setting up a computer workstation involves placing and adjusting the desk, chair, and lighting. The setup process follows well-established ergonomic principles and includes the following steps: placing the desk, adjusting the chair, laying out the work area, and optimizing the lighting.

Discussion

Step 1: Placing the Desk

Placing the desk is the physical action of locating and orienting the primary work surface in the work area. Normally, this step involves moving and orienting the desk that you will be using. The goal is to ensure an effective working location relative to traffic patterns, external light sources, power and network connectivity, and other required equipment. The most important considerations at this time are the location of windows and the primary traffic patterns in the workplace, since changing these aspects after the workstation is set up may be difficult. You must consider the angle and intensity of external light coming through the window (lighting will be addressed in greater detail in Step 4), and you must ensure an effective work area that can be traversed easily by those working in it. In other words, place your desk in a location where the exterior light will not impair your work, a location that enables you and others to move freely.

Caution

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Have adequate help on hand, and use proper lifting techniques to protect the desk, the floor beneath, and those doing the work.

Equipment

A cargo dolly might be required, depending on the weight and size of the desk and how far it must be moved.

1. Place the desk in the desired position. If you are locating the desk by an exterior window, orient the desk at a 90° angle to the window. That way, light from outside will come from the side, not from behind or in front, where it would be more likely to cause brightness and glare problems (see [Figure 7.3](#)).

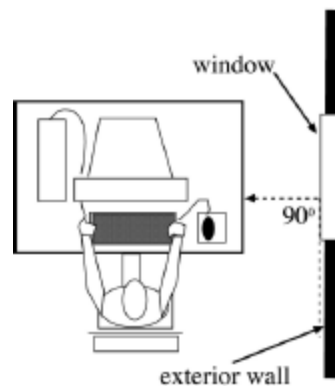


Figure 7.3

Desk placement relative to external window.

2. Make sure that the desk is located where adequate access to power, network, and if necessary, phone connections exist, and where you and others can move unhindered throughout the work area. Once the desk is in its final position with all power, network, and phone connections accessible, you can adjust the chair.

Step 2: Adjusting the Chair

Adjusting the chair is the physical action of optimizing seat, backrest, and arm positions. Sitting for long periods can be particularly hard on your back, legs, and feet and can impede blood circulation. You should optimize your chair's adjustments no matter what your size and shape. This step involves setting the seat height for proper forearm position, taking further actions to ensure the feet are resting firmly on the floor, and adjusting the backrest angle to provide proper back support.

Equipment

A chair with height and back support adjustments is a must for properly setting up a computer workstation. Most adjustable chairs do not require tools and can be adjusted with hand-operated knobs and levers.

1. Sit in the chair in front of the desk in your normal operating position. Make sure you and your chair fit easily under the desk. If you do not, make an initial height adjustment of the chair. Then check your arm position with your hands on the desk at the normal keyboarding position. Your forearms should be parallel to the floor with an elbow bend of 90° . Change the height adjustment so that your forearms are in the correct position. If you and your chair can no longer fit under the desk, you need a different desk or chair (see [Figure 7.4](#)).

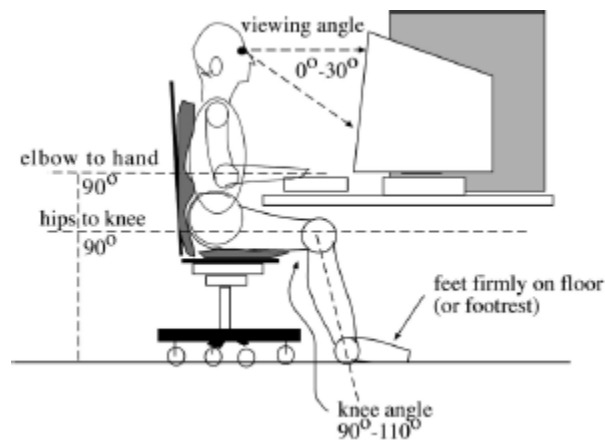


Figure 7.4

Chair adjustment.

2. Ensure your feet rest firmly on the floor when your knees are at a 90° to 110° angle. If you need a footrest, get one, but do not leave your feet dangling unsupported above the floor.
3. Set the backrest so that your lower back fits snugly against the back cushion, and you are either sitting upright or slightly reclining. Angle the bottom seat cushion so that your thighs are parallel to the floor with your knees and hips at approximately the same level. Once the chair is adjusted, you should lay out the work area.

Step 3: Laying Out the Work Area

Laying out the work area is the physical action of optimizing the location and position of the monitor, keyboard, pointing device, and other equipment. Improperly placed monitors can cause you to bend your neck backward to read, which in turn can lead to both neck and back disorders. Also, a large percentage of health problems associated with computer use relate to the keyboard and pointing device. Extended work on computers can cause repetitive stress injuries to the shoulders, elbows, forearms, wrists, and hands. Properly positioning and angling the keyboard and pointing device can help prevent these problems.

Equipment

A monitor, keyboard with adjustable angles, and pointing device are required. Wrist rests may also be needed for the keyboard and pointing device.

1. While you are seated in your normal work position, adjust the computer monitor so that you are the correct distance from the screen for reading with your neck in a relaxed position. Place your monitor so that the screen is 18 to 24 inches from your face. While you are sitting upright in a normal work position, if you can reach out and just touch the screen, your monitor is positioned correctly (see [Figure 7.4](#)).
2. Adjust the monitor's height and position so that you are looking down at a slight angle when reading the screen. For the typical monitor, the top of the screen should be at approximately your eye level. If you wear bifocals or progressive no-line glasses, you may need to adjust the screen level even lower so that you are not tilting your head backward when reading.
3. Adjust the keyboard and mouse. Place the keyboard so that, when you are keyboarding, your wrists are straight and your forearms are still parallel to the floor. Adding negative tilt (raising the front edge of the keyboard) or using a keyboard tray under the top surface of the desk may be necessary.
4. Position your mouse or other pointing device so that it is within easy reach and at the same height as the keyboard.

That way, you will not have to twist your body or lean your shoulder to use it.

5. Position wrist rests in front of both the keyboard and pointing device, but use them only when you are resting your hands and not actually keying or moving the mouse.

Once you have laid out your work area, you should adjust the lighting.

Step 4: Adjusting the Lighting

Adjusting the lighting is the physical action of optimizing the location, angle, and intensity of both exterior and interior illumination sources. Some of the most significant problems with computer workstations are visual fatigue, blurred vision, and headaches caused by glare and improper illumination.

External illumination sources tend to vary significantly in intensity and angle. In Step 1, you already placed your desk at a 90° angle to any external window to substantially reduce glare problems from external light. However, you might still need to install blinds, shades, or curtains to block out external light sources.

Internal light sources in offices are usually too bright for computer workstation use, with overhead lighting being the single biggest problem. One solution is to adjust the monitor's angle slightly to reduce the glare. Another is to reduce the intensity of the overhead lights by removing some of the bulbs, installing dimmer circuits where appropriate, or simply switching off banks of lights altogether. The contrast of the display can be increased, and glare screens such as neutral density and polarizing filters can be added in front of the monitor. Neutral density filters reduce the intensity of the glare relative to the intensity of the displayed information, while polarizing filters block the polarity of the reflected glare while permitting the displayed information to pass through. Monitor shields can be installed around the screen to block the source of the glare.

Equipment

The equipment you will need varies significantly with the amount of glare and the solutions required.

1. While you are seated in your normal work position, look at your computer monitor. If you see reflections, you have glare. First, try to remove it by slightly tilting the monitor. If that does not work, identify the source of the glare. Doing so is easy because you can see it reflecting in your monitor's screen.
2. If an external window is the problem, you can reduce the glare with shades, blinds, or curtains.
3. If internal lights are the problem, move the lights or reduce their intensity. In some cases, you can just turn them off. Use supplemental desk lighting to compensate for the loss of overhead lighting. If that does not work, install glare screens and monitor shields.

When you are able to sit in your normal work position and not see annoying reflections in your monitor's screen, the workstation's setup is complete.

Conclusion

Setting up a computer workstation is an ergonomics-based process designed to enhance the user's health, comfort, and productivity. Generally speaking, setting up a computer workstation involves optimizing the desk's location, adjusting the chair, arranging the work area, and adjusting the lighting. For more detailed information on setting up your computer, see "Computer Workstation Ergonomics: Self-Assessment Checklist."*

*National Institute of Health,
<https://www.ors.od.nih.gov/sr/dohs/Documents/Computer%20Workstation%20Ergonomics%20Self%20Assessment%20Checklist.pdf> (accessed December 30, 2020).

Now that you have read an example of a set of instructions, let us take a closer look at each of its text components and what they do (we will not discuss the figures in this section).

Disclaimer

It is worth noting that instructions often carry disclaimers because of potential risks and liabilities. One cannot buy a chainsaw without receiving inserts and brochures warning of the dangers of stopping the blade with your bare hands or a child's toy without warnings about small parts being a choking hazard or that plastic bags are choking hazards. So before getting into the specifics of setting up a computer workstation, it might be prudent to add a short disclaimer.

The following instructions for setting up a computer workstation summarize several ergonomic guidelines for the average person with a typical use profile. However, not all guidelines are included, and no single workstation arrangement can possibly accommodate everyone's ergonomic requirements. In addition, workstations designed for specialized, nonstandard functions may require unique configurations developed by a qualified ergonomics engineer.

Introduction

Following [Outline 7.1](#), begin with a definition of the process, describe the purpose, and then provide an overview of what happens:

Setting up a computer workstation is an ergonomics-based process designed to enhance the user's health, comfort, and productivity. Generally speaking, setting up a computer workstation involves placing and adjusting the desk, chair, and lighting.

Next, discuss any theory or principles **needed** by the reader to understand the setup process. In this case, we could address such topics as desk heights and surface glare, dynamic versus static sitting, lower back curvature support, and light intensity and color temperature. We could get into keyboard ergonomics and shoulder, elbow, forearm, wrist, and hand pathology. We also could look at repetitious static work habits and fatigue. But why? Is this information necessary for a layperson who just needs to set up a workstation? For the unskilled audience and the purpose at hand, a discussion of theory and principles is not required. We only need to mention that the process is grounded in ergonomic principles and then simply move on and list the steps.

The setup process follows well-established ergonomic principles and includes the following steps: placing the desk, adjusting the chair, laying out the work area, and optimizing the lighting.

Discussion

Step 1: Placing the Desk

At this point, we need to provide specific instructions for each step. Start with Step 1 by first defining the step. Then make an overview of what happens in the step and provide necessary information such as danger and caution notices and any needed equipment.

Placing the desk is the physical action of locating and orienting the primary work surface in the work area. Normally, this step involves moving and orienting the desk that you will be using. The goal is to ensure an effective working location relative to traffic patterns, external light sources, power and network connectivity, and other required equipment. The most important considerations at this time are the location of windows and primary traffic patterns in the workplace, since changing these aspects after the workstation is set up may be difficult. You must consider the angle and intensity of external light coming through the window (lighting will be addressed in greater detail in Step 4), and you must ensure an effective work area that can be traversed easily by those working in it. In other words, place your desk in a location where the light will not impair your work, a location that enables you and others to move freely.

Caution: Have adequate help on hand and use proper lifting techniques to protect the desk, the floor beneath, and those doing the work.

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Equipment: A cargo dolly might be required, depending on the weight and size of the desk and how far it must be moved.

Next, provide the specific execution instructions, describe the result, and transition to the next step. Notice the instructions are written in a snappy, imperative style where the subject **you** is understood. Also, notice that a brief explanation is provided as to why the action is needed since, given the audience and purpose, we did not provide a preliminary theoretical discussion.

1. Place the desk in the desired position. If you are locating the desk by an exterior window, orient the desk at a 90° angle to the window. That way, light from outside will come from the side, not from behind or in front where it would be more likely to cause brightness and glare problems.
2. Make sure that the desk is located where adequate access to power, network, and if necessary, phone connections exist, and where you and others can move unhindered throughout the work area.

Now transition to the next step.

Once the desk is in its final position with all power, network, and phone connections accessible, you can adjust the chair.

Step 2: Adjusting the Chair

Begin Step 2 by first defining the step. Then provide an overview of what happens in the step, along with necessary information and any needed equipment. Notice that, as before, a brief explanation is provided as to why this action is needed.

Adjusting the chair is the physical action of optimizing seat, backrest, and arm positions. Sitting for long periods can be particularly hard on your back, legs, and feet and can impede blood circulation. You should optimize your chair's adjustments no matter what your size and shape. This step involves setting the seat height for proper forearm position, taking further actions to ensure the feet are resting firmly on the floor (or a footrest), and adjusting the backrest angle to provide proper back support.

Equipment: A chair with height and back support adjustments is a must for properly setting up a computer workstation. Most adjustable chairs do not require tools and can be adjusted with hand-operated knobs and levers.

Next, provide the specific execution instructions, describe the result, and transition to the next step. Again, notice the instructions are written in a snappy, imperative style where the subject *you* is understood.

1. Sit in the chair in front of the desk in your normal operating position. Make sure you and your chair fit easily under the desk. If not, do an initial height adjustment of the chair. Then check your arm position with your hands on the desk at the normal keyboarding position. Your forearms should be parallel to the floor with an elbow bend of 90°. Change the height adjustment so that your forearms are in the correct position. If you and your chair can no longer fit under the desk, you need a different desk or chair.
2. Ensure your feet rest firmly on the floor when your knees are at a 90° to 110° angle. If you need a footrest, get one, but do not leave your feet dangling unsupported above the floor.
3. Set the backrest so that your lower back fits snugly against the back cushion, and you are either sitting upright or slightly reclining. Angle the bottom seat cushion so that your thighs are parallel to the floor with your knees and hips at approximately the same level.

Finally, transition to the next step.

Once the chair is adjusted, you should lay out the work area.

Step 3: Laying Out the Work Area

Deal with Step 3 in the same way you dealt with the first two steps. Define the step and then provide an overview of what happens, along with necessary information and any needed equipment.

Laying out the work area is the physical action of optimizing the location and position of the monitor, keyboard, pointing device, and other equipment. Improperly placed monitors can cause you to bend your neck backward to read, which in turn can lead to both neck and back disorders. Also, a large percentage of health problems associated with computer use relates to the keyboard and pointing device. Extended work on computers can cause repetitive stress injuries to the shoulders, elbows, forearms, wrists, and hands. Properly positioning and angling the keyboard and pointing device can help prevent these problems.

Equipment: A monitor, keyboard with adjustable angles, and pointing device are required. Wrist rests also may be needed for the keyboard and pointing device.

Next, provide the specific execution instructions, describe the result, and transition to the next step.

1. While seated in your normal work position, adjust the computer monitor so that you are the correct distance from the screen for reading with your neck in a relaxed position. Place your monitor so that the screen is 18 to 24 inches from your face. While you are sitting upright in a normal work position, if you can reach out and just touch the screen, your monitor is positioned correctly.
2. Adjust the monitor's height and position so that you are looking down at a slight angle when reading the screen. For the typical monitor, the top of the screen should be at approximately your eye level. If you wear bifocals or progressive no-line glasses, you may need to adjust the screen level even lower so that you are not tilting your head back when reading.
3. Adjust the keyboard and mouse. Place the keyboard so that, when you are keyboarding, your wrists are straight and your forearms are still parallel to the floor. Adding negative tilt (raising the front edge of the keyboard) or using a keyboard tray under the top surface of the desk may be necessary.
4. Position your mouse or other pointing device so that it is within easy reach and at the same height as the keyboard. That way, you will not have to twist your body or lean your shoulder to use it.
5. Position wrist rests in front of both the keyboard and the pointing device, but use them only when you are resting your hands and not actually keying or moving the mouse.

Now provide a transition to the next step.

Once you have laid out your work area, you should adjust the lighting.

Step 4: Adjusting the Lighting

Deal with Step 4 in the same way you dealt with the other steps. Define the step, explain why the step is needed, and then provide an overview of what happens, along with necessary information.

Adjusting the lighting is the physical action of optimizing the location, angle, and intensity of both exterior and interior illumination sources. One of the most significant problems with computer workstations is visual fatigue, blurred vision, and headaches caused by glare and improper illumination.

External illumination sources tend to vary significantly in intensity and angle. In Step 1, you already placed your desk at a 90° angle to any external window to substantially reduce glare problems from external light. However, you might still need to install blinds, shades, or curtains to block out external light sources.

Internal light sources in offices are usually too bright for computer workstation use, Page 77 with overhead lighting being the single biggest problem. One solution is to adjust the monitor's angle slightly to reduce the glare. Another is to reduce the intensity of the overhead lights by removing some of the bulbs, installing dimmer circuits where appropriate, or simply switching off banks of lights altogether. The contrast of the display can be increased, and glare screens such as neutral density and polarizing filters can be added in front of the monitor. Neutral density filters reduce the intensity of the glare relative to the intensity of the displayed information, while polarizing filters block the polarity of the reflected glare while permitting the displayed information to pass through. Monitor shields can be installed around the screen to block the source of the glare.

Equipment: The equipment you will need varies significantly with the amount of glare and the solutions required.

Next, provide the specific execution instructions, describe the result, and clarify that the process is complete.

1. While you are seated in your normal work position, look at your computer monitor. If you see reflections, you have glare. First, try to remove it by slightly tilting the monitor. If that does not work, identify the source of the glare. Doing so is easy because you can see it reflecting in your monitor's screen.
2. If an external window is the problem, you can reduce the glare with shades, blinds, or curtains.
3. If internal lights are the problem, move the lights or reduce their intensity. In some cases, you can just turn them off. Use supplemental desk lighting to compensate for the loss of overhead lighting. If that does not work, install glare screens and monitor shields.
4. When you are able to sit in your normal work position and not see annoying reflections in your monitor's screen, the workstation's setup is complete.

Conclusion

Finally, conclude the instructions by briefly summarizing the steps and providing the reader with sources for additional information.

Setting up a computer workstation is an ergonomics-based process designed to enhance the user's health, comfort, and productivity. Generally speaking, setting up a computer workstation involves optimizing the desk's location, adjusting the chair, arranging the work

area, and adjusting the lighting. For more detailed information on setting up your computer, see “Computer Workstation Ergonomics: Self-Assessment Checklist.”*

Example 7.2 Instructions for an Audience with Some Experience

Instructions for a Modestly Experienced Equestrian

Instructions should be geared specifically to the user’s skill level and knowledge. These instructions provide the process for putting a standing wrap on a horse’s leg.

Instructions for Putting a Standing Wrap on a Horse’s Leg

Disclaimer

The following instructions for putting a standing wrap on a horse’s leg include the best practices for the average horse-owner and a typical, well-behaved equine. However, not all best practices are included and there are circumstances in which an owner should not try to wrap a horse’s leg on their own.

Introduction

Putting a standing wrap on a horse’s leg is a hands-on process designed to prevent stocking up (swelling) when the horse will be standing for long periods (e.g., confined to a stall), to give warmth and promote circulation after a workout, or to protect the legs while trailering. Applying a standing wrap involves carefully wrapping a quilt

and a bandage wrap around a horse's lower leg, from just below to the knee or hock to below the fetlock. The standing wrap process follows well-established guidelines and includes the following steps: preparation, applying the quilt, and applying the bandage wrap. See [Figure 7.5](#) for an example of a completed standing wrap.



Figure 7.5

An example of a correctly applied standing wrap on a horse's front leg.

Discussion

Step 1: Preparation

Preparation is the physical process of getting the area, materials, and horse ready for Step 2. The goal is to ensure a safe and well-organized work environment so that you can complete the subsequent steps quickly and effectively. The most important requirements are a well-lit area

sheltered from wind and guarded against other distractions that could spook the horse, a clean place within easy reach to store the quilt and bandage wraps until you need them, and clean legs on the horse.

Equipment

A standing wrap requires two elements:

- **Quilts.** Usually a cotton/poly broadcloth blend for the outer layer, a flannel inner layer for softness next to the horse's skin, and 0.5" water-resistant, bonded fiberfill padding to keep moisture out of the bandage wrap and help put compression on the leg. The quilts' standard height can be 8"–20" (in increments of 2" with a length of 40.5"), but other sizes can be adapted for horses shorter or taller than the standard sizes allow.
- **Bandage wraps.** Typically made of synthetic fabric with two-way stretch and hook-and-loop closures. Height of 4" and 9' or 12' long, depending on the size of the horse.

Directions

1. **Prepare the area.** Sweep the area clean. Clear it of any obstacles that the horse can play with or move if it shifts its body.
2. **Roll the quilts** so that the cotton/poly broadcloth is on the inside of the roll and the flannel is on the outside. That way, when you roll the quilt onto the leg of the horse, the flannel side is touching the horse's skin.
3. **Roll the bandage wraps** so that the hook-and-loop closures are rolled back onto themselves and on the inside of the roll. That way, when you roll the bandage wrap around the outside of the quilt (which will be rolled around the horse's leg), the closure will be at the end of the bandage wrap to the outside. If the closure is to the inside (against the horse's leg), you will be unable to place the loop end onto the hooks and lock the bandage wrap in place.
4. **Make sure the quilts and bandage wraps are within reach** of the position you will be in when crouching down around the horse's leg.

5. **Clean the horse's legs** by brushing. Make sure that all dirt is removed to avoid causing skin injuries from dirt rubbing the skin raw underneath the bandage wraps.

Once everything is prepared, you can begin the process of putting a standing wrap on the horse's leg.

Step 2: Apply the Quilt

Applying the quilt is the physical process of firmly rolling the quilt around the horse's lower leg. The quilt must be applied, wrinkle-free and with even pressure, from just below the horse's knee (front leg) or hock (hind leg) to below the fetlock for the entire length of the quilt. When tight, the padding in the quilt creates the compression that reduces swelling, encourages blood flow, and protects against injury. The quilt must not be so tight that it constricts blood-flow or so loose that it can wiggle underneath the bandage wrap and create wrinkles.

Caution

Once you begin, you must be able to complete the process quickly as you will be crouching down around the leg of an ~1,200-pound flight animal whose instinct will be to run away from perceived danger. Even the most well-behaved horse has the potential to run into or over you, in response to fear-inducing stimuli (which are too many to mention—one old equine saying is that horses are only scared of two things: things that move and things that don't).

Equipment

Prepared **quilts** are required (see Step 1).

Caution

No matter which leg you are wrapping, you should always start medially and roll the quilt forward. Going back-to-front pulls against the tendons and ligaments at the back of the horse's legs and can cause permanent injury.

Directions

1. **Place the quilt in position.** With the bulk of the rolled quilt facing away from and anterior to the horse's leg, place the long edge of the quilt against the medial part of the leg (just posterior to the cannon bone). The top edge of the quilt should be just below the knee and the bottom edge of the quilt should be below the fetlock.
2. **Place your hand on the cannon bone** to hold the quilt in place and begin unrolling the quilt forward toward the horse's head, keeping a gentle, even pressure with the roll and removing wrinkles as you go (see [Figure 7.6](#)).



Figure 7.6

Placing the quilt on the horse's leg.

3. **Unroll the quilt around the leg** with continued gentle, even pressure. Continue to remove wrinkles and folds in the quilt, and

make sure to keep the top and bottom edges even for all layers of quilt around the leg. You may need to move your hands up and down, from the knee/hock to the fetlock and back, to ensure smooth, wrinkle-free pressure is applied.

Once the quilt is unrolled, you are ready to apply the bandage wrap.

Step 3: Apply the Bandage Wrap

Applying the bandage wrap is the physical process of firmly rolling the bandage wrap around the quilt on the horse's lower leg. Like the quilt, the bandage wrap must be applied wrinkle-free and with even pressure, from just below the horse's knee (front leg) or hock (hind leg) to below the fetlock. When the bandage wrap holds the quilt in place, it creates the compression that reduces swelling, encourages blood flow, and protects against injury. The bandage wrap must not be so tight that it constricts blood-flow or so loose that it can get caught on objects or worse, come free and wrap around the horse's other legs—all things that can seriously injure the horse.

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Equipment

Prepared **bandage wraps** are required (see Step 1).

Caution

No matter which leg you are wrapping, you should always start medially and roll the bandage wrap forward. Going back-to-front pulls against the tendons and ligaments at the back of the horse's legs and can cause permanent injury.

Once you begin, you must be able to complete the process quickly as you will be crouching down around the leg of an ~1,200-pound flight animal whose instinct will be to run away from perceived danger. Even the most well-behaved horse has the potential to run into, or over you, in response to fear-inducing stimuli.

Directions

1. **Place the bandage wrap** on the medial part of the horse's leg, flatly against the quilt, again with the bulk of the bandage wrap facing away from and anterior to the horse's leg. Put the edge of the bandage against the medial part of the leg (just posterior to the cannon bone). Ideally, the bandage will start one-quarter to one-half of the way down from the top of the quilt.
2. **Unroll the bandage wrap forward** toward the horse's head, moving downward first. Again, no matter which leg you are wrapping, always start medially and forward (see [Figure 7.7](#)).



Figure 7.7

Applying the bandage wrap.

3. **Use gentle and even pressure** as you unroll the bandage wrap down the leg, and apply a little more pressure with every pass around the front of the cannon bone—do not pull too

tightly and never pull so that you put pressure on the tendons and ligaments on the back of the horse's leg. Overlap the layers of bandage wrap as you unroll by half the width of the bandage wrap (if you have shorter bandage wraps, you might need to overlap less).

4. **Once you reach the bottom of the quilt** (below the fetlock), wrap around the fetlock once, keeping the lower edge of the bandage wrap even with or just above the lower edge of the quilt. Then move back up the leg, overlapping the bandage wrap and keeping a gentle, even pressure.
5. **Continue unrolling until you reach the top** of the quilt, keeping the upper edge of the bandage wrap even with or just below the upper edge of the quilt. The hook and loop closure should ideally finish on the outside of the horse's leg, but it is fine if it does not; just make sure it lines up even no matter where it ends.

The finished standing wrap should be smooth without any wrinkles or folds and the top and bottom edges aligned throughout the layers.

Conclusion

Putting a standing wrap on a horse's leg is a hands-on process designed to prevent swelling, promote circulation, and protect the legs. The standing wrap process includes preparation, applying the quilt, and applying the bandage wrap. Putting a standing wrap on a horse's leg takes a great deal of practice, but it is an essential skill that every equestrian should know how to do quickly and well.

More on Manuals

As mentioned earlier, manuals follow the same basic pattern as instructions for telling someone how to do something. However, manuals have a much larger scope and are more comprehensive.

Manuals provide instructions for many complicated tasks involving complex equipment. In some cases where systems are large and extremely

complex (such as an aircraft or automobile), many separate manuals are required and may, in fact, fill a small library. Smaller systems (such as a lawnmower) frequently require only a single, self-contained manual. Increasingly, manuals are provided in electronic format, including on websites as HTML interfaces or as downloadable documents like Adobe Portable Document Format files (PDF) or word processing files such as Microsoft Word (doc or docx). Here are the most common types of manuals and the kinds of tasks with which they are used:

- **Assembly manuals:** construction, alignment, calibration, testing, or adjusting a mechanism.
- **Owner or user manuals:** use of a mechanism or process, including routine maintenance and basic operation.
- **Operator manuals:** use of a mechanism and very minor maintenance.
- **Service manuals:** routine maintenance of a mechanism, including troubleshooting, testing, repairing, or replacing defective parts.
- **Technical manuals:** parts specifications, operation, calibration, alignment, diagnosis, and assembly.

Visuals and Instructions

When you are designing visuals such as diagrams, schematics, and photographs in support of a set of instructions, be sure to include only those visuals which illustrate the instruction's steps. Do not include visuals that do not match the instructions in subject matter, terminology, or directions. Also, avoid visuals that include too much or too little complexity for the level of explanation. See [Chapter 17](#).

- Visuals should relate specifically to the text discussion, and they should be referred to in the text before actually being used. The reader should never encounter a visual and wonder why it is there or what it is for. Always tell your reader specifically when to look at a visual, and try to do so *before* the point in the paper where the visual actually appears.

- Each visual should have an assigned sequence number and name, such as “Figure 6.5 QuadFINKEL.” These labels reference precisely the visuals that are included. It is a good idea to use compound numbers—to show both the section or chapter number and the sequence number. For example, in the label “Figure 6.5 QuadFINKEL,” the **6** indicates that the visual occurs in the sixth chapter of this book, and the **5** refers to the sequence in which the figure occurs. Numbering visuals in this way allows you to add or delete visuals in one section without having to renumber all the visuals in subsequent sections. Also, if possible, run separate sets of sequence numbers for each type of visual. For example, figures should have their own set, as should equations, tables, and photographs.
- Each visual must be included for a specific purpose. Do not add visuals to make the page look “pretty” or to “add visual interest.”
- Visuals for instructions need to show something happening and, as such, tend not to be static representations of something seen in a “slice of time.” Often you can show movement through space and time with a technique as simple as a well-placed arrow. At other times, you may need a coherent series of visuals to demonstrate your point (like [Figure 6.5 QuadFINKEL](#) has an a, b, c, d, e, and f). Alternatively, visuals for instructions could show the **before** and **after** of each step.

Conclusion

In conclusion, instructions describe processes so that your readers can accomplish the steps required to successfully complete a task. Your readers are end-users, or implementers; they will rely on the text to complete their work, often under stressful or potentially dangerous conditions. Because of this, you must write instructions with purpose and context in mind. Consider your audience’s need along with the context in which the instructions/manual will be used. Then make strategic decisions about the level of formality, the level of technicality (novice, skilled professional, etc.), the format (print, electronic file, phone application), the layout (font types and sizes, colors, and the many other visualization choices), the language (international, global English speakers, etc.), and the visuals to include. Manuals follow the same basic pattern as instructions, but are much broader in scope and more comprehensive.

Instructions Checklist

- ☐ Do I understand the process and the skill level of the intended audience?
- ☐ Have I defined the overall process and described its purpose?
- ☐ Have I explained any needed theories or principles?
- ☐ Have I listed the steps?
- ☐ Have I defined each step?
- ☐ Before I tell the reader to do something in each step, have I given an overview of what will happen?
- ☐ Have I provided needed information such as cautions, dangers, and required equipment?
- ☐ Have I transitioned to the next step, if there is one?
- ☐ Have I provided only visuals that relate to my text?
- ☐ Do the visuals show something happening (dynamic, not static) whether as an individual visual, a series, or before and after pairings?
- ☐ Have I referred to the visuals in my text before they are used by my audience?
- ☐ Have I given each visual an assigned sequence number and name?
- ☐ Have I run separate sets of sequence numbers for each type of visual?
- ☐ Does each visual have a specific purpose?

Exercise

Assess the following instruction set for using the Finkelkick martial arts technique to break a reinforced concrete wall.

- Is this example a realistic set of instructions?
- If so, is this topic appropriate for a technical instruction set?
- Should there be a disclaimer, and if so, what should it say?
- Should caution and warning notices be included anywhere in the instructions?
- How useful are the visuals? Would the individual graphics be better used individually, or grouped together as they are in this paper? Also, are the individual images properly keyed to, and well-integrated with, the text discussion?
- Is specifying the equipment useful here?
- Is specifying the result of the action useful here?
- Has any attempt been made to use layout to enhance this instruction set?
- Should the theoretical discussion of kinetic energy, force, and pressure be included? If so, should equations be added?

Instructions for Breaking a Reinforced Concrete Wall with the Finkelkick

Introduction

In martial arts, the *Finkelkick* is a specialized foot technique by which relatively large pressures are developed through the application of substantial kinetic energy to a limited target area on a reinforced concrete wall. The success of a Finkelkick depends on achieving adequate kinetic energy, which is the energy a martial artist possesses by virtue of their mass being placed into motion. Of course, the role of velocity is paramount because it is the variable with the single greatest influence

over the amount of energy generated. To fully exploit the contribution of velocity, you should rapidly accelerate the appropriate body part (your kicking foot) to achieve the greatest possible kicking force. Additionally, to break the concrete wall, you will need to maximize the pressure your foot actually applies to the wall. To do that, you must limit the contact area of the foot's impact surface. The following five steps are designed to optimize the energy created, the force applied, and the resulting pressure realized: *Positioning*, *Chambering*, *Extending*, *Retracting*, and *Resetting*.

Discussion

For the following discussion, refer to [Figure 7.8](#).

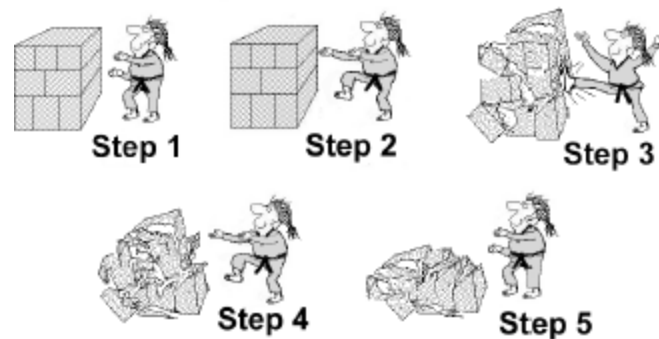


Figure 7.8

Finkelkick steps.

Step #1: Positioning

Positioning is the preparatory action of placing yourself in front of one of the long sides of the concrete wall. It does not matter which side. In this step, you will assume a traditional front stance approximately 3 feet from the target. Make sure your feet are approximately a shoulder-width apart, and that your right foot is approximately 2 feet in front of your left foot. The toes on both feet should be facing forward. Your front leg should be bent while the back leg should be straight.

Equipment: A reinforced concrete wall and a serviceable kicking foot are required.

1. **Assume a front stance approximately 3 feet in front of the wall.**

You should be facing the wall, with the toes of your right foot approximately 3 feet from the wall (see [Figure 7.8](#)). Once in the correct position, you are ready for chambering.

Step #2: Chambering

Chambering is a posturing action where you position the right (kicking) leg for the attack. A good chamber is essential if the kick is to contact the wall at the highest possible velocity and at the desired angle of 90 degrees. In this step, you will raise your right knee as high as possible and then hold your leg in that position with your toes still pointing forward toward the wall. You will also extend your hands toward the wall for stability.

Equipment: Same as Step #1.

2. Raise your right knee as high as possible, while extending both hands toward the wall.

You should now be balancing on your left foot, with your back straight and perpendicular to the ground, and with both hands extended toward the wall (see [Figure 7.8](#)). Note: Your right hand may be only a few inches from the wall's surface. Once in this position, you are ready to extend your foot into the wall.

Step #3: Extending

Extending is the front snapping action that drives your foot into the wall. In this step, you will thrust the ball of the kicking foot through the wall while raising your hands up and to the sides of your body. Your kicking leg should be at a 90-degree angle to the wall at the time of contact. Remember! You are kicking a structure that consists of tons of jagged concrete and rusty, reinforcing steel members. So, keep your mental focus on the wall and conquer your well-founded fears for your personal well-being.

Equipment: Same as Steps #1 and #2. The only additional equipment required for this step might be an ambulance staffed with paramedics.

3. Mentally focus deep into the wall, then accelerate your foot straight out to that point of focus as you raise your hands up and to the sides of your body.

If all goes well, you should hear sounds of the wall's structure failing catastrophically, and you should see multiple fault cracks rapidly forming (see [Figure 7.8](#)). If all does *not* go well, you should hear sounds of your foot's skeletal structure failing catastrophically, and you should feel intense pain and regret. In either case, you are ready to retract your foot from the wall.

Step #4: Retracting

Retracting is a de-posturing action where you pull your foot from the wall back into a chambered position. Retracting is normally done in competitive sparring to prevent your opponent from grabbing your leg. You do it in this wall-breaking demonstration as a matter of good form. In this step, you will simply pull your foot back from the broken wall and raise your knee as high as possible, while pulling your arms back to the chambered position. Again, your toes (or what is left of them) should be pointing toward the wall, and your hands should be in front of your body and extended straight at the target.

Equipment: Same as Step #3.

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4. Pull your kicking foot back toward your body, raising your right knee as high as possible, while extending your arms in front of you toward the destroyed wall.

You now should be facing what is left of the wall, with your right knee raised and both hands extended toward the destroyed wall (see [Figure 7.8](#)). Your right hand may be within a few inches of where the wall once stood. You are now ready for the final step of Resetting.

Step #5: Resetting

Resetting is the final action of placing yourself into your initial, starting position. In this step, you will reassume a traditional front stance approximately 3 feet from where the wall used to be. Again, make sure your feet are approximately a shoulder-width apart, and that your right foot is approximately 2 feet in front of your left foot. The toes (or what is left of them) should be facing forward. Your front leg should be bent, while the back leg should be straight.

Equipment: Same as Step #3.

5. Lower your right leg from its chambered position and assume a front stance.

Your right leg should be forward and bent, while your left leg should be to the rear and straight. You should be facing the wall, with your right foot approximately 3 feet from where the wall once stood (see [Figure 7.8](#)). Once in this position, you have successfully completed the Finkelkick.

Conclusion

Execution of the Finkelkick requires five steps: Positioning, Chambering, Extending, Retracting, and Resetting. Together, these steps result in a high kinetic energy technique capable of destroying a reinforced concrete wall.

Notes

- [1.](#) The Journal of Specialised Translation, https://jostrans.org/issue21/art_karwacka.pdf. Accessed December 30, 2020.
- [2.](#) IKEA homepage, <https://www.ikea.com/gb/en/>. Accessed December 30, 2020.
- [3.](#) Special thanks to Sharon Liebel McFarron, CPE, Ergonomics Engineer and Management Consultant, Humantech, Inc., Ann Arbor, Michigan, for her help with this example.

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CHAPTER EIGHT

Proposals



In both academia and industry, proposals are among the most important documents one can write. Persons and organizations that write effective proposals win grants, contracts, and jobs; persons and organizations that do not write effective proposals often just wind up “going away”—sometimes “far away.” If we are unable to write proposals that get accepted, we will not have the resources (time, money, or personnel) to do what we must do. Proposals are important because they, directly or indirectly, provide the income that keeps us warm, dry, and well fed.

What Is a Proposal?

Proposals are specialized, technical business documents that offer persuasive solutions to problems. All technical documents are designed to communicate ideas objectively and clearly. Your goal in writing a proposal, however, goes beyond just precise communication. A proposal must also sell the reader on some idea—usually that they (or their organization) need specific goods or services that you (or your organization) can provide. In other words, you are trying to persuade your audience, often consisting of a mix of decision-makers, advisors, and implementors, to give you resources.

To be successful, you normally must do at least three things in any proposal you write:

1. Describe, identify, or refer to a problem that must be solved. The reader sometimes already knows that they have a problem, but not always. Because the reader might not fully understand or appreciate the scope, magnitude, or complexity of the issues at hand, you must describe the problem. Additionally, by providing this description, you

establish credibility by showing your audience that **you** understand the problem.

2. Offer a viable solution to the problem. You must demonstrate to the reader that your proposed approach will, in fact, solve the problem effectively and efficiently.
3. Show that you can effectively implement this solution. Having an effective solution does not mean much if you cannot implement it. That is why, in any proposal, you must show that you have the necessary skills and ability to do what you are proposing. One of the best ways of demonstrating your capability is with successful prior performance. In other words, if you are proposing to design a web page for your company, it would be a definite plus if you could point to other successful web pages that you have designed. Showing that you have done the kind of thing you are proposing to do—and have done it well—can be persuasive.

Now, just for fun, think of a Poodle. You can probably imagine one immediately, but in your mind, are you seeing a standard, toy, or miniature Poodle? Does it have a show or puppy cut? Is it wearing a bow? Does it have fluffy ears or a ball of fur on the end of its tail? No matter what kind of Poodle you imagine, it is still a Poodle. The variety of Poodle types does not change that.

Much like the Poodle family, proposals come in **many** different shapes, sizes, haircuts, and colors (see [Figure 8.1](#)). Proposals have flexibility in terms of look, content, and size, but despite these variations, engineers, scientists, and managers will recognize a proposal when they see one. There are as many formats for proposals as there are Poodles on the planet. Some are very structured with required headers, font sizes, word counts, and content, like a proposal submitted in response to any government request for proposal—perhaps a construction bid for an overhead bridge to connect an interstate to a state highway. Some consist of a simple directive: “Send me an email,” like when an office manager mentions the need for an updated copier with scanning options. Regardless, they serve the same purpose: to convince someone to support your idea and often to provide resources: money, time, and/or political capital. Whether your Poodle is a full-size standard with a salon cut, or an itty-bitty miniature with out-of-control curls, it is still a Poodle. The same is true of proposals. Despite the

variation, everyone will recognize a proposal for what it is: a request for resources.



Figure 8.1

Proposals, just like Poodles, come in all different sizes, cuts, and colors. Generally, all proposals have the same purpose and basic structure, but the level of detail, length, level of language, and overall design differ based on audience need and expectations.

Formal and Informal Proposals

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Proposals are generally categorized as **formal** or **informal**. Formal proposals are normally large, comprehensive documents produced by a team of experts on behalf of an organization. They must often meet explicit deadlines and compete against others' proposals as well. Informal proposals are generally short documents of limited scope written by an individual. Informal proposals are often solicited via conversation, sometimes in a meeting or even in a hallway. A supervisor is interested enough in the

situation to ask you to send a short proposal; the request does not include any specific formatting, and often the deadline is “as soon as you can.”

Formal Proposals

A full discussion about formal proposals is well beyond the scope of this book. But what if your supervisor just told you to put together a formal proposal and you have no clue what to do, so you use this as your guide? Well, quite honestly, your supervisor is underestimating the task, and you are using the wrong book! Formal proposals are lengthy projects undertaken by skilled proposal writers functioning together in a well-structured, proposal-team environment. These documents are prepared in response to a formal **request for proposal** (RFP). Proposal teams take great care to respond to each RFP requirement. ([Chapter 21](#) provides a more thorough discussion of team writing in this context.) Team members often include a diverse group of experts, including both technical experts and managers from different business functions. When you are writing a formal proposal, you will often not know your audience personally. Doing some research online and/or through networking (see [Chapter 20](#)) might provide insight about their expectations. Formal proposals require that context and purpose be explicitly stated. They can take many forms, but a typical one might include the following:

- An **executive summary** that synthesizes the substance of the proposal. Executive summaries are written for senior decision-makers. The executive summary is the only part of the proposal that many audience members read, so it is very important. Executive summaries are discussed in greater detail in [Chapter 14](#).
- A **technical section** that lays out the proposed solution in detail. This section is often written by a team of engineers and scientists who are responsible for solving the problem.
- A **management section** that describes the organizational structure and key players who will implement the proposal, if accepted.
- A **cost section** that provides detailed analysis and data regarding the cost of implementing the proposed solution.

- A **resources section** that provides detailed analysis and data regarding both the human and the physical resources required to implement the proposed solution. In some cases, this information is included in the management section.

It is also common to organize a proposal using an executive summary, introduction, statement of work, and discussion section, which include combinations of technical and managerial plans. Proposal requirements often include a schedule like a Gantt chart and a table that summarizes estimated costs (see [Chapter 17](#) Visuals).

Formal proposals can take months to produce at a cost of tens of thousands of dollars. Writing these proposals is difficult, especially because these documents are evaluated in an extremely competitive environment. Winning or losing is based on the quality of the proposed solution, the credibility of the proposing organization, and the estimated cost of implementing the proposal. In the end, the proposal representing the best overall value normally wins the contract.

Government Proposals

Perhaps the best example of just how challenging proposal writing can be involves writing proposals for the U.S. government, where federal tax dollars are involved. Agencies that award grants to both businesses and academic researchers include the National Institute of Health (NIH),¹ the National Science Foundation (NSF),² the Department of Energy (DOE),³ the Department of Defense (DOD),⁴ and the Defense Advanced Research Projects Agency (DARPA),⁵ to name a few. To understand how complicated and rigorous the expectations are for these kinds of formal proposals, look at the “Proposal and Award Policies and Procedures Guide” for NSF. As of December 2020, this document is 185 pages long.⁶ The purpose of these extensive guidelines is to help technical writers provide their problem description, potential solution, and justification (the three things a proposal must include) in a standardized, “easy”-to-compare format.

Writing proposals for the government is truly an exacting, Page 90 unforgiving activity. The content, layout, and procedures are

precisely spelled out, and evaluation of proposals is rigorous and comprehensive. Source selection panels composed of technical, managerial, human resource, financial, and other experts evaluate each proposal. They aggressively seek out all discrepancies between tasks and skills, skills and costs, management philosophy and organizational structure, and so on. They score each proposal based on the effectiveness of the solution, the risks involved, the costs in both time and dollars, and the capabilities and past performance of the proposing company. The goal of the source selection process is to select the proposal that offers the best value to the government.

Interaction between proposal writers and proposal evaluators (i.e., between the entity seeking funding and the government) is strictly controlled. Even the appearance of impropriety can have serious legal implications. After preliminary interactions to evaluate the objectives of the project, the government issues an RFP that precisely spells out the problem to be solved and the parameters and constraints for any proposed solution. In some cases, proposals are presented as technical presentations (see [Chapter 18](#)), or they might be submitted electronically. Larger projects might require multivolume documents.

In most cases, government proposals have specific deadlines and delivery requirements and must precisely meet all format and content specifications. Over the years, many grants have been rejected for not following required formulation, including things like using the wrong font. In some cases, it is not even the proposal writers' fault that their document does not meet requirements. For example, in 2018, Dr. Joseph Schlesinger of Vanderbilt University made headlines when he had an 80+ page grant application rejected by the Veterans Administration for typography issues that were created in the conversion process from a PDF to a Word document.⁷ It is worth spending a few extra minutes to ensure that you will not encounter formulation issues that will negate your hard work without review.

One helpful resource if you are writing proposals for government agencies is www.Grants.Gov.⁸ Besides consolidating potential grant opportunities, this website also provides extensive advice and insight for writing successful government grant applications.

Informal Proposals

The informal proposal is the primary focus of this chapter. Written by individuals or small teams, informal proposals typically address a limited problem for which a relatively straightforward solution exists. Frequently, informal proposals take the form of a long letter or a short document.

Informal proposals can be either **solicited** or **unsolicited**. When you are asked to write a proposal in a course, it is solicited. A solicited proposal responds to a specific, often written, request. There are many occasions for solicited proposals in industry as well. A homeowner might ask a contractor for a proposal to replace the roof on a house, or a company might ask a lighting firm for a proposal to illuminate a manufacturing area. Your supervisor might email you and ask you to write a proposal for replacing a machine on an assembly line or changing the formula for a cleansing agent in a manufacturing process. With a solicited proposal, the problem has already been identified, and the decision to try to solve the problem has already been made. That said, however, your proposal must still earn the right to be resourced; if you are unable to clearly communicate a justified need and your ability to get the job done, it could be rejected.

Unsolicited proposals, on the other hand, are proposals that no one has requested and, perhaps, that no one wants. The recipient has not decided to solve a problem and might not even realize that a problem exists—or might not want to realize it. Unsolicited proposals can come from within or outside of an organization. For example, when an employee has an idea about how to improve a data collection process and emails you, this is an example of an unsolicited internal proposal. An example of an external proposal would be when a food manufacturer receives a letter from a customer suggesting that an ingredient be changed. Unsolicited proposals, as you might imagine, have less chance of being accepted.

Informal proposals can take many forms and be organized in Page 91 many ways (but they are still proposals—remember the many Poodles). Regardless of the variety, if you carefully consider audience expectations and needs, and clearly and concisely describe the context and purpose of your proposal, you will be successful. Context should include all aspects of the problem, and purpose should be met with your proposed solution. Let

common sense guide you, and follow any guidelines from your supervisor or the agency requesting the proposal as well.

If you do not have specific guidance, then [Outline 8.1](#) provides a fairly typical approach you can use to organize informal proposals. In the following example, we look at each element of this outline and examine how you might construct such a proposal.

Outline 8.1 Informal Proposals

Introduction

- | | |
|-------------------------|---|
| • Purpose | Describe the reason for writing this report. |
| • Background | Describe the problem that must be solved. |
| • Scope | Review what this report will and will not cover. |
| • Measurable objectives | Describe how you will measure the impact of this project. |

Discussion

- | | |
|---------------------|---|
| • Approach | Describe the proposed solution to the problem. |
| • Result | Show how the solution will solve the problem. |
| • Statement of work | List the tasks that will be performed as part of this solution. |
| • Deliverables | State the deliverables that this project will generate. |

Resources

- | | |
|------------------------|---|
| • Personnel | List those who will be doing the work and their qualifications. |
| • Facilities/equipment | List the physical resources required to do the work. |

Costs

• Fiscal	List the financial costs of implementing the proposed solution.
• Time	List the clock time and calendar time required to implement the proposed solution.
Conclusion	
• Summary	Highlight the benefits and risks of adopting this proposal.
Contact	Provide a contact for more information.
Sources Used	Provide all references used to develop the proposal.

Informal Proposal

As a hypothetical example, imagine that the International Olympic Committee (IOC) has asked your organization, XtremaLab, to provide computer modeling and analysis of the QuadFINKEl figure skating jump (the [Chapter 6](#) exercise). To respond to the IOC, your XtremaLab division must create an informal proposal to upgrade its computer systems. The following proposal, written for the experts at XtremaLab, is geared to an audience knowledgeable about modeling and analysis, the technology used to accomplish such modeling and analysis, and the overall resources available to XtremaLab.

Example 8.1 Informal Proposal

Proposal for Upgrading Modeling and Analysis Capability

1. Introduction

1.1 Purpose

This document proposes a general system upgrade for XtremaLab's Sports Analysis Measurement Division's (SPASM-D) computing capabilities. This upgrade is necessary to enable XtremaLab's response to the International Olympic Committee's tasking for modeling and analysis of the QuadFINDEL figure skating jump.

1.2 Background

The QuadFINDEL figure skating jump is so demanding that the IOC is considering the award of an extra-large gold medal to anyone successfully landing the jump in Olympic competition. The IOC has tasked SPASM-D, under IOC Contract IOC-135549, to accomplish advanced 3-D modeling, simulation, and analysis of the QuadFINDEL. This modeling and analysis would give the IOC the scientific basis for justifying the special award. This analysis must be completed within six months of final project approval.

Modeling this skating jump is a complex process due to the element's chaotic nature and high-speed dynamics. To accomplish this analysis in the specified time frame, SPASM-D, in the Technical Report TR-193345, has identified the need for a stand-alone, state-of-the-art graphics, modeling, and analysis capability within the laboratory area. The report identifies the requirement for quantum entanglement communication for various IOC activities. These stand-alone capabilities must allow interoperability to the other laboratories. TR-193345 also stipulates that the cost of the upgrade not exceed \$100,000.

1.3 Scope

This proposal addresses only the system upgrade of the SPASM-D Analysis Laboratory in support of the IOC tasking for modeling and analysis of the QuadFINDEL. This proposal does not include other graphics, modeling, and analysis tasks.

1.4 Measurable Objectives

This project, once implemented, will accomplish two objectives:

1. Decrease the computer analysis time for the QuadFINKEl jump to 5 minutes.
2. Increase the analysis accuracy of the QuadFINKEl model by 20%.

2. Discussion

2.1 Approach

SPASM-D proposes to the XtremaLab Technical Review Committee that the SPASM-D laboratory's capabilities be upgraded and augmented with a dedicated quantum local-area network (QLAN) composed of two Titanium Graphics 4000 Visual Workstations, one Titanium Graphics 4000A Modeling Station, and one XtremaLab SuperSWITCH 2000-X.

Titanium Graphics workstations provide a recognized standard of excellence in performing advanced graphics, modeling, and analysis tasks. Off the shelf, these systems have the required processing power, storage capacity, and network/device interfaces to function seamlessly on the existing network. By connecting these workstations with QLAN through an XtremaLab SuperSWITCH, then cascading this SuperSWITCH to the company's network backbone, these workstations can easily provide the needed stand-alone computational and storage capabilities. They will also meet all company and communication requirements.

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2.2 Result

The Titanium Graphics 4000A Modeling Station will provide the dedicated analytical capabilities required to thoroughly model and understand the various components of the QuadFINKEl. The two Titanium Graphics 4000 Visual Workstations will provide the 3-D graphics rendering required by the IOC tasking. The analysis accuracy of each QuadFINKEl jump will be increased by 20% while reducing the analysis time for each to 5 minutes or less.

2.3 Statement of Work

To achieve the goals of this proposal, the following tasks will be accomplished:

- Task 1: Acquire the necessary equipment; transport, unpack, assemble, and place in the work area. (8 hours)
- Task 2: Set up the operating systems and configure QLAN. (16 hours)
- Task 3: Install application software, test it, and run calibration and verification simulations. (16 hours)

2.4 Deliverables

The SPASM-D simulation and modeling staff will provide the following at the conclusion of this project:

1. Two fully implemented and updated workstations
2. One fully implemented and updated modeling station
3. Fully rendered QuadFINKEL graphics in SIM-IT
4. Full statistical analysis of QuadFINKEL simulations

3. Resources

3.1 Personnel

The SPASM-D simulation and modeling staff will analyze and model the QuadFINKEL using proprietary laboratory software. Additionally, XtremaLab graphics consultants and IOC figure skating experts will work closely with the simulation and modeling staff to ensure accuracy and effectiveness of all required 3-D renderings.

3.2 Facilities and Equipment

The SPASM-D facility includes adequate space for this effort. Suite 104 in Building 45 has the required fiber optic channel and is available for this project. The specific computer equipment required includes the following:

- Two Titanium Graphics 4000 Visual Workstations*
 - Integrated Visual Computing architecture with Mercury chipset
 - Intel Octagon III 5-gigahertz processors with 4G L-2 cache
 - 64-gigabyte ECC SDRAM
 - 8-terabyte Ultra2 drive
 - Two 64-bit PCI buses
 - Integrated QLAN
 - 4 USB-D ports
 - Winnex IN/IX Open Source OS

- Three-year warranty with one-year on-site service
- One Titanium Graphics 4000A Modeling Station
 - Dual Intel Octagon III 6-gigahertz processors with 4G L-2 cache
 - 64-gigabyte ECC SDRAM
 - 100-terabyte RAID disk array
 - Titanium Graphics 8000W 42-inch digital, flat-panel display
 - One XtremaLab SuperSWITCH 2000-X
 - QLAN cable with connectors

*Note: Existing XtremaLab display monitor resources are available to support these workstations.

4. Costs

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4.1 Fiscal

The proposed system upgrade includes the following equipment and installation costs:

Equipment:

- Two TGI 4000 Visual Workstations @ \$5,995 each = \$11,990
- One TGI 4000A Modeling Station @ \$9,090 each = \$9,090
- One SuperSWITCH X-2000 @ \$20,000 each = \$20,000
- 400 feet QLAN cable with connectors @ \$145/foot = \$58,000

Installation:

- 0.02 full-time equivalent (FTE) technician (40 hours) @ \$42,000/FTE = \$840

Total cost: = \$99,920

4.2 Time

Assuming availability of equipment and materials, and using fully qualified XtremaLab technicians, the entire upgrade can be completed in 40 hours. This estimate includes 8 hours to acquire, deliver, and set up the equipment; 16 hours to set up the operating system and network connectivity; and 16 hours to configure and test the applications. This upgrade schedule would provide adequate time to accomplish the modeling and analysis required by the IOC tasking. This project can be completed within three calendar weeks of its start.

5. Conclusion

5.1 Summary

The proposed Titanium Graphics QLAN upgrade will provide a viable, cost-effective solution for meeting the IOC QuadFINKEl modeling and analysis requirement. The following deliverables will be completed: 1) Two fully implemented and updated workstations; 2) One fully implemented and updated modeling station; 3) Fully rendered QuadFINKEl graphics in SIM-IT; and 4) Full statistical analysis of QuadFINKEl simulations. The upgrade will decrease the computer analysis time for the QuadFINKEl jump to 5 minutes and increase analysis accuracy by 20%. The entire system can be up and running within three weeks for a price that is within the cost guidelines. Given the quality of the system and reputation of its manufacturer for setting the standard for high-end analysis and modeling computing systems, the risks of this solution are minimal. In fact, the proposed system provides exactly the right capabilities at precisely the right time, with a cost and timeline that are well within the company's needs.

5.2 Contact

For more information regarding this proposal, please contact Erin R. Ronaldson, Ph.D., P.E., Director of Systems Engineering, SPASM-D, at ext. 445; or email: eronald@XtremaLab.com.

5.3 Sources

1. Home page, <https://XtremaLab.com>. Accessed November 26, 2020
2. International Olympic Committee RFP, <https://www.olympic.org/QuadFINKEl-RFP>. Accessed November 27, 2020
3. <https://QLAN-Distribution-Warehouse.com>. Accessed November 28, 2020

Now that you have read an example of a proposal, albeit a fictitious one, let us take a closer look at each of its components and what they do.

Writing the Introduction Section

Purpose

Tell the reader why you have written this document. The reader might not know or fully understand the purpose. Be specific, and if you are responding to a particular request, say so. For example, what if the purpose had been described as follows?

This document proposes a general system upgrade for XtremaLab's Sports Analysis Measurement Division (SPASM-D).

Pretty useless! This purpose description does not give the reader adequate specifics to put the proposal in its proper context. Everyone at XtremaLab is looking for a systems upgrade. In fact, everyone everywhere is looking for a systems upgrade. This proposal might wind up in a pile with everyone else's "wish list." To fix this problem, we need only add specifics:

This document proposes a general system upgrade for XtremaLab's Sports Analysis Measurement Division's (SPASM-D) computing capabilities. This upgrade is necessary to enable XtremaLab's response to the International Olympic Committee's (IOC) tasking for modeling and analysis of the QuadFINKEL figure skating jump.

Background

In this section, describe the problem that must be solved, adding any background necessary to clarify the requirement; in other words, provide context. Including specifics also demonstrates your understanding of the problem and adds to your credibility.

The QuadFINKEL figure skating jump is so demanding that the IOC is considering the award of an extra-large gold medal to anyone successfully landing the jump in Olympic competition. The IOC has tasked SPASM-D, under IOC Contract IOC-135549, to accomplish advanced 3-D modeling, simulation, and analysis of the QuadFINKEL. This modeling and analysis would give the IOC the scientific basis for justifying the special award. This analysis must be completed within six months of final project approval.

Modeling this skating jump is a complex process due to the element's chaotic nature and high-speed dynamics. To accomplish this analysis in the specified time frame, SPASM-D, in the Technical Report TR-193345, has identified the need for a stand-alone, state-of-the-art graphics, modeling, and analysis capability within the laboratory area. The report identifies the requirement for quantum entanglement communication for various IOC activities. These

stand-alone capabilities must allow interoperability to the other laboratories. TR-193345 also stipulates that the cost of the upgrade not exceed \$100,000.

Scope

Clarify exactly what your proposal covers. Remember, an accepted proposal could be considered a binding contract that obligates both parties. Be careful to spell out any exclusions in the scope section. For example, the following scope statement would effectively preclude any responsibility for “add-on” tasks that are not part of the original problem:

This proposal addresses only the system upgrade of the SPASM-D Analysis Laboratory in support of the IOC tasking for modeling and analysis of the QuadFINKEL. This proposal does not include other graphics, modeling, and analysis tasks.

Measurable Objectives

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Use measurable objectives to explicitly identify what your solution will accomplish. The structure for a measurable objective is formulaic. Measurable objective = Verb + Noun (+ optional details), where the verb is typically one of the following: increase, decrease, reduce, eliminate, create, establish, etc. These verbs are measurable; at the completion of the project, everyone can easily understand if the objectives were accomplished and by how much.

Verb Noun Details
↓ ↓ ↓

For example, you would say “increase first-time yield on Line 1” rather than “improve quality.” For the SPASM-D Analysis Laboratory project, you might list two measurable objectives:

This project, once implemented, will accomplish two objectives:

1. Decrease the computer analysis time for the QuadFINKEL jump to 5 minutes.
2. Increase the analysis accuracy of the QuadFINKEL model by 20%.

Writing the Discussion Section

Approach

Use this section to precisely describe your proposed solution to the problem. Provide enough details to clearly demonstrate that you have

researched the problem, that you understand it, and that you have developed an effective solution.

SPASM-D proposes to the XtremaLab Technical Review Committee that the SPASM-D laboratory's capabilities be upgraded and augmented with a dedicated quantum local-area network (QLAN) composed of two Titanium Graphics 4000 Visual Workstations, one Titanium Graphics 4000A Modeling Station, and one XtremaLab SuperSWITCH 2000-X.

Titanium Graphics workstations provide a recognized standard of excellence in performing advanced graphics, modeling, and analysis tasks. Off the shelf, these systems have the required processing power, storage capacity, and network/device interfaces to function seamlessly on the existing network. By connecting these workstations with QLAN through an XtremaLab SuperSWITCH, then cascading this SuperSWITCH to the company's network backbone, these workstations can easily provide the needed stand-alone computational and storage capabilities. They will also meet all company and communication requirements.

Result

Use this section to show what benefits will accrue from the proposed solution. In other words, if XtremaLab accepts the solution described in the approach section, this is what it will get for its money.

The Titanium Graphics 4000A Modeling Station will provide the dedicated analytical capabilities required to thoroughly model and understand the various components of the QuadFINKEL. The two Titanium Graphics 4000 Visual Workstations will provide the 3-D graphics rendering required by the IOC tasking. The analysis accuracy of each QuadFINKEL jump will be increased by 20% while reducing the analysis time for each to 5 minutes or less.

Statement of Work

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Use this section to describe the major tasks you will perform to implement the proposed solution. In this example, you might identify three major tasks:

To achieve the goals of this proposal, the following tasks will be accomplished:

- Task 1: Acquire the necessary equipment; transport, unpack, assemble, and place in the work area. (8 hours)
- Task 2: Set up the operating systems and configure QLAN. (16 hours)
- Task 3: Install application software, test it, and run calibration and verification simulations. (16 hours)

Deliverables

When you have completed all tasks for this project, what will your tangible deliverables include? This is anything that you can give to your client or implement for your client, either internal or external. Examples include a

new layout, a set of instructions, a formulation, a standard, a tool, updated training, a new machine installed, an upgrade to an existing piece of equipment, a new packaging design, a revised product specification, or any other output from your work.

The SPASM-D simulation and modeling staff will provide the following at the conclusion of this project:

1. Two fully implemented and updated workstations
2. One fully implemented and updated modeling station
3. Fully rendered QuadFINKEl graphics in SIM-IT
4. Full statistical analysis of QuadFINKEl simulations

Writing the Resources Section

Personnel

In this section, discuss who will be doing the work and why they are qualified:

The SPASM-D simulation and modeling staff will analyze and model the QuadFINKEl, using proprietary laboratory software. Additionally, XtremaLab graphics consultants and IOC figure skating experts will work closely with the simulation and modeling staff to ensure accuracy and effectiveness of all required 3-D renderings.

Facilities and Equipment

Here you should describe the physical resources that will be used to do the work. Be specific, accurate, and honest. Any gaps or missing information will work against your proposal:

The SPASM-D facility includes adequate space for this effort. Suite 104 in Building 45 has the required fiber optic channel and is available for this project. The specific computer equipment required includes the following:

- Two Titanium Graphics 4000 Visual Workstations*
 - Integrated Visual Computing architecture with Mercury chipset
 - Intel Octagon III 5-gigahertz processors with 4G L-2 cache
 - 64-gigabyte ECC SDRAM
 - 8-terabyte Ultra2 drive
 - Two 64-bit PCI buses
 - Integrated QLAN
 - 4 USB-D ports
 - Winnex IN/IX Open Source OS

- Three-year warranty with one-year on-site service
- One Titanium Graphics 4000A Modeling Station
 - Dual Intel Octagon III 6-gigahertz processors with 4G L-2 cache
 - 64-gigabyte ECC SDRAM
 - 100-terabyte RAID disk array
 - Titanium Graphics 8000W 42-inch digital, flat-panel display
 - One XtremaLab SuperSWITCH 2000-X
 - QLAN cable with connectors

*Note: Existing XtremaLab display monitor resources are available to support these workstations.

Writing the Costs Section

Fiscal

Be sure your cost estimate falls within the monetary requirements and constraints of the problem. In this case, the background discussion specified a maximum cost of \$100,000.

The proposed system upgrade includes the following equipment and installation costs:

Equipment:

- Two TGI 4000 Visual Workstations @ \$5,995 each = \$11,990
- One TGI 4000A Modeling Station @ \$9,090 each = \$9,090
- One SuperSWITCH X-2000 @ \$20,000 each = \$20,000
- 400 feet QLAN cable with connectors @ \$145/foot = \$58,000

Installation:

- 0.02 full-time equivalent (FTE) technician (40 hours) @ \$42,000/FTE = \$840

Total cost: = \$99,920

Time

Be sure your time estimate falls within the requirements and constraints of the problem. Note that there is a difference between **clock time** and **calendar time**, and often it is useful to address both. If there are 40 hours of actual work to do, that might require four weeks to accomplish because of the nature of the work. For example, you might need to spend one hour talking to an expert, but it takes one week to set up and have the meeting. Another example is that when you order supplies, it might take five hours to

determine necessary specifications and place the orders, but it takes 2 weeks (or 2 months!) to receive them. At a minimum, you should address calendar time; if you only include clock time in your proposal, you will set yourself and your audience/client up for disappointing, unmet expectations.

Planning for Hiccups

While you might want to check out a project management textbook or website, one thing we want to briefly address is the idea of including some “safety” or “padding” in the timing of your project plan. It is important to consider adding a little bit (maybe 20%?), but beware if your proposal goes “up the chain” and multiple managers continue to pad the timeline—you do not want to submit a proposal stating it will take you twice as long to do something than it really will. A little safety net is a good idea; too much safety net might result in your proposal being rejected.

In this case, the background discussion specified a maximum time of 6 months to complete the analysis. The schedule for the proposed upgrade must meet that requirement.

Assuming availability of equipment and materials, and using fully qualified XtremaLab technicians, the entire upgrade can be completed in 40 hours. This estimate includes 8 hours to acquire, deliver, and set up the equipment; 16 hours to set up the operating system and network connectivity; and 16 hours to configure and test the applications. This upgrade schedule would provide adequate time to accomplish the modeling and analysis required by the IOC tasking. This project can be completed within three calendar weeks of its start.

Writing the Conclusion Section

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Summary

Pay particular attention to this last section. It represents your final opportunity to sell the proposed solution by describing the benefits to be gained through your proposal. This section also allows you to demonstrate your competence by describing how your proposal addresses any risks inherent in the project.

The proposed Titanium Graphics QLAN upgrade will provide a viable, cost-effective solution for meeting the IOC QuadFINDEL modeling and analysis requirement. The following deliverables will be completed: 1) Two fully implemented and updated workstations; 2) One

fully implemented and updated modeling station; 3) Fully rendered QuadFINKEl graphics in SIM-IT; and 4) Full statistical analysis of QuadFINKEl simulations. The upgrade will decrease the computer analysis time for the QuadFINKEl jump to 5 minutes and increase analysis accuracy by 20%. The entire system can be up and running within three weeks for a price that is within the cost guidelines. Given the quality of the system and reputation of its manufacturer for setting the standard for high-end analysis and modeling computing systems, the risks of this solution are minimal. In fact, the proposed system provides exactly the right capabilities at precisely the right time, with a cost and timeline that are well within the company's needs.

Contact

Tell the reader whom to contact for more information. Be sure that this contact information is accurate and that the person specified understands the proposal and is available to answer questions.

For more information regarding this proposal, please contact Erin R. Ronaldson, Ph.D., P.E., Director of Systems Engineering, SPASM-D, at ext. 445; or email: eronald@XtremaLab.com.

Sources

Make sure to list all of the sources used to develop the proposal—and format the required documentation citation as expected by your audience.

1. Home page, <https://XtremaLab.com>. Accessed November 26, 2020
2. International Olympic Committee RFP, <https://www.olympic.org/QuadFINKEl-RFP>. Accessed November 27, 2020
3. <https://QLAN-Distribution-Warehouse.com>. Accessed November 28, 2020

Layout and Presentation

There is an unwritten rule that goes something like this: “Lousy work presented professionally often counts for more than decent work presented unprofessionally.” Many people present foolish ideas so smoothly that they convince others anyway. However, this “style over substance” approach is inappropriate and generally ineffective for technical writing.

You must present your substantive ideas as professionally as possible. Good ideas presented in an unprofessional manner are often interpreted as deficient—especially in technical proposals. To be persuasive, you must present valid ideas clearly and accurately in a document that is professional in every respect, including layout, style, and appearance.

The first page of a proposal is often a title page (see [Example 8.2](#)), which should be included whenever a transmittal document (email or letter) is used (see [Example 8.3](#)). Title pages for proposals vary considerably and often have their format and content specified by the requesting agency; however, they generally contain, at a minimum, the following information:

- The complete title of the report.
- The name(s) of the submitting organization(s), office(s), or author(s).
- The date of submission.
- The name(s) of the receiving office(s), individual(s), or organization(s).
- Title pages might also need to contain additional information, such as RFP numbers and dates, as well as corporate trademarks and designs.

Example 8.2 Title Page

Upgrading Modeling and Analysis Capability

A Proposal

By

XtremaLab SPASM-D/LXR

Submitted

on

March 31, 2021

For the

XtremaLab Technical Review Committee

Attachments and Appendices

Because proposals are persuasive documents designed to sell goods and services, you should not overlook the potential benefit of adding attachments to your report, but first be sure attachments are allowed by the requesting organization. Attachments can provide information that supplements or clarifies material in the report but that might be too detailed, extensive, or specialized to be included in the report itself. Attachments can be individually numbered and simply attached at the end of the report, or they can be included in a formal appendix section. If placed in an appendix section, attachments should be identified individually as Appendix A, Appendix B, and so on. If you include attachments or appendices, they should be referenced in the report body.

Attachments can provide a wide range of information. For [Page 101](#) example, you might use attachments to show specific delivery schedules, detailed cost analysis, or additional background information about you or your company. You could even include additional information that supports your proposal—perhaps a detailed listing of successfully completed projects to establish prior performance. Consider whether additional information would make your proposal more attractive to a potential client.

Transmittal Document

When submitting a proposal, especially to a recipient external to your organization, you should include a **transmittal document**. This document can be a **transmittal letter** (sometimes called a **cover letter**) or, more commonly for external recipients and when the recipient is internal to your organization, a **transmittal email**. Transmittal documents (also called **transmittals**) ensure that the proposal gets to the right place and is considered in the proper context. See [Chapter 19](#) for a more detailed discussion of transmittals.

Remember, however, that transmittals are not part of the proposal per se. Transmittal letters are attached to, or enclosed with, the proposal. In the case of a transmittal email, the text of the transmittal is in the email message, while the proposal (including title page, appendices, and attachments) is attached to the email, and the subject line identifies the attached document.

You should use transmittals and title pages when sending proposals to recipients external to your organization. When proposals are internal documents, as is the case with our XtremaLab example, they are often transmitted without a letter and often without a title page. Instead, these proposals are frequently sent as an attachment to an email message. Ensure that any documents shared electronically are in formats that your recipient can use (e.g., docx, PDF); you will have fewer issues with software incompatibility if you always save your documents as a PDF and attach them rather than trying to share a link from a cloud storage service or sending documents created in less commonly available software.

Here is an example transmittal email (see [Example 8.3](#)):

Example 8.3 Transmittal Email

Date: March 31, 2021
To: The XtremaLab Technical Review Committee
From: XtremaLab SPASM-D/LXR (E. R. Ronaldson)
Subject: Proposal for Upgrading Modeling and Analysis Capability, Contract IOC-135549; SPASM-D TR-193345

XtremaLab Technical Review Committee:

Attached is our proposal for upgrading SPASM-D modeling and analysis capabilities. This upgrade is required to accomplish advanced modeling and analysis of the QuadFINKEl figure skating jump in accordance with the referenced contract and technical report.

For additional information, please contact Erin Ronaldson, SPASM-D, ext. 445, email: eronald@XtremaLab.com.

Thank you for your consideration,
Erin

Additional Example 8.4 Topic Proposal

Student Proposal for a Class Project

The following example is a short, informal proposal that a student might submit to a professor to fulfill a proposal requirement in an engineering technical writing course. In this example, Ethan B. McFinkel (the fictitious cousin of the fictitious Sunny Misumoto in [Chapter 20](#)), is proposing a state-of-the-art research report to satisfy the final course requirement for a formal report on a topic in his major field of study. He is including cost data only because it is required for the course; in reality, like all the other students, he has no expectation of being paid anything. Notice the use of first person throughout this document. Such use might be preferred in short, informal documents; if McFinkel referred to himself in the third person, it would seem too formal. Also notice the expert level at which he is writing (his audience is, after all, an electrical engineering professor). Finally, notice how he numbered the sections and subsections. This type of compound-numbering scheme is frequently used in proposals because it allows for easy and precise reference to specific sections and elements.

A Proposal for a Research Report on the SuperXLTube

By Ethan B. McFinkel, EGR 335 student

1. Introduction

1.1 Purpose

I propose to research and produce a state-of-the-art report on the SuperXLTube transmitting tube to fulfill the course requirements of EGR 335, Technical Communication for Engineers and Scientists, Spring 2021.

1.2 Background

EGR 335 requires a formal report on a topic in my major field of study (electrical engineering). I am interested in high-power transmitting tubes that are widely used today in RF transmitters supporting TV, FM, and shortwave-broadcast activities as well as military applications in electronic warfare. In fact, high-power RF transmission is one of the few areas of modern electronics in which solid-state devices are not competitive in terms of either cost or efficiency (Anderson 2004, pp. 34–40). I am particularly interested in the SuperXLTube, which is the highest-power transmitting tube currently available in the commercial market, providing continuous RF power outputs of more than a megawatt (XL Tubes).

1.3 Scope

My research project will focus primarily on the technical aspects of the SuperXLTube used in high-power broadcast service. Other aspects of high-power broadcast systems, such as economic, environmental, and political considerations, will not be covered in this report. The measurable objective of this project is to create a literature review which can be referenced in future work.

2. Discussion

2.1 Approach

My proposed state-of-the-art report will begin with a brief review of power tube theory, followed by a short history of RF power amplifier tube design for high-power broadcast service. The report will then focus on the technical aspects of the SuperXLTube. The tube will be analyzed for both class C RF service up to 100 megahertz and AF service in high level amplitude modulation. Specific areas for investigation will include

heat dissipation requirements and techniques, ceramic design considerations, and functional performance characteristics.

2.2 Results

My proposed report will provide in-depth technical information on the functional performance of the SuperXLTube in a variety of broadcast services, and it will include both RF and AF applications in AM, FM, and TV modes. The proposed report will also fulfill the formal report requirement for EGR 335, Technical Communication for Engineers and Scientists.

2.3 Statement of Work

To achieve the goals of this research, I will accomplish the following tasks:

- Task 1: Briefly review basic theory of RF and AF vacuum tube amplifiers to provide the theoretical context for this report. (3 hours)
- Task 2: Research and document RF power amplifier tube design for high-power broadcast service since 1960 to provide a historical context for this device. (5 hours)
- Task 3: Research and document the design specifications and manufacturing techniques for the SuperXLTube that demonstrate the unique characteristics of this device. (5 hours)
- Task 4: Describe specific applications for which this device is used and provide examples of the transmission equipment employing this device. (2 hours)
- Task 5: Produce and deliver a technical presentation in class at a time and place to be determined. (4 hours)
- Task 6: Produce and deliver a formal report document in accordance with the course syllabus requirements. (6 hours)

2.4 Deliverables

I will submit two deliverables for this project:

1. Technical presentation
2. Formal report

3. Resources

3.1 Personnel

Consistent with the requirements of the course, I will be doing all the research and writing on this project. I am a senior majoring in Electrical Engineering with University Honors and have a solid background in electromagnetics and electrostatics. I also have a proven track record of success during the past four years producing written reports to meet academic requirements. Additionally, I have access to various professors and practicing engineers who can provide any guidance that might be required.

3.2 Facilities and Equipment

The University provides adequate research and document production facilities for this project. High-speed internet connectivity, multiple computer systems with all required software, and high-resolution printers are available for this project.

4. Costs

4.1 Fiscal

As provided in Section 2.3 (statement of work), 25 hours of my labor will be required to complete this project. Additionally, 60 miles of travel to and from research facilities will be required, along with the use of the University's computer systems and networks. Access to these networks is supported by the quarterly laboratory fee. The cost of the course itself is represented by tuition charges for 3 credit-hours. The line-item cost for the 5-week project is provided in [Table 8.1](#).

Table 8.1 Project cost data

Labor:	25 hours @ \$100/hour	=	\$2,500
Travel:	60 miles @ \$0.35/mile	=	48
Facilities:	Lab fee (\$150 pro-rated @ 50% / 5 weeks)	=	75
Course:	Tuition (\$1250 pro-rated @ 50% / 5 weeks)	=	625
Total cost		=	\$3,248

4.2 Time

I will accomplish all tasks during the last 5 weeks of the quarter and will provide all deliverables according to the following schedule:

- Presentation: At a time and place to be determined by the course instructor.
- Report: In the classroom, at 9:30 a.m. on Thursday, June 10, 2021.

5. Conclusion

5.1 Summary

EGR 335 requires a formal report document and formal report presentation on a topic in electrical engineering, my major field of study. The SuperXLTube transmitting tube is the highest-power RF amplifier available today and is used in a variety of broadcast and military applications. It represents the state of the art in high-power transmission devices and, as such, provides an appropriate topic for research in my major. Approval of this proposed report will allow me to fulfill the course requirements while learning more about high-power broadcast systems. This work will be completed by June 10, 2021.

5.2 Contact

For more information, please email me at mcfinkel@wxccs.edu, or phone (555) 867-5309.

5.3 Sources Used

Anderson, Warren S.: *Big Tubes—Big Power*. New York: RF Press, 2006. XL Tubes, <http://www.support.xltubes.com/SuperXLTube.html>. Accessed May 14, 2021.

- Have I thoroughly considered who my audience is, including understanding expectations and requirements?
- Do I understand the purpose of writing this proposal?
- Have I defined the problem in great enough detail to ensure that my readers will understand the context for this proposal?
- Have I described the background for this problem in great enough detail to clearly identify the variables driving my proposed solution?
- Have I defined in the scope section how I am limiting my proposal?
- Have I identified the measurable objectives that will be used to assess impact?
- Have I laid out my proposed solution in adequate detail? Do I have enough details to ensure that my solution is credible?
- Have I clearly described the benefits of my solution?
- Do I need to provide a statement of work? If so, have I adequately described the major tasks involved in implementing the proposed solution?
- Have I explicitly stated the deliverables that this project will produce?
- Have I clearly defined the resources required to implement my proposed solution, including people, facilities, and equipment?
- Have I provided the required budget for implementing this proposal, including the costs in both money and time? Have I broken out the costs in adequate detail for my audience? Are these cost estimates consistent with the financial constraints of the problem?

- Is my time estimate consistent with the tasks in my statement of work?
- Have I clearly summarized the proposal and provided a strong concluding argument for its adoption?
- Have I provided references for all the sources that I have used and/or acknowledged all the help I received?
- Have I provided an available and knowledgeable contact at the end of the proposal?
- As the last task and if required, have I written a one-page executive summary which encapsulates the entire story?

Exercise

Consider the following informal proposal for a wastewater treatment plant. How do you respond to each of the sections and the proposal as a whole?

- Do any particular sections stand out as obviously strong or deficient, and if so, why?
- Did you find any inconsistencies in the proposal?
- Which sections, if any, did you respond to positively—that is, you understood and agreed with what the proposal was saying?
- What will be measured in this project to assess success?
- Which sections, if any, did you respond to negatively—that is, the presentation just did not make sense, the support for the presentation was lacking, or you had a “bad feeling” about what you were reading?
- Since the success of a proposal often is based on the credibility of the proposing person, company, or agency, it is important to consider whether this proposal is credible. Which parts did you find credible and which did you not? To what extent does credibility in this proposal impact your assessment of it?

- What are the project deliverables?
- Finally, if you were the decision-maker with the responsibility for accepting or rejecting this proposal, what would you do and what would be the major factors influencing your decision?

Informal Proposal for a Wastewater Treatment Plant for Big City

Submitted in response to RFP #Y0643356 (March 26, 2021) by Acme Enterprises, Inc.

1. Introduction

1.1 Purpose

This report proposes a basic design concept for a wastewater treatment plant to be used for the primary sanitization and conditioning of Big City's aqueous wastes in response to Request for Proposal (RFP) #Y0643356, dated March 26, 2021.

1.2 Problem

In March 2020, in advance of an intense storm system, warm air advected into the Big Basin of the Great McFinkel River, rapidly melting the snowpack and flooding the Big City wastewater treatment facility adjoining the river. As the storm's cold front swept across the region, the Great Big City F5 Tornado of 2020 was spawned. This tornado hit the flooded wastewater treatment plant, turning the facility into debris, which was then carried as far away as Kansas.

As part of its emergency action response, the decision was made to discharge untreated water into the Great McFinkel River. Subsequent tests have shown that the discharged wastewater contained a variety of pollutants including dissolved ions of sodium and potassium, dissolved molecules from organic byproducts of plant and animal decay, dissolved human wastes, and other materials resulting from industrial and agricultural activities. Excessive amounts of suspended materials also existed, including *E. coli* and *Coliform* bacteria, floating debris (usually decaying plant and animal carcass material), and sand and dirt.

Additionally, a significantly higher infectious disease and mortality rate correlating to this discharge has been noted in the populated resort areas downstream. Both criminal and civil actions have resulted in court decisions requiring, among other legal remedies, that these pollutants be removed before the water is released by Big City into the Great McFinkel River. Big City's new Department of Environmental Safety's RFP #Y0643356 has requested a conceptual design proposal for a wastewater treatment facility that will be capable of removing these wastes.

1.3 Scope

This proposal is limited to presenting a preliminary concept of plant design. This document responds only to the requirements of RFP #Y0643356; consequently, it is not a final design proposal and will not present schedule or cost data. Additionally, it will not address the disposal of wastewater treatment residuals (e.g., sludge), as this is an operational concern appropriate for separate consideration.

2. Discussion

2.1 Approach

Acme Enterprises, Inc., proposes a wastewater-treatment facility and process that will provide conceptual guidance for developing a primary treatment facility to address the problem described in Section 1.2. The proposed wastewater treatment plant will be designed and constructed to achieve the following process steps: screening, pumping, aerating, sludge removal, scum removal, sanitization, and residual disposal. For a notional diagram of the proposed facility, see [Figure 8.2](#).

2.1.1 Screening

Screening is the mechanical process of removing floating debris. It is assumed that the water entering the proposed treatment facility will contain large pieces of floating debris such as tree branches, rocks, trash, and decaying animal carcasses. This debris must be removed before the water enters the treatment plant. The proposed solution includes a series of self-cleaning, water intake filtration screens that will accomplish this function. One problem with such an approach is the proclivity of zebra mussels to collect at the water intake screens and clog the intake. A time-released potassium permanganate (KMnO_4) compound will be used to

control the mussel population. Once this debris has been filtered, sediment in the water will be removed.

2.1.2 Sediment Removal

Sediment removal is the mechanical process of allowing suspended debris to settle out. It is assumed that suspended, insoluble particles (sand and dirt) will pass through the filter screen and will require a sedimentation tank. Here, the water will be allowed to sit so that heavier particles will naturally settle to the bottom. The sediment-free water can then be pumped from the top of the sedimentation tank to the aeration tank that will occupy the highest point of elevation at the waste treatment facility. This design will use gravity to move the water from the aeration tank through the rest of the system.

2.1.3 Aeration

Aeration is the mechanical process of re-oxygenating water by exposure to air. The water exiting the sediment tank will need to be moved and shaken to ensure wide exposure to air. This step also releases many of the unpleasant, dissolved gases (e.g., hydrogen sulfide) from the water, while replenishing oxygen that has been consumed by the oxidation of decaying material. A series of parallel tanks is proposed. Air will be pumped through the water as it flows from one parallel tank to another, replenishing the oxygen levels and preparing the water for sanitization.



Figure 8.2

Proposed treatment plant.

2.1.4 Sanitization

Sanitization is the chemical process of destroying bacteria and viruses. After aeration, the water flows into the sanitization tank, where chlorine is added to kill any bacteria that could pose a health risk. As the bacteria are destroyed, the chlorine is substantially neutralized. Any remaining chlorine will be neutralized by adding appropriate chemicals before the effluent is released into the Great McFinkel River.

2.2 Result

The proposed wastewater treatment system will ensure Big City's wastewater effluent will be safe for discharge into the Great McFinkel River.

2.3 Statement of Work

To achieve the desired outcomes, three tasks will be completed.

- Task 1: Clear the land (30 days)
- Task 2: Build the water treatment facility (300 days)
- Task 3: Turn on the water (5 minutes)

3. Resources

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3.1 Personnel

Acme Enterprises maintains a highly trained staff, all of whom are multifaceted and can easily perform adequately in a multitasking environment. Virtually all the required skills are available from one or more members of the Acme Enterprises team. For example, there's Jayden Jones, an accomplished steel walker before the accident, who will oversee structural design and construction; and we must not forget Cameron Smith, who, for many years as a cemetery worker, poured concrete internment vaults.

3.2 Facilities and Equipment

Acme Enterprises is justifiably proud of its superb heavy construction equipment capabilities.

4. Costs

4.1 Fiscal

The wastewater treatment facility recently constructed upriver at Williamstown costs \$135 million, including acquisition of all permits and licenses. Since Williamstown is about the same size as Big City, Acme Enterprises will do the entire job for the same price: \$135 million.

5. Conclusion

5.1 Summary

Big City's wastewater contains a variety of pollutants that must be removed before the water is released into the Great McFinkel River. The pollutants take the form of wastes including dissolved ions of sodium and potassium, dissolved molecules from organic byproducts of plant and animal decay, and dissolved human wastes and other materials resulting from industrial and agricultural activities. In addition, measurable amounts of suspended materials also exist, including *E. coli* and *Coliform* bacteria, floating debris (usually decaying plant and animal carcass material), and sand and dirt. Acme Enterprises proposes to construct a state-of-the-art waste treatment plant, in response to RFP #Y0643356 (March 26, 2021).

5.2 Contact

For more information, contact Jayden Jones at email: JJones@yourbestfriend.com.

Notes

1. National Institutes of Health (NIH), <https://www.nih.gov/>. Accessed December 20, 2020.
2. National Science Foundation (NSF), <https://www.nsf.gov/>. Accessed December 20, 2020.
3. Department of Energy (DOE), <https://www.energy.gov/>. Accessed December 20, 2020.
4. Department of Defense (DOD), <https://www.defense.gov/>. Accessed December 20, 2020.
5. Defense Advanced Research Projects Agency (DARPA), <https://www.darpa.mil/>. Accessed December 22, 2020.
6. "Proposal & Award Policies & Procedures Guide," NSF, June 1, 2020, https://www.nsf.gov/pubs/policydocs/pappg20_1/index.jsp. Accessed December 22, 2020.
7. Eric Boodman, "Font of despair: In the fierce competition for science funding, even a typeface glitch can be fatal," *Stat*, October 26, 2018, <https://www.statnews.com/2018/10/26/typeface-glitch-dooms-va-ptsd-funding/>. Accessed December 22, 2020.

8. Grants.Gov, <https://www.grants.gov/>. Accessed December 22, 2020.

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CHAPTER NINE

Progress Reports



If you have ever owned a pet animal, dog or otherwise, you have probably interacted with a veterinarian or animal clinic as part of caring for your creature. Sometimes pets get sick, but even if they are the picture of health, you keep them that way by ensuring they have necessary vaccines and other scientifically supported interventions (like flea prevention medication, for example!). Let us say that your favorite friend, Fluffy, goes to the vet for a checkup, and the doctor discovers a previously unknown health concern. The doctor prescribes a treatment that you must administer faithfully at home for the next two weeks, and informs you about possible side effects and what you should do if they occur. You say, great; you are glad this issue can be treated and head home, ready to help your faithful companion. On Day 2 of the treatment, the doctor calls you. “How is everything going? How is Fluffy doing?” You are happy the doctor has called; they want a progress report on Fluffy. You think to yourself, how **is** Fluffy doing? You must logically summarize what has happened since the doctor visit and relay it in a way that makes sense to your audience. The purpose is to inform them; there is typically no persuasion involved beyond reassuring your audience that you are on track with the plans (see [Figure 9.1](#)).



Figure 9.1

When it is time to provide a progress report, you want to be clear and efficient in your communication so that your audience clearly understands the current status.

This same idea is common in industry. Once a proposal has been Page 110 accepted and work begins on a project, you may be required to give the client (or your supervisor, or your professor, as the case may be) periodic reports on how well you are doing. A **progress report** (often called a **status report**) is the type of document commonly used to detail and document the status of the project. Writing a progress report usually means answering straightforward questions such as: Are you on schedule? Are you within budget? Are any risks evident? If so, how do you plan to control them? What problems have you encountered? What remains to be done? What is your plan for doing it? What is your overall assessment?

Does the progress report requirement indicate that your client or professor does not trust you? Maybe—but that is not the point. In the “real world,” normal business practice requires specific, written documentation, not abstract trust. Often your supervisor must update **their** supervisor about the progress being made. In some cases, continuation of the contract or

partial payments for work completed (or even a grade in a course) may depend on your writing and submitting satisfactory progress reports.

On larger projects, progress or status reports may also be known as **milestone reports**, because, like milestones on a road, they mark the passage from one point to another on a journey toward some final goal. In the real world, periodic “progress” payments may be contingent on your submitting successful milestone reports. Additionally, you may be asked by your supervisor periodically to submit an **activity report**. This type of project report provides the ongoing status of a project (or projects) for which you are responsible—kind of a “tell me what you have done for me lately” report.

Progress reports, status reports, milestone reports, and activity reports all do basically the same thing and contain the same kinds of information. To keep things simple, this chapter will focus only on progress reports.

What Is a Progress Report?

Progress reports document the status of a project. These reports describe the various tasks that make up the project and analyze the progress that has been made toward completing each task. Often a progress report is delivered informally: perhaps at a weekly departmental meeting or in a one-on-one conversation with your supervisor. Sometimes after an informal update, you will be asked to follow up with a short email which reiterates the progress you just verbally described. This is helpful for both you and your audience; by having a “paper trail,” which is, of course, usually electronic (ironic right?), everyone can access the facts as the project progresses and it does not devolve into a “you never...” or “they said...” situation. As with all other technical documents, if your audience has a required or preferred format for writing a progress report, an organization-specific form for example, you should use it.

If you have written a proposal that has been accepted, then you also have probably committed yourself (or your organization) to completing certain tasks by a particular time and for a specified amount of money. Generally, in a progress report, you must tell the reader three things: the problem you are solving, the solution you are implementing, and how well you are doing.

You will find that writing a progress report is more pleasant when you have some progress to report.

An impetus is a force that makes something happen. There are many ways to define it and it has many synonyms. You will frequently hear this word used in technical discussions.

Writing a progress report typically requires that you do three things: review, describe, and evaluate.

1. Review the problem that was the **impetus** for the original proposal. To do this, reference the original proposal by number or title, indicate when it was accepted, and then describe the problem that prompted the proposal in the first place. Often you can copy much of the problem description from the original proposal into the progress report.
2. Describe the solution offered in the original proposal, including the tasks involved in implementing this solution (usually listed in the statement of work) and the planned dates for completing each task.
3. Evaluate how well you are doing in terms of each task, and provide an overall assessment of your progress. In other words, lay out to what extent you are getting the job done within the time frame and cost constraints of the original proposal.

Progress Report Formats

Progress reports can take many forms, from a simple letter to a multipage document. Use the format that your supervisor or instructor tells you to use, or the one you are required to use by the client. If you do not have a specific format, [Outline 9.1](#) should do the trick.

Outline 9.1 Progress Reports

Introduction

Purpose

Review the reason for writing this report.

Background

Review the problem and your proposed solution.

Scope

Describe what this report will and will not cover.

Status

Tasks completed

For each relevant task (repeat this pattern for each task or activity completed).

- Provide a description of the task.
- Describe the things that have been accomplished.
- Describe how long it took to accomplish them.
- Describe the difficulties, if any, that were encountered.

Tasks remaining

For each relevant task (repeat for each task or activity remaining).

- Provide a description of the task.
- Describe the things that still must be accomplished.
- Provide the timetable and strategy for completing the task.
- Describe the risks and your approach to completing the task.

Conclusion

Summary

Provide an appraisal of the current status.

Evaluation

Provide an assessment of the progress made to date.

Forecast

Provide a forecast for completing this project.

Contact

Provide a contact for more information.

The following example of a progress report is a follow-up to [Chapter 8](#)'s proposal and uses [Outline 9.1](#): implementing a Titanium Graphics QLAN to analyze and model the QuadFINKEl figure skating jump.

However, before you read the progress report itself, let us talk about the context for it. Assume that the proposal was accepted and the project is under way. The XtremaLab Technical Review Committee has asked for a short progress report that the committee can forward to the International Olympic Committee for a meeting the day after tomorrow. No big deal! Producing this type of report is not difficult, especially if you have a copy of the original proposal handy. You will need that original proposal because it describes the context and tasks about which you will now report your organization's progress.

The entire progress report is assembled below as it might be submitted to the committee. Notice the addition of [Figure 9.2](#), a timeline Gantt chart showing the relative progress for the three tasks. For short, informal reports, a timeline chart normally is not required; one is included here only as an example. A more detailed Gantt chart is also provided in [Chapter 17 \(Figure 17.13\)](#).

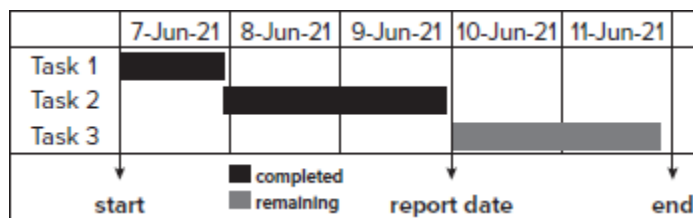


Figure 9.2

Project timeline for XtremaLab SPASM-D project.

Example 9.1 Progress Report

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1. Introduction

1.1 Purpose

This report documents the progress XtremaLab has made on upgrading the modeling and analysis capability pursuant to its accepted SPASM-D proposal of March 31, 2021.

1.2 Background

The QuadFINKEl figure skating jump is so demanding that the IOC has tasked SPASM-D, under IOC Contract IOC-135549, to accomplish advanced 3-D modeling, simulation, and analysis of the QuadFINKEl. Modeling this skating jump is a complex process due to the jump's chaotic nature and high-speed dynamics. To accomplish this analysis in the specified time frame, SPASM-D proposed to the XtremaLab Technical Review Committee that the SPASM-D laboratory's capabilities be upgraded and augmented with a dedicated quantum local-area network (QLAN). The new QLAN will be composed of two Titanium Graphics 4000 Visual Workstations, one Titanium Graphics 4000A Modeling Station, and one XtremaLab SuperSWITCH 2000-X. The following deliverables will be completed: 1) Two fully implemented and updated workstations; 2) One fully implemented and updated modeling station; 3) Fully rendered QuadFINKEl graphics in SIM-IT; and 4) Full statistical analysis of QuadFINKEl simulations. The upgrade will decrease the computer analysis time for the QuadFINKEl jump to 5 minutes and increase analysis accuracy by 20%. The proposal was accepted without change on June 1, 2021, with the total system upgrade to be completed on or before June 30, 2021.

1.3 Scope

This report provides the status of all tasks described in the accepted proposal's statement of work, which includes the following:

- **Task 1:** Acquiring the necessary equipment; transporting, unpacking, assembling, and placing in the work area. (8 hours)
- **Task 2:** Setting up the operating systems and configuring network connectivity. (16 hours)
- **Task 3:** Installing application software, testing the software, and running calibration and verification simulations. (16 hours)

2. Status

2.1 Tasks Completed

Task 1: Acquiring the necessary equipment; transporting, unpacking, assembling, and placing in the work area. (8 hours)

- All necessary equipment—including two Titanium Graphics 4000 Visual Workstations, one Titanium Graphics 4000A Modeling Station, and one

SuperSWITCH 2000-X—was purchased on June 2, 2021, within budget estimates, and delivered, unpacked, and placed in Suite 104, Building 45, on June 4, 2021. The entire task required approximately 7 hours. No difficulties were encountered.

Task 2: Setting up the operating systems and configuring network connectivity. (16 hours)

- All three workstations came with the Winnex IN/IX Open Source OS preinstalled. We successfully configured these machines and the switch to operate on the QLAN. Initially, the QLAN experienced an excessive number of packet collisions. The problem was isolated to the SuperSWITCH, which was subsequently replaced. We completed system setup and network configuration in 15 hours.

2.2 Tasks Remaining

Task 3: Installing application software, testing the software, and running calibration and verification simulations. (16 hours)

- We have just started this task. We are well ahead of schedule on implementing the statement of work, and we foresee no problems in completing this task and the entire project on time and within specified budget constraints.

3. Conclusion

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The SPASM-D technical staff is pleased with the progress made to date. We procured, transported, unpacked, assembled, and installed the equipment in its intended operating location in only 7 hours, thereby completing Task 1 ahead of schedule. We configured the operating systems and achieved full network connectivity in 15 hours, thereby completing Task 2 ahead of schedule (see [Figure 9.2](#)). We are now in the process of installing, testing, and calibrating the application software and expect to complete Task 3 within the scheduled 16 hours. Consequently, we at XtremaLab/SPASM-D assess the progress on this project as excellent and are confident that the entire project will be completed on time and within the proposed budget.

For more information on this project, please contact Erin R. Ronaldson, Ph.D., P.E., Director of Systems Engineering, SPASM-D, at ext. 445; or email: eronald@XtremaLab.com.

Now that you have read an example of a progress report, albeit a fictitious one, let us take a closer look at each of its components and what they do.

Introduction Section

Purpose

Following [Outline 9.1](#), first open the introduction by stating the purpose of this progress report. Think about the purpose carefully. Writing an effective report can be difficult if you cannot articulate why you are doing it.

This report documents the progress XtremaLab has made on upgrading the modeling and analysis capability pursuant to its accepted SPASM-D proposal of March 31, 2021.

Background

In the background section of the introduction, describe the context for the project on which you are reporting. The best approach is to reference the accepted proposal, note its date of acceptance, and describe the specific problem it addressed, along with the proposed solution.

The QuadFINKEl figure skating jump is so demanding that the IOC has tasked SPASM-D, under IOC Contract IOC-135549, to accomplish advanced 3-D modeling, simulation, and analysis of the QuadFINKEl. Modeling this skating jump is a complex process due to the jump's chaotic nature and high-speed dynamics. To accomplish this analysis in the specified time frame, SPASM-D proposed to the XtremaLab Technical Review Committee that the SPASM-D laboratory's capabilities be upgraded and augmented with a dedicated quantum local-area network (QLAN). The new QLAN will be composed of two Titanium Graphics 4000 Visual Workstations, one Titanium Graphics 4000A Modeling Station, and one XtremaLab SuperSWITCH 2000-X. The following deliverables will be completed: 1) Two fully implemented and updated workstations; 2) One fully implemented and updated modeling station; 3) Fully rendered QuadFINKEl graphics in SIM-IT; and 4) Full statistical analysis of QuadFINKEl simulations. The upgrade will decrease the computer analysis time for the QuadFINKEl jump to 5 minutes and increase analysis accuracy by 20%. The proposal was accepted without change on June 1, 2021, with the total system upgrade to be completed on or before June 30, 2021.

Scope

In the scope section of the introduction, describe the tasks or aspects of the project that this progress report covers. If the original proposal included a statement of work, you will probably report on some, or all, of the tasks listed. Specify what tasks this progress report will cover. If applicable, specify which tasks you have addressed in an earlier report, which tasks you

will report on in a subsequent report, and whether any tasks are recurring. If the original proposal did not include a statement of work, you must provide a more general description of the tasks on which you are reporting.

This report provides the status of all tasks described in the accepted proposal's statement of work, which includes the following:

- Acquiring the necessary equipment; transporting, unpacking, assembling, and placing in the work area. (8 hours)
- Setting up the operating systems and configuring network connectivity. (16 hours)
- Installing application software, testing the software, and running calibration and verification simulations. (16 hours)

Status Section

This section presents the status of each task listed in the scope section. Normally, you would treat completed tasks separately from the remaining tasks. The tasks-completed section includes tasks that have been concluded and closed out. The tasks-remaining section includes tasks that are still in progress and tasks that have not yet been started. Write the tasks-completed section first, discussing each task separately:

Tasks Completed

Task 1: Acquiring the necessary equipment; transporting, unpacking, assembling, and placing in the work area. (8 hours)

- All necessary equipment—including two Titanium Graphics 4000 Visual Workstations, one Titanium Graphics 4000A Modeling Station, and one SuperSWITCH 2000-X—was purchased on June 2, 2021, within budget estimates, and delivered, unpacked, and placed in Suite 104, Building 45, on June 7, 2021. The entire task required approximately 7 hours. No difficulties were encountered.

Task 2: Setting up the operating systems and configuring network connectivity. (16 hours)

- All three workstations came with the Winnex IN/IX Open Source OS preinstalled. We successfully configured these machines and the switch to operate on the QLAN. Initially, the QLAN experienced an excessive number of packet collisions. The problem was isolated to the SuperSWITCH, which was subsequently replaced. We completed system setup and network configuration in 15 hours.

Once you have finished the tasks-completed section, write the tasks-remaining section in the same way:

Tasks Remaining

Task 3: Installing application software, testing the software, and running calibration and verification simulations. (16 hours)

- We have just started this task. We are well ahead of schedule on implementing the statement of work, and we foresee no problems in completing this task and the entire project on time and within specified budget constraints.

Conclusion Section

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The conclusion of the progress report summarizes and appraises the progress to date and provides a forecast for the rest of the project (including any risks and plans for their mitigation). It also provides a contact in case more information is required. As in the proposal, the contact should be someone familiar with the project.

The SPASM-D technical staff is pleased with the progress made to date. We procured, transported, unpacked, assembled, and installed the equipment in its intended operating location in only 7 hours, thereby completing Task 1 ahead of schedule. We configured the operating systems and achieved full network connectivity in 15 hours, thereby completing Task 2 ahead of schedule. We are now in the process of installing, testing, and calibrating the application software and expect to complete Task 3 within the scheduled 16 hours. Consequently, we at XtremaLab/SPASM-D assess the progress on this project as **excellent** and are confident that the entire project will be completed on time and within the proposed budget.

For more information on this project, please contact Erin R. Ronaldson, Ph.D., P.E., Director of Systems Engineering, SPASM-D, at ext. 445; or email: eronald@XtremaLab.com.

Communicating the Progress Report

Once you have completed drafting, revising, and editing your progress report, you will need to formally transfer it to your audience. In this case, a short email is used to transmit the report. Note that the report does not include a title page or cover letter since it is an informal, internal document. If it were designed for use outside the company, it would need a transmittal document and title page (both in [Chapter 8](#)).

Transmittal Email

To: xltechreview@xtremalab.com

CC: sdjota@xtremalab.com

Subject: Progress Report for Proposal #CL-3478

XtremaLab Technical Review Committee:

Attached is our progress report for upgrading SPASM-D modeling and analysis capabilities. This upgrade is required to accomplish advanced modeling and analysis of the QuadFINKEl figure skating jump in accordance with the referenced contract and technical report. The project is on track to meet schedule and budget.

For additional information, please contact Erin Ronaldson, SPASM-D, ext. 445, email: eronald@XtremaLab.com.

Best regards,

Erin

Example 9.2 Progress Report for a Student Project Page 116

Student Progress Report

The following example is a short, informal student progress report required by an instructor for an assigned project. In this case, Ethan B. McFinkel is reporting on the status of his state-of-the-art report on the SuperXLTube transmitting tube (see his original proposal in [Chapter 8](#)).

Date: May 5, 2021

To: Albert J. Finnian, Ph.D., Instructor, EGR 335

From: Ethan B. McFinkel, EGR 335 student

Re: Progress report on EGR 335 class project

1. Introduction

1.1 Purpose

This report provides the current status of my EGR 335 research project.

1.2 Background

On April 5, 2021, I proposed a state-of-the-art research report on the SuperXLTube transmitting tube. You accepted my proposal as submitted on April 9, 2021. We agreed that my proposed report would provide in-depth technical information on the functional performance of the SuperXLTube in a variety of broadcast services, and include both RF and AF applications in AM, FM, and TV modes. The proposed report would also fulfill the formal report requirement for EGR 335, Technical Communication for Engineers and Scientists.

1.3 Scope

This status report provides my current assessment of the project as of Wednesday, May 5, 2021. The statement of work included the following tasks:

- Task 1:** Briefly review basic theory of RF and AF vacuum tube amplifiers to provide the theoretical context for this report. (3 hours)
- Task 2:** Research and document RF power amplifier tube design for high-power broadcast service since 1960 to provide historical context for this device. (5 hours)
- Task 3:** Research and document the design specifications and manufacturing techniques for the SuperXLTube that demonstrate the unique characteristics of this device. (5 hours)
- Task 4:** Describe specific applications for which this device is used and provide examples of the transmission equipment employing this device. (2 hours)
- Task 5:** Produce and deliver a technical presentation in class at a time and place to be determined. (4 hours)
- Task 6:** Produce and deliver a formal report document in accordance with the course syllabus requirements. (6 hours)

2. Status

2.1 Tasks Completed

Task 1: I successfully completed this review on April 12, 2021, without any problems.

Task 2: I successfully completed this task on April 16, 2021. Initially, I had difficulty finding adequate sources; however, I subsequently located the necessary information on an international manufacturer's website.

2.2 Tasks Remaining

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Task 3: This task is partially complete. I received design specifications from the manufacturer on April 21, 2021; however, I was advised that the manufacturing techniques are proprietary and would not be available. I now plan to research nonproprietary manufacturing techniques of similar tubes to address this task, and I expect to complete the task by May 10, 2021.

Task 4: I have not completed this task, although I have located adequate research materials and will complete this task by May 10, 2021.

Task 5: The presentation has been scheduled for May 19, 2021, but I will not produce the presentation slides until tasks 3 and 4 are complete. The current schedule allows me to delay this task and still present the briefing as scheduled.

Task 6: I have already drafted approximately half of this report and will complete the initial draft once I have finished Tasks 3 and 4. I will work on this task in parallel with Task 5 and will deliver the finished state-of-the-art report as scheduled on June 10, 2021.

3. Conclusion

3.1 Assessment

I am pleased with the project to date and assess my progress as excellent. Most of the major research has been completed, and I am making good progress toward filling in the gaps where information is still lacking. The balance of Task 3 and all of Task 4 should be completed by May 10, 2021. After that, I will complete Task 5 and deliver the presentation as scheduled on May 19, 2021. I also have more than adequate time to produce the final deliverable, state-of-the-art report, and I will deliver it as required by Thursday, June 10, 2021.

3.2 Contact

For more information, please email me at mcfinkel@wxccs.edu, or phone (555) 867-5309.

Progress Report Checklist

- ☐ Do I understand what the audience for this report wants and requires?
- ☐ Have I specified the purpose, background, and scope of this report?
- ☐ Is there any context that I need to understand regarding this report?
- ☐ Have I referenced the accepted proposal by name, number, and/or date?
- ☐ Have I reviewed the problem contained in that proposal?
- ☐ Have I reviewed the proposed solution to that problem?
- ☐ Have I specified the tasks that will be included in this report?
- ☐ Have I properly discussed the tasks completed and tasks remaining?
- ☐ Have I provided an appraisal and forecast in the conclusion?
- ☐ Have I formatted my report according to audience expectations and to make it easy for my audience to read?
- ☐ Have I planned how I will formally transfer my progress report?

Exercise

The following progress report concerns the wastewater treatment facility proposed for Big City in the [Chapter 8](#) Exercise. Assume that somehow, through corruption or some inexplicable set of events, the Acme Enterprises

proposal was accepted and Acme Enterprises was awarded the contract to build the wastewater treatment plant. The contract calls for a progress report 180 days after the contract was signed, which was April 30, 2021. For the purpose of this exercise, assume this document is the progress report that is being provided by Acme Enterprises to Big City pursuant to the terms of the contract.

- What does this document do well, if anything? Review it in terms of the progress report checklist.
- Specifically, what parts, if any, of this progress report seem to be well done and give you a “warm, fuzzy feeling” that this project is going well?
- Specifically, what does this document do poorly, if anything? In other words, what parts of this progress report give you a “cold, hollow feeling” that something about this project is very wrong? Also, focus on “disconnects” between the original requirement and what is being reported.
- After reading the report, what concerns would you have, if any, regarding the progress of this project?

Progress Report

Introduction

Purpose

This status report is provided pursuant to, and as required by, the referenced contract and, as such, constitutes the semiannual progress report requirement for the period of May 1, 2021–October 31, 2021.

Problem

In March 2020, in advance of an intense storm system, warm air advected into the Big Basin of the Great McFinkel River, rapidly melting the snowpack and flooding the Big City Wastewater Treatment Facility adjoining the river. As the storm’s cold front swept across the region, the Great Big City F5 Tornado of 2020 was spawned. This tornado hit the

flooded wastewater treatment plant, turning the facility into debris, which was carried as far away as Kansas.

As part of its emergency action response, the decision was made to discharge untreated water into the Great McFinkel River. Subsequent tests have shown that the discharged wastewater contained a variety of pollutants including dissolved ions of sodium and potassium, dissolved molecules from organic byproducts of plant and animal decay, and dissolved human wastes and other materials resulting from industrial and agricultural activities. Excessive amounts of suspended materials also exist, including *E. coli* and *Coliform* bacteria, floating debris (usually decaying plant and animal carcass material), and sand and dirt. Additionally, a significantly higher death rate correlating to this discharge has been noted in the populated resort areas downstream. Big City's Department of Environmental Safety contracted with Acme Enterprises on April 30, 2021, under contract #C0643356 to design and construct a wastewater treatment facility that will be capable of removing these wastes. To process this wastewater, Acme Enterprises proposed a facility designed to accomplish the following four-step process: **screening, sediment removal, aeration, and sanitization** with the following statement of work (SOW):

Task 1: Clear the land (30 days)

Task 2: Build the water treatment facility (300 days)

Task 3: Turn on the water (5 minutes)

Status

Completed Tasks

Task 1: Clear the land

This task involved removing the existing vegetation and leftover debris from the proposed site. Due to weather conditions and other physical exigencies, Acme Enterprises was unable to clear the land within the first 30 days. However, by using controlled fires on Day 41, we were able to destroy all of the vegetation with only modest impact to vegetation outside the area. Because of minor release errors and unpredicted wind fields, undefined and unproven health risks may have been associated with the normal lifestyle of a few nearby residents. Between Day 42 and Day 81, the site was bulldozed and prepared, with all debris either being

buried in the local landfill or discharged into the Great McFinkel River, where, hopefully, it was properly processed by wastewater treatment plants of the downstream communities.

Tasks Remaining

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Task 2: Build the water treatment facility (300 days)

Acme Enterprises has the equipment and the skills, and work on the treatment plant is in progress. We have pretty much laid out the location of the various facilities and have a good plan to build them.

Task 3: Turn on the water (5 minutes)

Once Task 2 has been completed, we will locate the relevant valve and turn it on.

Conclusion

Summary

Everyone at Acme Enterprises agrees that this is the most successful project we have ever undertaken. We have all the pieces and we are putting them together!

Evaluation

Acme Enterprises evaluates progress on this project as **excellent**.

Forecast

Acme Enterprises will definitely complete this project in the future.

Contact

For more information, contact Jayden Jones by email at: jjones@yourbestfriend.com.

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CHAPTER TEN

Feasibility and Recommendation Reports



Imagine this scene: you have a report to write and you are procrastinating (that never happens, right?). You grab the remote control and turn on the television. You have already binge-watched every episode of your favorite series multiple times and you wonder what other options exist. You flip through the channels and discover a dog show. Maybe it is Westminster, the National Dog Show, or Crufts—you see many well-dressed handlers trotting around the ring with some beautiful animals. If you happen to catch the final event called “Best in Show,” perhaps among the competitors you see a Standard Poodle with exceptionally styled ears and tail, a Welsh Pembroke Corgi with no tail and legs almost as short, a sleek Doberman Pinscher that looks like it means business, a Whippet that looks like it could grace an ancient monument, and a Havanese fluff ball (see [Figure 10.1](#)). How does a judge compare these dogs to each other? How do the short legs of a Corgi compare to the medium legs of a Doberman? How does the long, fluffy coat of a Havanese compare to the short, slick coat of a Whippet and the styled hair of a Standard Poodle? The only way a judge can make this determination is by assessing each dog based on multiple criteria and choosing the best dog from among the alternatives. There is no scientific or mathematically correct answer, but the judge can justify a best answer based on predetermined and assessable criteria.

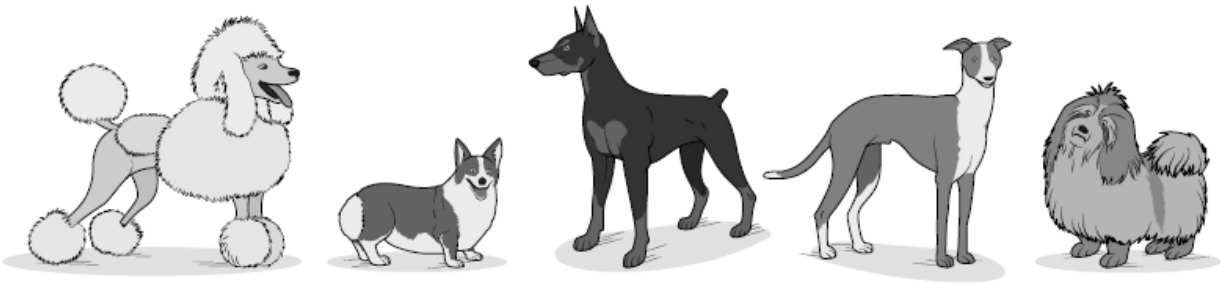


Figure 10.1

Just like deciding which is the “Best in Show” from very different dog breeds, making any complex decision often requires comparing difficult-to-compare criteria that might also be weighted unequally. In a work setting, you often have multiple alternatives that must be compared; one might be the least expensive, another might be the fastest to implement, a third one might require the least amount of training, and so on. How do you justify your recommendation in a way that logically appeals to your audience?

Like a Westminster judge, as an engineer or scientist, you will be called upon as a technical expert to produce justification for your “best” answers (notice, not “perfect” answers). **Feasibility reports** and **recommendation reports** are objective documents that identify and evaluate potential solutions to problems. In technical writing, these reports address subjects that have well-defined parameters, including a problem, or multiple problems, that can be precisely described; and a solution, or multiple solutions, that can be objectively and empirically tested.

You will often write these reports for an audience that includes decision-makers and advisors, who will use your reports to make decisions or determine a course of action. They might have a preferred format; if so, you should use it. Before you do the research and write the report, you need to understand how your audience will use it—that is, know the purpose of the report. You also need to know any/all context that will affect the decision to be made. For example, if your supervisor wants to choose a parts supplier and there is a soon-to-be-announced corporate directive to give internal suppliers priority, it is best if you know this information from the onset. Sometimes managers will not remember to volunteer context; you should always ask about it.

You should also become familiar with what your audience Page 122
values and how strongly they feel about each value, as this knowledge will help you develop the assessment criteria and determine how much weight each criterion should have in your analysis. For example, imagine that you

have been asked to evaluate the feasibility of building and maintaining a day care on your organization's campus. You know that cost and employee productivity are both significant concerns, so you assess alternatives based on those two criteria alone and choose option 1. However, unknown to you (because you did not do your due diligence, i.e., ask a lot of questions) is that employee satisfaction is of great value to your organization's owner and CEO, and that employees are much happier with the design in option 2. That missing criterion could dictate a completely different outcome in your analysis.

You must also determine how much your audience knows about the problem and provide enough context so that they can better understand your assessment. Consider the state public health official who has been asked by the local school board to evaluate face-to-face, hybrid, and online learning options and recommend the best solution during a pandemic caused by a virulent respiratory virus. The school board understands the need for safety, but they might need additional knowledge about many things: how long the virus can live outside the body, the viral load needed to develop symptoms, the efficacy of current prevention measures, the percentage of asymptomatic carriers, the virus incubation period, the size of classrooms, the maximum capacity prepandemic, the current student-teacher ratio, classroom airflow patterns, the air exchanger efficiency, the mortality rate of those with comorbidities, the percentage of individuals who likely have comorbidities but don't yet know it, the percentage of students who rely on school lunch for their main daily meal, the percentage of students without computers or reliable internet, and more. There will not be a perfect answer, but there can be a best answer.

Feasibility and recommendation reports are unbiased evaluations, although their conclusions and recommendations can be—and frequently are—used to promote ideas and sell goods and services. Ideally, however, only someone who is totally impartial should write these reports. The author should have no stake in the outcome and should not care whether any or all of the solutions are adopted. That said, you should assess each situation; if objectivity is critical and you cannot provide it, consider recusing yourself. If your supervisor **wants** your opinion about a best choice for something that will affect you, do your best to provide a fair and justifiable analysis.

How Do Feasibility and Recommendation Reports Differ?

Feasibility reports and **recommendation reports** are similar, and the terms are often used synonymously. Both reports define a problem and objectively evaluate solutions based on a set of criteria—but with one small difference. Feasibility reports, on one hand, determine the feasibility or viability of solving a problem in one particular way. In other words, feasibility reports consider a single solution to a problem and determine whether, or to what extent, the proposed solution is feasible. Recommendation reports, on the other hand, look at several approaches for solving a problem and recommend the most feasible approach. Both feasibility and recommendation reports, then, basically do the same thing: they objectively evaluate the feasibility of solutions. [Outline 10.1](#) provides a typical approach for organizing a feasibility report. [Outline 10.2](#) does the same thing for a recommendation report.

Establishing and Assessing Measurable Criteria

Before we can explain how feasibility and recommendation reports are structured, we must first discuss the idea of **criteria**. In your early science and engineering courses, you frequently practice solving what are called “closed-ended” problems. Closed-ended problems have **correct** answers. Examples of closed-ended problems include finding the derivative of $\tan x$, the future value of an investment of \$1,000 at 5 percent interest for one year, the current in a circuit given voltage and resistance, and the chemical reaction and resulting products when you combine hydrochloric acid and sodium hydroxide. Repeatedly solving these kinds of problems is very necessary practice for future engineers and scientists. However, as you continue to progress through your college curriculum, you will probably observe that there are fewer closed-ended problems (in other words, less rote practice), and more open-ended problems the closer you get to graduation. Open-ended problems do **not** have a correct answer; they have

a **best** answer. Just like the judge at the Westminster Dog Show, you identify this best answer based on comparing multiple alternatives and choosing the best one for the situation at hand. To compare alternatives, you must establish a method for comparison. We use the term **criteria** to refer to the items of comparison.

Criterion or Criteria?

Criterion is singular. **Criteria** is plural. For example:

- There is only one criterion needed to make this decision: cost.
- There are multiple criteria needed to make this decision: quality, simplicity, and cost.

A single criterion is typically not enough to make a good decision for an open-ended problem. For example, if you were going to purchase a car, you most likely would not make that decision based solely on cost. You also would not make that decision based solely on safety rating. Nor would you make that decision based solely on fuel efficiency. Instead, you might consider all three of these criteria (and maybe more), comparing alternatives to make your decision.

Similarities between Solving a Technical Problem and Writing about a Technical Problem

When you are solving an open-ended problem, you must consider many things: who has the problem (audience), what kind of solution is needed and why (purpose), and what are the realistic constraints for the given situation (e.g., Is there a budget? Is there a time constraint for implementing a solution? Does a new solution require training people? Are there environmental, societal, or political impacts that must be considered? Etc.—all of this provides context). Sound familiar? There is a reason for the similarities. While maybe not as technically challenging as the scientific and engineering problems that you are solving, writing a technical document is still a solution for an open-ended problem.

The same thing is true in industry. You must understand the problem to be solved and why, who is affected by the problem, as well as who could be affected by each potential solution, how soon a solution is needed, how much money is available to address it, and often many other considerations. Each of these is called a criterion. The keys to identifying relevant criteria for solution alternative comparison include several things:

1. **Communicate with everyone affected by the problem.** This means people at all levels who are or could be affected, both internally and externally. A design engineer who is solving a product design problem should talk with the process engineer who is responsible for ensuring manufacturability, the quality engineer who is responsible for ensuring product quality, the operator who must assemble the new design, the test engineer, the packaging engineer, the folks in supply management who communicate with the shippers, the customers, and of course, managers. If you try by yourself to identify all the relevant criteria, you will fall short more often than not. Sometimes you are confident that you know what matters, but until you communicate with others, the completeness of your understanding is not guaranteed. Besides gaining insight from others' perspectives about the problem, you will also learn how knowledgeable people are about the problem and can gauge the level of technical vocabulary that they can understand.
2. **Limit how many criteria you will evaluate.** Trying to assess everything can get unwieldy. Consider keeping the number of criteria to a manageable number (perhaps 3–10). Start by listing all possible criteria and then prioritizing and choosing those that are most significant and relevant to the problem at hand. For example, perhaps you need to purchase a new machine and care about size, capacity, how soon it is available, and several other things including the color of the machine. Your facility has a color scheme for equipment and getting a new machine in baby blue would be nice, but it is not a deal breaker because the machine color can be addressed later. If you already have a long list of criteria to assess, prioritize and reduce the number of criteria to a manageable number.
3. **Ignore criteria that have the same assessment value for all of your alternatives.** For example, if there is no difference in cost between alternatives 1, 2, and 3, then cost does not matter and will

not be affected by your decision (assuming that a solution alternative must be chosen from among these three).

4. **Make sure that the criteria you choose can be assessed.** If you are buying a tool off the shelf, you will be able to determine cost, which you can assess and compare. Likewise, you can assess the weight of the tool, its loudness in decibels when it is in operation, and whether it requires batteries, electricity, or a thermonuclear power source. What you might not be able to assess is the likelihood of the operator who is not yet part of the department (but unbeknownst to you, will be next month) to like the tool brand, or how long the tool will last in your production environment.

While we do not explain in detail the different ways to visually organize or conduct a criteria assessment (sometimes called weighted criteria matrices or decision matrices), there are many online sources with explanations and templates, such as the description provided by the American Society of Quality (ASQ)¹ (an example of a completed weighted criteria matrix is shown in the exercise at the end of this chapter in [Table 10.3](#)). When assessing multiple criteria, because you are solving an open-ended problem, there is no correct solution—only a best solution. Assuming you have multiple criteria that are each measured in their own way (product cost, implementation cost, implementation time, training time, color, size, weight, environmental impact, safety risk, capacity, operator preference, maintenance requirements, etc.), you have a true “apples to oranges” set of comparisons to make. Alternative 1 costs 10 percent less than alternative 2, but alternative 2 can be installed in less than half of the time that it will take for alternative 1. How does one compare incomparable qualities? Using a weighted criteria matrix allows you to create a weighted value for each alternative being compared, and then find a total score for each alternative, so that you can choose the best one for your situation. It is not a perfect method—and is best employed using some sensitivity analysis as well—but we will leave the details of that process to another textbook.

Outline 10.1 Feasibility Report

Introduction

- *Purpose* Describe the reason for writing this report.
- *Problem* Describe the problem that needs to be solved.
- *Scope* Describe the proposed solution and list criteria in the order they will be discussed.

Discussion

- *Criterion 1*
 - Explanation Describe the criterion, why it was selected, and how it is used.
 - Data Provide the findings (data) for this criterion.
 - Interpretation Interpret the data for this criterion relative to the solution.
- *Criteria 2–n* (Treat each remaining criterion in the same manner as criterion 1.)

Conclusion

- *Summary* Review the data and interpretations.
- *Conclusions* Present your conclusions based on the data and interpretations.
- *Recommendation* Recommend or reject the proposed solution.

Contact

Provide a contact for this report.

Documentation

List the sources and references used.

Appendix

Include materials needed for additional information, but not for understanding the report.

Outline 10.2 Recommendation Report

Introduction

- **Purpose** Describe the reason for writing this report.
- **Problem** Describe the problem that needs to be solved.
- **Scope** Describe the proposed solutions and criteria in the order they will be discussed.

Discussion

- Criterion 1
 - Explanation Describe the criterion, why it was selected, and how it is used.
 - Data Provide the findings (data) for this criterion for each solution.
 - Interpretation Interpret the data for this criterion relative to each solution.
- Criteria (Treat each remaining criterion in the same manner as criterion 1.)
- 2–n

Conclusion

- **Summary** Review the data and interpretations.
- **Conclusions** Present your conclusions based on the data and interpretations.
- **Recommendation** Recommend the best solution based on these conclusions.

Contact

Provide a contact for this report.

Documentation

List the sources and references used.

Appendix

Include materials needed for additional information, but not for understanding the report.

Writing Feasibility and Recommendation Reports

As Outlines 10.1 and 10.2 show, you need to do several things when writing either a recommendation or a feasibility report:

1. Define a problem that needs to be solved.

The difficulty here is that we are often solution-oriented; in many cases, we skip the problem and go directly to the solution. Suppose a friend came to you and said, “I have a problem: I need to buy a computer, but I do not know which one to purchase.” Your friend thinks that their problem is not knowing what computer to purchase. But saying “I need to buy a computer” states a solution, not a problem. You ask your friend why they need a computer, and they reply that they need one because they must record a video presentation for a course. You continue to question them, and they reveal to you that they could make the video using their phone if they want—there is no restriction on the platform or software to be used for making the video. When your friend realizes that they have a smartphone that can be used to make the video, they also realize that they do not need to buy a computer. The problem is not what computer to buy; the problem is that they need access to video software. Once they define the problem, alternatives can be identified; in this case, potential solutions might include using their own phone, using a lab computer, or buying a computer.

Once we determine there is a problem, then we must identify what we call **root cause**. The root cause is what is actually causing the problem. Only after we know this can we begin to identify potential solutions, often called alternatives. Feasibility and recommendation reports work the same way. You cannot evaluate a solution to a problem that is not clearly defined.

2. Identify one or more potential solutions.

This process can be tough; sometimes many more solutions exist than you will have the time or capability to evaluate. If you need a computer to surf the internet, how many choices do you have? This is almost like asking how many stars are in the Milky Way galaxy. Identifying just a few viable solutions can be challenging. If you consider all realistic constraints, you can typically narrow the list. Maybe you will buy only from an approved, local vendor; or you will shop only within a 5-mile radius of your home; or you will consider using only a certain supplier that gives you a rebate.

3. Develop a set of criteria by which to objectively evaluate the potential solution(s).

The key here is objective. As we discussed previously, find meaningful measures that relate to the problem you have defined, and then identify valid methods for applying them. For example, when looking for a computer to surf the internet, you might use criteria such as cost, processor speed, monitor size and quality, reliability and warranty, bundled software, and included peripherals. These can be objectively described and measured. The attractiveness of the case would be a less objective (and maybe relevant) assessment, so that might not be a good criterion; computer case attractiveness cannot easily be objectively described and measured.

4. Collect and interpret data for each criterion as it relates to each potential solution.

Decide how to weight the importance of each criterion; we recommend using a weighted criteria matrix to do this.² While the criteria can sometimes be weighted equally, in many situations, some criteria are more important and need to count more in the final decision. For example, what if you wanted a computer to surf the internet using a satellite link from the back of an all-terrain vehicle in the upper Amazon region? In this case, reliability and maintainability might be far more important than, say, processor speed. However, if you planned to use the computer to do serious number-crunching in the office, processor speed would be more important.

5. Draw conclusions and make recommendations regarding the feasibility of the potential solutions based on your interpretations.

The primary requirements here are objectivity and clear thinking. Look at the interpretations you have made for each criterion and consider the relative weighting of the criteria. Always base your conclusions on this information—never on other information or considerations that are not fully treated in the report. It is common to have a team do the scoring of alternatives; this adds much-needed perspective to the problem-solving process.

Recommendation Report

Both feasibility and recommendation reports are organized and written in the same straightforward, logical manner. To simplify this discussion, we provide a recommendation report example first, since it includes all the elements of a feasibility report. This recommendation report example identifies and evaluates, in the context of astronautics, two alternative transfer maneuvers for moving a fictitious signals reconnaissance satellite (codenamed “Big Ears”) from one earth orbit to another. This type of recommendation report might be used as an internal **position** or **background** paper, which is also referred to frequently as a **white** paper.

Why Is It Called a White Paper?

The term “white paper” is an old term that gets its name from British government documents and is based on the report cover color. There are many online sources for guidelines about writing white papers. If your client, supervisor, or other intended audience has a preferred white paper format, as always, use it!

Here is the recommendation report, assembled, formatted, and modified to include supporting diagrams and a summary table of data. The extensive calculations that would comprise the appendices have not been included in this chapter because they are not needed for the purposes of this example.

Recommendation Report on Transfer Maneuvers for the Big Ears Satellite

Introduction

Purpose

The purpose of this report is to recommend the best orbital transfer maneuver for moving a satellite from a low equatorial orbit to a high geosynchronous orbit.

Problem

NASA recently launched a Big Ears reconnaissance satellite from Cape Canaveral and injected it into a 300-kilometer near-earth orbit (NEO) over the equator (0° inclination). The satellite has been successfully configured and checked out, and now it needs to be moved to its operational position in geosynchronous earth orbit (GEO) at an altitude of 38,786 kilometers with an inclination of 0° . This position will enable the satellite to remain stable over the earth's surface so that it can survey those portions of the earth required by its mission profile.

The National Intelligence Authority requires that the satellite be on station by 0500 GMT on June 1, 2021. To allow time for full calibration on orbit, an appropriate transfer maneuver must be selected and executed at least 72 hours prior to this time. Since the satellite is already in an equatorial orbit, transferring the satellite from the current orbit to the required orbit involves only an altitude change within the same plane. Two possible maneuvers are being considered to accomplish this transfer: the Hohmann transfer (HT) and the one-tangent burn (OTB). The HT is a double-tangent transfer ellipse that requires two tangent burns—one at NEO and one at GEO ([Figure 10.2a](#)). The OTB requires a single-tangent burn at NEO and a second nontangent burn at GEO ([Figure 10.2b](#)).

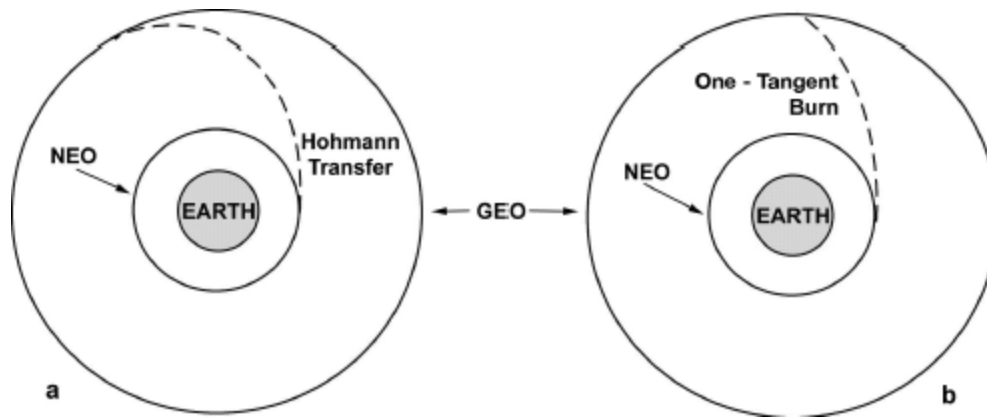


Figure 10.2

(a) Hohmann transfer and (b) one-tangent burn.

Scope

This report compares the HT to the OTB, using three criteria: accuracy, time of flight, and required fuel.

Discussion

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Accuracy

Explanation

Accuracy refers to the relative precision with which the satellite is placed into its target orbit. If a satellite is not accurately placed, additional onboard fuel will be required to adjust the orbit. Normally, a transfer is considered accurate if the radius of the target orbit is within ± 0.001 DU (1 DU = 0.5 earth diameter, or 6378.145 kilometers). Because the problem is relatively short term, no special weighting is assigned to this criterion.

On-board thrusters can correct for insertion errors; however, the use of fuel may have long-term implications.

Data

Using this criterion, and in terms of past performance, the HT has achieved an accurate transfer 85.5 percent of the time. The accuracy for the OTB is at 60.2 percent (Orbitus 2004, pp. 224–228).

Interpretation

In terms of past performance, the HT provides higher accuracy than the OTB. Using on-board thrusters, either maneuver can solve the current problem; however, the greater accuracy of the HT has positive long-term implications for system viability on-station. The energy to precisely position a satellite on station is provided by a finite supply of on-board fuel, which also will be used to power the on-board thrusters for later station-keeping activities on orbit. More accurate insertion means less fuel required for positioning, thus conserving more fuel for subsequent station-keeping functions. Since higher accuracy translates to longer operational life for the satellite, the HT is preferred.

Time of Flight

Explanation

Time of flight (TOF) is the elapsed time between the initial firing of the satellite's retrorockets at a tangent to the initial NEO and the point when the satellite intersects the radius of its target GEO. Because the problem does not provide time-critical constraints, this criterion is given no special weighting.

Data

Using the HT, the satellite requires a TOF of 24.10 TU (1 TU = 13.44686457 minutes). The OTB provides a TOF of 23.45 TU to transfer the satellite from NEO to GEO. (See Appendix A for the calculations supporting these results.)

Interpretation

The OTB requires less time than the HT to transfer the satellite from NEO to GEO. Because time-sensitive constraints do not exist in this case, and because the differences in TOF for both options are relatively small, no preference exists for either choice.

Required Fuel

Explanation

As mentioned earlier, the energy to move the satellite from NEO to GEO is provided by on-board fuel, which also powers the on-board thrusters for station-keeping activities on orbit. Fuel not required for the orbital transfer would then be available for future on-orbit maneuvers.

The satellite moving in NEO possesses considerable kinetic energy, as shown in Equation (1).

$$E_k = \frac{1}{2}mv^2 \tag{1}$$

While the mass m of the satellite is considered to be constant, the change in velocity v requires an increase in energy, which, in turn, requires more fuel.

Data

The total change in velocity for the HT is 0.498 DU/TU (1 DU/TU = 7.90536828 km/s). The total change in velocity for the OTB is 0.594 DU/TU. (See calculations in Appendix B.)

Interpretation

The HT requires a smaller change in velocity, thereby requiring less fuel compared to the OTB. In terms of fuel savings, the HT is preferred.

Conclusion

Summary

This study evaluated both the HT and OTB based on three criteria: accuracy, time of flight, and fuel required. The HT’s performance was better in terms of accuracy and required fuel. The OTB’s performance was better in terms of TOF (see [Table 10.1](#)).

Table 10.1 Summary of results for HT vs OTB

Maneuver	Accuracy	TOF	Change in v
HT	85.5%	24.10 TU	0.498 DU/TU
OTB	60.2%	23.45 TU	0.594 DU/TU

Conclusions

The HT is preferred based on accuracy and fuel requirements. The OTB performed better in terms of time of flight, but the advantage is not considered significant given the constraints of the problem.

Recommendation

The HT is recommended as the transfer maneuver for moving the Big Ears satellite from NEO to GEO.

Contact

For more information, contact Kathryn Baker, Ph.D., NIA/OR3, at (202) 444-9999.

Source

Orbitus, Delta V.: *The Joy of Orbital Transfer*. Dayton, Ohio: Astro Press, 2004.

Appendices

Calculations are not included for the purposes of this example, but this is where they would go.



Dissecting Example 10.1

Now that you have read an example of a recommendation report, let us take a closer look at each of its components and what they do.

Introduction

Purpose

Describe the purpose of this report. This part is straightforward. What do you hope to achieve with this document? To what requirement does this document respond? Your purpose statement might look something like this:

The purpose of this report is to recommend to NASA the best orbital transfer maneuver for moving a satellite from a low equatorial orbit to a high geosynchronous orbit.

Once you have stated the purpose, you can define the problem addressed in this report.

Problem

You cannot write a feasibility report unless you have a problem. Solutions without problems are not useful, and neither are solutions that are not geared specifically to solving problems. To define the problem in this section, you should provide the needed background and describe specific requirements that any viable solution must meet. Consider the following problem statement:

The National Intelligence Authority needs to move a satellite from its present near-earth orbit to a geosynchronous orbit.

This problem statement is not very useful because it does not include adequate information about the problem. How can anyone possibly identify and evaluate a solution if the problem is not properly defined in the first place? In fact, in this case, moving the satellite is actually part of the solution. The “what,” “when,” “why,” and “how” are important aspects of the problem that need to be addressed. To describe the problem, then, you need to specify the kind of satellite that needs to be moved, why it needs to be moved, how much time is available to get it where it needs to go, and how accurate the transfer maneuver needs to be. A better problem statement might look something like the following:

NASA recently launched a “Big Ears” reconnaissance satellite from Cape Canaveral and injected it into a 300-kilometer near-earth orbit (NEO) over the equator (0° inclination). The satellite has been successfully configured and checked out and now needs to be moved to its operational position in geosynchronous earth orbit (GEO) at an altitude of 38,786 kilometers with an inclination of 0° . This position will enable the satellite to remain stable over the earth’s surface, so that it can survey those portions of the earth required by its mission profile. The National Intelligence Authority requires that the satellite be on station by 0500 GMT on June 1, 2021. To allow time for full calibration on orbit, an appropriate transfer maneuver must be selected and executed at least 72 hours prior to this time. Since the satellite is already in an equatorial orbit, transferring the satellite from the current orbit to the required orbit involves

only an altitude change within the same plane. Two possible maneuvers are being considered to accomplish this transfer: the Hohmann transfer (HT) and the one-tangent burn (OTB). The HT is a doubly tangent, transfer ellipse that requires two tangent burns—one at NEO and one at GEO. The OTB requires a single-tangent burn at NEO and a second non-tangent burn at GEO.

Scope

The scope section lists the alternatives to be considered, along with the criteria that will be used to test each solution. First ensure that any alternative you consider is at least apparently viable. Some solutions may look good initially; however, when they are closely examined in terms of the problem to be solved, these same ideas may appear silly. Take a preliminary look at each potential solution to ensure it makes sense. Also list the criteria by which you will evaluate each alternative. These criteria, driven by the problem, can include almost anything from cost, to reliability, to ease of use. Normally, in a technical paper, you use only criteria that can be objectively and empirically quantified and tested. In the case of our example problem, we know that the solution must be robust, reliable, economically feasible, and efficient. Based on these requirements, we might develop a scope section that looks like the following:

This report compares the HT to the OTB, using three criteria: accuracy, time of flight, and required fuel.

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Discussion

The discussion section is the heart of the feasibility and recommendations reports. In this section, you discuss the criteria, apply them to the identified alternatives, generate data, and interpret the results. Notice that this section is organized by criteria, not solutions. That is extremely important. You want to compare solutions for each criterion, not vice versa. If instead you organize the discussion around solutions, you will have to go through all your criteria for each solution. By the time you get through all the solutions, you will find it difficult to conceptualize and compare the solutions effectively because you will not remember which data pertained to which solution. To begin the discussion section, here is the kind of treatment you might give the first criterion, *accuracy*.

Accuracy

Explanation

First, in the explanation section, define the criterion, explain why you selected it (if necessary), and note any special weighting it will have in the evaluation process. If it has special weighting, explain how you chose your weighting system.

Explanation: Accuracy refers to the relative precision with which the satellite is placed into its target orbit. If a satellite is not accurately placed, additional on-board fuel will be required to adjust the orbit. Normally, a transfer is considered accurate if the radius of the target orbit is within ± 0.001 DU (1 DU = 0.5 earth diameter, or 6378.145 kilometers). Because the problem is relatively short term, no special weighting is assigned to this criterion. On-board thrusters can correct for insertion errors; however, the consequent use of fuel may have long-term implications.

Data

In the data section, discuss the findings, using this criterion for each alternative solution. Be sure to properly document the source for the data being used.

Data: Using this criterion, and in terms of past performance, the HT has achieved an accurate transfer 85.5 percent of the time. The accuracy for the OTM is at 60.2 percent (Orbitus 2004, pp. 224–228).

Interpretation

In the interpretation section, discuss your interpretation of the data for this criterion for each alternative solution.

Interpretation: In terms of past performance, the HT provides higher accuracy than the OTB. Using on-board thrusters, either maneuver can solve the current problem; however, the greater accuracy of the HT has positive, long-term implications for system viability on-station. The energy to precisely position a satellite on station is provided by a finite supply of on-board fuel, which also will be used to power the on-board thrusters for later, station-keeping activities on orbit. More accurate insertion means less fuel required for positioning, thus conserving more fuel for subsequent station-keeping functions. Since higher accuracy translates into longer operational life for the satellite, the HT is preferred.

Next, in a similar manner, deal with the second criterion, *time of flight*.

Time of Flight

Explanation

Describe the criterion, explain why you selected it, and note any special weighting it will have in the evaluation process.

Explanation: Time of flight (TOF) is the elapsed time between the initial firing of the satellite's retrorockets at a tangent to the initial NEO and the point when the satellite intersects the radius of its target GEO. Because the problem does not provide time-critical constraints, this criterion is given no special weighting.

Data

Next, provide the data.

Data: Using the HT, the satellite requires a TOF of 24.10 TU (1 TU = 13.44686457 minutes). The OTB provides a TOF of 23.45 TU to transfer the satellite from NEO to GEO. (See Appendix A for the calculations supporting these results.)

Use Significant Digits

Just because software can give you decimal places that go out 8–10 digits does not give you license to include them. Confirm that you are sharing significant (or at least reasonable!) digits in your calculations or you run the risk of looking silly, or worse, incompetent.

Interpretation

Discuss your interpretation of the data for this criterion for each alternative solution.

Interpretation: The OTB requires less time than the HT to transfer the satellite from NEO to GEO. Because time-sensitive constraints do not exist in this case, and because the differences in TOF for both options are relatively small, no preference exists for either choice.

Required Fuel

Handle this criterion similarly to that for the first two.

Explanation: As mentioned earlier, the energy to move the satellite from NEO to GEO is provided by on-board fuel, which also powers the on-board thrusters for station-keeping activities on orbit. Fuel not required for the orbital transfer would then be available for future on-orbit maneuvers.

The satellite moving in NEO possesses considerable kinetic energy, as shown in Equation (1).

(1)

$$E_k = \frac{1}{2}mv^2$$

While the mass m of the satellite is considered to be constant, the change in velocity v requires an increase in energy, which, in turn, requires more fuel.

Data: The total change in velocity for the HT is 0.498 DU/TU (1 DU/TU = 7.90536828 km/s). The total change in velocity for the OTB is 0.594 DU/TU. (See calculations in Appendix B.)

Interpretation: The HT requires a smaller change in velocity, thereby requiring less fuel compared to the OTB. In terms of fuel savings, the HT is preferred.

If you have visuals of your alternatives, include them here to help your audience better understand their options. If you decide to create and use a weighted decision matrix as part of your decision process, include a visual of the table or figure to summarize your discussion and help your audience follow your logic for comparing alternatives. See [Chapter 17](#) for more on good visual design.

Conclusion

In the conclusion section, summarize the data and interpretations for all criteria and solutions. Do not include new information in this section; it should deal only with information already presented in the paper.

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Summary

Briefly summarize the report and its findings.

This study evaluated both the HT and OTB based on three criteria: accuracy, time of flight, and fuel required. The HT's performance was better in terms of accuracy and required fuel. The OTB's performance was better in terms of TOF.

Conclusions

Provide any conclusion you can draw from the summary above.

The HT is preferred based on accuracy and fuel requirements. The OTB performed better in terms of time of flight, but this advantage is not considered significant given the constraints of the problem.

Recommendation

Make your recommendation.

The HT is recommended as the transfer maneuver for moving the Big Ears satellite from NEO to GEO.

Contact

Finally, provide any needed contact information.

For more information, contact Kathryn Baker, Ph.D., NIA/OR3, at (202) 444-9999.

Example 10.2 Feasibility Report

Feasibility Report

The following example is a feasibility report on using the TygerMic-1000 microphone in the new Arctic Ponies' Music Production Facility. Note that, unlike the prior recommendation report example, this feasibility report example considers only a single possibility and determines the feasibility of this possible solution.

Feasibility Report on Using the TygerMic-1000 Microphone for the Arctic Ponies' New Music Production Facility

Introduction

Purpose

The purpose of this report is to assess the feasibility of using the TygerMic-1000 microphone in the new Arctic Ponies' music production facility (MPF).

Problem

The Arctic Ponies, a successful rock band, has recently purchased a former missile control bunker from the U.S. Air Force. The band plans to

use this bunker within 30 to 45 days as a new music production facility since the thick walls and remote location make it ideal for this type of studio application. In the past, the Ponies have relied on professional grade dynamic microphones. These microphones, which use a moving coil attached to a diaphragm, are starting to prove inadequate for the increasingly dynamic range of the Arctic Ponies' music. The frequency response of the dynamic microphone lacks the required linearity across the Ponies' extraordinary range. Prior laboratory analysis has indicated this dynamic range to be 22 to 19,500 Hz. Furthermore, given existing equipment, proper signal processing of Pony music requires a microphone with an equivalent noise level of 19 dB and a sensitivity of 12.5 mV/Pa. The layout of the studio would indicate a preference, not a necessity, for a near-omni-directional (binaural) pickup pattern over omnidirectional (hyper cardioid) or unidirectional (super cardioid) pickup patterns.

As a practical solution, the Ponies have decided to outfit the new MPF with condenser microphones, which use a thin conductive film to form a moving plate of a variable capacitor. This thin film is more responsive and linear than the diaphragm and coil found in dynamic microphones and would better meet the group's requirements. A local electronics dealer, AudioStuff, has offered to sell the Ponies 20 preowned, TygerMic-1000 professional condenser microphones for \$890 each for a total of \$17,800. This report will address the feasibility of both this microphone and \$890-per-unit AudioStuff offer for the Ponies' new MPF.

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Scope

The TygerMic-1000 is an advanced condenser microphone that is widely used in the music industry. This microphone will be evaluated using the following criteria: *cost*, *availability*, and *performance*.

Discussion

Cost

Explanation

Cost is the “street” price of a single TygerMic-1000 professional condenser microphone in U.S. dollars when acquired in bulk lots of 20 or more.

Data

A leading comparison-shopping retail website provided a range of \$899.97 to \$1,999 for various vendors with online sales of this microphone.*

Interpretation

The offer of \$890 received from AudioStuff is below the low end of web-based offers and is judged to be competitive.

Availability

Explanation

Availability is the time required for delivery of the product once a sales contract has been consummated. The new facility must be available to the Ponies within 30 days, including all of the required microphones.

Data

AudioStuff has all the required microphones in stock in its local warehouse and can deliver all of the microphones with 24 hours’ notice.

Interpretation

The required microphones are available in the required time frame.

Performance

Explanation

Performance includes the microphone’s frequency range, sensitivity, and signal/noise ratio, as well as its pickup pattern.

Data

[Table 10.2](#) provides performance results for this microphone compared to the performance requirements of the Ponies. [Figure 10.3](#) provides the pickup pattern for the TygerMic-1000 microphone.

Table 10.2 Required vs. Documented Performance of the TygerMic-1000

	Required**	Actual***
Dynamic range	22 to 19,500 Hz	20 to 20,000 Hz
Equivalent noise level	19 dB	19 dB
Sensitivity	12.5 mV/Pa	10.2 mV/Pa

*Comparison price data obtained from www.professmics.com (Accessed June 15, 2021).

**Procurement Report PD061123, "Analysis of Microphone Requirements," Procurement Division, Arctic Ponies, Inc., 2006.

***Technical Report LD78823, "Performance of the TygerMic 1000," Laboratory Division, Arctic Ponies, Inc., 2006.

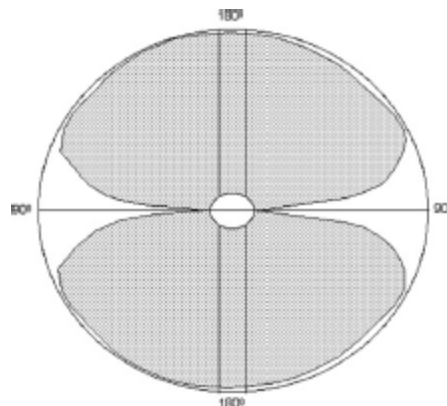


Figure 10.3

TygerMic-1000 Pickup Pattern.

Source: Arctic Ponies Technical Report LD78823

Interpretation

As [Table 10.2](#) shows, the TygerMic-1000 meets or exceeds all Pony requirements in terms of dynamic range, equivalent noise level, and sensitivity. As [Figure 10.3](#) shows, the TygerMic-1000 also provides a binaural pickup pattern, which also meets the Arctic Ponies' requirements.

Conclusion

Summary

The data presented demonstrate that the TygerMic-1000 meets all the dynamic range, equivalent noise level, and sensitivity requirements of the Arctic Ponies for their new MPF. Additionally, the pickup pattern of the microphone is consistent with pickup pattern requirements.

Conclusions

The TygerMic-1000 is a feasible alternative for use in the new MPF.

Recommendation

The Ponies should accept the AudioStuff offer of 20 TygerMic-1000 professional condenser microphones for use in the new MPF.

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Documentation

1. Gostable, James S., Ph.D., "Specification of Microphone Requirements." Procurement Report PD061123, Procurement Division, Arctic Pony Enterprises, May 2021.
2. Williams, William W., Ph.D., "Performance of the TygerMic-1000." Technical Report LD78823, Laboratory Division, Arctic Pony Enterprises, May 2021.

Feasibility and Recommendation Report Checklist

- ☐ Do I understand what my audience wants, needs, and values?

- ❑ Do I understand how this report will be used—in other words, its purpose?
- ❑ Do I have all of the context needed to define the problem accurately?
- ❑ Have I defined the problem with the necessary level of detail to ensure the requirements for solving it are clear?
- ❑ Have I selected a manageable number of potential solutions that appear to be viable?
- ❑ Have I developed criteria that relate to the problem?
- ❑ Have I explained all criteria, including why they were selected and how much weight each is being given?
- ❑ For all criteria, have I collected information (data) that is objective and meaningful?
- ❑ Have I provided useful interpretations of this information (these data)?
- ❑ Have I summarized this data clearly?
- ❑ Have I included visualizations of the alternatives?
- ❑ Have I included a conclusion based on these interpretations?
- ❑ Have I made a recommendation based on this conclusion?
- ❑ Have I included a contact who can provide more information about this report?
- ❑ Have I documented the sources I used for my information?
- ❑ Have I included any necessary supporting information in an appendix?

Exercise

Use the feasibility and recommendation report checklist to review the following recommendation report. What is done well? What could be improved?

Recommendation Report for Raakiba Hammad's Job Search

Introduction

Purpose

The purpose of this report is to recommend the best full-time job for Raakiba Hammad to accept.

Problem

Ms. Hammad needs to accept a job after graduating with her industrial engineering degree. She has \$2,000 saved from her undergraduate teaching assistant job, and her parents have generously offered to lend her up to \$1,000 for her transition to her new place of employment. Monetary limitations are considered absolute and cannot be increased.

Scope

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The three alternatives from which Ms. Hammad has offers and can choose include a hospital in Waterloo, Iowa; an automobile parts manufacturing facility near Chicago, Illinois; and a financial institution in Seattle, Washington. This report will evaluate Ms. Hammad's alternative job offers using the following criteria: type of job, salary, benefits, cost of living, proximity to home, city vs. country, ease of travel, opportunities for advancement, working environment.

Discussion

Type of Job

Explanation

This criterion considers the type of job that was offered including the expected work responsibilities.

Data

The hospital is creating a new process improvements department. Ms. Hammad's responsibilities would include building and supporting a Lean culture using process improvement techniques. The manufacturing facility is focused on establishing a Total Productive Maintenance program and wants someone to drive this including building a robust Overall Equipment Effectiveness measurement system. The financial institution is focused on building a standard work library and reducing process times. Ms. Hammad's position would be an internal consultant.

Interpretation

Because Ms. Hammad's degree is quite flexible across industries, she is prepared to handle all three of the opportunities offered to her. Her preference is for an experience in manufacturing, but she is open to any industry.

Salary

Explanation

This criterion includes the current offer that was made based on annual salary. Neither signing bonus nor yearly bonus were considered.

Data

The annual salary for the hospital is \$65,000; the annual salary for the automobile parts manufacturer is \$80,000; and the annual salary for the financial institution is \$75,000.

Interpretation

The hospital offers the lowest annual salary, and the manufacturing facility offers the highest annual salary; the difference between the two is approximately 23 percent. The difference between the manufacturer and the financial institution is only \$5,000 (~6 percent), which could potentially be overcome because of possible yearly bonuses from the financial institution.

Benefits

Explanation

This criterion considers the benefits that are included as part of the offer, including health insurance and vacation.

Data

All three positions offer health insurance that is relatively equal in terms of cost and coverage. All three positions also offer vacation and holidays. The differences in quantity of vacation days and when it is available to use are negligible.

Interpretation

Because there is no significant difference in health insurance or vacation between the three companies, this criterion is moot and does not affect the final decision.

Cost of Living

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Explanation

This criterion includes the current offers that were made based on annual salary. Neither signing bonuses nor potential yearly bonuses were considered.

Data

Equivalent salaries were determined using CNN's Money (<https://money.cnn.com/calculator/pf/cost-of-living/>, accessed December 28, 2020). The annual salary for the hospital is \$65,000; the annual salary in Chicago is \$95,527; and the annual salary for the financial institution is \$126,222.

Interpretation

While the hospital offers the lowest annual salary, it is also located in the city (Waterloo, Iowa) with the lowest cost of living. The Chicago salary is approximately 20 percent below the equivalent cost of living, which would be manageable. The Seattle salary is approximately 70 percent below the equivalent cost of living, so Ms. Hammad might need to find a roommate, live near work and take public transportation, and/or find other ways to reduce living expenses.

Proximity to Home

Explanation

This criterion considers how close the job location is to Ms. Hammad's parents. She considers this first job a good opportunity to live "somewhere else" for a while. She grew up in Naperville, Illinois, and is very familiar with the Chicago area.

Data

The distance from Waterloo, Iowa, to Naperville, Illinois, is approximately 300 miles and 4.5 hours by car. The distance from Chicago, Illinois, to Naperville, Illinois, is approximately 30 miles and 0.75 hours by car. The distance from Seattle, Washington, to Naperville, Illinois is approximately 2100 miles and 29 hours by car. There are direct flights from Waterloo to Chicago and from Seattle to Chicago.

Interpretation

If Ms. Hammad prefers to live farther away from home than "drop in" distance from her parents, she should consider either Waterloo or Seattle.

City vs. Country

Explanation

Ms. Hammad would like to live in a slower-paced environment; big-city life is not appealing.

Data

Waterloo, Iowa, is a very small city (or large town) of approximately 70,000 (<https://www.census.gov/quickfacts/waterloocityiowa>, accessed December 28, 2020), surrounded by rural farm country. Chicago, Illinois, is very large metropolitan city with an estimated population of 2.7 million

(<https://www.census.gov/quickfacts/fact/table/chicagocityillinois/PST045219>, accessed December 28, 2020). Seattle, Washington, is a large, metropolitan city of approximately 750,000 residents in Seattle proper (<https://www.census.gov/quickfacts/fact/table/seattlecitywashington,US/PST045219>, accessed December 28, 2020).

Interpretation

Waterloo has a small-town feel and in fact, one can actually live in the countryside and still have a short commute to work. Chicago is definitely not a "country" feel; in fact, trying to live far enough away from the city to find a slower pace requires a significantly longer commute. Depending

on where in Seattle one lives and works, it is possible to have a more “country feel,” but this also requires having a vehicle.

Ease of Travel

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Explanation

Ms. Hammad’s favorite hobby is travel. She wants easy access to all forms of transportation, including planes, trains, and automobiles.

Data

Waterloo has an airport with one airline that has daily flights to Chicago. It is located at the intersection of U.S. Highway 20, which spans the North American continent east-west, and U.S. Highway 218 (the “Avenue of the Saints”), which connects St. Paul and St. Louis. The closest passenger train service is St. Paul, Minnesota. Chicago has two major airports, including O’Hare and Midway. It has extensive road access, including U.S. Interstates 80, 90, and 94, along with many U.S. routes and toll roads. It is a hub for many airlines with extensive international flights. It is also a hub for Amtrak. Seattle’s airport has many domestic and international carriers and flights, and is a hub for travel across the Pacific. It has extensive road access, including U.S. Interstates 5 and 90, and many U.S. routes. It is served by Amtrak with both an east-west route and a north-south route. It is also a hub for cruises in the Pacific.

Interpretation

Travel from Waterloo will require an extra leg of flying or driving to a larger airport like Chicago. Chicago has easier access to flights going to Europe, while Seattle has easier access to flights going across the Pacific. Both Waterloo and Chicago are located centrally in the United States, making driving to the central and eastern portions of the United States more feasible than Seattle. Seattle offers the potential for convenient cruise travel, which is appealing to Ms. Hammad.

Opportunities for Advancement

Explanation

Each organization has explained the likelihood of promotion and to what level, assuming that Ms. Hammad performs as well as her internship record indicates.

Data

The hospital in Waterloo has made it clear to Ms. Hammad that they are just beginning to appreciate how important adding an industrial engineering team to their facility is, and since she is part of the initial effort, she will be a candidate for promotion if she performs as well as they expect. The hospital is part of a health care system with facilities in other Iowa and Illinois cities. She could move from one location to another as she is promoted (“big fish in a small pond”). The manufacturer in Chicago is a large, family-owned single facility which supplies the automobile industry with small but necessary parts. Ms. Hammad could rise to the level of manager in different roles, but promotion within the organization would be limited to one or two times over the course of 5 years (“medium fish in a small pond”). The financial institution in Seattle is part of an international company and includes many engineering functions like consulting. The opportunity to be promoted multiple times and up to the level of CTO exists (“big fish in a big pond”).

Interpretation

Because Ms. Hammad does not intend to work for the same company for 30 years, the question of promotion is more about personal short-term benefits. She would prefer not to move in the short term if promoted, so the hospital opportunities are less appealing. The manufacturing facility promotion opportunities would be very acceptable in the short term. The financial institution promotion opportunities would be acceptable considering that they would either be located in Seattle or would involve taking an overseas assignment, which is appealing.

Working Environment

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Explanation

This criterion considers the overall environment of each company that Ms. Hammad experienced during her interviews and on-site visits.

Data

Overall, the people with whom Ms. Hammad interacted at all three companies were very friendly and helpful. The hospital requires that engineers wear scrubs and maintain a “clean-room” environment. The manufacturer requires steel-toe shoes with metatarsals, and part of its

operations are located in a facility without controlled climate (i.e., it gets really hot in the summer, and really cold in the winter). The financial institution is located in Seattle. It rains. A lot.

Interpretation

Ms. Hammad did not experience any “red flag” moments when interviewing with any of the three organizations (e.g., aggressive or passive aggressive behavior, discriminatory language or tone, etc.). All three organizations are located in communities that have the types of social structures and support systems (e.g., religious institutions, charitable organizations, youth clubs, etc.) that Ms. Hammad is interested in joining. The only differentiating factor is related to physical comfort when both working and not working. Ms. Hammad is not a fan of cold or rainy weather. For that reason, none of the three options available to her are perfect.

Conclusion

The data for this decision are summarized in [Table 10.3](#), which shows a weighted criteria matrix. (For another example of a weighted criteria matrix, see [Chapter 18](#) Presentations.) In this matrix, Ms. Hammad’s relevant criteria are listed in the leftmost column. Notice that this column does NOT include the criterion called “benefits.” Even though this criterion is important to her, because there was no significant difference between the three alternatives, they have no impact on the decision and were therefore not included. Each criterion was weighted according to priority, and each option was scored on a scale of 5 to 1, with 5 being the best possible score. The total score for each option was summed at the bottom.

Table 10.3 Weighted criteria matrix of R. Hammad’s full-time offers for employment.

Weighted Criteria Matrix

Criteria	Value	Alternative 1	Score	Alternative 2	Score	Alternative 3	Score
		Hospital, Waterloo, IA		Manufacturing Facility, Chicago, IL		Financial Institution, Seattle, WA	
Type of Job	20	Lean hospital	4 80	TPM/OEE Manufacturing	5 100	Financial internal consultant	4 80
Salary	20	\$65K	3 60	\$80K	5 100	\$75K	4 80
Cost of Living	20	Low	5 100	High	3 60	Crazy High	1 20
Proximity to Home	5	Perfect	5 25	Too close	3 15	Too far	1 5
City vs. Country	5	Too rural	1 5	Too city	3 15	Perfect	5 25
Ease of Travel	10	Less convenient	2 20	Ideal	5 50	Very good	4 40
Opportunities for Advancement	10	Moving required	3 30	Limited opportunities long term, but okay short term	4 40	Extensive opportunities; could include international	5 50
Working Environment	10	Scrubs. Winters are cold and snowy.	4 40	Steel toes/no climate control on shopfloor. Winter is a thing.	3 30	Rains. A lot.	2 20
	100		360		(410)		320

The manufacturing position, while a little too close to home (Ms. Hammad might have to manage some unexpected “drop-in” visits from her loving and well-meaning parents), a little uncomfortable in terms of environmental working/living conditions, and not quite as lucrative as the hospital job based on cost of living, scores the highest of the three offers. Ms. Hammad prefers a manufacturing role, and because she plans to travel extensively during her free time, living in Chicago provides the most economical opportunities for life outside of work.

Recommendation

Ms. Raakiba Hammad should accept the manufacturing position located in Chicago, Illinois, for her first full-time industrial engineering position.

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Notes

- <https://asq.org/quality-resources/decision-matrix>. Accessed December 27, 2020.
- “What is a Decision Matrix?,” ASQ, <https://asq.org/quality-resources/decision-matrix>. Accessed December 27, 2020.

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CHAPTER ELEVEN

Laboratory and Project Reports



Remember those old horror movies in which the insane scientist transplanted human brains from one body to another in the laboratory? Or the more recent Disney animation of a dog brought back to life in *Frankenweenie*? The labs were always impressive. All around one could see and hear high-voltage arcs traveling up Jacob's ladders, corona discharges zapping off spherical electrodes, or impressive uses of lightening. What those movies did not show was the scientist typing up a laboratory or project report after the experiments were concluded. We must say that one does not successfully transplant a brain from one body to another or resurrect the neighborhood's deceased pets and not write a report (see [Figure 11.1](#)).



Figure 11.1

While we highly oppose attempting it, if a lab created a FrankenDog, they should document it.

Tom Harper Photography/Shutterstock

Virtually anyone working in engineering and the sciences will be tasked, in one fashion or another, with producing a laboratory or project report. These documents present information that relates to the controlled testing of a hypothesis, theory, or device using test equipment (the apparatus) and a specified series of steps employed to perform the test (the procedure). The emphasis in a laboratory or project report is on documenting the design and conducting of the test, how the variables are controlled, and what the resulting data show.

What Are Laboratory and Project Reports?

In their purest form, laboratory reports are research-oriented documents, meaning that they start with a hypothesis or theory that must be applied and tested under highly controlled conditions. For example, suppose you are an aeronautical engineer hypothesizing that your new wing design could be used to generate higher lift at hypersonic speeds with increased flight stability. To test that hypothesis in a laboratory, you would need an apparatus—in this case, a hypersonic wind tunnel and a model of your wing design. You would also need a procedure for using that wind tunnel to test your wing design model. You could then use the procedure to collect data from the wind tunnel tests and interpret the data to see whether your new wing design generated higher lift with increased stability under hypersonic conditions. Finally, you could assess whether the original hypothesis was supported and, if so, probably recommend that more research be done.

Laboratory reports can also take the form of project reports, which are commonly used in teaching laboratories. Instead of testing a hypothesis or theory, you would focus on fulfilling specific requirements that normally come from your instructor in the form of a project assignment. The goal of the project might be to demonstrate the application of a theory or set of theories by using available technology. For example, you and several other students in an engineering course might be asked to build a motorized vehicle with a homeostatic control system that would track accurately along a 100-yard course. The course could be marked with a white line and could contain six 90° curves. Your group's vehicle would also have to avoid several obstacles placed in its path along the way while completing the course within a specified time.

As students, you would design and build an operating vehicle within the cost and equipment constraints prescribed by the teacher. Your group would probably apply basic control theory, using optical and ultrasonic sensors, several electromagnetic servos, and an embedded microprocessor. In effect, your vehicle and the test track would be the apparatus, and the process used to test your vehicle's performance would be the procedure. As part of the class, your group would make several test runs of the vehicle, collect data relating to speed and accuracy, and then develop a project report to document how well your vehicle met the requirements.

Laboratory reports and project reports are similar and generally follow the same pattern, except for a few minor differences. In both cases, you must understand who your audience is and what they need. Most likely your

audience will be familiar with the science and technology that you are using in your process, so this informs the vocabulary you can use. The purpose of these reports is mostly informational. They are factual reports of the experiment or process, and while the outcome might be persuasive in terms of supporting recommendations, that is not the primary purpose of a laboratory or project report. When you write these reports, you must include enough context for your audience to understand the conditions of the experiment or proved process. As always, if your audience has specified a required formulation, use it. However, if they do not have specific formatting requirements, you can use the following suggested outlines to structure your reports. [Outline 11.1](#) provides a model for a research-oriented laboratory report. [Outline 11.2](#) provides a slightly different model for the kind of project report frequently required as part of a teaching laboratory. In our first example, [Outline 11.1](#) is used to develop a research laboratory report on the SuperXLTube transmitting tube. Following that, [Outline 11.2](#) is used to produce a project report for an undergraduate civil engineering course.

Outline 11.1 Laboratory Report

Introduction

- **Purpose** Describe the reason for writing this report.
- **Problem** Describe the context and hypothesis(es) for this report.
- **Scope** Describe the limitations of this report.

Background

- **Theory** Review the theoretical basis of this research.
- **Research** Review prior research relevant to this research.

Test and Evaluation

- **Apparatus** Describe device(s) used to do this research.

• <i>Procedure</i>	Describe procedure(s) used to do this research.
Findings	
• <i>Data</i>	Review the results of the test.
• <i>Interpretation</i>	Provide your interpretation of the results.
Conclusion	
• <i>Assessment</i>	State whether, and to what extent, the hypothesis is supported.
• <i>Recommendations</i>	Provide your recommendation(s), if any.

Outline 11.2 Project Report

Introduction

- *Purpose* Describe the reason for writing this report.
- *Problem* Describe the context for this report, including project requirements.
- *Scope* Describe the limitations of this report.

Background

- *Theory* Review the theoretical basis for responding to requirements.
- *Research* Review prior research relevant to requirements.

Test and Evaluation

- *Apparatus* Describe device(s) used to accomplish the task.
- *Procedure* Describe procedure(s) used to accomplish the task.

Findings

- *Data*
- *Interpretation*

Review the results of the test and evaluation.

Provide your interpretation of the results, that is, to what extent requirements were met.

Conclusion

- *Assessment*
- *Recommendations*

State your conclusions based on the interpretation(s).

Provide your recommendation(s), if any.

Example 11.1 Laboratory Report

Laboratory Report

Our first example illustrates a research-oriented laboratory report written in third person and with sources, charts, and a table. This report documents preliminary laboratory testing on the SuperXLTube, assessing the device's performance in audio service.

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Preliminary Report on the SuperXLTube in Audio Service

Introduction

Purpose

This report documents the preliminary testing and analysis of the SuperXLTube transmitting tube to determine its effectiveness when used

as an audio-frequency power amplifier for rock concerts held in and around large arenas.

Problem

The Arctic Ponies, a rock group known for its high-amplitude, megabass music, is planning an extended concert series in a large stadium (more than 80,000 seats). The audio system must produce low-frequency pressure waves with amplitudes capable of disorienting audience members through vestibular displacement anywhere within the stadium or its immediate surrounding area (approximately 20 acres). To meet the low-frequency audio requirement, the sound system will use the enclosed stadium as a resonant baffle, with four arrays at each corner containing 25 woofers rated at 1 kilowatt each. With a cumulative power requirement for the four arrays of 100 kilowatts, and assuming a nominal loss of 10 decibels, the amplifier must produce at least 1 million watts of audio to provide the necessary drive to the four woofer arrays.

Solid-state amplifiers are not as feasible as vacuum tubes at this power level. This power level would require hundreds of thousands of power transistors operating in parallel in small groups through the use of cascaded combiner transformers to equal the output of a single vacuum tube. Additionally, electron tube amplifiers are being used increasingly as power amplifiers in demanding audio environments because they might provide more linear sound reproduction with a richer blend of harmonics. It is therefore hypothesized that the SuperXLTube will meet the audio requirements of the Arctic Ponies stadium concert.

Scope

This preliminary test and evaluation of the SuperXLTube in a power audio amplifier application includes only an analysis of power, distortion, and noise. Furthermore, this report is limited to the technical performance of the device and does not include analysis of its cost–benefit ratio or marketability in audio applications.

Background

Theory

Vacuum tubes were first developed in the early 1900s to control the flow of electronic current and were used in virtually all audio amplifier equipment until the early 1960s. At that time, solid-state devices quickly replaced tubes in audio applications because of their smaller size and higher efficiencies, except in very high-power applications where solid-state devices were not as feasible.

Additionally, in recent years, there has been a noted resurgence in audio amplifier applications using vacuum tubes for reasons other than power. Apocryphal data indicate that many audiophiles, audio engineers, and professional musicians prefer the tube amplifier's "warm and softer" sounds. Some attribute the tube's comparatively better sound, vis-à-vis solid state, to some or all of the following: better power management, less distortion, improved signal-to-noise ratios, better linearity, less clipping, and fewer feedback problems. This preliminary analysis focuses on the first three variables—power, distortion, and noise.

The SuperXLTube is a 1.2-megawatt transmitting tube designed primarily as a wideband, radio-frequency (RF) power amplifier running in Class C mode across the spectrum of 0.5 kilohertz to 300 megahertz. This laboratory research is designed to test the performance of the SuperXLTube in audio-frequency (AF) service, running as a power amplifier in Class A across the spectrum from direct current (dc) to 20 kilohertz.

Research

Limited research has been conducted on using RF power tubes in AF applications, although Williams did investigate the limited use of low-power transmitting tubes in audio applications (Williams 2000, pp. 347–350). This investigation indicated that while the tube could function effectively at audio frequencies and provide solid performance at its rated RF output power, the cost factors were not favorable for such applications. This analysis predates the current resurgence in tube amplifier applications. It also did not address high-power applications that lend themselves to the state-of-the-art technology capabilities of the SuperXLTube.

Test and Evaluation

Apparatus

The SuperXLTube is capable of producing a continuous output of 1.2 million watts of audio into large arrays with an input signal of 1,000 volts peak to peak. This power is more than adequate for demanding environments such as rock concerts; and it will even meet the controversial pressure-wave requirements of the Arctic Ponies, that is, disorienting audience members through vestibular displacement anywhere within the stadium or its immediate surrounding area.

Local zoning ordinances and environmental protection considerations would not permit laboratory evaluation of the SuperXLTube at its maximum rating. Consequently, for test purposes, the SuperXLTube was used in a single-ended audio amplifier, running in Class A. The tube was driven with a multistage, high-gain voltage amplifier with a nominal output power of 100 watts.

The voltage and power amplifiers were combined into the amplifier test unit (ATU). The ATU was constructed on a steel chassis with proper external and interstage shielding, and the power amplifier stage employed appropriate cavity and coaxial socket interfaces for the SuperXLTube. A calibrated signal generator was used to drive the amplifier. The output was fed into a purely resistive 8-ohm load, where it was analyzed by wide-bandwidth, multitrace oscilloscopes, as well as high-end computers running proprietary analysis software.

Procedure

A pure, distortion-free, sine wave signal of 1 volt peak to peak was applied to the voltage amplifier stages of the ATU. This level was the maximum input signal amplitude possible without saturating these stages. The frequency of the signal was varied across a range from dc to 20 kilohertz in 1-kilohertz increments. The output was analyzed in terms of power management, distortion, and noise.

Findings

Data

The data gathered fell into these three categories: power in watts, distortion as a percentage of total signal, and signal-to-noise ratio.

Power

Power is the product of voltage and current output from the amplifier. Constancy of power across the full frequency range is essential for properly driving complex speaker systems in demanding environments. The power output of the SuperXLTube was observed across the full range of frequencies tested. Although the output varied slightly across the test range and clearly dropped off above 18 kilohertz, overall, the output remained relatively constant. This constancy was especially true in the primary audio frequencies between dc and 15 kilohertz (see [Table 11.1](#) and [Figure 11.2](#)).

Table 11.1 Laboratory data

<i>kHz</i>	<i>P</i>	<i>dis</i>	<i>SN</i>
1	10.22	.00	.15
2	10.24	.00	.15
3	10.26	.01	.10
4	10.30	.01	.10
5	10.29	.01	.09
6	10.29	.01	.09
7	10.30	.01	.09
8	10.30	.02	.09
9	10.29	.09	.10
10	10.29	.01	.10
11	10.31	.02	.10
12	10.30	.01	.11
13	10.31	.02	.11
14	10.33	.01	.11
15	10.21	.01	.12
16	10.30	.03	.12
17	10.29	.03	.13
18	10.00	.03	.14
19	10.00	.03	.14
20	10.00	.04	.15

kHz = frequency in kilohertz; P = power in watts; dis = distortion in percent; SN = signal-to-noise ratio

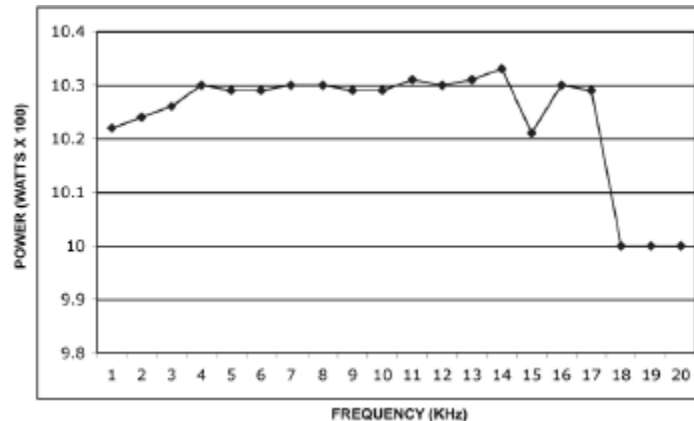


Figure 11.2

Power and frequency.

Distortion

Distortion is the unwanted characteristic of an amplifier to modify the nature of a signal while amplifying it. Although solid-state amplifiers are often credited with having lower distortion values than tube amplifiers, the SuperXLTube exhibited harmonic distortion levels across its total operating range of less than 0.05, which is as good as, if not better than, most solid-state amplifiers. The distortion that did occur was at the upper limit of the range, well above the threshold of human hearing (see [Table 11.1](#) and [Figure 11.3](#)).

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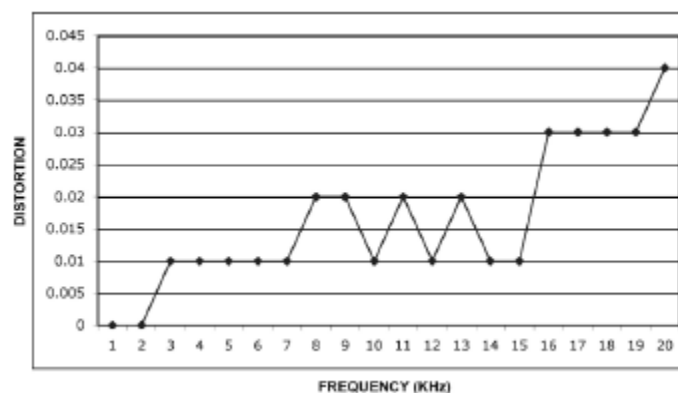


Figure 11.3

Distortion and frequency.

Noise

Noise includes spurious output signals that were not part of the original input signal. Good solid-state amplifiers typically have a signal-to-noise ratio (SN) of 0.15. By comparison, the SuperXLTube provided a typical SN of 0.09 to 0.15 across the primary audio spectrum (see [Table 11.1](#) and [Figure 11.4](#)).

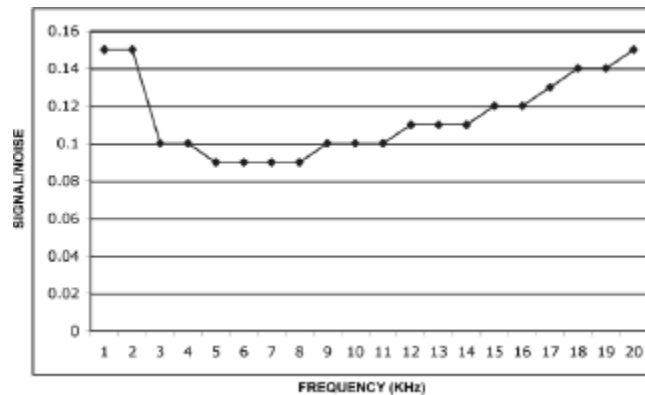


Figure 11.4
Signal-to-noise ratio and frequency.

Interpretation

In terms of the preliminary tests involving power, distortion, and noise, the SuperXLTube performed well in an audio-frequency, power-amplifier mode. Although the high-power levels could not be duplicated in the laboratory setting, the low-power tests demonstrated excellent results ([Table 11.1](#)).

Conclusion

Assessment

Initial laboratory analysis, based on the SuperXLTube's measured performance, supports the hypothesis; the tube seems to be a viable candidate for the sound system supporting the Arctic Ponies stadium concert.

Recommendation

The Arctic Ponies should consider the SuperXLTube a technically suitable option for audio power amplifier applications.

Sources

Williams, Robert E. “Investigation of Transmitting Tubes for Audio Applications,” *Technical Tidbits*. New York: SoundProducts International, 2000, pp. 345–354.

Dissecting [Example 11.1](#)

Now that you have read an example of a laboratory report, let us take a closer look at each of its components and what they do.

Introduction

Purpose

Start the introduction by briefly and succinctly describing the purpose of the report.

This report documents the preliminary testing and analysis of the SuperXLTube to determine its effectiveness when used as an audio-frequency (AF) power amplifier for rock concerts held in and around large arenas.

Problem

Next, provide the reader with enough information on the problem so that they can understand what the report is about and put the information that follows in the proper context. Remember that your focus here is on what your reader needs. For this example, the problem involves our fictitious rock group, the Arctic Ponies, whose act depends on high-amplitude, low-frequency “galloping” sound waves to ensure a truly measurable impact on the audience.

The Arctic Ponies, a rock group known for its high-amplitude, megabass music, is planning an extended concert series in a large stadium (more than 80,000 seats). The audio system must produce low-frequency pressure waves with amplitudes capable of disorienting audience

members through vestibular displacement anywhere within the stadium or its immediate surrounding area (approximately 20 acres). To meet the low-frequency audio requirement, the sound system will use the enclosed stadium as a resonant baffle, with four arrays at each corner containing 25 woofers rated at 1 kilowatt each. With a cumulative power requirement for the four arrays of 100 kilowatts, and assuming a nominal loss of 10 decibels, the amplifier must produce at least 1 million watts of audio to provide the necessary drive to the four woofer arrays.

Solid-state amplifiers are not as feasible as vacuum tubes at this power level. This power level would require hundreds of thousands of power transistors operating in parallel in small groups through the use of cascaded combiner transformers to equal the output of a single vacuum tube. Additionally, electron tube amplifiers are being used increasingly as power amplifiers in demanding audio environments because they might provide more linear sound reproduction with a richer blend of harmonics. It is therefore hypothesized that the SuperXLTube will meet the audio requirements of the Arctic Ponies stadium concert.

Round out the introduction with a statement of scope regarding what is being tested and any significant limitations of these tests.

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Scope

Specify the limitations of this report—i.e., what it will cover, and what it will not cover.

This preliminary test and evaluation of the SuperXLTube in a power audio amplifier application includes only an analysis of power, distortion, and noise. Furthermore, this report is limited to the technical performance of the device and does not include analysis of its cost-benefit ratio or marketability in audio applications.

Background

In this section, provide the information necessary for the reader to understand and appreciate the test report and the findings that will follow. Again, make sure your focus is on what the reader needs. This section should review any relevant theory and past research that the reader needs to know.

Theory

Vacuum tubes were first developed in the early 1900s to control the flow of electronic current and were used in virtually all audio amplifier equipment until the early 1960s. At that time, solid-state devices quickly replaced tubes in audio applications because of their smaller size and higher efficiencies, except in very high-power applications where solid-state devices were not as feasible.

Additionally, in recent years there has been a noted resurgence in audio amplifier applications using vacuum tubes for reasons other than power. Apocryphal data indicate that many audiophiles, audio engineers, and professional musicians prefer the tube amplifier's

“warm and softer” sounds. Some attribute the tube’s comparatively better sound, vis-à-vis solid state, to some of or all the following: better power management, less distortion, improved signal-to-noise ratios, better linearity, less clipping, and fewer feedback problems. This preliminary analysis focuses on the first three variables—power, distortion, and noise.

The SuperXLTube is a 1.2-megawatt transmitting tube designed primarily as a wideband, radio frequency (RF) power amplifier running in Class C mode across the spectrum from 0.5 kilohertz to 300 megahertz. This laboratory research is designed to test the performance of the SuperXLTube in audio-frequency (AF) service, running as a power amplifier in Class A across the spectrum from direct current (dc) to 20 kilohertz.

Research

Review significant prior research that has been done on this topic to establish a baseline for your research. Here is where you show that you are knowledgeable about the subject and have done your due diligence.

Limited research has been conducted using RF power tubes in AF applications, although Williams did investigate the limited use of low-power transmitting tubes in audio applications (Williams 2000, pp. 347–350). This investigation indicated that while the tube could function effectively at audio frequencies and provide solid performance at its rated RF output power, the cost factors were not favorable for such applications. This analysis predates the current resurgence in tube amplifier applications. It also did not address high-power applications that lend themselves to the state-of-the-art technology capabilities of the SuperXLTube.

Test and Evaluation

In this section, describe the physical **apparatus** used in the test and the **processes or procedures** for doing the testing. The apparatus includes the device being tested and the equipment used to do the testing, while the procedure would include the steps in the test.

Apparatus

The SuperXLTube is capable of producing a continuous output of 1.2 million watts of audio into large arrays with an input signal of 1,000 volts peak to peak. This power is more than adequate for demanding environments such as rock concerts, and it will even meet the controversial pressure wave requirements of the Arctic Ponies—that is, disorienting audience members through vestibular displacement anywhere within the stadium or its immediate surrounding area.

Local zoning ordinances and environmental protection considerations would not permit laboratory evaluation of the SuperXLTube at its maximum rating. Consequently, for test purposes, the SuperXLTube was used in a single-ended audio amplifier, running in class A. The tube was driven with a multistage, high-gain voltage amplifier with a nominal output power of 100 watts.

The voltage and power amplifiers were combined into the amplifier test unit (ATU). The ATU was constructed on a steel chassis with proper external and interstage shielding, and the

power amplifier stage employed appropriate cavity and coaxial socket interfaces for the SuperXLTube. A calibrated signal generator was used to drive the amplifier. The output was fed into a purely resistive 8-ohm load, where it was analyzed by wide-bandwidth, multitrace oscilloscopes as well as high-end computers running proprietary analysis software.

Procedure

A pure, distortion-free sine wave signal of 1 volt peak to peak was applied to the voltage amplifier stages of the ATU. This level was the maximum input signal amplitude possible without saturating these stages. The frequency of the signal was varied across a range from dc to 20 kilohertz in 1-kilohertz increments. The output was analyzed in terms of power management, distortion, and noise.

Findings

In this section, present the data that the tests yielded and provide an interpretation of these data. As you saw earlier, a table and various graphs can be used to display the data.

Data

The data gathered fell into these three categories: power in watts, distortion as a percentage of total signal, and signal-to-noise ratio.

Power

Power is the product of voltage and current output from the amplifier. Constancy of power across the full frequency range is essential for properly driving complex speaker systems in demanding environments. The power output of the SuperXLTube was observed across the full range of frequencies tested. Although the output varied slightly across the test range and clearly dropped off above 18 kilohertz, overall, the output remained relatively constant. This constancy was especially true in the primary audio frequencies between dc and 15 kilohertz.

Distortion

Distortion is the unwanted characteristic of an amplifier to modify the nature of a signal while amplifying it. Although solid-state amplifiers are often credited with having lower distortion values than tube amplifiers, the SuperXLTube exhibited harmonic distortion levels across its total operating range of less than 0.05, which is as good as, if not better than, most solid-state amplifiers. The distortion that did occur was at the upper limit of the range, well above the threshold of human hearing.

Noise

Noise includes spurious output signals that were not part of the original input signal. Good solid-state amplifiers typically have a signal-to-noise ratio (SN) of 0.15. By comparison, the SuperXLTube provided a typical SN of 0.09 to 0.15 across the primary audio spectrum.

Interpretation

In terms of the preliminary tests involving power, distortion, and noise, the SuperXLTube performed well in an audio-frequency power amplifier mode. Although the high-power levels could not be duplicated in the laboratory setting, the low-power tests demonstrated excellent results, as the following data show.

Table 11.1 Laboratory data

<i>kHz</i>	<i>P</i>	<i>dis</i>	<i>SN</i>
1	10.22	.00	.15
2	10.24	.00	.15
3	10.26	.01	.10
4	10.30	.01	.10
5	10.29	.01	.09
6	10.29	.01	.09
7	10.30	.01	.09
8	10.30	.02	.09
9	10.29	.09	.10
10	10.29	.01	.10
11	10.31	.02	.10
12	10.30	.01	.11
13	10.31	.02	.11
14	10.33	.01	.11
15	10.21	.01	.12
16	10.30	.03	.12
17	10.29	.03	.13
18	10.00	.03	.14
19	10.00	.03	.14
20	10.00	.04	.15

kHz = frequency in kilohertz; P = power in watts; dis = distortion in percent; SN = signal-to-noise ratio

Conclusion

Finally, provide your overall conclusions related to the original purpose of this study, and make any specific recommendations that you believe are warranted by the results.

Assessment

Initial laboratory analysis, based on the SuperXLTube’s measured performance, supports the hypothesis that the tube seems to be a viable candidate for the sound system supporting the Arctic Ponies stadium concert.

Recommendation

The Arctic Ponies should consider the SuperXLTube a technically suitable option for audio power amplifier applications.

Source

Williams, Robert E.: “Investigation of Transmitting Tubes for Audio Applications,” *Technical Tidbits*. New York: SoundProducts International, 2000, pp. 345–354.

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Example 11.2 Project Report

Student Project Report

The following is a hypothetical example of a short project report based on [Outline 11.2](#). It is written in first person. The report is required by a fictitious civil engineering course where small groups of students are tasked to design a system for measuring the average height and slope of a road surface. The height and slope data are necessary for repaving an existing asphalt road.

Acquiring Height and Slope Data for Paving Operations

Introduction

Purpose

This report documents the analysis required for the course project assigned to Group 2 to fulfill in part the requirements of CE 400, Civil Engineering: Technology and Methods.

Problem

Our group consists of four seniors majoring in civil engineering. The group was assigned the task of designing a system that would permit road pavers to obtain average height and slope data of a 70-foot segment of two-lane road. The requirements included being able to supply data to pavers in “real time” to allow them to lay down an asphalt mat that is precise in thickness and smoothness and that fills in surface irregularities, whether severe or gradual in nature. The system must be constructed from off-the-shelf components, be electronic in nature and operation, and be designed and fabricated within 7 weeks at a total cost not to exceed \$200 (see course tasking, Appendix A).

Scope

This report documents the approach, test, and evaluation of Group 2’s laboratory project for CE 400, Fall 2021. We noted that the course tasking requires only demonstration that the data generated by the system could be used for paving operations. We assumed, therefore, that no requirement exists to test our solution in actual paving operations.

Background

Research

Paving an asphalt road with precise mat thickness and smoothness requires an averaging system that can track variations in the height and slope of a road’s surface, and provide this information in real time to construction equipment operators conducting paving. Traditionally, obtaining this kind of data requires either a mechanical or electronic approach. Mechanical systems typically use grade-leveling devices similar to “skis” that extend in front of and behind the paver. The front ski senses irregularities in the road, while the trailing ski provides data on the smooth surface that has just been paved. These skis provide the average height and slope for the road, which are then matched in real

time by the *screed*, which controls the height and slope of the asphalt mix being applied. Mechanical skis can be effective, but they are easily damaged by road hazards, require high labor costs to set up, and wear rapidly because of friction with the asphalt surface (Koko 2003, pp. 58–60).

Electronic approaches to providing paving data use emitted radio-frequency, acoustic, or light energy that reflects off the road's surface and is then sensed and read over time and distance to map this surface. Often called *time-of-flight* systems, these devices map the road's surface by measuring the time required by the emitted energy to travel to the surface and back. Several approaches using emitted energy exist, including (but not limited to) ultrasonic, laser, pulsed laser, active triangulation, and continuous RF wave that is either amplitude- or frequency-modulated (Tyger 2005, p. 34).

To respond to the course tasking criteria, especially those placing severe constraints on time and cost, our group chose the ultrasonic solution. Ultrasonic transducers are widely available and relatively inexpensive, and the apparatus required to function at acoustic frequencies is easier to fabricate than those operating in optical or radio-frequency spectra. Typically, off-the-shelf ultrasonic range finders costing less than \$150 are capable of measuring distances up to 30 feet (Peesher 2003, p. 136).

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Theory

Ultrasonic range finders usually operate using a pulsed method, where a ping is emitted and its echo is measured. The sound waves of the ping propagate from the transducer, bounce off the road's surface, and echo back to the receiver. Since the speed of sound can be determined accurately based on air temperature, the time between ping and echo can be converted to precise distance measurements. The constant mapping of distance from a moving vehicle can provide an accurate sampling of irregularities in, and slope of, the road's surface over which the vehicle is moving.

To determine the average height of the road, several tasks must be accomplished: (1) Select a Cartesian coordinate system with a precisely defined origin. (2) Create a system of xy plots from the perspective of the

origin by transforming a synchronized set of sensor-to-road measurements. (3) Calculate the best-fit line between these points. (4) Adjust the application of asphalt by the paver to match this line. Our group developed an algorithm to accomplish tasks 1 through 3, a limitation that is consistent with the scope of this project (Appendix B).

Test and Evaluation

The group's approach was to construct an ultrasonic range finder that could be mounted on a moving vehicle. This approach employed a computerized recording device to track variations in distance, process this information based on our averaging algorithm, and produce a physical profile of the road's surface that could be used to control the application of asphalt.

Apparatus

Our group acquired, at a local flea market for \$100, a Poladucer 500 transducer commonly used in consumer devices. We designed a range-finding circuit (Appendix C) that samples with a controlled, narrow beam directed at a 90° angle to the road's surface. Temperature sensors mounted at the transducer and near the road's surface provided the thermal gradient data necessary to properly estimate the propagation delay of the acoustic energy. We used a Pentium Gold 640 5U laptop computer obtained from an electrical engineering lab to do the necessary I/O functions and processing, employing interfaces and code produced by our group (Appendix D). A 3.2-megapixel digital camera was used to create photographic sequences of the road's surface.

Procedure

To test our surface-measuring device, we mounted it on the trailer hitch of a pickup truck. The sensor looked straight down at a right angle to the road 2.5 feet below. The computer was operated in the cargo bed of the truck. To sample a smooth strip of road for test purposes, we drove the truck down the well-kept driveway to the president's house on the university campus. To gather data from a rough road surface, we conducted the same test on the decaying driveway leading to the on-campus student apartments. For both tests, we stopped the truck every 10

feet and physically measured the distance to the road's surface. We flagged the data taken at that point and compared it with the physical measurements. We also produced a cross-sectional profile of the road's surface and visually compared that computer-generated surface plot to sequenced photographs of the actual road surface (Appendix E).

Findings

Data

The test data are provided in Appendix F, along with the profiles taken of the road surfaces. Generally, the data tracked well with the physical measurements, with accuracy of 0.2 to 1.5 percent of the total range. Additionally, the profiles constructed closely matched the visual appearance of the road's surface. However, we did note a substantial reduction in accuracy (2.5 to 5 percent of total range) where surface features changed suddenly. For example, our system was not able to handle the sharp vertical offsets associated with potholes encountered on the road to the student housing area.

Interpretation

Our system appears to meet the assignment criteria of the course. Arrays of overlapping ultrasonic transducers could be fitted onto traditional paving vehicles, thereby replacing the front and rear mechanical skis. A computer on board the paver could be programmed to sample all the devices in the arrays. Together, these devices would provide a much wider and higher-resolution view of the road than the single, narrow strip our test apparatus sampled. This improved view would enhance the capability to deal with sudden changes or irregularities in the road's surface, and would enable better control for the application of asphalt in paving operations.

Conclusion

Assessment

We assess the performance of our group as *excellent*. We have developed a simple yet effective electronic device that can measure the height and

slope of a road's surface and translate these measurements 1) to control signals from a computer, and 2) to manage in real time the application of asphalt from a paver.

Recommendations

For obvious reasons involving cost and safety, our group could not obtain the services of a paver, construction access to a deteriorated road that needed to be paved, or the expertise and materials necessary to actually pave it. We recommend that the university adapt our device for an operational paver and conduct a true field test by repaving the driveway leading to student housing.

References

- Siamesus F. Koko, "The Problem with Mechanical Skis," *Paving Quarterly* (2003), pp. 50–65.
- Elmer S. Tyger, "Time of Flight Can Do It Right," *Paving Quarterly* (2005), pp. 30–38.
- Stephen B. Peesher, "Range Finding Distances: The Efficacy of Evolving RF Systems," 22nd ISMEAX Conference, Anchorage, Alaska, June 2003, pp. 131–159.

Visuals in Laboratory and Project Reports

When you are designing visuals such as data tables, diagrams, schematics, and photographs in support of a laboratory or project report, be sure to include only information that directly relates to the report and is specifically keyed to the report's content. Do not include visuals that do not match the report in subject matter or terminology. Also, avoid visuals that include too much or too little complexity for the level of discussion in the paper.

- Visuals should relate specifically to the text discussion, and they should be referenced in the text before actually being used. The

reader should never encounter a visual and wonder why it is there or what it is for. Always tell your reader specifically when to look at a visual and try to do so *before* the point in the paper where the visual actually appears.

- Each visual should have an assigned sequence number and name. Also, if possible, run separate sets of sequence numbers for each type of visual. For example, figures should have their own set, as should equations and tables.
- Each visual must be included for a specific purpose. Do not add visuals to make the page look “pretty” or to “add visual interest.”
- Process visuals usually need to show something happening and, as such, tend not to be static representations of something seen in a “slice of time.” Often you can show movement through space and time with a technique as simple as a well-placed arrow. At other times you may need a coherent series of visuals to demonstrate your point or before-and-after pairings of visuals.

For more information on visuals, see [Chapter 17](#).

Laboratory and Project Report Checklist

- ☐ Do I understand who my audience is for this report?
- ☐ Does my audience have specific formulation requirements that I should use?
- ☐ Have I clearly defined the purpose of this report?
- ☐ Have I clearly described the problem that requires this report?
- ☐ Have I clearly explained the limitations of this report?
- ☐ Have I discussed any theory necessary for the reader to understand the report?
- ☐ Have I reviewed relevant prior research?

- ❑ Have I described the apparatus I used to collect the data?
- ❑ Have I described the procedure I used to collect the data?
- ❑ Have I clearly provided the results of the test?
- ❑ Have I properly interpreted these results?
- ❑ Have I made proper conclusions from these interpretations?
- ❑ Have I made a recommendation based on my conclusions?
- ❑ Have I provided only visuals that relate to my text?
- ❑ Have I referred to the visuals in my text before they are used by my audience?
- ❑ Have I given each visual an assigned sequence number and name?
- ❑ Have I run separate sets of sequence numbers for different types of visuals?
- ❑ Does each visual have a specific purpose?

Exercise

Read the following laboratory report and apply the checklist above. Pay particular attention to the amount of detail and relevance of each section and whether anything is missing. Also, are there any obvious inconsistencies that could undermine the credibility of this lab report?

Evaluation of the Rapid Manufacturing and Prototyping System for the Production of Spectacle

Lenses in Canis Optics Network Vision Centers

Introduction

Purpose

This report provides a preliminary assessment of the use of the Rapid Manufacturing and Prototyping System (RAMPS) for the production of corrective vision lenses for spectacles in the Canis Optics Network (CONE).

Problem

CONE is a constellation of independently franchised vision centers operating in all 50 states and in many overseas locations. CONE provides integrated optometry and optician services, including eye exams and spectacle production at all locations. Normally, new eyewear is available within 60 minutes; however, Canis senior management believes that substantially greater revenues could be realized by switching from traditional lens-grinding methods to RAMPS. RAMPS, which uses high-speed lasers and advanced thermodynamic methods, is capable of producing a perfect set of prescription lenses for normal vision requirements, including astigmatism correction, in 1.25 seconds. This system could substantially accelerate lens production throughput with a consequent increase in revenues.

Scope

This report will consider only the production of standard-index plastic lenses for use in bifocal or trifocal spectacles. Progressive “no-line” and high-index polycarbonate lenses will not be considered in this report.

Theory

Correcting deficient vision requires the introduction of optical refraction to combine with the lens and cornea of the eye to elongate or shorten the focal length of the image being viewed, such that the image is precisely focused on the retina. In this way, myopia (nearsightedness) and hyperopia (farsightedness) can be effectively addressed by the introduction of a spherical power commonly referred to in spectacle prescriptions as **sphere**. Positive (+) powers address hyperopia, while negative (−) powers address myopia. In addition to myopia or hyperopia, many people also suffer from astigmatism, where the sphericity of the

eye does not match the refractivity of the cornea and lens. In effect, the eyeball has an oblong cylindrical shape distorting the basic “ball” curves of the eye. To correct this problem, an additional cylinder power is added to the lens, modifying the spherical power in the axis of toricity. This correction is called **cylinder**, and its alignment is called **axis**. See [Figure 11.5](#) for a sample prescription for corrective lenses.

Test and Evaluation

Apparatus

The devices used in this laboratory procedure included a base-model RAMPS 2000 and 100 lbs. of polycarbonate optical lens material.

Procedure

The prescription provided in Figure 11.5 was loaded into the “To Do” Buffer, the polycarbonate material was loaded into the Raw Materials Bin, and the command <MAKE LENSES /3/4.5/ RX1/LOOP> was entered at the control console. The operation was timed and the lenses produced were inspected for optical clarity and purity, as well as correctness of the prescription.

Findings

The RAMPS was allowed to operate until the raw lens polycarbonate material was exhausted, at which point the number, quality, and correctness of lenses were ascertained. [Table 11.2](#) provides the results.

Canis Optics Center
2345 Executive Way
Beavercreek, Ohio 45430
Tanisha Wu, M.D., O.D., FACS

Name Nicholas Miller

Address 4881 Wed. St., Beavercreek, OH

Date: May 8, 2021 ☒ Bifocal ☐ Trifocal ☐ Progressive

Distance OD	Sphere	Cylinder	Axis	Prism
	-3.25	+2.50	032	

Distance OS	Sphere	Cylinder	Axis	Prism
	-3.50	+3.00	150	

NearAdd OD	Sphere	Cylinder	Axis	Prism
	+2.25			

NearAdd OS	Sphere	Cylinder	Axis	Prism
	+2.25			

Remarks *Axis must be exact.*

SIGNED: Tanisha Wu, MD, OD

Figure 11.5

Corrective lens prescription.

Table 11.2 Laboratory data

Amount of polycarbonate used/wasted	99.99 lbs. used /02 grams wasted
Number of lenses produced/unit of time	1200 lenses/second
Percent of lenses that were optically perfect	99.99
Percent of lenses with a correct prescription	100

Data

Interpretation

The RAMPS' performance was really good!

Conclusion

Assessment

The RAMPS performed as expected and has a significant potential for increasing the revenue stream of the Canis Optics Network Vision Centers.

Recommendation

Canis Optics Network Vision Centers should be equipped with the RAMPS as soon as possible.

==

CHAPTER TWELVE

Research Reports



Research reports resemble the **research papers** that every student has written at one time or another. You remember—papers with titles such as “The Modern Marvel of CRISPR,” “The Humanity of Lady Macbeth,” “The Benefits of the Modern Mosquito,” or “GMOs Are Changing the World.” You probably wrote one of your very first research papers in elementary school, perhaps on dogs. It included some facts, a few pictures, and a timeline. Research papers ask that you conduct research for the purpose of arguing a point (“CRISPR is good.” “Lady Macbeth was ruthless.” “Dogs are better than wolves.”) (see [Figure 12.1](#)). The research you conduct provides arguments and evidence in support of your own evaluation, position, or opinion. This is how research **reports** are different research **papers**; in technical writing, instead of supporting opinion, research reports are focused, objective inquiries into technical subjects.



Figure 12.1

A research report describing how dogs evolved from wolves might include visuals showing their physical similarities and differences.

What Are Research Reports?

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Research reports describe the discovery, analysis, and documentation of knowledge obtained through some type of investigation. In technical writing, these reports are specifically geared to the purpose at hand, the readers who will use them, the clients who will read them, and whatever limitations have been placed on the scope of the project. Technical research reports frequently focus on new, evolving, sometimes purely hypothetical technologies, in which case they are called **state-of-the-art reports**. In some cases, research reports might focus on past technology, in which case they are called **historical reports**.

One distinguishing characteristic of research reports is the extensive research and documentation required. The research might consist of a variety of research strategies, including library, internet, and laboratory

research; interviews and focus groups; surveys and questionnaires; various types of corporate technical reports; and trade journal articles. Unlike laboratory reports, however, research reports often do not involve doing the actual research being reported; they frequently present the findings of research that has already been done.

The organization of a research report is straightforward, as shown in [Outline 12.1](#). However, what goes in the discussion section depends on the topic and the specific requirements for the research. If the purpose is to report on how we got to where we are in developing a certain technology, the discussion will be primarily historical. But if the purpose is to describe new, evolving technologies, the discussion might be geared more toward future implications. For example, if the purpose is for a product engineer to research the feasibility of introducing a new axle housing design into an existing product line, the discussion might be geared more toward torque standards, required hardness, thread adhesive, torque specifications, etc., along with the potential impacts of making such a change.

Research reports are usually comprehensive documents that often extend beyond the scope of this book. However, to illustrate the kinds of things that go into a research report, this chapter provides two abbreviated examples. The first example is a state-of-the-art report on a fictitious but theoretically correct piece of technology, the quantum computing chip called the Quantum Central Processing Unit (QCPU). The second example is a factually correct, historical report on the Pegasus Computer, which was retired in 2014. Both examples follow [Outline 12.1](#), but the first example is oriented toward the future, while the second example reviews the technology of the past.

Outline 12.1 Research Report

Introduction

- **Purpose** Describe the reason for writing this report.
- **Problem** Provide a brief overview or introduction for the topic.
- **Scope** Describe the limitations of this report.

Background

- **Theory** Review any theory needed to understand the topic.
- **History** Provide any necessary historical perspective.

Discussion

Provide the main section of the report—content varies significantly depending on the purpose of the report, the audience need and expectations for the report, and the broader context in which the report is being written.

Conclusion

- **Summary** Summarize the discussion section.
- **Recommendation** Provide suggestions based on the summary.

References

- **Sources Cited** List sources consulted and used in the report.
- **Sources Not Cited** List sources consulted but not specifically used.

Appendices

Include supporting material if desired; it might not be required for audience understanding.

Example 12.1 State-of-the-Art Research Report

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This first example is the state-of-the-art report on the Quantum Chips Corporation's (QCCorp) Quantum Central Processing Unit (QCPU). This quantum processor chip exploits the tremendous potential of quantum computing architectures to effectively increase, by several orders of magnitude, the high-end computing power currently available. (Quantum computing is actually an area of serious scientific research among leading physicists and engineers. It is a dynamic theoretical field.)

The following report was written for a knowledgeable technical reader, and it includes visuals and references. To conserve space, an appendix is not

provided with this example, although a research report on this topic would have one. Note, too, that the documentation for this fictitious report is also fictitious, but it models how to do it.

State-of-the-Art Report on Quantum Chips' Quantum Central Processing Unit

Introduction

Purpose

The purpose of this report is to provide a state-of-the-art investigation of the Quantum Chips Corporation's (QCCorp) Quantum Central Processing Unit (QCPU), including a theoretical review of the premises of its operation.

Problem

Traditionally, computing power has been enhanced by increasing CPU speeds, primarily through decreasing the size of conductors and solid-state devices used in chip fabrication. Decreasing size, however, has finite limitations, such as those associated with reducing the dielectric constants of the required materials. There has also been a move toward increasing the number of instructions executed for each clock cycle, especially with reduced-instruction-set computer (RISC) processors.

QCCorp has transcended this traditional paradigm by developing the QCPU. The QCPU exploits and manipulates the quantum nuclear spin states of atoms. The QCPU is capable of performing large numbers of advanced computational tasks simultaneously, using the superimposition of multiple values encoded into the respective spin states of individual atoms. The resulting (effective) CPU speed is equal to or better than 500 gigahertz (Josephson 2004, p. 291). This effective speed makes the QCPU an ideal chip for processing-intensive tasks such as cryptographic

factoring of large numbers, DNA sequencing in genetic research, and interactive, three-dimensional holographic imaging in advanced virtual reality systems.

Scope

The QCPU is built using proprietary information owned and protected by QCCorp. Consequently, this report will be limited to the general theoretical approaches underlying the QCPU architecture; it will not investigate the actual methods used by the QCPU to implement these theoretical approaches.

Background

Theory

Quantum computing theory applies the knowledge of quantum physics to exploit subatomic phenomena of common elements to perform extremely complex computational tasks. When properly exploited, these phenomena provide a truly unprecedented ability for massively parallel processing (Ardvark 2004, pp. 446–448). Several options exist to exploit quantum phenomena in this regard. One is to equate binary values to the ground and excited states of an atom. Another is to use traditional nuclear magnetic resonance (NMR) techniques to read induced spin states of atoms. A third is to polarize photons in an optical chamber. QCCorp has applied the second option in the QCPU, using NMR techniques to read specifically induced spin states in carbon, hydrogen, and other atoms (Josephson 2004, p. 301).

To manipulate carbon and hydrogen atoms, radio-frequency (RF) energy is applied to each atom at its specific resonant frequency. This RF energy is applied to the atom while it is in a fixed magnetic field. Because the atom remains in a fixed position, the position can serve as its memory address. The nucleons of these atoms spin predictably while in this magnetic field. If an atom lines up with the direction of the magnetic field, it is considered to be in a “spin up” orientation. If it lines up in a direction opposite to the magnetic field, it is considered to be in a “spin down” orientation. Different spin states have

different energy signatures for different atoms at different magnetic field magnitudes. These differences can be read by NMR sensors.

Discussion

Genesis of the QCPU

In 1998, George Yamaslute demonstrated that different RF signals cause predictably different spins for certain atoms. He also showed that the spin of these atoms can be altered by the application of different RF signals. These altered spin states then can be used symbolically to represent different values. The spin states of these atoms effectively store the values encoded by these RF signals. These values are then read by traditional NMR techniques, thereby creating a machine memory capability (Yamaslute 1998, pp. 200–210).

Besides memory capabilities, manipulation of the spin states of atoms can be used to perform various logic operations. Early experiments with molecules containing carbon and hydrogen atoms demonstrated such theoretical feasibility. The carbon and hydrogen atoms can be manipulated independently by varying RF signals applied at different resonant frequencies while these atoms are held in a fixed magnetic field. By making both atoms spin up, or spin down, or alternately spin up and spin down, the atoms can constitute a 2-bit truth table ([Table 12.1](#)). Adding more types of atoms and using intermediate spin states substantially increase the range of logical operations. Such manipulation provides the quantum logic capabilities of the QCPU (Yamaslute 1998, p. 240).

Table 12.1 Spin state truth table.

RF signal applied	Carbon	Hydrogen
Frequency 1	U	U
Frequency 2	U	D
Frequency 3	D	U
Frequency 4	D	D

The QCPU

The Quantum Chips QCPU is proprietary technology used today only in highly classified government projects. Although unclassified information is limited, some scientists believe that the QCPU is now providing the computational power for the genetic manipulation of bacteria. The specific goal is to create virulent strains of Naomi-B bacteria that are capable of eating through the armored titanium and steel hulls of enemy submarines at depths exceeding 10,000 feet (Mierson 2005, p. 50).

The QCPU assembly consists of the quantum molecular matrix (QMM), the magnetic field coil (MFC), the nuclear magnetic resonance (NMR) sensor, and the RF assembly (RFA) (see [Figure 12.2](#)). The MFC provides the fixed magnetic field engulfing the QMM. The QMM provides atomic structures that have unique spin-up and spin-down characteristics. The RFA provides the RF energy needed to change the spin characteristics of each atom to reflect specific values (Bearkins 2006, p. 91). The energy is radiated by a phase directional array, but the exact design or method of control is proprietary. The NMR sensor is the means of reading these altered spin states.

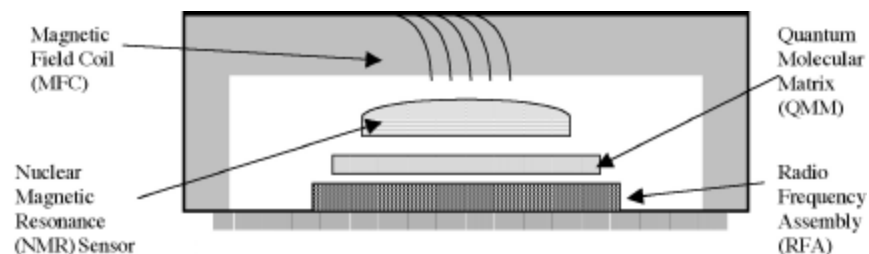


Figure 12.2

QCPU assembly.

Source: George S. Yamaslute, "Magnets and Bits," *Journal of Applied Magnetic Resonance* 4:9 (September 2002), p. 260.

Conclusion

Summary

Quantum Chips' Quantum Central Processing Unit (QCPU) has made quantum computing a reality. By controlling and reading the spin states

of selected atoms, using applied RF and NMR techniques, the QCPU effectively uses quantum phenomena to store data and conduct logical operations on those data. Quantum computing, by exploiting the possibilities of data manipulation and storage at the atomic level, provides the power to accomplish parallel processing at far greater amplitudes than traditional approaches.

The actual design and implementation of the QCPU is not only proprietary but also highly classified for national security purposes. Little information is available about the specifics of the QCPU's design implementation.

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Ardvark, Wilma J.: "NMR and Beyond," *Quantum Computer Quarterly* 9:6 (August 2005), pp. 460–495.

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Josephson, Maxine E.: "Technical Report 1999-35." Quantum Chips International, November 2007.

Mierson, Wilhelm F.: "The Dreaded Threat of Naomi-B," *Journal of Truth in Science* 12:2 (February 2005), pp. 460–462.

Yamaslute, George S.: "Magnets and Bits," *Journal of Applied Magnetic Resonance* 4:9 (September 1998), pp. 150–265.

Sources Consulted and Not Used

Baker, Joe-Bob.: *A View of That Which We Cannot See*. New York: Acme Press, 1999.

Dissecting Example 12.1

Now that you have read an example of a research report, let us take a closer look at each of its components and what they do.

Introduction

Purpose

As in other documents, the first section of the research report is the introduction. Start with the purpose statement to explain why you are writing the report:

The purpose of this report is to provide the results of a state-of-the-art investigation of the Quantum Chips Corporation's (QCCorp) Quantum Central Processing Unit (QCPU), including a review of the theoretical premises of its operation.

Problem

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Next, state the problem that the report addresses. In a research report, note that the problem is really more of a general background statement that expands on the topic and gives a brief context for what the report will investigate. Also note the inclusion of a source citation to support the assertion regarding processing speed. This type of documentation is required in research reports and is explained more thoroughly in [Chapter 16](#).

Traditionally, computing power has been enhanced by increasing CPU speeds primarily through decreasing the size of conductors and solid-state devices used in chip fabrication. Decreasing size, however, has finite limitations, such as those associated with reducing the dielectric constants of the required materials. There has also been a move toward increasing the number of instructions executed for each clock cycle, especially with reduced-instruction-set computer (RISC) processors.

QCCorp has transcended this traditional paradigm by developing the QCPU. The QCPU exploits and manipulates the quantum nuclear spin states of atoms. The QCPU is capable of performing large numbers of advanced computational tasks simultaneously, using the superimposition of multiple values encoded into the respective spin states of individual atoms. The resulting (effective) CPU speed is equal to or better than 500 gigahertz (Josephson 2004, p. 291). This effective speed makes the QCPU an ideal chip for processing-intensive tasks such as cryptographic factoring of large numbers, DNA sequencing in genetic research, and interactive, three-dimensional holographic imaging in advanced virtual reality systems.

Scope

In any research paper, you cannot possibly research everything about your topic. Human knowledge is not that simple or easy, and there is too much of

it. You must limit your paper by including only certain aspects of your topic. To complete the introduction, provide a scope statement that addresses this limitation. This section tells your readers what you are including in the paper and why, and it articulates the rationale for the limitations you are imposing.

The QCPU is built using proprietary information owned and protected by QCCorp. Consequently, this report will be limited to the general theoretical approaches underlying the QCPU architecture; it will not investigate the actual methods used by the QCPU to implement these theoretical approaches.

Background

In the background section, discuss the theoretical and historical aspects of the topic, as appropriate. Given the purpose here, this example will focus on only the theoretical aspects of Quantum Chips' QCPU technology. The background should start with a brief discussion of quantum computing theory because this theory is not common knowledge for the audience; consequently, the theory discussion is essential to understanding the rest of the paper. Note again the inclusion of source citations throughout this discussion.

Theory

Quantum computing theory applies the properties of quantum physics to exploit subatomic phenomena of common elements to perform extremely complex computational tasks. When properly exploited, these phenomena provide a truly unprecedented ability for massively parallel processing (Ardvark 2004, pp. 446–448). Several options exist to exploit quantum phenomena in this regard. One is to equate binary values to the ground and excited states of an atom. Another is to use traditional nuclear magnetic resonance (NMR) techniques to read induced spin states of atoms. A third is to polarize photons in an optical chamber. QCCorp has applied the second option in the QCPU, using NMR techniques to read specifically induced spin states in carbon, hydrogen, and other atoms (Josephson 2004, p. 301).

To manipulate carbon and hydrogen atoms, radio frequency (RF) energy is applied to each atom at its specific resonant frequency. This RF energy is applied to the atom while it is in a fixed magnetic field. Because the atom remains in a fixed position, the position can serve as its memory address. The nucleons of these atoms spin predictably while in this magnetic field. If an atom lines up with the direction of the magnetic field, it is considered to be in a “spin up” orientation. If it lines up in a direction opposite to the magnetic field, it is considered to be in a “spin down” orientation. Different spin states have different energy signatures for different atoms at different magnetic field magnitudes. These differences can be read by NMR sensors.

History

This report on the QCPU is a **state-of-the-art** report, which means, if you remember from the beginning of this chapter, that it will focus on “new, evolving, sometimes purely hypothetical technologies.” The history section of the Background is for “any necessary historical perspective,” and as the author was writing about a “new, evolving, purely hypothetical” technology, they decided not to include a section on history, which makes sense here. Making decisions about what not to include in a research report is entirely the author’s prerogative. However, avoid making such decisions for reasons like it is too hard, you do not have enough time, the report is already too long, or other similar reasons when your audience needs and/or expects the information.

Discussion

As an example of the kinds of discussion material this type of report might contain, some information is also provided on the genesis of the QCPU. This kind of discussion would be useful for topics that deal with radically new technologies that vary significantly from traditional methods. Quantum computing is clearly such a topic.

Genesis of the QCPU

In 1998, George Yamaslute demonstrated that different RF signals cause predictably different spins for certain atoms. He also showed that the spin of these atoms can be altered by the application of different RF signals. These altered spin states then can be used symbolically to represent different values. The spin states of these atoms effectively store the values encoded by these RF signals. These values are then read by traditional NMR techniques, thereby creating a machine memory capability (Yamaslute 1998, pp. 200–210).

Besides memory capabilities, manipulation of the spin states of atoms can be used to perform various logic operations. Early experiments with molecules containing carbon and hydrogen atoms demonstrated such theoretical feasibility. The carbon and hydrogen atoms can be manipulated independently by varying RF signals applied at different resonant frequencies while these atoms are held in a fixed magnetic field. By making both atoms spin up, or spin down, or alternately spin up and spin down, the atoms can constitute a 2-bit truth table. Adding more types of atoms and using intermediate spin states substantially increase the range of logical operations. Such manipulation provides the quantum logic capabilities of the QCPU (Yamaslute 1998, p. 240).

Finally, the discussion should include a brief description of the QCPU device and its functional application because the chip is the primary focus of this report.

The QCPU

The Quantum Chips QCPU is proprietary technology used today only in highly classified government projects. Although unclassified information is limited, some scientists believe that the QCPU is now providing the computational power for the genetic manipulation of bacteria. The specific goal is to create virulent strains of Naomi-B bacteria that are capable of eating through the armored titanium and steel hulls of enemy submarines at depths exceeding 10,000 feet (Mierson 2003, p. 50).

The QCPU assembly consists of the quantum molecular matrix (QMM), the magnetic field coil (MFC), the nuclear magnetic resonance (NMR) sensor, and the RF assembly (RFA) (see [Figure 12.2](#)). The MFC provides the fixed magnetic field engulfing the QMM. The QMM provides atomic structures that have unique spin-up and spin-down characteristics. The RFA provides the RF energy needed to change the spin characteristics of each atom to reflect specific values (Bearkins 2000, p. 91). The energy is radiated by a phase directional array, but the exact design or method of control is proprietary. The NMR sensor is the means of sensing or reading these altered spin states.

Conclusion

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The conclusion section of a research report normally summarizes the report and might provide a recommendation. Any recommendation must be supported and justified by information in the discussion section. Given the nature of this sample report's theoretical discussion, a specific recommendation is not supported or justified, and one is not provided. In this example, the conclusion summarizes only the information provided on the QCPU. We also note that in our example, [Figure 12.2](#) (referenced in the discussion), was placed after the reference but in the conclusion due to paging constraints.

Summary

The Quantum Chips' Quantum Central Processing Unit (QCPU) has made quantum computing a reality. By controlling and reading the spin states of selected atoms and using applied RF and NMR techniques, the QCPU effectively uses quantum phenomena to store data and conduct logical operations on those data. Quantum computing, by exploiting the possibilities of data manipulation and storage at the atomic level, provides the power to accomplish parallel processing at far greater amplitudes than traditional approaches.

The actual design and implementation of the QCPU is not only proprietary, but also highly classified for national security purposes. Little information is available about the specifics of the QCPU's design implementation.

References

Finally, include a list of the references used in the report. Always list all references that were actually cited in the report, both *sources consulted and used* and *sources consulted and not used*. As a courtesy to your reader, list sources that you consulted but did not specifically use because these sources might have influenced your thinking or might provide additional information for further exploration of the topic. In this example, five sources were actually cited in the paper, and one source was consulted but not used. For a more detailed discussion of documentation, see [Chapter 16](#).

Note about Length

Research reports are usually comprehensive documents and, as such, you should prioritize breadth and depth. Make sure that you have provided all the information your audience needs; document length is secondary.

The examples provided in this chapter are **abbreviated** examples due to limitations of space in this textbook. Unless there are length limitations for your report, give your audience the full breadth and depth of information they need; err on the side of too much rather than not enough detail.

Appendices

By the way, although not provided here, this example could also include an appendix containing additional, useful information that is not necessary for understanding the report. For example, this material might include the manufacturer's press releases on the QCPU, more information on the use of NMR in materials research, or detailed discussions of atomic spin states.

Obtaining the Necessary Details

What is the secret to obtaining all the details that you need for a research report? Start looking! There are two main types of research: primary and secondary. Primary research gathers new information, usually research done by you (the researcher) and includes things like laboratory data collection, surveys, questionnaires, interviews, and focus groups. Secondary research

collects information that someone else has previously gathered, and which you (the researcher) summarize and synthesize. Secondary research includes things like corporate technical reports, trade journals, and library or online research.

The types of research you need will depend on your context, purpose, and audience. It might seem odd because it goes contrary to the numerals, but **secondary research** actually happens before **primary research**. You must first learn what others have researched and determined before you can continue with your own primary research. Before an immunologist can do primary research on a new type of vaccine, they must know what types of vaccines have been tried before, including those that work, those that do not work, and why there are differences. Before a new factory is designed and built, the process engineer must know what factory floor designs have been used, which layouts facilitate efficient flow, and which layouts hinder flow and cause inefficiencies. The same is true for research reports. If you anticipate needing primary research for your report, you must first seek already-published sources to learn what has been done and documented.

Secondary Research

Consequently, let us discuss secondary research first. A hypothetical secondary research situation would occur when your supervisor (perhaps a faculty member for whom you are doing undergraduate research) asks you to identify different ways to measure human response to stimulus in a classroom situation. They want this information because they know that in the next two months, they will be applying for a National Science Foundation (NSF) grant, and they want to be fully informed when the time comes. Let us say that your supervisor does not specify the format of the report they want, but they share an example report with you (perhaps on measurement techniques for surface finish of ductile iron castings) and ask you to follow the same formulation. (Of course, the content in the surface finish report would not be relevant to the current request, but when your audience provides preferences or requirements about format, it is best to use them.) In this case, you would collect as much information as you could find about electrodermal activity (EDA) and any other techniques you could identify used for measuring human responses.

The Skewed Results of Search Engines

While we cannot live without search engines, depending on the one we like to use, we must learn how to live with them. Some search engines use algorithms which return results based on personal information about us, including our physical location in the world, previous searches that we have done, our social media posts, and the type of device we are using (e.g., computer vs. cell phone). You have probably noticed the phenomena: you were looking online for bunny slippers to purchase, and all of a sudden, you see many ads for bunny slippers. While this might be helpful when it comes to online shopping, it also indicates that your online search results include a biased list in the first few pages of hits. For example, the biologist who is doing research on the impact of habitat on smallmouth bass fat will get a very different set of returns from the keyword search of “fat bass” than the sound engineer who does a search on the same two words. Consider this when you are conducting technical research; you can do a search on how to overcome this for your favorite search engine (inception!)—they are all different.

As part of your secondary research, you might obtain this previously documented information from online searches, journal articles, and/or books at the library. To use library databases most effectively, it is often most efficient to take advantage of the library’s resources, including workshops and librarian expertise. Librarians are specialists, and in many cases, they have studied extensively, which puts them in an excellent position to help you. For online search engines, use keywords for your searches. Start broadly and slowly narrow down the information that you would like; consider it a game of **Twenty Questions**, but instead of getting “yes” or “no” answers, you are getting potential source material. Do not assume that the first sources returned in your searches are the best ones; scroll through multiple pages of options, read, investigate their references, look for new (and more precise) search terms, and look for patterns of information.

Primary Research

For primary research, you will most likely need to develop your own measures and protocols for the type of procedure you plan, like surveys, questionnaires, interviews, focus groups, and lab work. Each of these research strategies is a complex project with multiple considerations,

including best practices and/or required techniques. We recommend starting with secondary research to learn about each strategy and consulting with your instructor and/or supervisor.

Once you are ready to conduct primary research, you will collect data. For example, if you were trying to establish whether student engagement could be measured using electrodermal activity (EDA), you could set up an experiment (after approval through an institutional research board) in which students would wear monitors that record their EDA responses to their class activities. Your data might show that students have high EDA when working with partners on in-class quizzes, and low EDA when the instructor is droning on about how to count the balance of payment in economics for a given nation by the apex financial institution (note to our friends in Economics—we are not saying this topic is boring; only that our fictitious faculty member is a little sleep-inducing when it comes to PowerPoint lectures!). If you collected data for different students in different situations and then statistically analyzed this data, it could inform your research. Depending on your audience, you might also need other research details to provide the desired recommendations. Perhaps your research group is partnering with industry on a National Science Foundation project; in this case, your audience would have extensive, specified requirements about what the report must include. This situation would be considered primary research.

Evaluating Sources

As you find and collect potential source material, it is important that you critically evaluate its quality. Your work could be used to make important, sometimes potentially life-altering, determinations by decision-makers, advisors, and implementors, so it is vital that the information you use be accurate.

A common source evaluation text developed by Sarah Blakeslee and colleagues at Meriam Library at California State University (Chico) that you might already be familiar with is the **CRAAP Test**.¹ The test asks you to evaluate every potential source on five criteria:

1. Currency
2. Relevance
3. Authority
4. Accuracy
5. Purpose

Each criterion has a list of questions to consider before using the potential source in your own work (see [Figure 12.3](#)).

Evaluating Information – Applying the CRAAP Test

Meriam Library California State University, Chico

When you search for information, you're going to find lots of it . . . but is it good information? You will have to determine that for yourself, and the **CRAAP Test** can help. The **CRAAP Test** is a list of questions to help you evaluate the information you find. Different criteria will be more or less important depending on your situation or need.

Key: ■ indicates criteria is for Web

Evaluation Criteria

Currency: *The timeliness of the information.*

- When was the information published or posted?
- Has the information been revised or updated?
- Does your topic require current information, or will older sources work as well?
- Are the links functional?

Relevance: *The importance of the information for your needs.*

- Does the information relate to your topic or answer your question?
- Who is the intended audience?
- Is the information at an appropriate level (i.e. not too elementary or advanced for your needs)?
- Have you looked at a variety of sources before determining this is one you will use?
- Would you be comfortable citing this source in your research paper?

Authority: *The source of the information.*

- Who is the author/publisher/source/sponsor?
- What are the author's credentials or organizational affiliations?
- Is the author qualified to write on the topic?
- Is there contact information, such as a publisher or email address?
- Does the URL reveal anything about the author or source?
examples: .com .edu .gov .org .net

Accuracy: *The reliability, truthfulness and correctness of the content.*

- Where does the information come from?
- Is the information supported by evidence?
- Has the information been reviewed or refereed?
- Can you verify any of the information in another source or from personal knowledge?
- Does the language or tone seem unbiased and free of emotion?
- Are there spelling, grammar or typographical errors?

Purpose: *The reason the information exists.*

- What is the purpose of the information? Is it to inform, teach, sell, entertain or persuade?
- Do the authors/sponsors make their intentions or purpose clear?
- Is the information fact, opinion or propaganda?
- Does the point of view appear objective and impartial?
- Are there political, ideological, cultural, religious, institutional or personal biases?

The CRAAP Test gives us questions to use for evaluating sources to ensure they are quality source material.

Sarah Blakeslee, Meriam Library, California State University, Chico

Incorporating Source Material

When you incorporate source material into your report, you must summarize or paraphrase the content of your sources and synthesize. A summary is basically a shortened or condensed version of a source, and it should provide a summary of the source's main idea and its content. While a summary can be as short as a few sentences or longer than a page, summaries incorporated into research reports tend to be on the short side (for more on how to write a summary, see [Chapter 14](#)). Paraphrases put only part of a source in your own words. Paraphrases allow you to focus on an idea expressed in the original text without including all of the other content. There are some guidelines to follow when you summarize and paraphrase:

- **Use your own wording.** Do not copy and paste wording from the original source into your summary. Read the source, close/hide the print text or browser, and type the idea in your own words.
- **Cite the source.** Just because you put the idea into your words does not mean the idea is yours; it still belongs to the original source. Always, **ALWAYS** cite ideas and information that you saw in your source material, and use the documentation style required or expected by your audience. (For more on documentation, see [Chapter 16](#).)
- **Do not misconstrue or take out of context.** When you summarize or paraphrase (or quote for that matter), never, **EVER** take the information out of context or imply that the summary or paraphrase means something different than what it means in the original source.

Summarizing and paraphrasing is not enough; you must also strategically put everything together. Writing a research report means combining the ideas across multiple sources into a single, cohesive document; that is, synthesizing sources. You are not writing a series or listing of separate ideas, but combining them together to create something new. Consider a tie-

dye pattern (see [Figure 12.4](#)). Each color remains distinct, like a source's information or idea, but colors are combined in unique ways, which are your report's main ideas expressed in paragraphs and sections. The source information and ideas remain the same, but how they are connected to each other and expressed to the audience is determined by you.

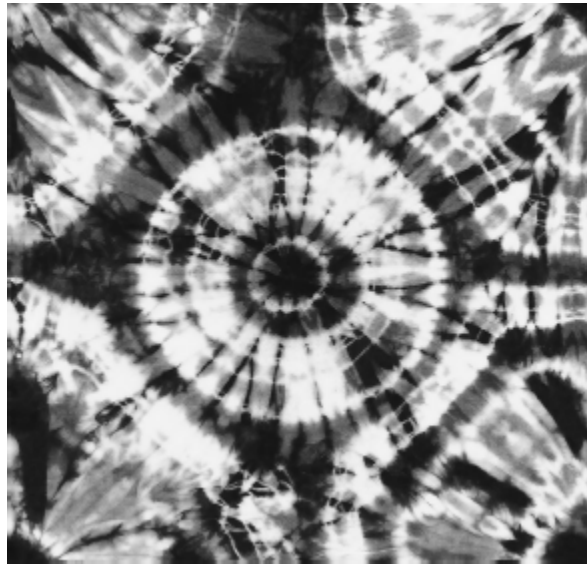


Figure 12.4

Source material are like the colors of a tie-dye pattern. They are separate but integrated, blended by the author to form a cohesive message.

Alena Tselesh/Shutterstock

Example 12.2 Historical Research Report

This second example, a short historical research report on one of the first production computers, is located at the opposite end of the research report spectrum from the state-of-the-art report. The historical report's focus is on where we have been, not where we are going. Unlike the first example, this example, which deals with the Ferranti Pegasus Computer, is totally factual. It also assumes the reader has a basic level of computer engineering literacy.

Research Report on the Ferranti Pegasus Computer

Introduction

Purpose

This research report provides a historical review of the Ferranti Pegasus Computer (Pegasus).

Background

Pegasus was one of the pioneering achievements of postwar computing technology. Developed by the British company Ferranti Ltd., the first Pegasus was placed in service in March 1956. (1) It was designed to deal with numeric calculations, since the use of a computer to process non-numeric information was not even considered at that time. In 1949, only four computers existed in the world: one in the United States at Eckert-Mauchley Computer Corporation, one in Australia at the Commonwealth Scientific and Industrial Research Organisation, and two in the United Kingdom at Cambridge and Manchester Universities. When research on the Pegasus project began in 1951, only about a dozen computers existed in the world. (2)

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Pegasus 1 was delivered in 1956, while a follow-on version, Pegasus 2, was delivered in 1959. Altogether, 40 Pegasus machines were sold between 1956 and 1962. (3) [Figure 12.5](#) shows the 25th Pegasus machine produced. It is now on display in the Science Museum of London. (3)



Figure 12.5

Last operational Pegasus on display in London at the Science Museum, November 2005.

Dittrich/Agencja Fotograficzna Caro/Alamy Stock Photo

Scope

This report focuses on the vintage technologies employed by Pegasus, as well as general operating profiles and technical specifications. It provides only a brief, general overview.

Discussion

Historical Development of Computers

At the end of World War II, several computing projects were undertaken. At first, electromechanical computation equipment was used, and it generally was programmed from punched cards or paper tape. The IBM Selective Sequence Electronic Calculator in 1948 used 23,000 electromechanical relays and 13,000 vacuum tubes. The U.S. Army's Electronic Numerical Integrator and Computer (ENIAC) in 1946 used 18,000 tubes and 1,500 relays, weighed over 60,000 lbs., and was a pioneer in computer hardware evolution. Additionally, the British code-breaking series of Colossus electronic machines in the 1940s provided experience with high-speed, logical, and conditional branching operations, as well as high-speed operations on parallel bit-streams. Together, the ENIAC and Colossus set the stage for stored-program computers like the Pegasus. (1:5-7)

Applications of the Pegasus Computer

Unlike the early massively complex systems such as ENIAC and Colossus, the Pegasus Computer was designed to be a reliable, user-friendly system for numerical computation that could store programs and be marketed to business and research establishments. With modular construction, faults could be quickly located and repaired. Pegasus was also designed to be produced in quantity and sell for approximately £45,000 (about \$80,000). (4)

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Usage

Many Pegasus Computers were used in materials science and engineering, mainly for stress calculations in aircraft design. In fact, after several planes crashed due to metal fatigue in the 1950s, twelve of the 40 machines were used exclusively for aircraft structural design. Other machines were used in the design of bridges and large buildings, guided missiles, project management for businesses, and wind tunnel experiments. (5)

Technical Specifications

Because the Pegasus Computer predated solid state devices, it incorporated a novel thermionic, valve-based design using modular circuit boards ([Figure 12.6](#)); in other words, it employed vacuum tubes as switches. As mentioned earlier, this modular design enabled easy maintenance on the computer, which was a key innovation. The ECC81 (a reliable, dual triode tube) provided much of the switching capability for this machine.

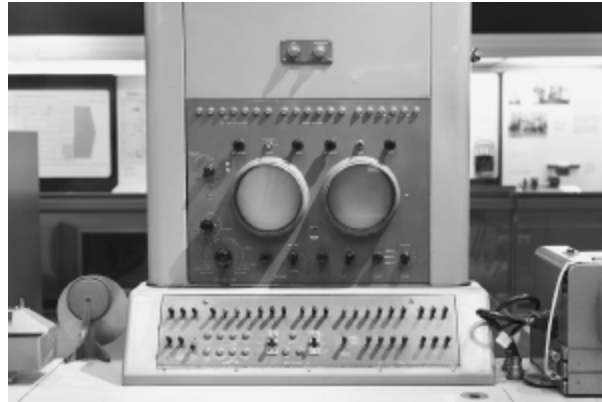


Figure 12.6

Modular circuit board design with three ECC81 tubes.

Alex Segre/Alamy Stock Photo

Random access memory (RAM) storage technology was a challenging task in the 1950s. While magnetic drums, punched cards, and paper tape provided nonvolatile mass storage, fast memory required another approach. In the Pegasus, RAM storage was accomplished with a delay line package. Each package contained a loop of coiled nickel wire approximately 20 inches long. A transducer at one end would send a pulse down the coiled wire, while a second transducer at the other end picked up the pulse after the propagation delay provided by the coiled wire. This particular device could store approximately 1,000 to 2,000 bits of information, allowing Pegasus to add two numbers in $1/3,000$ of a second and multiply two numbers in $1/500$ of a second. (1:5) A summary of major technical specifications is provided in [Table 12.2](#).

Table 12.2 Specifications.

Number of tubes (valves)	1,200
Clock speed	333 kHz
Word length	39 bits
Memory (drum storage)	200 kbits
Memory (RAM – delay line)	2 kbits
Input (punched paper tape)	200 cps
Output (punched paper tape)	33 cps
Output (teleprinter)	7 cps
Power consumption	15 kW

Source: "Technical Specifications," Display materials, Science Museum, London, November 15, 2005.

Conclusion

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Pegasus was a key element in the evolution of modern digital computers. Evolving out of the ENIAC and Colossus projects during and after World War II, Pegasus used a valve-based design in a modular system. It was designed to combine the computational capability of dealing with complex numeric calculations with a system that was affordable, reliable, and easily maintained. Pegasus was used primarily for business and engineering design applications.

Sources

1. Lavington, Simon, *The Pegasus Story: A History of a Vintage British Computer*. London: The Science Museum, 2000.
2. "Pegasus (computer)," www.wikipedia.org (Accessed August 24, 2006).
3. "Pegasus—a vintage British computer," <http://cswww.essex.ac.uk/staff/lavington/Pegasus/index.htm>.
4. "Pegasus Computer, 1959," Display materials, Science Museum, London, November 14, 2005.
5. "What was Pegasus used for?" Display materials, Science Museum, London, November 14, 2005.

Documenting Sources

Documentation gives formal credit to a person, organization, or publication for an idea or information that either is not original or is not common knowledge of the field. It represents an acknowledgment of your indebtedness to the source. All quotations, summaries, paraphrases,

concepts, ideas, etc. that you get from a source must be documented. Accurate and comprehensive documentation builds your credibility and ethical behavior as an author, as well as demonstrates your understanding of academic and legal requirements of source attribution.

Documentation is an exercise in precision—down to the spacing and placement of punctuation—and the format of the documentation can vary significantly. For example, you might have noticed that the two examples from this chapter used quite different documentation styles. Consider the differences for in-text references:

In-Text Reference for the QCPU Research Report

These values are then read by traditional NMR techniques, thereby creating a machine memory capability (Yamaslute 1998, pp. 200–210).

In-Text Reference for the Pegasus Research Report

Other machines were used in the design of bridges and large buildings, guided missiles, project management for businesses, and wind tunnel experiments. (5)

And notice the differences between the lists of references at the end of the reports:

Reference at the End of QCPU Research Report on QCPU

Ardvark, Wilma J.: “NMR and Beyond,” *Quantum Computer Quarterly* 9:6 (August 2005), pp. 460–495.

Reference at the End of Pegasus Research Report

1. Lavington, Simon, *The Pegasus Story: A History of a Vintage British Computer*. London: The Science Museum, 2000.

Documentation Styles

There are too many documentation styles to provide a comprehensive list, but here are a few common examples: Modern Language Association, Chicago Style, American Psychology Association (APA), Institute of Electrical and Electronics Engineers (IEEE), American Chemical Society (ACS), American Institute of Physics (AIP), Council of Science Editors (CSE), and more. The best approach is to use the style guide specified by your employer, teacher, or field. Conferences, journals, and government grant submissions will usually require a certain format and will indicate this in their author’s guide. If a style guide is not specified, you can use any consistent form of documentation (that is, stays consistent throughout your document)

that provides the necessary information, including the simplified method provided in [Chapter 16](#).

How your in-text and end-of-text references should look depends Page 176 on the documentation style that your audience is expecting or requiring. Deviating in any of the details from the required or expected documentation style can result in loss of credibility as an author and can, fairly or unfairly, lead the reader to mistrust the information in your report.

For a more detailed explanation of documentation, refer to [Chapter 16](#).

Visuals in Research Reports

When you are incorporating or designing visuals to include in a research report, be sure to include only those which are absolutely necessary. Do not include visuals that do not match the report in subject matter or terminology. Also, avoid visuals that include too much or too little complexity for the level of explanation.

- Visuals should relate specifically to the text discussion, and they should be referenced in the text before actually being used. The reader should never encounter a visual and wonder why it is there or what it is for. Always tell your reader specifically when to look at a visual, and try to do so *before* the point in the paper where the visual actually appears.
- Each visual should have an assigned sequence number and name, and if possible, run separate sets of sequence numbers for each type of visual. For example, figures should have their own set, as should equations and tables.
- Each visual must be included for a specific purpose. Do not add visuals to make the page look “pretty” or to “add visual interest.”
- Visuals should be static or dynamic depending on need. If, for example, you intend to show a change of movement, make sure the visual is dynamic.

For more information on incorporating visuals, refer to [Chapter 17](#).

Formatting a Research Report

The formatting of your research report should follow the requirements and expectations of your audience. Formatting guidelines are often part of documentation guidelines. For example, if your audience requests APA documentation style for your in-text and end-of-text references, then you should format your document according to APA document formatting guidelines. If your audience requests MLA documentation style for your in-text and end-of-text references, then you should format your document according to MLA document formatting guidelines. Formatting a research report is not the time to release your creativity or be lazy (“this will be good enough”). Imagine an instructor reading through 30+ research reports that are perfectly formatted in APA documentation style (see [Figure 12.7](#)). Things are going well, and the instructor has been very pleased with the content, organization, style, and expression of all reports so far. Then the instructor picks up this paper to evaluate (see [Figure 12.8](#)). What impression does the incorrect formatting give? What effect could that impression have?

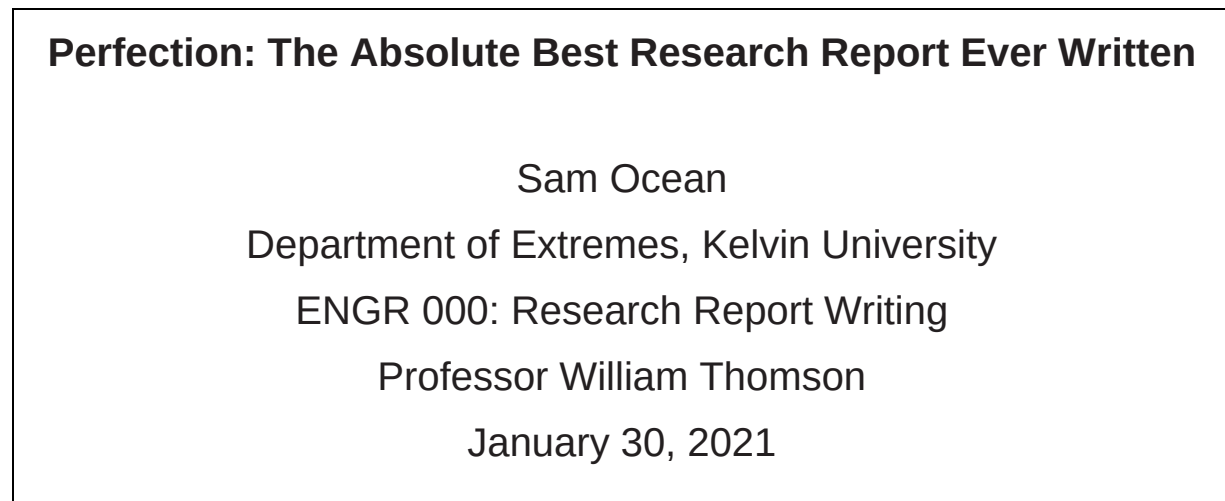


Figure 12.7

Correctly formatted title page.

Perfection: The Absolute Best Research Report Ever Written

By Sam Ocean

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Figure 12.8

Incorrectly formatted title page.

In the professional world, incorrectly formatted documents can cost more than just a bad impression. Many companies have very strict internal formatting standards and not following those standards can mean your work doesn't get read. Government agencies have extremely strict guidelines; for example, as of December 2020, the NSF guide for proposals is 185 pages long.² Formal research reports are significant documents and can require a cover page (contact information for author, list of all persons involved, abstract, dates, etc.), a full table of contents, sections, titles, and referencing. Submissions that do not precisely follow the guidelines are rejected immediately.

Most internal reports are informal and do not require the additional items that a formal report does; however, there can be instances where an internal document should be formatted like a formal report. Always ask about the submission expectations so you know how to format your report.

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Transmitting a Research Report

When submitting a research report, especially to a recipient external to your organization, you should include a **transmittal document**. This document can be a **transmittal letter** or, more commonly, a **transmittal email**. Transmittal documents (also called **transmittals**) ensure that the research report gets to the right place and is considered in the proper context. See [Chapter 19](#) for a more detailed discussion of transmittals.

Remember, however, that transmittals are not part of the report, not even if you are sending a formal report with a cover page. Transmittal letters are attached to, or enclosed with, the report. In the case of a transmittal email, the text of the transmittal is in the email message, while the report (including title/cover page, appendices, and attachments) is attached to the email, and the subject line identifies the attached document.

You should use transmittals and formal formatting when sending reports to recipients external to your organization. When reports are internal documents, they are often transmitted without a letter and often without a title page. Instead, these reports are frequently sent as an attachment to an email message. Ensure that any documents shared electronically are in formats that your recipient can use (e.g., docx, PDF); you will have fewer issues with software incompatibility if you always save your documents as a PDF and attach them rather than trying to share a link from a cloud storage service or sending documents created in less commonly available software.

Here is a general format for a transmittal email (see [Example 12.3](#)):

Example 12.3 Transmittal Email

From: Your email (the report's author)
To: Email address of your audience Date: Month ##, ####
Cc: Email addresses of those who need to know you have sent it
or who need to see but not respond
Subject: Identification of Topic (ex: Name of Report)

Greeting and/or audience's name:

Attached is *[identify what is attached]*. *[Briefly explain who requested the report and why.]*

[Offer to answer questions and provide contact information.]

[Closing],

[Your name]

[Your signature information, e.g., title, organization, phone number, etc.]

Conclusion

In conclusion, research reports describe the discovery, analysis, and documentation of knowledge obtained through some type of investigation; they are focused, objective inquiries. In technical writing, research reports are specifically geared to the purpose at hand, the readers who will use them, and any limitations that have been placed on the scope of the project. Consequently, the content of a research report requires comprehensive knowledge of audience, purpose, and context. Research reports require data, which can be collected using secondary and/or primary research, as long as the research strategy will provide the data needed for the report's audience, purpose, and context. All source material should be critically evaluated before use and meticulously cited and documented, regardless of if it is used in the final report. As always, the inclusion of visuals should be purposeful and strategic, and the format of your research report should adhere to audience expectations or established guidelines. Finally, whether the report is formal or informal, careful attention should be paid to how it is transmitted to the audience.

Research Report Checklist

- ☐ Do I understand who will be the audience for this report?
- ☐ Have I clearly stated the purpose of this report?
- ☐ Have I introduced the topic with a brief overview of the problem or background?

- ☐ Have I discussed how I limited the report and my rationale for doing so?
- ☐ Have I provided adequate background for my reader to understand the report?
- ☐ Have I provided substantive, well-documented information in the report?
- ☐ Have I included necessary visuals and data?
- ☐ Have I summarized my research in the conclusion?
- ☐ Have I made a recommendation; and if so, is it supported by the discussion?
- ☐ Have I cited sources where necessary in the text?
- ☐ Have I listed the sources I cited?
- ☐ Have I included in an appendix any relevant material not necessary for understanding the paper?
- ☐ Have I provided only visuals that relate to my text?
- ☐ Have I referred to the visuals in my text before they are used by my audience?
- ☐ Have I given each visual an assigned sequence number and name?
- ☐ Have I run separate sets of sequence numbers for each type of visual?
- ☐ Does each visual have a specific purpose?
- ☐ Have I formatted my research report in the exact requested format?

Exercise

Consider the following excerpt from the “Theoretical Considerations Section” for a research report entitled “Magnetic Resonance Imaging.” Assume the report is being considered for use by an introductory science class at the undergraduate level.

- Is the report written at an appropriate level? If not, why not? And what would you change?
- Is the audience level consistent throughout this discussion? If not, where does it vary?
- Is the visual relevant and useful? If not, what kind of visual(s) would you include?

Magnetic Resonance Imaging (MRI) Theoretical Considerations

Magnetic resonance imaging (MRI) technology generates tomographic “slices” of the human body using a magnetic field and radio frequency (RF) signals. The part of the body to be imaged is positioned in a strong, uniform magnetic field. The nuclear magnetic moments (spin states) of water protons in this magnetic field are forced into one of two possible orientations, **spin up** or **spin down**. At this point, the anatomy to be imaged is illuminated by pulsed, polarized radio-frequency (RF) energy at the resonant frequency of these water protons. As the protons absorb these pulses of energy, their spin states are altered; however, between pulses, the spin states snap back into alignment with the magnetic field. These transitions create a second RF signal that is unique to the resonant frequency of the anatomical material being imaged, and these signals are detected by a receiving coil.

The tomographic slices are created by applying magnetic field gradients in such a way as to create a resonant condition only for those protons located in a predetermined slice. Two-dimensional Fourier

Transforms are then used to extract positional information. The resulting data matrix contains each NMR signal from a single volume element (voxel) within a tomographic slice. This matrix is then converted into a two-dimensional display (see [Figure 12.9](#)) that can be read by a radiologist.

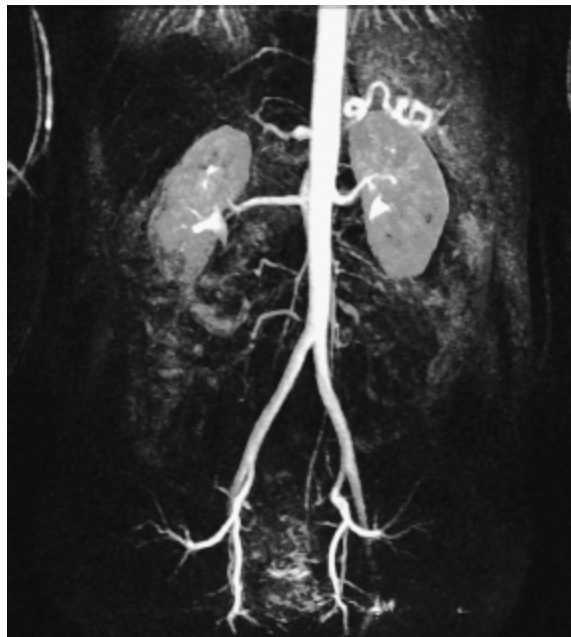


Figure 12.9

MRI arteriogram (MRA) of kidneys with contrast. Note the suspected occlusion in the left renal artery.

Miriam Maslo/SPL/Getty Images

Notes

1. “Evaluating Information—Applying the CRAAP Test,” Meriam Library California State University, Chico, <https://library.csuchico.edu/sites/default/files/craap-test.pdf>, Accessed December 31, 2020.
2. “Proposal & Award Policies & Procedures Guide,” NSF, June 1, 2020, https://www.nsf.gov/pubs/policydocs/pappg20_1/index.jsp, Accessed December 22, 2020.

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CHAPTER THIRTEEN

A3 Reports



Companies from around the world have benchmarked Toyota as an example of how best to run a business. Toyota's history is long and remarkable, with their global success increasing dramatically over the decades, particularly since the 1950s. Their business philosophy, called **The Toyota Way**, and their approach to manufacturing, called **The Toyota Production System**, have both been renamed "Lean" around the world. While many companies profess to having a Lean philosophy or business, including using the tools which Toyota has successfully fine-tuned and consistently implemented, they often fall short of implementing the same culture that Toyota fosters.¹ The A3 report was popularized by Toyota as a tool and as part of their philosophy. It is fundamental to their visual management. If you work for a company which believes it is Lean, they might very well ask you to create A3 reports as a primary method of both problem solving and efficient communication.

The essence of an A3 is this: tell the entire story of a problem on a single A3 size sheet of paper. For those not familiar with standard European paper sizing, it is approximately 11×17 or roughly the same size as two sheets of printer paper laying side by side. There are many critical Toyota Way concepts built into the A3 process, including **nemawashi** (consensus building), C. Edwards Deming's **Plan-Do-Check-Act** (PDCA) systematic problem-solving process,¹ and **genchi genbutsu** (go and see for yourself). A3 reports, when built and used appropriately, increase the efficiency of effective problem solving in an organization. This chapter cannot do justice to the Toyota history or organizational culture that is required for the successful use of A3 reports, but it can show you the general content equation and layout.

What Is an A3

Big things come in small packages. When you think of high impact stemming from a small and unassuming source, you might think of Yoda, Napoleon, lemons, Tinkerbell, or Tyrion Lannister. An A3 report is just such a communication device. In the dog world, it is the Pembroke Welsh Corgi, or Corgi for short (and shorter!). A Corgi is a visual dog both in looks and primary functionality (see [Figure 13.1](#)). Their majestic “floof” on their broad chest, bright eyes, and dog smile convey a sense of “large and in charge,” even though they are one of the shortest breeds in the dog world. When humans see a Corgi, they immediately respond positively (“Ohhhhh, a Corgiiii!”). For Corgis, short is their advantage. As cattle herders, they are able to herd beasts literally more than a hundred times their size without getting kicked *because* of their short stature. They are fast, nimble, powerful, and visual full-size dogs in small and unassuming packages.

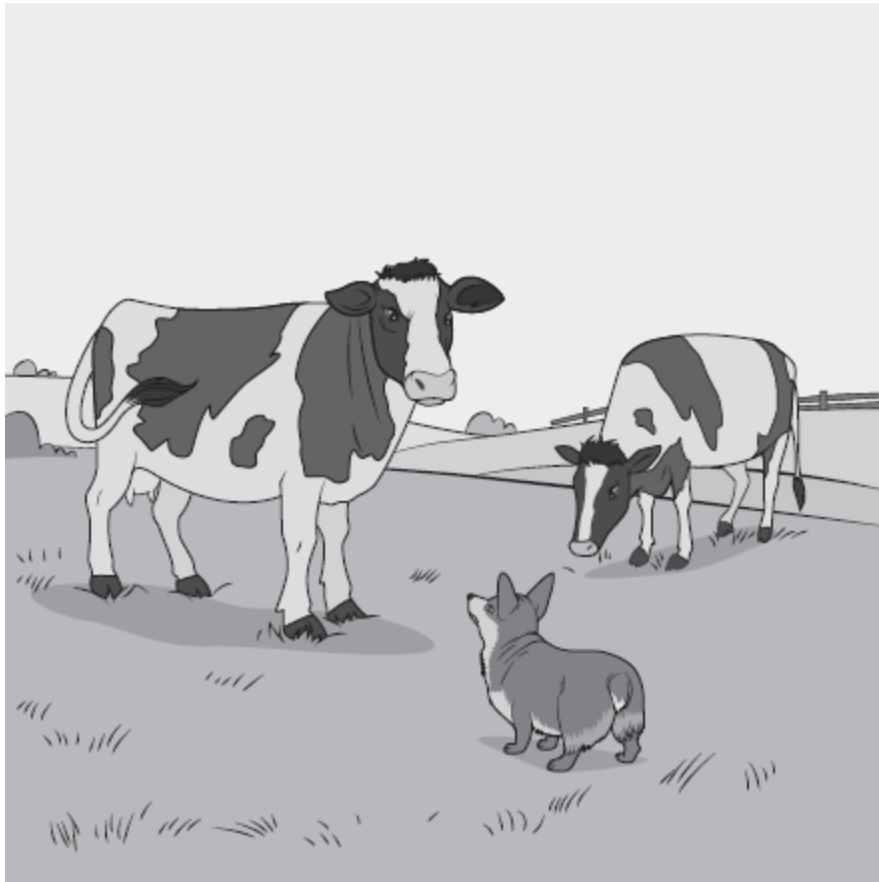


Figure 13.1

Like a Corgi, an A3 is able to pack a lot into a small package.

In the business world, scientists and engineers discover information and solve process problems. Those problems range from testing hypotheses about vaccine efficacy to increasing the efficiency of airplane boarding processes to ensuring that a manufacturing line is safe for workers. Any problem that must be solved can be managed and communicated through an A3. Teams of people work together to clearly and concisely pare down the information needed to convey the situation, including current and desired future state, what must be done to accomplish the desired change, and who is responsible for what by when. The document often includes graphical representation of relevant data, helping the team and everyone else concerned to quickly understand the situation. The A3 is both a **means** and an **end**. By working through the process of what to include on the A3, a team must build consensus. By the time the A3 is completed, everyone involved should be onboard with its content; consequently, approving the plan to move forward takes very little conversation or convincing. There is no better example of “a picture is worth a thousand words” than an A3 report.

Outline of an A3 with Variations

There are many different templates for the content that should be included on an A3. Many companies have their own template. You can also find them for free online.^{[2,3,4](#)} There is no “right” way to construct an A3, but thorough templates will include versions of the following:

- Problem or need identified
- Current state description
- Desired future state description (also known as a proposal)
- Root cause analysis
- Alternative countermeasures with methodical decision for choosing
- Assignments to individuals

- Cost summary
- Timeline, including controls

One **possible** way of providing content on an A3 (remember, Page 183 there is not a “right” way) is based on Deming’s PDCA cycle (see [Figure 13.2](#)):

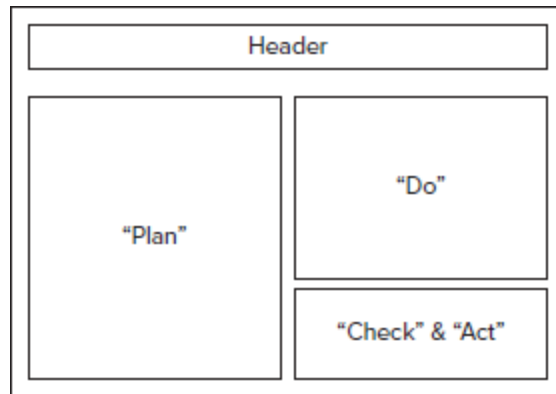


Figure 13.2

Example layout of an A3 document.

- Header (theme/topic, process, company, date, and your team members’ names)
- Left side—“Plan”
 - Current state
 - What you are trying to improve
 - The gap between the current and your target conditions
 - Background and supporting data
 - Root cause analysis
- Right side—
 - “Do”—Actions that need to be taken
 - What?
 - Who will do them?
 - By when?
 - “Check” and “Act” (bottom right)

- What will be measured?
- Follow-up
 - What?
 - By whom?
 - When?

It is important to remember that the A3 serves **two** purposes:

1. As a way of helping a team agrees on the critical aspects of the problem (and solution development) at hand (the **means**).
2. As a way of efficiently and effectively communicating the plan to others (the **end**).

The document must be thorough, yet concise. All audience types who will see it should be able to understand the story and support its logical progression. The context will be different for every project, so be sure to consider it.

Example

In the following hypothetical example in [Figure 13.3](#), a problem was identified. It seems that instructors are sometimes forgetful (shocking!); moving from one classroom to another causes the involuntary separation of mouse from laptop, causing instructors and teaching assistants to make subsequent unwanted laps around campus trying to retrieve said hardware. The following A3 uses the general format previously described (see [Figure 13.2](#)). The header information includes a descriptive title, the date of completion, and the team/team members doing the problem solving. The left side of the document succinctly describes the situation. Supporting data are visualized (see [Chapter 17](#)). A root cause analysis is shown using an **Ishikawa** (or “fishbone”) diagram. Note that there are many other effective ways of doing root cause analysis, including Five Whys, Pareto charts, Failure Mode Effects Analysis (FMEA), etc.—for more on root cause analysis tools, there are many good sources like the American Society for Quality.⁵ In this section of the A3, the problem is clearly defined with the

root cause identified. It includes the impetus (why do we care?), the potential impact (e.g., is it a quality, safety, customer, or cost issue?), and relevant visualization of supporting data.

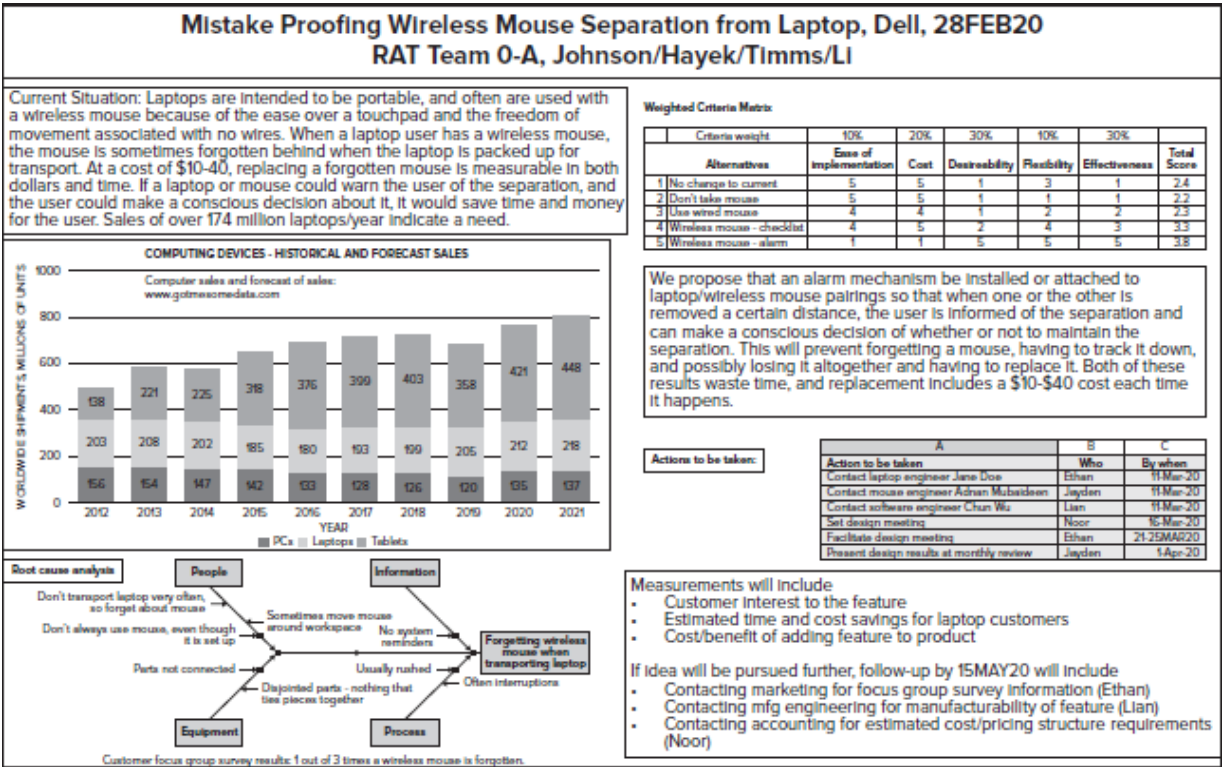


Figure 13.3
Example of an A3 report.

In the “Do” section of the document, the developed alternatives are described. While this can be done in text or drawings, in the case of this example, it is accomplished by including a weighted criteria matrix (also addressed by ASQ,2 but there are extensive online resources for decision analysis tools) which lists the alternatives. Managers want to know that multiple options were considered and want objective assessment for choosing the “best” alternative of those identified. In other words, as this example shows, there should be an explanation of “we will do this and why.”

After everyone agrees on the solution, it must be implemented. Page 185
This often requires tasks that must be completed, and to ensure accountability, it is necessary to assign them to specific people with due

dates. We can check our progress against this plan and correct if necessary. We must be clear about how improvement will be measured, keeping in mind that measurements drive behavior. After team members are all in agreement about measurements, assignments, and due dates, then stretch goals or activities can be included.

We repeat, there is no “correct” way to write an A3. However, if your company or manager has a preferred formulation, you should use it. If they do not have a preference, you have a lot of flexibility. Following are two examples of actual A3 reports from industry: the first is from the health care industry (see [Figure 13.4](#)) and the second is from manufacturing (see [Figure 13.5](#)). A3 documents can be used in any industry or organization by any employees who do problem solving.

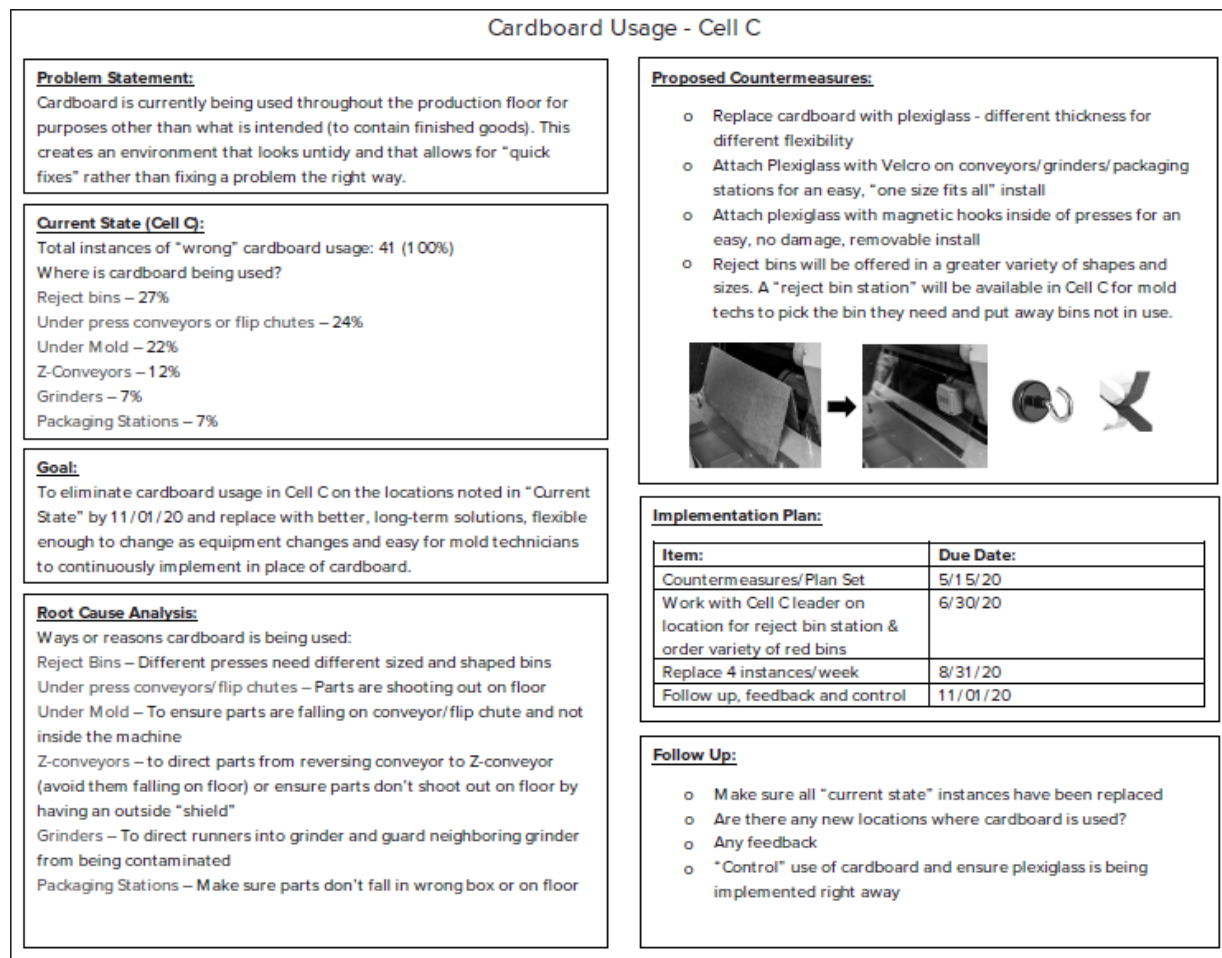


Figure 13.5

Example of an A3 report from manufacturing.

Disclaimer

Using A3 problem solving and documents does **not** make a company "Lean." While it does contribute to effective and efficient problem solving if used correctly, an A3 is simply one of many tools that is most beneficial when part of a Lean culture. Company culture is an entirely different animal . . . but one worth understanding.

Problems to Avoid When Creating an A3

- Dysfunctional teams or teams who have not yet reached consensus being forced to accept an A3 as complete when there is no agreement
- Missing critical pieces of the story
- No assignments, due dates, or required follow-up
- No visuals
- Getting hung up on the actual size of paper or medium of communication
- Spending too much time making things perfect when perfection isn't necessary
- Not spending enough time understanding the problem and identifying root cause

Summary

An A3 report can be very useful for process problem solving. There is no standard format, but the general rule of thumb is to include all information necessary to tell the full story in a single A3 size of paper. Using an A3 does not make a company Lean, but Lean companies often use A3s. The benefit of the A3 report format as a **tool** is that it is quick and easy for all levels of an organization to read and understand:

- It is concise—a single sheet of paper.
- It is visual—a picture is worth a thousand words.
- It tells the whole story, from problem to action plan.

The benefit of the A3 as part of a communication and business philosophy is that it creates a culture of teamwork:

- Teams develop the problem and plan together.
- Teams concur as they move through the problem-solving approach.
- Teams communicate proactively in developing the A3, making implementation more efficient.

In other words, big results come in a small package, much like the Pembroke Welsh Corgi!

A3 Checklist

- ☐ Who is the audience and how will they use the A3?
- ☐ What is the situational context of the problem and use of the tool?
- ☐ Have we identified the problem or need including the symptoms that are cause for concern (the impetus)?
- ☐ Have we identified the root cause using a methodical, standard tool like Five Whys, Ishikawa, Pareto, FMEA, etc.?
- ☐ Have we described the current state?
- ☐ Have we included visuals or data to support the current state?
- ☐ Have we described the future state we desire?
- ☐ Have we considered and listed multiple alternative countermeasures?
- ☐ Have we explained/shown a methodical decision for choosing a best solution?
- ☐ Have we made assignments to individuals with due dates?
- ☐ Have we included a cost summary or impact statement?
- ☐ Have we included a timeline for completion, including controls?
- ☐ Does everything fit on one A3 size page or approximately the same size?

Exercise

You are a manager for a dog treat factory. You have a quality engineer, a process engineer, a chemist, and a marketing person who put together the following A3 document and have now approached you for final approval of their plan (see [Figure 13.6](#)). They have consulted with several focus group members (Roofus, Spike, Marley, Toto, Lassie, and Fluffy) and are feeling very confident about their problem-solving efforts.

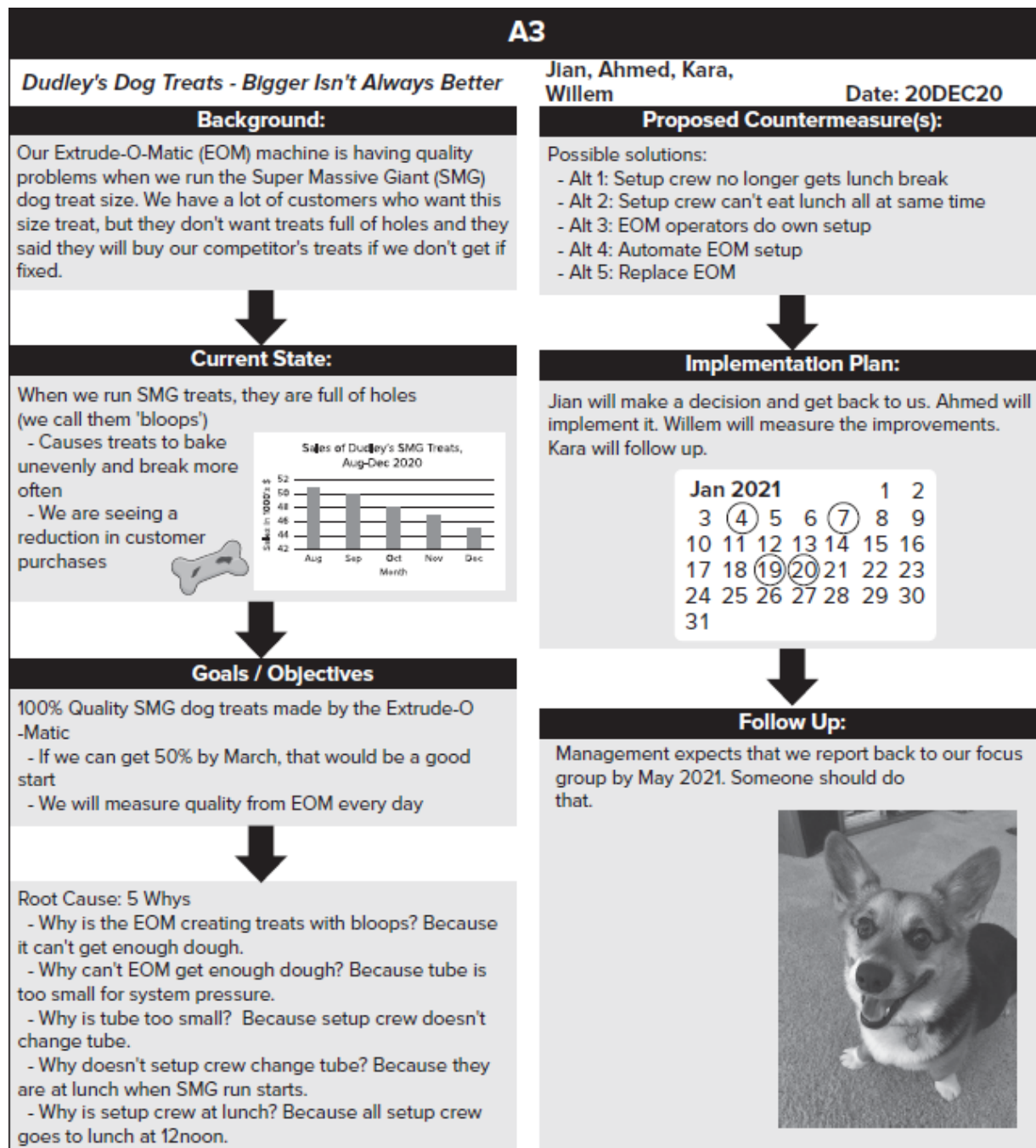


Figure 13.6

A3 document for analysis.

- What does this document do well, if anything? Review it in Page 189 terms of the A3 checklist.
- Specifically, what parts, if any, give you a “warm, fuzzy feeling” that this project is going well?

- Specifically, what does this document do poorly, if anything? In other words, what parts of this report give you a “cold, hollow feeling” that something about this project is very wrong?
- After reading the report, what concerns would you have, if any, regarding the problem-solving plan in this A3?
- Who is the expected audience for this communication?
- What is the purpose of this A3 document?

Notes

1. For more on the history, read *The Toyota Way* by Jeffrey Liker. *The Toyota Way*. New York: McGraw Hill Education, 2020.
2. “What is an A3?” *Go Lean Six Sigma*. Accessed December 4, 2020.
<https://goleansixsigma.com/a3/>.
3. “A3.” *UNC Institute for Healthcare Quality Improvement*. Accessed December 4, 2020.
<https://www.med.unc.edu/ihqi/resources/a3/>.
4. “Toyota A3 Template Download.” *Shmula.com*. Accessed December 4, 2020.
<https://www.shmula.com/toyota-a3-template-download/>.
5. American Society for Quality, <https://asq.org/>. Accessed December 3, 2020.

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CHAPTER FOURTEEN

Abstracts and Summaries



When you have finished writing your technical paper or report, take a deep breath. While you have completed the project, you might still not be finished writing. Often your instructor, advisor, supervisor, manager, potential grantor, or client will also require that you add a special summary of your report, called either an **abstract** or a **summary**. Abstracts and summaries both summarize information in your document, but not in the same way. Abstracts are more commonly used in academic or professional society publications. They are usually shorter than summaries and generally come in two forms: **descriptive** and **informative**. There are mainly two kinds of summaries used in technical writing: **academic** and **executive**. Academic summaries are used to concisely describe an academic text, and as their name implies, used often in academia. Executive summaries are typically used in business settings.

The first rule with all technical documents is to follow the requirements given to you. Every organization decides what they want, and successful interactions with organizations occur when people meet the organization's needs. We provide the following generally accepted structures of abstracts and summaries in this chapter, which you can use as the default for when an organization does **not** provide specific detailed requests. If, for example, you are applying for a grant and they specify a format, which they happen to call an abstract but which does not seem to match our explanation of an abstract, just provide what they request.

As we discussed in the Preface of this book, there are three kinds of audience: decision-makers, advisors, and implementors. When you are assessing audience for an abstract or summary, you are typically writing for decision-makers and/or advisors. The author of an abstract or summary is

often an implementor. This is why it is so important to understand audience for each genre **and** the sections of each genre. In the case of a research report for Dudley's Dog Treats, the audience for the Executive Summary is the management team, while audience members for the methods section include the engineers who will revise the product formula, the production process, or the safety protocols. These two sections require different approaches to audience to be successful.

Abstracts and executive summaries are the smallest, stand-alone genres of the technical writing world. They must pack a full punch, though, and writing less requires more time to do it well. In the dog world, we can think of them as Chihuahuas. At a whopping six pounds, the Chihuahua is the proverbial "pocket dog." You see them in the purses of Hollywood stars and in backpacks at the grocery store. As a full-fledged member of the dog family, they still get the job done with their charm and "big-dog" attitude. They are a complete dog, despite their small size, though they do require some special care . . . because they are small, they need extra attention when it comes to being their best. It is critical to pay extra attention to abstracts and summaries to ensure that they convey the full story concisely, often in a single paragraph (abstracts) or a single page (executive summaries). Do not discount the power of these genres! Just because they are small, they are still exceptionally important and deserve the time needed to get them right (see [Figure 14.1](#)).

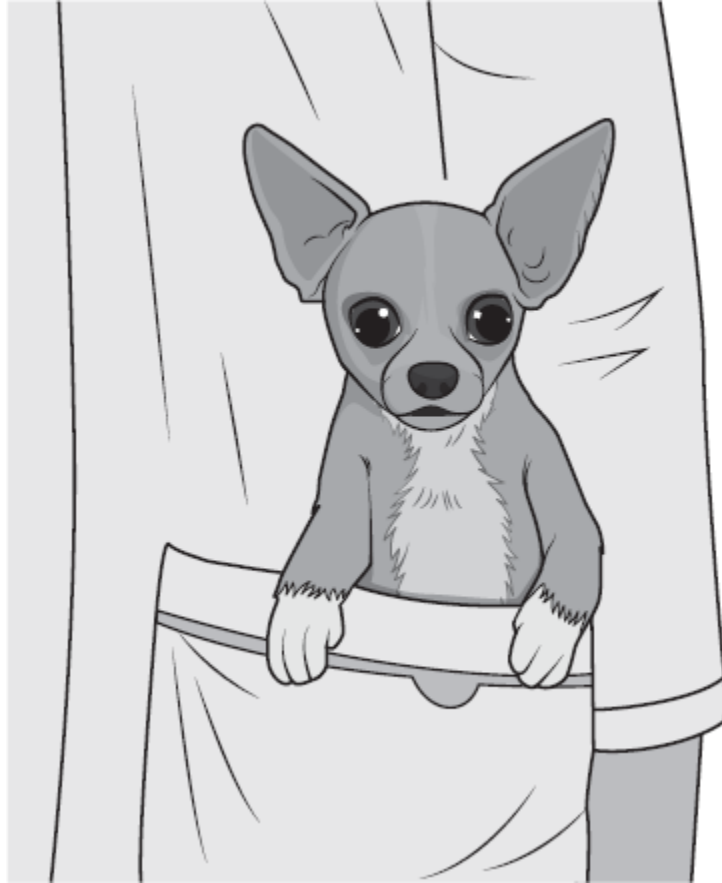


Figure 14.1

While abstracts and summaries are short, when they are done well, they are extremely effective. Similarly, while Chihuahuas are very small, they are extremely effective at their jobs.

Abstracts

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When writing an abstract, you must answer the ever-present questions about audience, purpose, and context. The audience of academic and professional society papers is, not surprisingly, often people from academia and professional societies. The purpose of an abstract is to assist these audience members in their decision of whether to read your entire paper. They must quickly assess if your paper is relevant to them and if it will help them with the work they are doing. Searching abstracts is a much more efficient way to decide this than searching titles (not enough information to make a decision) or full papers (too broad and inefficient to narrow the scope of the search). For academic authors, then, the abstract is, in some ways, a sales

pitch. This is because in academia, you want others to value your work, and citations of your work are a measure of your professional contributions to society. In other words, the only way others will use and cite your work is if they read and use it. So how do others know if they want to read what you have written? They determine this based on your abstract.

You should make conscious decisions about your word choices in an abstract. Every word should add value. In some cases, potential readers will do a search on the entire abstract; in others, they will only search using keywords. If their keywords do not match the keywords you have identified, they might not find your paper. Abstracts also play an important role earlier in the academic process when your paper is being evaluated for publication. Editors, track chairs, and grant directors often form opinions quickly based on your abstract. According to Dr. Sarah Ryan, Editor-in-Chief of *The Engineering Economist*, editors are also more frequently being assisted by electronic algorithms, which automatically assess submitted abstracts and keywords. The algorithms suggest possible reviewers for your work.^{1,2}

Decision-Maker Roles

Editor: The person in charge of making decisions about what articles or papers will be included in media. They often assign submissions to area editors, and area editors find reviewers.

Track Chair: The person in charge of a category or subset of papers for a conference. They often assign reviewers for papers and have final say over acceptance/declination.

Grant Director: While this isn't necessarily the term used by all organizations which award grants (for example, at NSF this role is called Program Director), the general idea is that this person assigns submissions to a review panel and has authority to approve grant recommendations.

Recognizing that every abstract type and its expected components will vary from organization to organization, there are still some general guidelines and tips. While abstracts must be concise, meaning that they omit many details, abstracts should include a sentence about each major section of the paper: the problem and why it matters, your methods, your

results, and the impact of your work. Limitations should also be clearly addressed.

Abstracts should **not** be longer than the allowed word count limit, should **not** include citations of other works, and should **not** include definitions. They should **not** consist of sentences that are copied verbatim from the paper; the abstract should be a stand-alone document. Abstracts should **not** include information that does not appear in the full text, either; an abstract can only summarize what is presented in the paper.[3.4.5](#)

What Are Descriptive Abstracts?

Descriptive abstracts (also called **limited abstracts**) summarize the structure of a report, but not its substance. In other words, descriptive abstracts basically present the table of contents in paragraph form. They refer to the title and the author and may briefly sketch out the purpose, problem, and scope of the document. They also describe the major topics covered by the report. A typical descriptive abstract contains approximately 50 words.

Writing Descriptive Abstracts

Writing descriptive abstracts is simple because you do not have to get into the substance of the report. Here is a descriptive abstract for the sample research report on Quantum Chips Corporation's Quantum Central Processing Unit (QCPU) in [Chapter 12](#):

Descriptive Abstract

Quantum Chips Corporation's Quantum Central Processing Unit (QCPU)

This research report provides a state-of-the-art investigation and theoretical review of Quantum Chips' QCPU device. The report begins with a description of its purpose, problem, and scope. Next it provides theoretical background on the device's operation. The main discussion deals with the genesis of the QCPU and provides nonproprietary information on the main parts and basic operation of the device.

Notice how this abstract provides no substantive details about the report except for general topic areas. For example, you know from the abstract that the report discusses the purpose, problem, and scope—but you do not know what the purpose, problem, and scope are. You also know that the report addresses the theoretical background regarding the chip's operation, but the abstract gives no information on this theory. In some descriptive

abstracts, a brief statement providing additional details about the purpose and scope may be included.

What Are Informative Abstracts?

Informative abstracts (also called **complete abstracts**) actually summarize the substance of your report, not just the structure. They provide a condensed discussion of the important points. In other words, informative abstracts tell the reader not only the major topics of the report, but also, in a nutshell, what you said about those topics. Informative abstracts may be designed as stand-alone documents.

If you intend for the abstract to be part of your report, you need not include the title or author of the report in the abstract. However, if you intend for the abstract to replace the report for some readers, like in published conference proceedings, you should include the title and author, if appropriate, at the beginning of the abstract. In either case, informative abstracts normally contain 100 to 200 words.

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Writing Informative Abstracts

Writing informative abstracts is a little more involved than producing descriptive abstracts because you need to summarize the substance of the report. That means you need to understand the substance of the report. Never try to abstract a report that you do not fully understand. Also, decide early how you are going to organize the abstract. Normally, informative abstracts are developed around the main topics of the report. For each of these topics, choose what information to include. Here is an informative abstract of [Chapter 12](#)'s report on Quantum Chips' QCPU.

Informative Abstract

Quantum Chips' Quantum Central Processing Unit (QCPU)

This research report provides a state-of-the-art investigation and theoretical review of Quantum Chips' Quantum Central Processing Unit. Quantum Chips Corporation has transcended traditional CPU designs by using quantum nuclear spin states of atoms to store and manipulate large amounts of binary data in parallel. CPU speeds equal to or better than 500 gigahertz have been realized with this chip. The device has been developed using quantum computing theory to exploit subatomic phenomena of common elements to perform extremely complex computational tasks, resulting in massively parallel processing.

The QCPU uses nuclear magnetic resonance (NMR) techniques to read specifically induced spin states in carbon, hydrogen, and other atoms. Data are stored and manipulated by using radio-frequency (RF) energy to alter the spin states of these atoms while they are trapped in a fixed magnetic field. Different spin states have distinct energy signatures for specific

atoms at varying magnetic field magnitudes. These differences are read by the NMR sensor, thereby providing an accessible memory for the data stored as spin states. Besides this memory function, manipulation of the spin states of atoms also can be used to perform various logic operations.

Building on previous quantum CPU successes with carbon and hydrogen atoms, the QCPU uses proprietary technology and is built around the following: a Quantum Molecular Matrix, which provides the atomic material; an NMR sensor, which reads the spin states of the atoms; an RF assembly, which provides phased array RF illumination of individual atoms; and a magnetic field coil, which establishes the required fixed magnetic field.

Notice the substantive details included in the informative abstract. Also, notice how some materials, such as those related to George Yamaslute and the genesis of the device, have been omitted from the abstract. The fact that Yamaslute demonstrated the feasibility of quantum computing in 1998 was not deemed important enough to be included in the abstract's summary of the report's key points. Writing these kinds of abstracts often requires some tough decisions about what stays and what goes.

Summaries

Once again, we start with who is the audience, what is the purpose, and what is the situational context? The audience for a summary depends on the purpose and context, but basically it is a shortened or condensed version of a text. If you are a student in a class or a professional in a technical role trying to understand the essence of a text, knowing you will reference that information later, you might choose to write an academic summary. Likewise, if you are student writing an annotated bibliography or paper that needs a summary of a source, you will likely write an academic summary. If you are a student in a class or a professional in a technical role summarizing a report for an instructor or manager, however, you will want to use an executive summary.

What Are Academic Summaries?

The audience for an academic summary includes those seeking the summary of a text, but not a stand-alone document with an introduction and conclusion. An academic summary can be as short as a few sentences or a couple of pages long, depending on the audience, the purpose, and the length of the text you are summarizing. Audiences needing academic

summaries are often people in academia (student, instructor, or advisor), but can also be people in industry. The purpose of an academic summary is to provide a summary of a source's main idea and supporting content. These summaries are often completed to assist with academic exercises such as reviewing for an exam, writing an essay, or compiling an annotated bibliography. They are also used to prepare for a discussion or to prepare a supervisor about a topic or issue. In such cases, you might write a report that includes multiple academic summaries from a variety of sources across multiple perspectives. An academic summary typically includes more information than an abstract, but most definitely less than the entire original text. Its structure is usually similar to that of the original text, but is written in the summarizing author's own words.

Academic summaries should **not** include interpretations of the Page 195 original text being summarized. They should also **not** include phrasing copied from the original text. Academic summaries should **not** include discussion or conclusions from the original text.

Writing Academic Summaries

As an example of an academic summary, we summarize the chapter you are reading now—and, in the process, provide you with a ready-made set of review notes on abstracts and summaries. What is the strategy for reducing all this material into a few paragraphs?

- Start with a statement of the source's main idea—its argument, its thesis—and include the author and title
- Provide a brief overview of the content of the source, including the sections of the source
- Identify the support used by the source for making its points
- Cover all the key concepts or ideas (note that “key” does not mean “all”)
- Organize ideas in the summary in the same order they were in the original, but avoid a play-by-play (first, second, next, then, etc.)
- Put the entire summary in your own words

Obviously, not everything can be included. To determine which ideas are “key,” a good idea is to go back to the purpose of the academic summary. In

this case, we must analyze [Chapter 14](#) material in terms of how much it contributes to achieving the goal of instructing science and engineering students and professionals on how to write and edit abstracts and summaries. Using this focus, we developed the following academic summary by synthesizing the chapter.

Example of an Academic Summary

In [Chapter 14](#) “Abstracts and Summaries” from *Technical Writing for Engineers and Scientists*, the authors Finkelstein, Aune, and Potter explain that in many cases, technical documents are not complete without a separate section that summarizes the structure or substance of the document, and that there are instances where technical writers need to provide summaries of texts for other uses. Abstracts, academic summaries, and executive summaries provide this function—they summarize a text within a set of requirements (like word counts) set by the audience, normally the decision-makers and/or advisors. Abstracts and summaries are short, which means they are more difficult to get right.

The opening identifies the source (book chapter) and authors.

The opening line also states the main idea, or thesis, of the original text.

Audiences use abstracts to determine whether or not a full text is relevant to the work they are doing, so you must ensure that every word is of value. Abstracts should always be under the word count, be in your own words, not include definitions, and stand alone (that is, should be understandable without context). Abstracts can be descriptive or informative. Descriptive abstracts inform the audience about the topic of the text. Descriptive abstracts summarize the overall structure of a text, but do not include the content of the text. They are often approximately 50 words and do not provide any detail. Informative abstracts, however, do require summarizing the structure and substance of the original text, usually in 100-200 words. Informative abstracts provide a condensed discussion of the important points (note, not **all** the points).

These lines provide an overview of the original text, each idea of which will be taken up in more detail later.

Audiences use summaries, which are shortened or condensed versions of the original text, to familiarize themselves with the content of the text and may even use the summary instead of the original text to make informed decisions. Academic summaries summarize the main idea and supporting ideas of a source and are used when the audience needs a summary of a text’s content but not in a stand-alone document. In fact, academic summaries are usually part of a larger document, like an annotated bibliography or a position paper, or used by a writer to prepare for a meeting, discussion, or exam. Executive summaries typically summarize lengthy reports and are stand-alone documents. In other words, executive summaries are documents with an introduction and conclusion of their own and which correspond to the original text. Executive summaries provide both the structure and substance of the original text and are used by decision-makers to make decisions.

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The ideas from the original chapter are taken up in the same order (summary uses the same structure as the original) with selected supporting detail (summary prioritizes which details are most important to the audience).

Note that an academic summary does not:

- provide an introduction or conclusion
- use quotations
- run a play-by-play of the original text (first, second, next, then, etc.)
- list topics (“talks about,” “is about”).

Academic summaries summarize the substance of the original text.

What Are Executive Summaries?

An **executive summary** is often requested to summarize lengthy technical reports, such as formal proposals and other fully developed business or technical documents. While some sources still subscribe to the idea that executive summaries can be up to 10% of the length of the original document which they are summarizing, the prevalent view in most businesses today is that an executive summary should never be longer than one page. This is critical. While most managers (decision-makers) expect that their engineers and scientists (implementors) will do all the work needed to make justifiable decisions, if they are confident in their employees, they do not actually want to read the full documentation. For that reason, executive summaries should be written as stand-alone documents. As such, they have both informative and descriptive characteristics. They contain both the substance and the structure of the report. In fact, an executive summary often substitutes for the full report.

An executive summary is analogous to the class review notes that teachers post online which summarize the course text. Of course, teachers say that these notes are an insufficient substitute for the real book—but, as we all know, they can work to some extent depending on the class and the teacher. Similarly, executive summaries are designed to provide management (decision-makers) and staff (advisors) with enough information about what is in a report that managers can make informed decisions without reading the entire document. They can always go back and read the document, if warranted—or, more likely, have experts on staff (advisors and implementors) read and analyze the complete report. Because these summaries often take the place of the report for key decision-makers, executive summaries can take on critical importance and must be well written.

Writing Executive Summaries

Writing executive summaries is a challenging undertaking. Your task is to capture as much of the full report's substance as is reasonable, or feasible, in a fraction of the full report's space. To do so, you need a good strategy, especially if the report you are summarizing is complex and extensive. You cannot possibly include everything, so you must think through exactly what to include and what to leave out of the summary. A good approach for doing that is to go back to the original purpose of the report. Evaluate everything in the report in terms of (1) how much it contributes to achieving that purpose and (2) how extensive the treatment would have to be if you were to include it. Then include only those portions that contribute the most to the goal of the report and that can be handled effectively in a summary. Start by structuring your executive summary to match the same structure as the full report. For example, if you are summarizing a 50-page report that includes Introduction, Methods, Results, and Impact sections, you would include these same sections in your summary. You might choose to use subtitles or not; this is personal preference, though of course, if it is part of the requested formulation, you want to include it.

In the following example executive summary, we have noted points of interest in the document. We start by reiterating that this is a stand-alone document; we haven't provided the original 50-page report to you, because as the decision-maker audience, you don't need it. What you want is to know if we, the engineers (and implementors), have done a satisfactory job of addressing the problem from start to finish. You are telling a story from start to end, and this story should be understandable as soon as you distribute your report, as well as six months or six years from now. In other words, it must include all of the necessary context to stand alone.

The opening paragraph clearly establishes the “who, what, where, and why.” It removes any guesswork about the impetus and establishes the scope of the project.

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Example of an Executive Summary

Waggin' Tails Dog Hospital (WTDH), with four locations in Finecity, State, is committed to their strategic philosophy of providing quality canine healthcare by focusing on four Keys of Excellence (KoE): Dogs, Owners, Quality, and Service. Waggin' Tails' vision is to ensure a “healthy dog, happy owner, every time.” Their KoE and vision statement require that WTDH have high-quality biomedical equipment readily available when needed. At the flagship location, three full-time WTDH technicians repair broken equipment and perform preventative maintenance (PM) on over 200 types of equipment to ensure machine availability and quality.

The problem is justified using data.

The individual, team, department, or whoever is responsible for the work should be identified. Note that this is written in first person. First person allows the author to write as concisely and precisely as possible (see [Chapter 1](#) Introduction).

The engineering department determined that approximately 40% of the technicians' time is spent on repetitious repairs and PM that involves numerous, daily non-value added (NVA) tasks. These NVA tasks consist of searching for spare parts, tools, and test equipment throughout the shop. NVA tasks reduce the time available for the technicians to complete required repairs and PM, and decrease equipment availability. To increase equipment availability, WTDH is faced with the choices of improving the current process or purchasing additional equipment. The current process also results in overtime costs of \$5,200 annually, and according to WTDH management, shop disorganization contributes to technician headaches, stress, and fatigue. Because of this, employee satisfaction surveys were completed company-wide, and technicians have a 35% level of satisfaction with their work environment. Waggin' Tails values their employees and must address these concerns. Engineering was asked to lead the evaluation and drive improvements. We identified three measurable objectives:

Measurable objectives start with a verb. Are you increasing, decreasing, eliminating, adding, etc.? This allows you to actually measure the change resulting from your efforts. Note the parallel construction (see [Chapter 15](#) Style and Mechanics).

This paragraph summarizes the methods used by the engineers. It explains why each tool and technique were used, giving confidence to the audience that the implementors used good judgment while addressing this project. It also sets up the bulleted list of deliverables.

- Decrease the amount of time technicians spend on unnecessary NVA tasks
- Increase equipment availability
- Increase employee morale

Engineering tools were used to generate a recommended layout that reduces the amount of time the maintenance technicians spend on NVA tasks. Initial 5S methods were used in the shop to eliminate unnecessary items. Workstations which increase technician safety were redesigned using NIOSH guidelines. Overall Equipment Effectiveness (OEE) metrics were put in place for the top 20 pieces of equipment experiencing availability issues so that we can track improvement. With the technicians' input, a new process for machine tracking was developed using TRIZ (Theory of Inventive Problem Solving). An implementation plan was developed and mapped to a Gantt chart, requiring approximately nine weeks to fully implement. The engineering methods used have resulted in the following deliverables for WTDH:

Note that a bulleted list works well in this case because it allows the audience to quickly assess the accomplishments of the project team (see [Chapter 15](#) Style and Mechanics).

- A recommended shop layout for the maintenance shop
- Sorted and organized technician shared areas
- Workstation redesign
- OEE metrics and process

- Machine tracking process
- Standardized checklists
- Implementation plan with Gantt chart

Results from the implemented 5S process, obtained using a follow-up survey, showed a 56% increase in employee morale. The reduction of NVA tasks resulted in an expected overtime wage reduction of \$4,800 annually. With an expected implementation cost of \$1,200, this resulted in a 0.25-year payback. Though not calculated, the additional savings from creating a safer work environment, increasing employee morale, and increasing equipment availability are also beneficial to WTDH.

When your audience is decision-makers, they must always know two things: 1) how much will this cost (and save), and 2) how soon can it be completed. By including these key pieces of information in an executive summary, you can be sure to check two of the most important boxes.

The full implementation will be completed by 14FEB21. These economic and strategic results will contribute to Waggin' Tails Dog Hospital's Keys of Excellence and help make it an even greater place to work.

Conclusion

Abstracts and **summaries** summarize texts. Abstracts can be either descriptive or informative. **Descriptive abstracts** summarize the structure of a report, but not its substance. **Informative abstracts** summarize the substance of a report, not just the structure. Descriptive abstracts are often approximately 50 words in length, while informative abstracts usually run approximately 100 to 200 words.

Summaries can be described as either academic or executive. **Academic summaries** provide a shortened or condensed version of the original source. More specifically, they provide a summary of the source's main idea, or thesis, and its support for that main idea. **Executive summaries** provide a comprehensive, detailed synthesis of a large technical report. This type of summary is intended as a stand-alone document that can substitute for the report itself. An executive summary is designed to provide enough information about what is in the report to allow informed decisions without the need to read the report.

Finally, we also note that while we did not include A3 reports as a form of summary (see [Chapter 13](#)), they are in fact, just that. An A3 does not summarize another document, but it does summarize the process of problem

solving for a given situation. It accomplishes this by reducing the process to a combination of text and visuals that tell the full story and allow others to understand the progression from problem identification to implementation tasks and follow-up.

Checklist for Abstracts and Summaries

General Checklist

- ☐ Do I understand who my audience is for this abstract or summary?
- ☐ Do I understand the purpose of this abstract or summary?
- ☐ Do I know the context for which I am writing this abstract or summary?
- ☐ Have I been given specific formulation requirements?
- ☐ Am I using the formulation that is most effective for my audience, purpose, and context?

Checklist for Abstracts

Have I:

- ☐ Met the word count requirement?
- ☐ Structured the abstract so that it meets either descriptive or informative expectations?
- ☐ Included author and title information if required?
- ☐ Included all main ideas and limitations from the paper?
- ☐ Included carefully chosen keywords that will inform both evaluators and computer algorithms?

- Proofread for readability?

Checklist for Summaries

Have I:

- Identified and met any formulation requirements such as word count, page count, or formatting?
- Determined if the summary is for academic or business purposes?
- Included author and title information if required?
- Included all main ideas needed for audience and purpose?
- Written the summary with original phrasing (no quotations)?
- For an academic summary, explained how the main argument/idea is supported?
- For an academic summary, organized ideas in the summary in the same order they were in the original, but avoided a play-by-play (first, second, next, then, etc.)?
- For an executive summary, told the whole story in a stand-alone document?
- For an executive summary, written in first person?

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Notes

- [1.](#) Dr. Sarah Ryan, Editor – *The Engineering Economist*, personal interview, December 10, 2020.
- [2.](https://publons.freshdesk.com/support/solutions/articles/12000047303-how-does-publons-reviewer-connect-work-) <https://publons.freshdesk.com/support/solutions/articles/12000047303-how-does-publons-reviewer-connect-work-> Accessed December 11, 2020.
- [3.](http://users.ece.cmu.edu/~koopman/essays/abstract.html) Koopman, Philip. “How to Write an Abstract.” October 1991. Accessed December 10, 2020. <http://users.ece.cmu.edu/~koopman/essays/abstract.html>.

4. "Abstracts." *The Writing Center*. University of North Carolina at Chapel Hill. Accessed December 10, 2020. <https://writingcenter.unc.edu/tips-and-tools/abstracts/>.
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CHAPTER FIFTEEN

Style and Mechanics



Be calm! This chapter is not intended to bring back the “good old days” when stern English teachers demanded that you look up how to spell a word from a (gasp!) hardcover dictionary or practice verb conjugation on worksheet after worksheet. Rather, the goal is to provide you with a practical guide to common stylistic considerations and the mechanics that support those choices in technical writing—and specifically, to help you identify and fix problems when you are reading and editing.

Language is inherently powerful because it defines meaning. Jack Lynch discusses the idea of power and argues that being able to use appropriate grammar enables access to power:

... standard edited English—with all its stupid rules and irrational regulations—is still the form that’s used in corridors of power. Refusing to use it will exclude you from those corridors, even though the exclusion is often for dumb and prejudicial reasons.¹

When we write, our conscious decisions affect both the **meaning we intend to convey** as well as the **meaning that is received** by our audience. As technical writers, our goal is to reduce the gap between the giving and receiving ends as much as possible. Our conscious choices affect the **style**, and consequently the **tone**, of our writing. For example, we choose how formal our tone is by whether we use slang words (shade, lit), colloquialisms (pie in the sky), lowercase font (mr. smith), acronyms (LOL), phrasal verbs (stand down), and emojis (☺). We choose how sophisticated our message is by the level of technical wording, phrasing, and sentence complexity that we incorporate. We also choose whether to proofread our work or not; when an audience reads our text and it has errors in the mechanics and/or inappropriate stylistic choices, we convey a “style” of incompetence or indifference. Neither of these styles is helpful when trying to persuade someone to invest time or resources into your project, company, or education.

Style and mechanics are inextricably linked. While style might seem to be a higher level of concern and mechanics a basic building block, we really cannot discuss one without the other. Style—the flair, tone, and flourish of your writing—should be purposefully created on a solid foundation of hardworking, yet sometimes underappreciated, mechanics.

Consider the Siberian Husky, which is one of the oldest dog breeds on the planet and essentially the origin of all dog genetics. As one of the closest links to wolves, Huskies represent the basic building blocks of all dog breeds. But the Husky plays double duty in the canine world; not only does it provide the foundation of dog DNA, it does so with incredible flair. Huskies “look” and “do” with style: their own unique eye color combinations and beautiful coats, in addition to their incredible work ethic of long, hard days contribute to their beauty. They are team players, working together to accomplish a goal (see [Figure 15.1](#)). Much like Siberian Huskies, style and mechanics are the heroes and hard-working team members of writing, though maybe slightly less revered than they should be. However, when a comma is the subject of headlines because it makes the difference in court cases, resulting in settlements of millions of dollars, it earns some respect (more on that later.). And like Siberian Huskies, it turns out that with a little effort and practice, mechanics CAN be mastered, predictable, and support writing style.



Figure 15.1

Mechanics and style are the building blocks of good writing, much like Huskies represent the building blocks of all dog breeds.

Stylistic Considerations

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Style is a general concept that refers to the way we say something. Style encompasses many things, including word choice, order, and selection; and it actually includes all other distinctive features of the way we express ourselves in language. So what does style have to do with technical writing? Besides being grammatically correct, technical writing must be geared to communicating precise information in a straightforward and unambiguous way. It is easy to be ambiguous and imprecise while still being grammatically correct. For technical writing to be effective, however, it must use a style of expression characterized by **economy** and **precision**.

Economy

One of the most important things you can do in technical writing is to say what you want to say with the fewest words possible. Consider the following sentence from a technical proposal cost section.

With the availability of all necessary equipment, along with required materials and a student who can perform required tasks at the level of competency appropriate for the tasks involved, we believe that all satisfactory and unsatisfactory options can be fully and completely considered, and a reasoned and effective recommendation provided within the time and cost constraints of the proposal, given the past, present, and future uses of the technology.

Look at the style of this sentence. What a disservice to the reader! What about **all satisfactory and unsatisfactory options**—as opposed to what other kinds of options? No other kinds of options exist. How about a **reasoned and effective recommendation**, as opposed to what?—an unreasoned and ineffective recommendation? And how about **past, present, and future uses**? That pretty much includes everything, does it not?

This kind of writing is ridiculous. Much of this paragraph is pure padding, and most of the words add absolutely nothing to the meaning. In fact, this kind of writing actually confuses or dilutes the point

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the author is trying to make. We could easily rewrite this paragraph and say the same thing in a much more straightforward way:

With the required equipment and materials, and a competent student, we can fully consider all options and make a recommendation within the proposal constraints.

Get to the point as quickly as possible. Look at the following example, a letter from higher management sent to the technical staff of a company:

To All Technical Staff:

Centuries ago, a generalized philosophy of the earth and the cosmos constituted scientific thought and did not require exactness. The very essence of science and technology today relies on a foundation of reproducible standards, and since 1866, this country has specified the International System of Units (SI), a.k.a. the *metric system*, as the only legal system for electricity and illumination. Standardized nomenclature is frequently victimized by the conflicting approaches of SI and the English system, even though SI provides the standard prefixes required for dealing with a wide range of units. Consequently, all technical staff will use only SI in company documentation.

Of course, we could shorten this paragraph by just getting to the point:

To All Technical Staff:

Use the metric system in all company documentation.

There is no need to “soften” the employees with an esoteric and irrelevant discussion of SI. It is unlikely that the company’s technical staff will fall to the ground in despair when told up front that they should use the metric system.

Precision

Because technical writing must be precise, we must select words that are precise in meaning. Look at this example:

The X-49 is a very large chip that costs a lot of money and has a lead time of about three days. It has unprecedented power; in fact, its predecessor, the X-45, is just a faint shadow of the X-49.

So, how big is **very large**? Is the X-49 larger than the solar system? That certainly would be very large unless, of course, we were comparing the solar system to the Milky Way galaxy. Then it would be **really small**. And what exactly does **unprecedented power** mean? Is that more than a **whole**

lot of power? Also, if the X-45 is just **a faint shadow** of the X-49, does that mean it is **a lot smaller**, or maybe just **somewhat smaller**? You get the idea. Avoid words such as **very, awfully, extremely, really, a lot, and kind of** in technical writing. Such words may sound good in some cases, but in reality, they have no precise meaning. Generally speaking, avoid all adverbs in technical writing; adverbs imply your personal opinion, and in technical writing, information should be presented objectively, allowing the audience to form their own opinions. You should also use the word **approximately** for estimates rather than **about** in technical communication.

The sentence above could be rewritten precisely as follows:

The X-49 is an Ardvark1000-socketed chip that costs \$9,650 per unit to manufacture and has a lead time of approximately three days. Its performance benchmarks are two orders of magnitude greater than those of its predecessor, the X-45.

Style Guides

Style requirements will often be imposed upon you in technical writing. The formulation is dictated by the audience. Your supervisor might require a file format or page length, your customer could specify a medium such as an email or a formal letter, your potential grantor might mandate the font type of a proposal, or your instructor could require the use of APA citation style. In all cases, it is to your benefit to meet the requirements. Remember, you are communicating with a purpose—to inform, persuade, and/or establish good will. Your audience will be much more receptive to your message if you adhere to all required formulation. Examples of frequently required style considerations include

- Page length (e.g., 1 page maximum, 300–500 words)
- Font size and type (e.g., Calibri size 12, minimum size 10 font sans serif)
- Title and subtitle/structure formatting (e.g., boldface, italic)
- Citation method (e.g., Chicago, APA, MLA)

It is always prudent to inquire about requirements for a requested communication of any type, and it is better to determine this earlier in the

writing process than later. When it comes to mandated requirements, choose wisely—adhere to them.

Grammar: What Is It and Why Is It a Big Deal?

Grammar is nothing more than a large set of rules—commonly accepted standards for assembling words so that, together, they make sense and convey meaning. We acknowledge that “commonly accepted” is also “commonly imposed.” As societal sub-cultures evolve over time, so, too, do language norms and rules. Sometimes we don’t even realize that what was previously culturally acceptable is no longer. For example, in recent years in American English culture, the use of plural pronouns referencing singular individuals has come to be grammatically acceptable (more on that in a bit). As we noted earlier, style is reflected in the choice of words and the way we apply the rules of grammar in our writing. In traditional grade school curricula, grammar and style are the subjects of much attention. What is important in technical writing, however, is not the ability to recite obscure grammatical rules; rather, it is being able to write correctly and effectively using the most informed societal language rules available. In fact, grammar is important in technical writing for only two reasons:

1. Incorrect or improper grammar can change the meaning of what you are trying to say or, at least, make your meaning hard to decipher. That is fundamentally opposed to the goal of technical writing, which is precision in meaning.
2. Incorrect grammar says something about you and the quality of your thinking. Poor grammar in a technical report can communicate to the reader that you are not terribly meticulous or that you lack the required education or professional attention to detail. Right or wrong, true or false, fair or unfair, poor grammar can seriously undermine your credibility. If you don’t believe us, check out Weird Al Yankovic’s “Word Crimes” video on YouTube, where you can feel judged, be entertained, and experience the most complete style and mechanics lesson possible in under four minutes.^{[2](#)}

In a technical document, you are judged to some extent based on the quality of the document's writing and presentation. When a reader is attempting to understand what you have written, you are not there to defend yourself, you are not available to explain what you really meant, and you have no opportunity to fix your errors.

Fortunately, most technical writers who make errors in grammar Page 205 do so in relatively few areas. This chapter focuses on the most frequent mechanics (spelling, punctuation, and grammar) mistakes in technical writing and provides some straightforward solutions. Here are the most common problem areas:

- Spelling
- Punctuation errors
- Sentence fragments
- Misplaced modifiers
- Passive voice
- Verb agreement
- Pronoun agreement
- Pronoun reference
- Case
- Noun clauses
- Compound adjectives
- Phrasal verbs
- Parallel construction
- Bullets vs. paragraphs
- Proofreading

Spelling Errors

Because grammar is affected by both spelling and punctuation, we begin with spelling. Most people assume that an educated professional can spell words correctly. Spelling errors make your reader doubt the validity of your writing. So check and recheck your writing for spelling errors.

Consider how these errors would look in a technical communication:

Correct word	Wrong word	Misspelled word
↓	↓	↓
It's important for our company two understand the equipment and it's capabilities.		

The expression **two understand** should be **to understand**, and **it's** (the contraction for **it is**) should be **its** (the possessive pronoun). Adding an apostrophe to **it** does not make the word possessive; rather, it makes the word into a contraction. The correct sentence reads as follows:

It's important for our company **to** understand the equipment and **its** capabilities.

Which sentence will give your supervisor confidence in your recommendation?

A Word about Homonyms

All the errors in the sentence above involve homonyms, which are words that sound alike but have different meanings. Substituting **to** for **two** or **it's** for **its**, as was done in the sentence above, is a homonym error that computer spell checkers often miss, but which can look really stupid in a technical report. The problem is that the English language has many homonyms that lend themselves to these kinds of errors. How many writers have used **bare** (meaning naked) instead of **bear** (the big furry animal), or **oral** (something involving the mouth) instead of **aural** (something you hear)? Many technical writers also routinely confuse words that sound almost alike. For example, while not technically homonyms because they sound slightly different, consider the words **effect** and **affect**. **Effect** can be either a noun meaning “the result of some cause,” or a verb meaning “to cause something to happen.” **Affect** can be a noun meaning “mood or emotion” or a verb meaning “to have an influence on something.” Typically, though, effect is used as a noun and affect is used as a verb in the majority of usage.

Keeping homonyms straight can be challenging. Just consider Page 206 the editor who marked **carets** on her proofreading copy, who wore a 2-**carat** diamond in a ring setting made out of 18-**karat** gold, and who ate

a **carrot** every day for lunch. The bottom line: Stay alert for homonym errors; see [Figure 15.2](#) for a diagram showing the relationships between homographs, homophones, and synonyms.

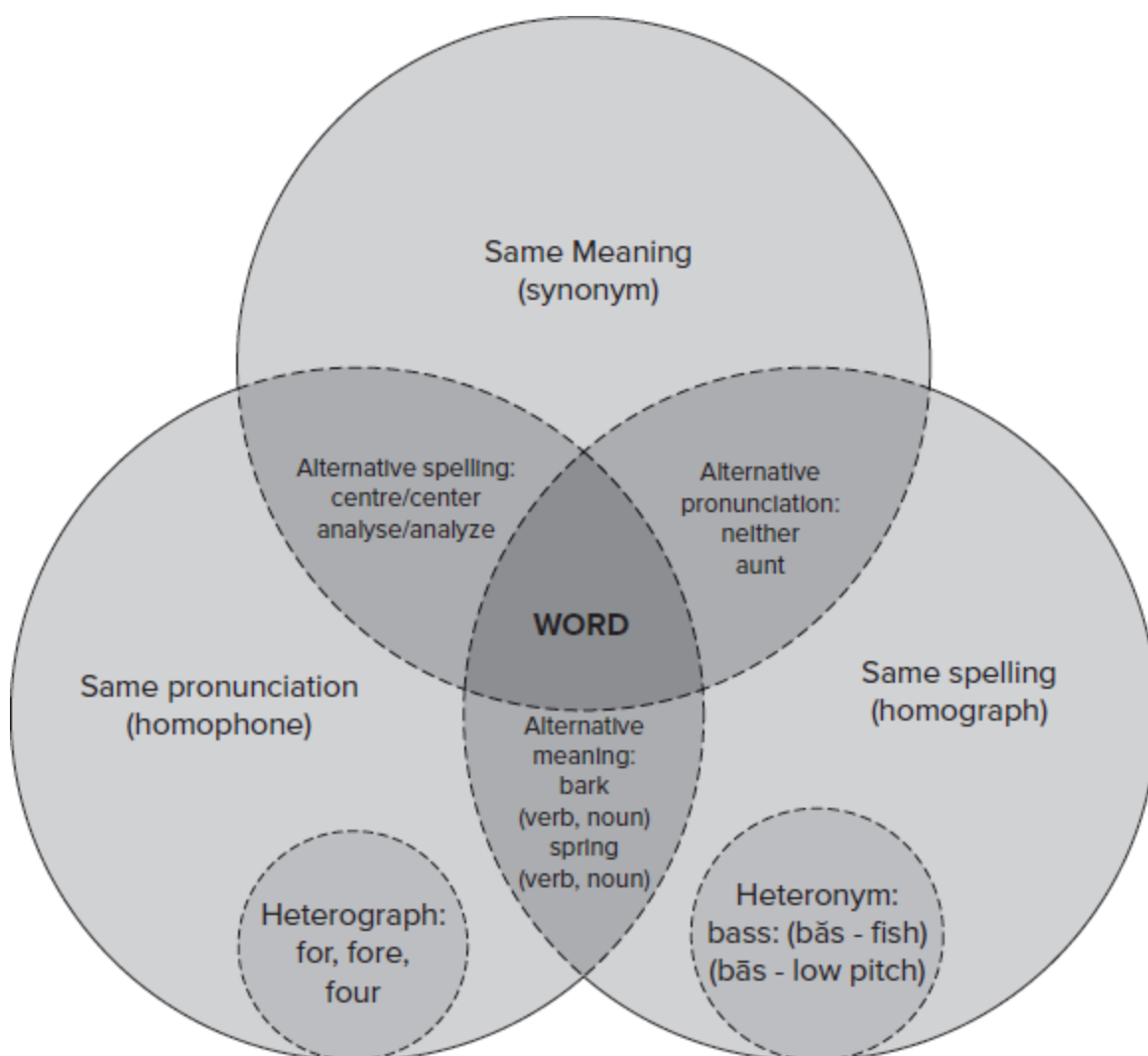


Figure 15.2

Venn diagram of the relationships between the different types of homonyms.

Spelling Numbers

An additional spelling consideration worth mentioning involves **numbers**. One of the most common errors is to start sentences with a numeral:

9,192,631,770 hertz is the spectral line frequency of cesium 133.

Correcting this kind of error can be difficult, but you must replace that leading number because it is grammatically incorrect, unattractive, and can be confusing in the middle of a document. The worst fix would involve spelling out the number:

Nine billion, one hundred ninety-two million, six hundred thirty-one thousand, seven hundred and seventy hertz is the spectral line frequency of cesium 133.

An easier solution would be to simply insert an appropriate modifier before the number:

Exactly 9,192,631,770 hertz is the spectral line frequency of cesium 133.

Or simply invert the sentence:

The spectral line frequency of cesium 133 is **9,192,631,770** hertz.

When two-digit integers start a sentence, you can just spell them out:

Thirty-two degrees Fahrenheit is the freezing point of water.

A good rule of thumb for when to write out numbers using letters in the middle of sentences is that integers which do not have units of time, money, etc. and are less than 20 should be spelled using letters, while integers of 20 or higher should be written in numerals. If units of measure are included, it is good to use numerals:

We can travel 6 feet in 3 seconds for a cost of \$5,000.

Capitalization

Finally, pay attention to which words you capitalize. Normally, in technical writing, some type of convention or style sheet will tell you what you should capitalize (see the previous Style Guides section). Many style guides exist, along with lots of differences regarding the use of capital letters. Always follow the style guidelines for your organization or activity.

A good general approach is to avoid unnecessary capital letters. Consequently, if you are going to capitalize a word, have a specific reason for doing so. Here are some common reasons for capitalizing words:

- Capitalize names of specific persons, places, or things (proper nouns). These include the names of specific people (Albert Einstein); cities or street names (Dayton, Ohio; First Street); historical documents and religions (the Declaration of Independence, Buddhism); titles of papers, books, films, software packages, and trademarks (*Star Wars*, Microsoft Office); and periods of time or geographic regions (Cambrian Period, the West).
- Capitalize abbreviations or acronyms (ATM, Ph.D.).
- Capitalize titles that precede a person's name, but not those that follow (Professor John Smith or John Smith, the professor).
- Capitalize the first word of every sentence and the pronoun **I**.

Note that the prevalence of texting and other informal social media communication has found its way into people's daily communication habits, but this should be avoided in technical writing. FYI, save the acronyms, lower case type, and emojis for your next text to your significant other or your mother—LOL TTYL. G2G asap ☺.

Punctuation Errors

Punctuation is the use of symbols, marks, and signs to separate or connect certain parts of a sentence and to clarify meaning. Writing did not always use punctuation. Medieval manuscripts, for example, do not all use punctuation (or spacing, for that matter!). Try reading the first paragraph of this textbook without any punctuation:

Technical writing is a fundamental skill for virtually everyone working in science and engineering and that includes a broader range of people than just scientists and engineers most science and engineering activities require technical communication whether in written electronic or some other form research development finance manufacturing and a host of technical commercial services rely on precise communication to relay complex information to a wide range of audiences for many purposes technical writing is the means by which such communication is produced

Now imagine the entire book written this way—what a disaster! Lynne Truss, author of *Eats, Shoots & Leaves* says, “Punctuation marks are the traffic signals of language: they tell us to slow down, notice this, take a detour, and stop.”³ Without proper punctuation, technical writing crashes

and burns (or at the very least, the audience gives up trying to understand the author’s meaning). Misusing punctuation can obscure, confuse, and mislead your audience, and can compromise your credibility as a writer.

The two most common punctuation errors are comma splices and fused sentences, and they will be treated in detail here. We also address the serial comma, or as it is famously known, the Oxford comma. However, many other marks exist and their proper use is critically important to having a quality document. [Table 15.1](#) provides an overview of the various marks of punctuation, along with their functions in English and examples of their proper use.

Comma Splice

Comma splices occur when we join one sentence with another sentence by using a comma instead of a conjunction. This mistake is easy to make and easy to correct. Here is an example:

This comma improperly splices or joins two independent clauses together.

The circuit operates at dc, Ohm’s law applies.

Here we have two independent clauses: “The circuit operates at dc” and “Ohm’s law applies.” The comma that follows “dc” is splicing these two sentences together, which is not something commas are supposed to do. To fix the problem, you have three choices: since the two clauses are closely related, 1) you can join them with a semicolon, which functions as a “soft” conjunction in this instance; 2) you can use a comma and a conjunction; or 3) you can use a semicolon, an adverbial conjunction, and a comma.

1) Use a semicolon instead of that splicing comma.

The circuit operates at dc; Ohm’s law applies.

2) Or follow that splicing comma with a conjunction.

The circuit operates at dc, and Ohm’s law applies.

3) Or add a semicolon, an adverbial conjunction, and a comma.

The circuit operates at dc; therefore, Ohm’s law applies.

Fused Sentence

A **fused sentence** is a comma splice without the comma. In other words, two sentences are fused without any mark of punctuation:

Point of fusion
↓
The workstation was not designed ergonomically it leaves much to be desired.

Again we have two sentences: “The workstation was not designed ergonomically” and “It leaves much to be desired.” Note how they just run together at the indicated point of fusion. The solution, as with a comma splice, is easy:

Insert a semicolon.
↓
The workstation was not designed ergonomically; it leaves much to be desired.

Or add a comma
and a conjunction.
↓
The workstation was not designed ergonomically, and it leaves much to be desired.

Or add a semicolon, an adverbial
conjunction, and a comma.
↓
The workstation was not designed ergonomically; consequently, it leaves much to be desired.

As we teased in the opening paragraphs of this chapter, punctuation matters to more than just English teachers. It is quite possible that there is not a more expensive punctuation mark than the serial (often called the Oxford) comma. Multiple legal rulings have imposed significant settlements because of the changed meaning associated with having/not having a serial comma.

- In March 2017, a court’s ruling hinged on a missing comma: “Delivery drivers for local milk and cream company Oakhurst Dairy have been tussling with their employers over whether they qualify for overtime. On March 13, a US court of appeals determined that certain clauses of Maine’s overtime laws are grammatically ambiguous. Because of that lack of clarity, the five drivers won their appeal and were found eligible for unpaid overtime.”⁴
- In 2006, “The dispute between Rogers Communications of Toronto, Canada’s largest cable television provider, and a telephone company in

Atlantic Canada, Bell Aliant, is over the phone company's attempt to cancel a contract governing Rogers' use of telephone poles. But the argument turns on a single comma in the 14-page contract. The answer is worth 1 million Canadian dollars (\$888,000).”⁵

Yikes! What is this serial comma and how do we avoid putting ourselves in the same situation? It is actually easy to identify; look for the conjunction in a list of items. The serial comma is the comma that goes **in front of the conjunction** for a list of items. One of the most common examples is structured as so:

I would like to thank my parents, Carl Sagan and Mother Nature.

Notice the missing comma in front of the conjunction **and**. Without it, the reader can rightly interpret this sentence as someone thanking their parents, who happen to be Carl Sagan and Mother Nature. The only way to avoid this misinterpretation is to include the serial comma.

I would like to thank my parents, Carl Sagan, and Mother Nature.

By using the serial comma, it is crystal clear that this is a list of three distinct entities, none of whom are related to each other. If you want to avoid expensive legal settlements, you can avoid ambiguity by including the serial comma when your list items do not go together.

Punctuation

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See [Table 15.1](#) for a summary of the various marks of punctuation and examples of their proper use.

Table 15.1 Quick Guide to Punctuation

	MARK	FUNCTION	EXAMPLE
Introducers	colon	Introduces with emphasis a word, phrase, clause, or series that follows.	<i>Please do the following: get up, get going, and get a job.</i>
	dash	Introduces with emphasis a parenthetical idea or new thought within a sentence. Considered more informal.	<i>If this invention works—and the lab thinks it will—we will be rich.</i>
Separators & connectors	comma	Separates items in a series.	<i>Soldering involves cleaning, heating, flowing, and cooling.</i>
		Separates nonrestrictive phrases and clauses.	<i>Because we care, we are donating this equipment.</i>
		Separates phrases and clauses with meaning that is different from what comes before.	<i>Do not install the new patch, not unless you're ordered to do so.</i>
		Separates main clauses with a conjunction.	<i>He secured the blockhouse, and she started the countdown.</i>
	semicolon	Separates items in a series when any one item contains a comma (sometimes called a 'super-comma' in this case). Separates two closely related independent clauses.	<i>Please get up; get going, if you can; and get a job.</i> <i>She smiled; I frowned.</i>
	slash	Separates options.	<i>The process resulted in a true/false output.</i>
	hyphen	Divides words at the end of the line. Connects certain written whole numbers and fractions. Combines multiple descriptive words into complex adjectives.	<i>Please meet me at the al-ternate location.</i> <i>Forty-five percent is less than one-half interest.</i> <i>She worked at a military-industrial complex.</i>
Containers	quotation marks	Contains direct quotations.	<i>He said, "You have the job."</i>
		Contains minor titles such as chapters, articles, episodes.	<i>My favorite Tolkien chapter is "The Steward and the King."</i>
	parentheses	Contains parenthetical material incidental to the main thought.	<i>We were relieved when the ion chamber worked (it never had before).</i>
	brackets	Contains explanations or editorial corrections. Contains a second level of parentheses within a level of parentheses.	<i>Scientists tend to [be] critical thinkers.</i> <i>The operating system (the OS upgrade [Mac OS Big Sur]) worked well.</i>
Terminators	period	Terminates a sentence that is neither a question nor an exclamation. Used with most abbreviations.	<i>I'll be back.</i> <i>He has a Ph.D., experience, and solid references.</i>
	question mark	Terminates a sentence that asks a question.	<i>Can you fund my project?</i>
	exclamation point	Terminates a sentence that makes an exclamation. Punctuates an interjection.	<i>I will not take a pay cut!</i> <i>No! I will not take a pay cut.</i>

Indicators	apostrophe	Indicates possession.	<i>Phyllis's artwork</i>
		Indicates an omission in words or numbers.	<i>His career in '08.</i>
		Pluralizes letters, abbreviations, symbols, and numerals.	<i>I need more 0's before the decimal point in my bank balance.</i>
	ellipsis	Indicates omitted material.	<i>I need more 0's . . . in my bank balance.</i>

Sentence Fragments

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For a sentence to be complete, it must contain a verb. Fragments usually occur when the writer substitutes something for the verb or leaves out the verb altogether:

Testing may look like a verb, but it is actually a noun—in this case, a gerund, which is a verb used as a noun. No verb exists in this sentence; consequently, it is a fragment.



Tensile testing the specimen carefully with high levels of precision.

This example is a fragment because it does not contain a real verb (it also does not make sense). The word **testing**, derived from the verb **to test**, does not act as a verb here; rather, it is a gerund, which is a verb used as a noun. To fix this problem, you need to add a verb:

Tensile testing the specimen carefully with high levels of precision is necessary.



A verb has been added here.

Sentence fragments also occur when we put a subordinator before an otherwise perfectly good independent sentence:

Subordinator added here.



Because the transformer could not take the load.

The subordinator **because** makes this clause dependent on something else, but the “something else” is not there. To fix the problem, you can remove the subordinator:

Subordinator removed.

↓
~~Because~~ The transformer could not take the load.

Or you can add the “something else”:

Independent clause has been added.

↓
Because the transformer could not take the load, the system quickly failed.

Misplaced-Modifier Errors

English relies on word order or placement for meaning. In effective technical writing, modifiers must be close to the words they are supposed to modify. Sentences in which modifiers are misplaced may be grammatically correct, but often they will not mean precisely what the writer intended. Here is an example of a misplaced modifier (often called a **dangling participle**):

Modifier should refer to **ignorance**, but here it refers to **society**.

↓
Ignorance of science is a phenomenon in society that must be destroyed.
↑

The writer is advocating destroying the ignorance of science in society. Because of the misplaced modifier, however, the sentence may be interpreted as proposing the destruction of society. To fix this problem, move the modifier so that it relates more directly to what it is supposed to be modifying:

Modifier now indicates that the **phenomenon**,
not **society**, must be destroyed.

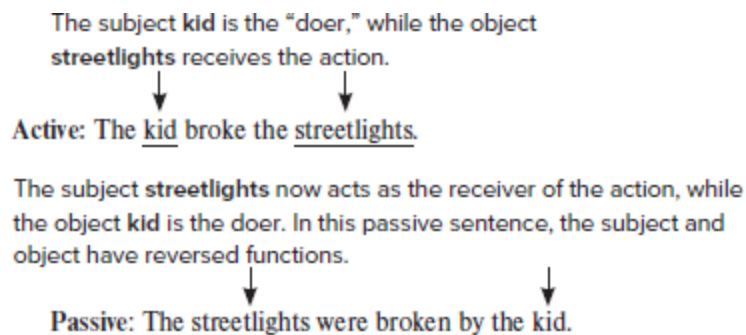
↓
In society, ignorance of science is a phenomenon that must be destroyed.
↑

Passive Voice Problems

Passive voice and active voice refer to the movement of action through the sentence. In an active sentence, the subject is the “doer” and comes first, the

verb or “action word” follows, then the object receives the verb’s action. In a passive sentence, the subject’s function changes and it becomes the receiver of the verb’s action, while the object, if it even shows up at all, takes on the function of the “doer.”

Consider the following active and passive sentences:



In the active sentence, the subject is the kid. The kid’s action involved breaking, and the object of this action was the streetlights. The second sentence, the passive one, says exactly the same thing, but the subject now has taken on the traditional function of the object—it is receiving the action, while the object now has taken on the traditional function of the subject because it is doing the action. The passive sentence is longer (7 words for the passive vs. 5 words for the active), and its construction is often considered weaker. In addition, notice that the verb in the passive sentence has the auxiliary verb **were**—a form of the verb **to be**. Instead of just **broke**, the verb now consists of the verb **were** and the past participle **broken**. Passive constructions always pair a form of the verb **to be** with the past participle of a verb. That is a good way to spot passive voice, although such pairings are not always passive and this method is not foolproof. The final test for passive voice is whether the subject is receiving the action of the verb.

To Be Verbs

have
has
had
is
are

were
was
am

Passive constructions are weak, so why do people write in the passive voice? One reason is that it allows them to hide responsibility for their actions. Consider this modified passive voice sentence:

Modified passive: The streetlights were broken.

What is missing? The **doer**—the kid who broke the streetlights. The sentence may be grammatically correct, but it leaves out an important piece of information. We no longer know who is responsible.

People sometimes use the passive voice to hide their culpability in something that is bad; then, in the same sentence, they switch to active voice to take credit for something that is good. Consider this compound sentence:

Passive Active
↓ ↓
Your medical records were lost, but I found them.

Here the passive voice in the first clause hides who is responsible for losing the medical records, but the writer has switched to the active voice in the second clause to take credit for finding them. For any true bureaucrat, this approach sounds much better than saying, “I lost your medical records, but I found them.”

When to Use the Passive

Active voice is often, but not always, preferred in technical writing because it is more direct, it is clearer, and it provides the most information with the fewest words. In terms of economy, it beats the passive voice. However, passive voice does have its place in technical writing. It can be useful when the doer of the sentence is unimportant or obvious, or when the receiver of the action is the primary focus. It is often preferred because it “sounds” more objective and can also be a useful way of breaking the pattern of sentence structure to keep the reader from falling asleep. Lab reports are

often written in passive voice. Also, as mentioned earlier, passive voice may be useful when one wants to hide responsibility. Finally, passive voice is appropriate when some higher authority expects it. For example, if you are writing a technical article for a journal that prefers passive voice as a matter of style, then use the passive voice.

Verb Agreement Errors

Verbs must agree with their subjects in person and number. If the subject is in the first person, the verb has to be a first-person verb. That is why “I **is** smart” is wrong and “I **am** smart” is correct. Many verbs in the English language are not person-sensitive, and for them, person does not matter.

In addition, if the subject is singular, its verb has to be singular. This requirement is the source of most verb agreement errors because, unlike agreement in person, some agreement-in-number errors do not sound wrong.

Here is an example:

This may sound correct, but **were** does not agree with **implant**.

↓ ↓

The implant, along with its associated circuits, **were** inserted into the patient's chest cavity.

The subject of this sentence is **implant**, which is singular. There is only one implant, but the verb **were** is plural. The verb does not agree with the subject, so a verb agreement error exists. This sentence may sound correct because the qualifier, **along with its associated circuits**, comes between the subject and the verb and makes the subject sound plural; however, the verb must agree with the subject, not the qualifier, and the subject is **implant**, which is singular.

You can fix this sentence either by making the subject plural or [Page 214](#) by making the verb singular. Changing the subject **implant** to its plural form might be grammatically correct, but technically it is not accurate—unless, of course, the surgeon is really putting two or more implants into the patient’s chest cavity. If we assume that only one implant is being inserted, one option is to make the verb singular, as follows:

The verb now agrees with its subject.
The implant, along with its associated circuits, was inserted into the patient's chest cavity.

Another option, and sometimes the best option, is to rewrite the sentence:

The implant and its associated circuits were inserted into the patient's chest cavity.

By using a plural noun, a plural verb is now correct. Restructuring the sentence is often the better choice, because the purpose is to convey the message effectively and efficiently. If our sentence structure interrupts the reader as they read, we have not accomplished what we want. If you find yourself struggling with how to accomplish noun-verb agreement, you might consider a new sentence altogether.

Pronoun Agreement Errors

Historically, it was expected that pronouns must agree with their antecedents (the words they replace) in person, number, and gender. This rule is somewhat similar to the subject-verb agreement rule. Consider the following sentence where the pronoun does not agree with its antecedent in number:

Plural pronoun does not agree with singular antecedent.
Each person in the lab must replace their radiation badges.

Person is singular and the pronoun **their** is plural. Trying to make **their** agree with **person** has traditionally been considered a grammatical error. The solution was to make the antecedent plural or the pronoun singular. Consequently, either of these two sentences would be considered correct:

The pronouns his and her now agree with their antecedent person.
Singular: Each person in the lab must replace his or her radiation badge.
or . . .
The pronoun their now agrees with its antecedent people.
Plural: All people in the lab must replace their radiation badges.

As we introduced earlier, societal language norms evolve, and it is now quite common to see what was once considered a plural pronoun used to refer to a singular person. Some people might dismiss this as political correctness, but there is growing acceptance. This is because plural pronouns in the English language are not gender-specific. That means you do not have to worry about their gender. It is more inclusive to avoid specific gender references by using plural pronouns. **Their** is not gender-specific and, consequently, represents an inclusive approach. If the antecedent refers to only one individual and you know their gender identity, you can use gendered pronouns. If you do not know the person's gender identity, then use **their**.

Another important point to make about pronouns is that many professionals and organizations encourage people to list their personal pronouns as part of their email signatures and during introductions in meetings. This is a result of focused attention on inclusiveness, with a desire to avoid assumptions about people's gender identity. If you are aware that an individual uses specific pronouns, it is wise to use them in your communication with and/or about them.

Pronoun Reference Errors

The other common pronoun error in technical reports involves the use of a pronoun whose antecedent is unknown or unclear. In technical writing, pronouns must refer clearly and without question to specific antecedents. Here is an example:

The coolant leak impaired the CPU's heat dissipation, resulting in an erroneous reading at the most critical part of the process. This had a cascading effect on the system.

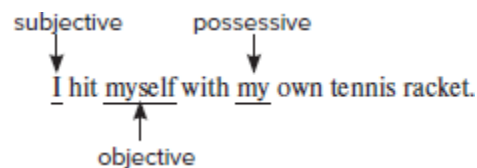
↑
What is the antecedent?

To what does **this** at the beginning of the second sentence refer? It could refer to the coolant leak, to the erroneous reading, or to both. Such ambiguity can be problematic in technical writing, especially when you are using the pronoun **this**. In fact, a good rule is to always include a noun after the pronoun **this**. You could fix this sentence as follows:

The coolant leak impaired the CPU's heat dissipation, resulting in an erroneous reading at the most critical point in the process. This coolant leak had a cascading effect on the system.

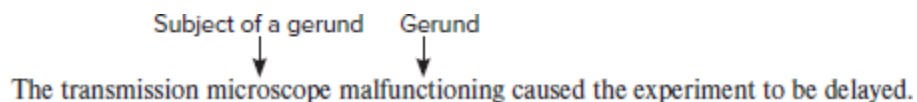
Case Errors

Case errors involve putting a noun or pronoun in the wrong case. The three English cases are **subjective**, **objective**, and **possessive**. The subjective case (also called the **nominative case**) is the form of a noun or pronoun used as the subject of a sentence. The objective case is the form of a noun or a pronoun used as an object in a sentence. The possessive case shows possession. Here is an example:



The first pronoun, **I**, is in the subjective case because it is the subject of the sentence. The second pronoun, **myself**, is in the objective case because it is receiving the action (myself is actually the reflexive form of **me**). The third pronoun, **my**, is in the possessive case because it shows possession of the tennis racket. If it were written as, "I hit myself with **me** own tennis racket," the **me** would represent a case error.

The cases of pronouns such as **who** and **whom** are a cause of frequent errors. **Who** is subjective (**Who** stole my watch?), while **whom** is objective (From **whom** was my watch stolen?). Case errors can also get a little tricky in some instances:



In this example, the word **malfunctioning** is a gerund, or a verb used as a noun. The subject of a gerund is always in the possessive case. In this sentence, the subject of this gerund is **microscope**, which should be possessive, as in the following sentence:

Subject of the gerund is now possessive



The transmission microscope's malfunctioning caused the experiment to be delayed.

A better way of writing this sentence would be to put the gerund first as the subject.

The malfunctioning of the transmission microscope caused the experiment to be delayed.

Noun Clauses

You can think of a noun clause as a short sentence used in its entirety as a noun in a longer sentence. These constructions can be troublesome for technical writers because the individual elements of a noun clause are treated as elements of a complete sentence, while the entirety of a noun clause is treated as a single element in another sentence. Confusing? You bet it is. Consider the following two versions of the same sentence:

Version 1: The supervisor decided whom will go to the conference.

Version 2: The supervisor decided who will go to the conference.

At first glance, it might seem that version 1 with **whom** is correct, since **whom** is the object of the verb **decided** and should be in the objective case. But it is not that simple. The pronoun **whom** is not the object of **decided**. It is only part of the object. The entire clause that follows **decided** is, in fact, the object. Since **whom** is the subject of that clause, it needs to be in the subjective case, as in version 2, and should be **who**. Also, since the entire clause **who will go to the conference** acts as the object of the verb decided, the entire clause functions as a noun and is therefore considered to be a noun clause.

Compound Adjectives

Compound adjectives are often helpful and necessary descriptors used in technical writing, but they can be confusing when it comes to hyphenation. The simple rule is that if you use a compound adjective before the noun being described, it is hyphenated. After the noun, a compound adjective is not hyphenated.

Before the noun

We have on-site support.

Jane is a part-time employee.

After the noun

We have support on site.

Jane works part time.

As the authors were writing this textbook, they spent a lot of time trying to determine which words use hyphens. For example, the word “nonhyphenated” appears in one online dictionary, while “non-hyphenated” is used by another. Similarly, on the next page you will see the word “Nonparallel.” Or is it non-parallel? It depends on your source. These are “real-time” examples of the changing nature of language, and you should always follow your organization’s style guide.

Phrasal Verbs

Shut up! Do not look down on phrasal verbs. Just know that they tick off some people—at least when it comes to technical writing. A phrasal verb is a verb that requires another element, typically an adverb or a preposition, to make sense. For example, when you want someone to be quiet immediately, you do not yell “SHUT!” You yell, “SHUT UP!” It does not make sense without the preposition. But not all verbs need the prepositions often combined with them. We can tell you, “Stand,” and you know what to do. You do not need us to say “up.” (This, of course begs the question, what is the context? Are you talking to someone who is sitting on a frog, or are you talking to someone who is holding a weapon? In the first case, “stand” suffices. In the latter, you would probably say, “Stand down.”)

All that said, while phrasal verbs are grammatically correct, you have probably been taught somewhere along the way that you should never end a sentence (or phrase) with a preposition. That instruction (sorry, Mrs. Boeyink!), was incorrect. However, because most of the English-speaking world (check out that compound adjective!) believes that this is true, it is better to avoid ending a phrase with a phrasal verb. As we said before, perception is reality, and if people think you are wrong, you are wrong, even if you are right.

Parallel Construction

When writing lists, either in sentence or bullet form, it is important that each item of the list be constructed using the same pattern of speech. We call this parallel construction. Think of this like parallel parking—for all cars to maintain safety expectations, each car must follow the expected distance from the curb and distance from other cars. When we write a list, if the first entry starts with a verb, then all entries in the list should start with a verb of the same tense. If we start with a noun, then all entries should start with a noun. There is no rule about **which** part of speech you must use to start an entry, only that they consistently be the **same** part of speech.

If we do not park our car well and according to expectations, we have a fender bender. In writing, if we do not use parallel construction, we cause our reader's intake of information to be interrupted. Sometimes it is just a momentary pause, but in doing so, the reader loses some ability to efficiently process the information being shared, jeopardizing the likelihood that they will understand the meaning the author intended to convey.

Nonparallel constructed list:

- By the operators
- Safety protocol
- Saves \$5 million/year

Parallel construction of the same list:

- Operators will design the process
- Safety protocol will be implemented
- Savings will be \$5 million/year

This same list can be written in sentence form. “For this project, three things will happen: the operators will design the process, the safety protocol will be implemented, and the savings will be \$5 million/year.”

Bullets vs. Paragraphs

The economy of writing is especially relevant when it comes to format. There are many paragraphs in the world which could be effectively written as bulleted lists, and many lists which could be written as useful paragraphs. How and when does one decide which is more appropriate? Bulleted information is very effective at conveying meaning when the list is a collection of unrelated or disjointed bits. A good example of this is when you list your assumptions for a project. You **can** put these in a paragraph, but the flow suffers because they are usually completely unrelated. For example, we might assume the following three things:

- Employees understand English
- Cloud access to the software will be stable
- The grant for which we applied is awarded by January

If we write those three items in a paragraph, it is harder to absorb efficiently, because our brains waste time trying to find a connection which does not exist:

We made the following assumptions for this project, including that employees understand English, that the cloud access to the software will be stable, and that the grant for which we applied is awarded by January.

Another appropriate use of bulleted lists is when you have a set of instructions or tasks that must be performed sequentially. Work instructions are almost always written as a numerically bulleted list.

In general, paragraphs are better for connecting ideas, logic, and information. They are helpful for guiding the reader to a rational conclusion. A bulleted list is not the best tool for that purpose, but can be effective within a narrative.

English as a World Language

English is a world language; it is the native language of 335 million people, spoken in 110 countries, and has 1,500 million learners.⁶ There are native English speakers on every continent and in countries including India, Australia, Great Britain, Canada, and Uganda. That said, English in India is different from English in Canada, and again from English in Uganda.

Language is always evolving, and much like dog breeds around the world, the English language has and continues to evolve in distinctive ways wherever it is prevalent. No one English is better than another, just different, which is something to keep in mind when communicating across cultures. English is frequently used as the language of global business; even in countries where English is not the official or de facto language, you will often find that business meetings are conducted in English.

The United States and Great Britain have a very connected history, but even so, American English and British English have evolved in their own ways over the past two centuries. The biggest differences are in vocabulary, spelling, and punctuation. Considering that vocabulary is regionally specific in much smaller geographic areas than full countries, we simply note that authors should do their best to choose words which do not have inappropriate meanings for a global audience. If you are not sure, consider asking a co-worker or peer from the intended-audience country to provide feedback.

Using typically British spellings of words is not incorrect, but it can interrupt your non-British audience as they read. **Colour**, **grey**, and **labour** are common examples. For an American audience, you would have better flow in your text if you used **color**, **gray**, and **labor**.

When it comes to punctuation, if you are writing for a British audience, use British-English punctuation. This includes periods and commas on the outside of quotation marks. If you are writing for a mostly American-English audience, periods and commas go inside of quotation marks.

Proofreading

While software can automatically identify many writing/typing issues, it cannot find everything. It also depends on you as the author to recognize the concerns that it does identify and make conscious decisions about them. You should always run spell check and grammar check, but it is up to you to proofread your writing in its entirety, including spelling, punctuation, and grammar. If a document is critical, it is wise to have others proofread it as well (assuming they have legal/ethical permission to do so). **Proofread** does not just mean “read it again.” We have all experienced our own

inability to find errors in our own writing. There is science behind why (that is another book), but your brain sees what it intends to see, not what it actually sees. However, there are some tricks you can use to fool yourself, which will help you find your own errors.

- Reduce the width of the text window in which you are typing so that the paragraphs also change width. This causes words to move to new locations and allows your brain to find missing words and identify style issues.
- Cut and paste text from one software application to another (e.g., from email to a text editor). This has the same effect.
- Print what you have written on paper and read it.
- Read paragraphs in reverse order.
- Read your text slowly and out loud.
- Wait two hours (or two days) before proofreading.

Exercise 1

The following paragraph is mechanically correct, but the author made some very unwise stylistic decisions for a technical document.

- Can you find imprecise word choices?
- Can you find wordy (uneconomical) phrasing?
- If the requestor of the communication was the author's supervisor, and that individual required a clear and simple message, has the author done what they need to do to be successful?
- Who is the audience?
- What is the purpose?
- Do you need more context?

Recommendations for New Safety Protocol
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As a result of the very recent incident in which an employee experienced an unfortunate cut to the finger, and consequently had to miss several days of work, requiring about four hours of interviews with the local OSHA representative, we, the company will be implementing a new safety protocol for the power supply. Besides needing around three employees to help with the transition between fuel cells, we can never put too much water in the nuclear reactor.

Question: does this mean we SHOULD put a lot of water in the reactor, or SHOULD NOT put a lot of water in the reactor? Uh oh . . .

Exercise 2

Read the following short description of a piston assembly that has been written for the average technical audience.

- How many grammar errors can you find? Hint: there are more than fifteen and less than 25!
- Is this description written in an economical style? Are any sentences too long and is passive voice used inappropriately anywhere?
- Are any areas unclear?
- Does the description have adequate precision for the audience?

The Function of a Piston Assembly

An internal combustion engines piston assembly converts thermal energy into mechanical energy. It's fuel is a volatile fuel-air mixture. An explosion in the top of the piston cylinder produced by Acme Enterprises workers. They are a great company who knows about power and explosions, etc. In fact, one of their advisory board did extensive work in weak acid cation (WAC) resins. Because of the costs, he developed a

process to regenerate hydrochloric acid and sodium hydroxide using exotic allows. It is no wonder that It has been said that they are the best at what they due.

The piston assembly, along with its associated subsystems, consist of a cylinder, spark plug, two valves, a piston, and a connecting rod. Its operation includes four stroaks: *intake*, *compression*, *power*, and *exhaust*. The fuel-air mixture is compressed to sum fraction of its original volume, at which point a 15,185 volt electrical charge ionizes the air within the plugs gap, causing a conductive path through which a spark travels, thereby igniting the volatile fuel-air mixture, thus causing it to burn rapidly, producing exhaust gasses that occupy 2000 times the original volume, creating a downward pressure on the piston head, and in the process, moving the connecting rod to turn a cam shaft, thereby producing torque. This entire stroke work together in the process of its operation.

Notes

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1. Jack Lynch, *The Lexicographer's Dilemma: The Evolution of 'Proper' English, from Shakespeare to South Park*. New York, Walker Publish, p. 275.
2. "Weird AI" Yankovic, "Word Crimes," <https://www.youtube.com/watch?v=8Gv0H-vPoDc>, posted 15JUL14, accessed 06DEC20.
3. *Eats, Shoots & Leaves*, Lynne Truss.
4. "O'Connor v. Oakhurst Dairy, No. 16-1901 (1st Cir. 2017)," <https://law.justia.com/cases/federal/appellate-courts/ca1/16-1901/16-1901-2017-03-13.html>. Accessed December 6, 2020.
5. "The Comma That Costs 1 Million Dollars (Canadian)," *The New York Times*, <https://www.nytimes.com/2006/10/25/business/worldbusiness/25comma.html>. Accessed December 6, 2020.
6. "A World of Languages." *South China Morning Post*. 27 May 2018. Accessed December 6, 2020. <https://multimedia.scmp.com/culture/article/SCMP-printed-graphics-memory/lonelyGraphics/201505A51.html>

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CHAPTER SIXTEEN

Documentation



Your technical writing teacher has just assigned a large research report that must have at least ten sources, not including Wikipedia. To save time and effort, you obtain everything you need from (where else?) Wikipedia. Then, using that information, you whip up a complete research report. Next, through quick searches on Google, DuckDuckGo, or Bing; Google News; and book summaries from your library's website, you identify multiple sources that somehow relate, even tangentially, to the topics about which you have already written. These become the “sources” that you reference in your paper to keep your teacher happy.

The Skewed Results of Search Engines

Some search engines use algorithms which return results based on personal information about us, including our physical location in the world, previous searches that we have done, our social media posts, and the type of device we are using (e.g., computer vs. cell phone). You have probably noticed the phenomena: you were looking online for bunny slippers to purchase, and all of a sudden, you see many ads for bunny slippers. While this might be helpful when it comes to online shopping, it also indicates that your online search results include a biased list in the first few pages of hits. For example, the biologist who is doing research on the impact of habitat on smallmouth bass fat will get a very different set of returns from the keyword search of “fat bass” than the sound engineer who does a search on the same two words. Consider this when you are conducting technical research; you can do a search on how to overcome this for your favorite search engine (inception!)—they are all different.

The problem with this approach—besides leading to a paper with inferior content and throwing all standards of individual integrity and academic ethics into the dumpster (see Ethical Considerations in [Chapter 2](#))—is that it

compromises the role of documentation in technical writing. It ignores the need for complete, accurate source citations in the most serious form of engineering and scientific writing.

Consider this: when you get a dog, documentation is critical. Whether you are adopting a rescue puppy from a shelter or purchasing a purebred puppy from a breeder, you need accurate information about the puppy's background and health. If the puppy has been given any vaccines or had any illnesses, you need to know. Honest, complete documentation is crucial to its survival. Sometimes puppies also have pedigrees—family trees that show the puppy's parents and grandparents and whether they have won any awards. If you are planning to show your dog, this information is also essential and must be complete and accurate. For most people, a family dog and even a show dog are not make-or-break business, but technical writing often involves big business, lots of money, and a competitive, unforgiving environment. Proper documentation is an essential element of technical writing—an element that can have serious legal, ethical, and credibility implications for those who fall short of the mark. Documentation requirements are something that any technical writer must take seriously (see [Figure 16.1](#)).

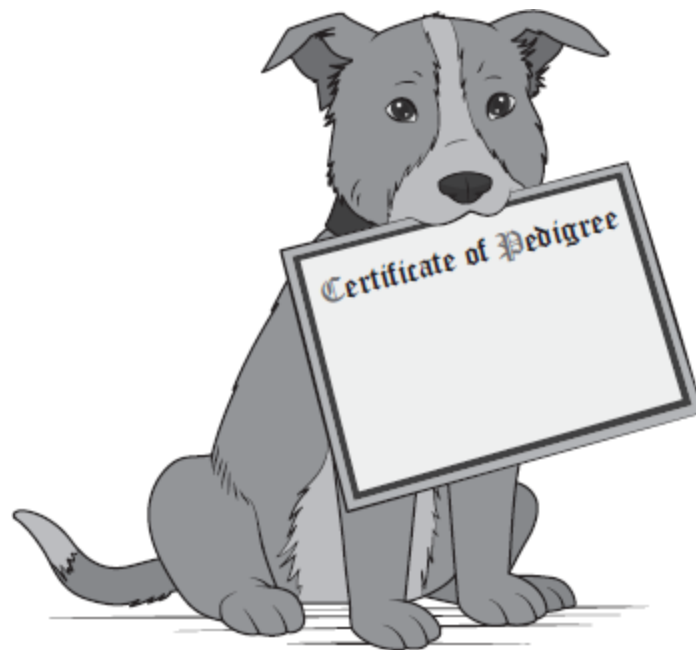


Figure 16.1

Accurate and complete documentation is critical to success.

What Is Documentation?

In its most general meaning, **documentation** refers to creating almost anything recorded or “documented,” whether electronic or on paper. All the documents described in this book are forms of technical documentation. However, this chapter focuses on the kind of documentation that involves referencing sources. In this sense of the word, documentation gives formal credit to a person, organization, or publication for an idea or information that either is not original or is not common knowledge of the field. It represents an acknowledgment of your indebtedness to the source.

For example, if you need to quote the performance specifications for, say, the Big City Water Treatment Plant, you must document the technical report that contains that information. Those specifications represent ideas that are not original—in other words, they are not your ideas. On the other hand, if you use Ohm’s law to show that 10 volts across a 1-ohm load produces 10 amperes of current, you do not need to document your source. Even though $I = E/R$ is not your idea, it is common knowledge in the field of electronics and, as such, does not need to be referenced.

Whether something is common knowledge of the field is often a judgment call. Normally, we think of something as being common knowledge when the average skilled person in the field should already be familiar with it. However, the best approach is to document any source when you are in doubt. Not only does this ensure that those who deserve credit receive it, but also it usually enhances the credibility of your writing by adding some authority to the argument. An example of this is shown in [Chapter 3](#) Note-taking where the authors included a lengthy quote from Christopher Proskey, who is an attorney at law and provided legal insight about the topic. By quoting him rather than rewriting his contributions, it provides professional authority for the chapter.

Your goal in providing documentation should be to give enough information about the sources you have used to enable your reader to find and consult those sources conveniently and independently. Sources typically include electronic media such as websites, newsgroups, newspapers, forums, journals, conference proceedings, and audiobooks; lectures and interviews; and traditional (read that as old-fashioned!) storage media such as CDs and DVDs, along with print media such as books,

periodicals, and dissertations. Some sources that are typically published electronically used to only be available in print media, including books, journals, etc. If you are looking for older documents and cannot find them online, you might have to look for the print version at a library.

Documentation Styles

There are many excellent and comprehensive online resources for documentation. One example is The Purdue Online Writing Lab (OWL).^{[1](#)} Other colleges and universities also have writing resources, and most of them will include links to external style guides, of which there are many. Here are a few: *MLA Handbook for Writers of Research Papers*,^{[1](#)} *The Chicago Manual of Style*,^{[2](#)} the *Government Printing Office Style Manual*,^{[3](#)} the *APA Publications Manual*,^{[4](#)} the *IEEE Reference Guide*,^{[5](#)} *American Chemical Society's Manual for Authors and Editors*,^{[6](#)} the *American Institute of Physics Style Manual*,^{[7](#)} and *Scientific Style and Format* by the Council of Science Editors.^{[8](#)} The best approach is to use the style guide specified by your employer, your teacher, or your field. Conferences, journals, and government grant submissions will usually require a certain format and will indicate this in their author's guide. If a style guide is not specified, you can use any consistent form of documentation that provides the necessary information, including the simplified method provided in this chapter.

When to Document Sources

As a technical writer, you should document sources for any of the following reasons.

To Meet Legal Requirements

Legally, when using copyrighted sources, you are required to document these sources. Federal copyright statutes control the reproduction of original works, including books, music, drama, computer programs, databases, videos, sculptures, and virtually any other media. Although copyright laws

do not specifically protect the ideas contained in these works, they do protect the expression of ideas by these works. Also, some works, such as photographs, contain ideas that are intertwined with expression and cannot be separated easily. You may use copyrighted material, either with permission of the copyright holder or without permission if your use is within the scope of the Fair Use Doctrine. Fair Use provides for limited reproduction of copyrighted material without the permission of the owner for noncommercial, teaching, and research purposes. Fair Use also requires that the original work be fully documented (referenced). If you do not provide this documentation, then it is not Fair Use—it is a violation of copyright law.⁹

To Meet Academic Standards

Academic standards require that you document any nonoriginal ideas, except those that represent common knowledge of the field. You must document not only direct and indirect quotations, but also paraphrases or any other discussions that specifically refer to or include original ideas that are not yours.

To Establish Credibility

You should support your original assertions and conclusions, which are not based on common knowledge, when you can show complementing positions on the part of authoritative sources. The purpose here is to establish the credibility of your position. Avoid making unsupported assertions in technical documents. If your assertions are consistent with the ideas of a recognized authority or previous work, then document (reference) that authority or previous work to establish your credibility.

What Happens When You Do Not Sufficiently Document Sources

As we discussed in [Chapter 2](#) Ethical Considerations, different levels of negative impact from not citing sources fully and correctly can occur. Your college or university might have a process for academic dishonesty that includes warnings or worse. Your employer might react on a spectrum that

ranges from you being warned to being passed over for promotion to being fired and/or sued if you cause harm to the organization.[ii](#), [iii](#)

How to Document Sources

Generally speaking, two different approaches exist for documentation: **notational** and **parenthetical**. There are many bibliography software applications available that make managing references efficient, including EndNote, Zotero, and BibTeX. If you use an automatic bibliography generator, do not assume that it is correct; you must make sure that all sources are correctly referenced in the final product.

- **Notational documentation** places footnote or endnote superscript numbers in your paper at the point where you need to document a source. You include the actual source citation either as a footnote at the bottom of the page, or as an endnote at the conclusion of the report (or a subsection of the report). In this chapter of the textbook, you can see how both footnotes and endnotes are used.
- **Parenthetical documentation** places a source citation in parentheses in your paper at the point where you need to document a source. You include a list of references at the end of the report (or major subsection). The citations are keyed to that list of references by either number or author's last name. Most technical documents use parenthetical documentation because it is simple and effective. Parenthetical documentation also gives the reader more information at the point of the citation.

Parenthetical documentation consists of two parts: the list of references at the end of the report or subsection and the parenthetical references cited within the text of the paper.

- The **list of references** (also called **list of sources**, **sources**, **references**, **notes**, **works cited**, or, frankly, whatever sounds reasonable) is essentially a bibliography that provides specific information (author, title, publication, date) about the works used or considered by the writer. You can list the references alphabetically by the first significant

word, which is usually the author's last name. Or you can sequentially number your sources. If you list your references alphabetically, you can reference the source within the body of text using the author's last name. Numbering your sources will make things even easier because you can reference each source in the paper by its number in the list.

- The parenthetical references are inserted into the text of the paper as the source citations. In the parentheses, you first identify the source by either source number or author's last name along with the date of publication, and then you add the specific pages being referenced (if applicable—some sources do not have page numbers). For example, consider the following paragraph:

The SuperXLtube is a high-power transmitting tube normally used in class C radio-frequency applications. However, as research commissioned by the Village Bigs so clearly demonstrates, the tube can be used successfully as a power amplifier in class A audio applications as well. Its power, distortion, and signal-to-noise performance specifications are quite impressive in this regard (3, pp. 22–23) or (Yinburg 2020, pp. 22–23).

This particular paragraph needs a reference to the Village Bigs' commissioned research. Either of the two sample citations will work. The first example uses a source number, whereas the second example uses the source's last name—in this case Dr. Roberta W. Yinburg's last name because she is Big Enterprises' chief scientist who wrote the referenced report. Also notice that the year **2020** is included after Yinburg's last name. Including the year of publication provides the reader with more information at the point of the citation; the year can also help differentiate between references by the same author in the same list of references.

As mentioned, the sources themselves are listed at the end of the report. This list of references might look something like the following:

List of References

1. Pradeep Misra, "Order Recursive Gaussian Elimination," *IEEE Transactions on Aerospace and Electronic Systems*, AES-32 (January 1996), pp. 396–401.
2. Thomas S. Kuhn, *The Structure of Scientific Revolutions*, 2d ed. Enlarged. Chicago: The University of Chicago Press, 1970.

3. Roberta W. Yinburg, “Preliminary Report on the SuperXLtube in Audio Service,” TR-05-0022. Big Station, N.Y.: Big Enterprises, September 2020.

The parenthetical documentation in our example was either (3, pp. 22–23) or (Yinburg 2020, pp. 22–23). Notice that both point to the same source in our list of references. Whichever parenthetical format you decide to use, be consistent throughout the entire report. Do not list a source by number in one section and by author’s last name in another.

You might also want to add sources that you consulted but did not specifically use. The idea is not to pad your reference list, but to acknowledge sources that may have influenced your thinking even though you did not specifically cite them. These latter sources should be included in a separate list of references clearly indicating that they were consulted but not used. In other words, you might end up with a “List of Sources Consulted and Used” and a “List of Sources Consulted but Not Used”—or perhaps “Sources Cited” and “Sources Consulted.”

Again, it is best to use the style guide specified by your organization, your supervisor, your teacher, or your field. However, if you do not have a specified format, the simple approach provided in this chapter should be adequate for documenting the most commonly used sources in technical writing.

Of course, several different types of sources exist, and each type is handled differently in the list of references. The following examples will show you how to handle the most common types of sources.

Online Media

The following examples provide a general guide for documenting online media. Documenting online media, especially sources that exist solely in cyberspace, creates challenges for the technical writer. Sources range from specialized databases (e.g., Lexis-Nexis), to online data services (e.g., AWS), to forums and news groups (e.g., USENET), and to the mass of material available to everyone on billions of websites. In addition, there are wide variations in styles for documenting online media. You should always ask your teacher or employer if a specific style format is required.

When you document web sources, keep in mind the differences between cyberspace and traditional sources. For example, an internet address does more than just indicate where a source is located. It also provides the detailed electronic directions, including the precise path, for accessing the source. In effect, a cyberspace address not only shows you where the source is located, but it also takes you there.

Another problem with cyberspace documents is the lack of key information. The goal when you are documenting an online source (in a perfect world) is to provide the reader with all the relevant information about that source. But in cyberspace, the world is far from perfect. For example, the author's name may not be included on the web page, and the title may be missing or only included in a /title tag in the HTML code. Additionally, the universal resource locator (URL) or web address may be redirected from one page to another without your knowledge or any obvious indication. You may not be getting the information from where you think you are.

Finally, cyberspace addresses tend to be unreliable over time. Page 226
Websites come and go, and material on websites is “here today and gone tomorrow.” On some sites, web addresses are dynamically assigned to time-sensitive materials, then reassigned once the materials are archived. In other words, an address that works today may not work tomorrow—or even later today! Generally, the best advice when you are documenting a cyberspace source is to do the best you can do with the information available. Remember, your goal in documentation is to provide your reader with enough information so that they can independently locate the source. When all else fails, even providing partial information—perhaps a topic or an author's name—might be enough with modern search engines to track down the source. When documenting electronic sources, however, you are obligated, to the maximum extent possible, to include all the relevant information. In this case, the information would include the author, title, database record identifier (file name and path), medium, and date of posting (or date of access, if the posting date is unknown).

Example Documentation

Documentation is an exercise in precision, from the order of the content, to the number of spaces between items, to the placement of punctuation. We start with some examples of electronic media sources.

Large, Complex Website

Include author (if known), article title, host agency or organization, department or office (if relevant), web URL, and date of access:

1. “SOI Tax Stats – Corporation Complete Report.” Internal Revenue Service, <https://www.irs.gov/statistics/soi-tax-stats-corporation-complete-report>, accessed December 21, 2020.

In this example, the author is not known, so the reference begins with the title.

University Website

Include author (if known), article title, document title (if relevant), web URL, and date of access:

2. Kirkpatrick, Dan, “ISU Study Indicates Diet May Help Reduce Cognitive Decline,” Office of the Vice President for Research, Iowa State University, <https://www.research.iastate.edu/news/isu-study-indicates-diet-may-help-reduce-cognitive-decline/>, accessed December 21, 2020.

Online Forum

Include author (or topic), title, medium, site owner, complete network address, and date:

3. First Robotics Challenge, “Problems with FRC Packages using a 2019 Project.” <https://forums.firstinspires.org/forum/general-discussions/first-programs/first-robotics-competition/competition-discussion/programming-aa/java-ad/91713-problems-with-frc-packages-using-a-2019-project>, accessed December 21, 2020.

Source Code Repository

Include author (if known), URL, and access date. List the version if relevant, including the commit hash, if that level of specificity is required:

4. <https://github.com/wren-lang/wren>, accessed December 23, 2020.

Journals

Include the author(s), article title, journal, volume (and number if necessary), date, and inclusive pages. If the article is available online, include the web address and access date as well:

5. Brad Bryant and Marian K. Kazimierzczuk, "Modeling the Closed-Current Loop of PWM Boost DC-DC Converters Operating in CCM with Peak Current-Mode Control," *IEEE Transactions on Circuits and Systems*, Vol. 52, No. 11, November 2005, <https://ieeexplore.ieee.org/document/1528686>, accessed December 21, 2020.

Conference Papers

Include the author(s), paper title, conference or transactions information, date, and inclusive pages:

6. Leslie Potter, Richard Stone, and Devna Fay Popejoy-Sheriff. "Increasing Graduate School Enrollment of Female Industrial Engineers through CUREs." *2019 ASEE Annual Conference & Exposition, Tampa, Florida, 2019, June*. ASEE Conferences, 2019, <https://peer.asee.org/32960>, accessed December 21, 2020.

Computer Local Storage Media

(Computer Flash Drive, External Hard Drive, or Other Local Storage Device)

Include file name, medium title, medium type, version, series or ID number, and date:

7. "Readme," Apple Hardware Test—iMac, CD, version 2.0, 691-4199A, 2003.

Print Media Examples

The following examples provide a general guide for documenting the most commonly used forms of print media.

Books (Print or Audio)

Include the author(s), title, edition, city of publication, publisher, and date of publication:

8. T. Sudkamp, *Languages and Machines: An Introduction to the Theory of Computer Science*, 3d ed. New York: Addison-Wesley, 2006.

Encyclopedias

Include article, encyclopedia, edition, place of publication, publisher, date, and inclusive pages:

9. "Cancer, Ethical Issues Related to Diagnosis & Treatment," *Encyclopedia of Bioethics*, 3rd ed., New York: Macmillan Reference USA, 2004, p. 341–349.

Newspapers

Include author(s) if known, article title, newspaper name, date, section, and page(s):

10. James Gorman, "A Wolf Pup Mummy From the Ancient Arctic" *The New York Times*, December 21, 2020, sec. Science - Trilobites.

Note: In this case, the author is listed. If no author is listed, you would alphabetize the entry by the first significant word of the article title.

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Nonjournal Entries

Include author, article title, publication, date, and inclusive pages:

11. Harry Goldstein, "Irradiation Nation," *IEEE Spectrum*, August 2003, pp. 24–29.

Technical Reports

Include author, title, number, agency, place, and date:

12. C.K. Charles Spaniel, "Appropriate Protein Treats for Exceptional Canine Behavior – A Final Report," Dudley's Dog Treats, Big City, Iowa, March 2021.

Dissertations and Theses

Include author, title, degree level, school, and date:

13. Shih Tzu, "Assessing Early Intervertebral Disk Disease in the Toy Breeds," Ph.D. dissertation, University of Illinois at Urbana-Champaign, 2020.

Other Examples

Interview

Interviews include most situations where you pose questions to a source and receive answers. Interviews do **not** have to occur face to face or in real time. Include interviewee, method, topic, affiliation, place, and date:

14. Leo Finkelstein, Jr., Personal Interview, Topic: “The Ethics of Using the FinkelKICK for Self-Defense.” Office of the Dean, College of Engineering and Computer Science, Wright State University, Dayton, Ohio, April 21, 2002.

Lecture

Include lecturer, occasion, topic, location, and date:

15. Sigurdur Olafsson, IE 413 class lecture, Topic: “Continuous-Time Markov Chains,” Industrial and Manufacturing Systems Engineering, Iowa State University, Ames, Iowa, October 19, 2020.

Although we have provided examples of documentation, always ensure that you are following the documentation style requested by your audience or required by your organization.

Checklist for Documentation

Have I

- ☐ Determined if my audience required a specific citation style?
- ☐ Used source citations throughout the text keyed to my list of references?
- ☐ Documented all uses of copyrighted material?
- ☐ Documented all nonoriginal ideas that are not common knowledge?
- ☐ Referenced authoritative sources that support assertions I have made that are not otherwise supported?
- ☐ Used the prescribed method and form of documentation (if applicable)?
- ☐ Used consistent methods and forms of documentation?

Notes

- [1.](#) Modern Languages Association (MLA), *MLA Handbook*, 8th edition, 2016.
- [2.](#) *Chicago Manual of Style Online*, 17th edition, 2017, <https://www.chicagomanualofstyle.org/home.html>. Accessed December 21, 2020.
- [3.](#) Government Publishing Office (GPO), *Style Manual: An official guide to the form and style of Federal Government publishing*, 2016, <https://www.govinfo.gov/content/pkg/GPO-STYLEMANUAL-2016/pdf/GPO-STYLEMANUAL-2016.pdf>. Accessed December 21, 2020.
- [4.](#) American Psychological Association (APA), *Publication Manual of the American Psychological Association, Seventh Edition*, 2020.
- [5.](#) Institute of Electrical and Electronics Engineers (IEEE), *IEEE Reference Guide*, 2018, <https://ieeauthorcenter.ieee.org/wp-content/uploads/IEEE-Reference-Guide.pdf>. Accessed December 21, 2020.
- [6.](#) *The ACS Style Guide: A Manual for Authors and Editors*. Washington, D.C.: American Chemical Society, 1986.
- [7.](#) American Institute of Physics (AIP), *AIP Style Manual*, 4th edition, 1990, https://web.mit.edu/meu-office/communication/aip_style_4thed.pdf. Accessed December 21, 2020.
- [8.](#) Council of Scientific Editors (CSE), *Scientific Style and Format*, 8th edition, 2014, <https://www.scientificstyleandformat.org/Home.html>. Accessed December 21, 2020.
- [9.](#) For more information, see the United States Copyright Office: <https://www.copyright.gov/>. Accessed December 21, 2020.

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- [i.](#) The OWL provides free writing resources for students and instructors. <https://owl.purdue.edu/>. Accessed December 21, 2020.
 - [ii.](#) Abby Ohlheiser, "A CNN News Editor Was Fired for Plagiarism," *The Atlantic*, May 16, 2014, <https://www.theatlantic.com/national/archive/2014/05/cnn-editor-fired-for-plagiarism/371045/>. Accessed December 21, 2020.
 - [iii.](#) "Elias A. Alsabti vs. Board of Registration in Medicine," 1988, <https://law.justia.com/cases/massachusetts/supreme-court/volumes/404/404mass547.html>. Accessed December 21, 2020.

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CHAPTER SEVENTEEN

Visuals



When you are waiting—perhaps for class to start or while you are at the doctor’s office—how do you pass the time? Most adults pull out their phones and start scrolling. If you are looking at headlines, TikTok, or Instagram, what catches your attention the fastest? Very often it is a visual: a photograph, chart, or other visualization that represents the topic of interest. After seeing the visuals, you might be enticed to “dig deeper,” maybe reading the full story or watching the entire video.

What Are Visuals?

Visuals are presentations of ideas that exploit our sense of sight to communicate a large amount of data quickly and efficiently. You look at the pictures first because of their information bandwidth; you can quickly get more information from them than from reading the text, especially with the limited time available while you wait. You also look at pictures first because of their greater interest and (perceived) higher credibility.

Technical writing deals with complex topics in precise ways. It is not surprising that one of the most important tools for a technical writer is visuals, which includes equations, formulas, figures, diagrams, drawings, illustrations, graphs, schematics, maps, photographs, and tables. Whether you are showing an exploded view of a mechanism or plotting the regression curve from an experiment, you will find that visuals are absolutely essential for technical report writing.

Does the picture in [Figure 17.1](#) make you think of Lassie? You might never have seen a Lassie movie, but a picture of a Collie might still make you think, “Oh no, Timmy has fallen in the well!” With any visual, your mind immediately goes to the connotations associated it, whether it is

Bruiser, Elle's Chihuahua in *Legally Blonde*; the St. Bernard in *Beethoven*; the yellow Labrador in *Old Yeller* or *Marley & Me*; the pit bull in *John Wick*; the French Mastiff in *Turner and Hooch*; or even Clifford the Big Red Dog. Wherever these dogs appear, they get an instant reaction from their audience, which is exactly what you are wanting with your technical writing visuals as well.



Figure 17.1

When you see a picture of a Collie, do you think of Lassie? While it's cliché, an effective visual **can** save you 1000 words.

General Guidelines for Using Visuals

There are some general guidelines for using visuals.

- Include visuals in a technical paper only when you have a reason to do so. If you do not know why you are putting a visual into a paper (“to make it look pretty” or “add visual interest” are not valid reasons to include a visual), you probably do not need it.

- If you include a visual, you must refer to it in the text discussion **prior** to its placement in the report. If the visual precedes its reference, the reader will wonder why it is there. This interrupts the reader and causes the message to get lost.
- Number and title all visuals. Tables have their titles shown at the **top** of the visual. Everything else has its title at the **bottom** of the visual. A good mnemonic for remembering this is “Tables on top; figures follow.”
- Ensure all visuals directly clarify or otherwise enhance the Page 232 text discussion. You must integrate them into your report, not just stick them somewhere. That means the labels and captions used in a visual should match the text discussion that refers to the visual. For example, if you are describing the negative terminal of a diode, do not call it the **negative terminal** in the text and the **cathode** in the visual.
- Document your visuals when they contain copyrighted information or represent borrowed ideas. Because visuals often get separated from the report, do not rely solely on notational or parenthetical documentation; include a source line with the visual itself, usually under the visual’s number and title.

Guidelines for Design of Visuals

Reproducibility

Design each visual expecting two things:

1. That it will eventually be shared in grayscale. If your report will be printed or duplicated in a single-color ink or toner, consider that fact when you are developing graphs and diagrams, which are particularly susceptible to confusion caused by the absence of color. Be especially wary of different colors that might look great on your computer monitor but could print with exactly the same shade of gray or even blend into the color of the paper on which the report is printed. Also consider that a significant proportion of the population are “color blind,” meaning they have deficiencies in their ability to distinguish

colors.¹ Because of these things, a safe approach for visuals is to use pattern fills in combination with or instead of colors. Consider also how specific colors will work in different mediums. While yellow might be an effective print-medium color, it can be very ineffective in a projected presentation format (see [Chapter 18](#) Presentations).

2. That it will be shared without any written context. Whenever you create a visualization of data, include enough information within the visual itself so that it can stand alone without your contextual information surrounding it. For example, be sure to include a title within the visual itself, label axes, include units, and ensure that color references are not required to interpret the visual.

Simplicity

Remember that the purpose of visuals is to supplement and clarify the information you are presenting. Some concepts (such as precisely modeling the entire supply and distribution process of a large retail organization) are so complex that they do not lend themselves well to detailed visual presentation. Other concepts might need to be divided into smaller components for effective presentation. For example, to explain the operation of a laser printer, you would probably want to include separate visuals showing each part of the printer process (charging electrode, electrostatic plate, scanning laser, toner reservoir, transfer roller, fusion roller, cleaning pad, and paper transport mechanism), as opposed to a single visual of all these components.

Accuracy

Ensure that every visual in your technical communication accurately portrays the information being presented. We briefly addressed photo alteration in [Chapter 2](#) Ethical Considerations; most people are very aware when they go out of their way to alter a photo that they are changing the factual representation. However, many people do not realize that software makes it very easy to distort data without even realizing one is doing so. Consequently, it is critical to understand that just because software allows you to create juxtaposed graphs, altered axes, and 3-dimensional effects or allows you to replace people in photographs (e.g., replacing the image of a white person with a Black person in a sea of white faces to imply diversity

where there is none) does **not** mean these are good choices. Many available visualization options are never good choices.

Assuming that you would never purposely try to distort your message without clearly stating what you were doing and why (refresher—[Chapter 2](#) Ethical Considerations),² this chapter addresses several important things to consider so that you don't **unintentionally** distort your message. Once again, there are other excellent, detailed sources that are worth consulting if you need more information than we address here, including Tufte's landmark, *The Visual Display of Quantitative Information*.³ While Tufte is a source for what to do and how to do it, you can also quickly find examples of what not to do by entering the words "bad graph" into any search engine. In fact, there is a sub-culture of bad graphs and other bad visualizations on Reddit and other social media platforms. There are also parodies of bad visualizations, such as the *Journal of Irreproducible Results*.⁴ We do not want your visuals to be mocked on these websites!

Discussing the accuracy of visuals is a little bit of a chicken-and-egg thing; to show examples of accuracy issues, we must show you examples of specific visuals that we have not yet discussed, but we cannot overemphasize the importance of ensuring that your data appear undistorted. Consider the following two basic graphs of the same data in [Figures 17.2](#) and [17.3](#): a comparison of the costs of two alternatives for process changes on a production line at Dudley's Dog Treats.

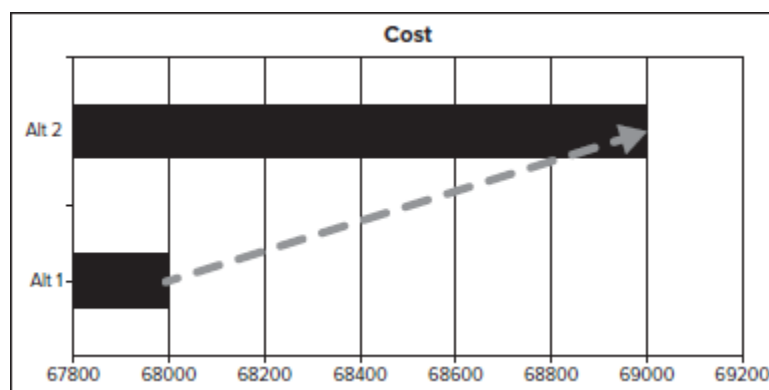


Figure 17.2

Process change alternative costs for Dudley's Line 1.

What Is Wrong with This Visual? (Hint: What Isn't?)

1. Truncated x-axis distorts difference between options
2. Both alternatives use same “color” and no pattern
3. Title insufficient for when visual pulled out of context
4. No units indicated
5. Trendline is nonsensical for categorical data
6. Font sizes are small
7. Delineations of x-axis are hard to use
8. Source of visual not included

See the huge difference in the costs in [Figure 17.2](#)? Alternative 1 is clearly better because the costs are much lower, right? Except...wait. Look closer. There is only a \$1000 difference between the two options (<1.5% difference). The data are distorted in [Figure 17.2](#) because of the truncated x-axis. If the comparison is made using an x-axis starting at 0 as in [Figure 17.3](#), we can see that the two options are very close to equal. This comparison does not yet consider the **context** of the situation, either. What if you are told that implementing Alternative 1 will take six years, while implementing Alternative 2 can be done by next week with 50% higher quality than Alternative 1. Suddenly, if we look at [Figure 17.3](#) and compare the cost of the two alternatives with the appropriate context, that “huge” difference is negligible at best—and deceptive at worst.

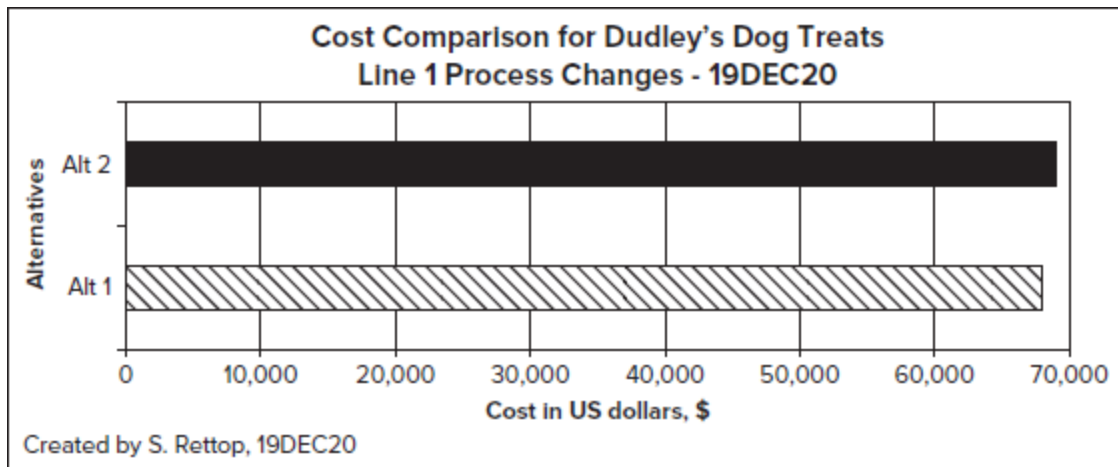


Figure 17.3

Process change alternative costs for Dudley's Line 1.

Let us examine another way that data are frequently and unintentionally distorted. While we will suggest later in this chapter that you should avoid pie charts altogether, you sometimes will find that marketing or management wins that decision and pie charts **will** be used. In this case, you **must** be willing to draw the line: **never include 3-dimensional pie charts**. With the advent of Microsoft Excel and its convenient and accessible graphing options, 3-dimensional (3-D) charts are all too commonplace. But these quick visuals often tell their audience the wrong message. Note in [Figure 17.4](#) how significant the front wedge of size 23 is. Clearly it is larger than the back wedge of 24. But how can that be? And the front wedge of size 12 is at least double if not more than the size of the wedge behind it which is size 7. Neither of these observations makes any sense, but this is what happens with 3-D charts. One way to avoid distorting data is to never use 3-D charts of any kind. If you look at [Figure 17.5](#), the relative sizes of the wedges appear to visually match their labels; this is a better choice if you must use a pie chart—we will discuss the usefulness of pie charts in a bit.

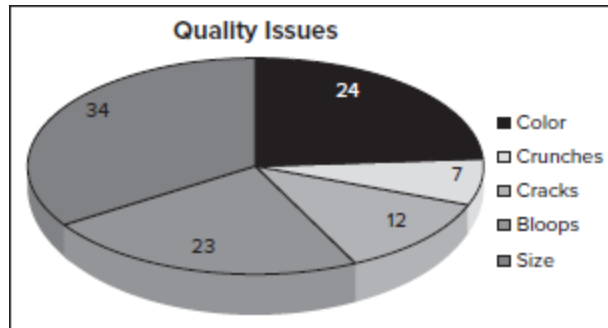


Figure 17.4

Quality issues for Dudley's Super Massive Treats on Line 1

What Is Wrong with This Visual (Hint, What Isn't?)

1. Distorted data representation by 3-D
2. Color value differences are insufficient to distinguish in black and white
3. Numbers in the graphic are difficult to read
4. Legend is hard to match
5. Title is insufficient when visual is pulled out of context
6. Source of visual not included

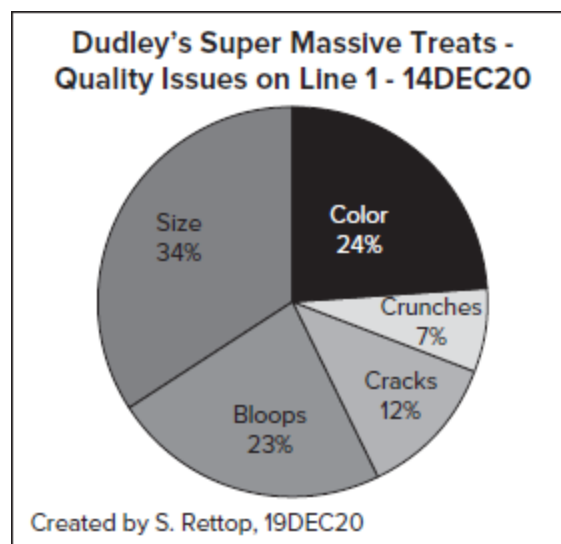


Figure 17.5

Quality issues for Dudley's Super Massive Treats on Line 1. While this is a better pie chart than that in [Figure 17.4](#) because the data are not distorted, you should still avoid pie charts as much as possible.

In general, do not exaggerate the impact of data by manipulating scales or misrepresenting relative sizes. However, in certain documents, when you are clearly presenting a biased point of view, some visual enhancement of information is acceptable. For example, in a proposal, where you are selling your skills or your organization's ability to do a particular task, it is expected that you might highlight a certain part of a chart or enhance the detail of a particular feature. But that does not mean it is acceptable to lie about, or misrepresent, your strengths or weaknesses.

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Types of Visuals

Visuals generally fall into one of the following categories:

- Equations and formulas
- Diagrams
- Graphs
- Schematics
- Tables
- Images
- Fonts (not visuals by themselves, but essential for readability)

The following sections provide a brief discussion of each of the types of visuals. Note that we use the synonyms of **graph**, **chart**, and **plot** interchangeably in this textbook.

Equations and Formulas

In technical writing, equations and formulas fall into one of two categories: **mathematics** and **chemistry**.

Equations and Formulas in Mathematics

Equations and formulas in mathematics are symbolic expressions that represent a rule, statement, principle, or a relationship between two

expressions. Equations and formulas follow the same formatting rules. They are frequently written on a separate line and numbered sequentially. The symbols in the expressions should have been previously defined or should be defined immediately below where the equation or formula appears (Example 17.1). If not too large, equations and formulas can be integrated into the text (Example 17.2).

By convention, all variables used in equations are italicized; however, abbreviations, such as logarithmic or trigonometric functions (e.g., log, tan, cos, and sin) are **not** italicized in equations and formulas.

Numerals are also not italicized, while **any** term or symbol that is being defined **is** italicized.

Example 17.1 Mathematical Equations and Formulas

When working with antennas, you can calculate the additional loss caused by standing waves using Equation 1:

$$L_A = 10 \log [(a^2 - p^2)/a(1 - p^2)] - \text{dB} \quad (1)$$

Where

L_A = additional loss

a = 10dB/10

p = (SWR-1)/(SWR+1)

dB = line loss in dB when perfectly matched

SWR = standing wave ratio at the load

Example 17.2 In-Text Mathematical Equations and Formulas

When working with antennas, you can calculate the additional loss caused by standing waves using the equation, $L_A = 10 \log [(a_2 - p_2)/a(1 - p_2)] - \text{dB}$, where L_A is the additional loss caused by standing waves, a is equal to $10\text{dB}/10$, SWR is the standing wave ratio at the load, p is $SWR - 1$ divided by $SWR + 1$, and dB is the loss in dB when perfectly matched.

Terms like **corollary**, **definition**, **theorem**, **rule**, and **proposition** are often set in large and small caps when they introduce a specific corollary, definition, theory, rule, or proposition, as shown in [Example 17.3](#). The text of the corollary, definition, theory, rule, or proposition is italicized.

Example 17.3 Format of Terms

Rule 1. *The teacher is always right.*

Law 2. *Entropy will increase in any closed system.*

Definition 3. *RADAR is the use of electromagnetic energy to detect and track reflective objects.*

Formulas in Chemistry

Formulas in chemistry are symbolic representations of atoms, molecules, and/or chemical reactions, where the equal sign or arrow separates the symbols or quantities of reactants from the products (Example 17.4). Normally, numerals and elements in chemical expressions are not italicized.

Example 17.4 Formulas in Chemistry

In fuel cells, hydrogen (H) ions are combined with oxygen (O) ions to generate free electrons, and in the process, water (Equation 2).



Notice that in this example, the individual elements were defined in the referring text. They also could have been defined immediately after the expression as follows:



where

H = hydrogen

O = oxygen

H₂O = water

In [Example 17.5](#), the chemical formulas for glucose and water have been totally integrated into the text of a sentence.

Example 17.5 In-Text Chemistry Formulas

To rehydrate the patient suffering from gastrointestinal malaise, an IV of glucose (C₆H₁₂O₆) and water (H₂O) was started.

Diagrams

Diagrams are drawings that show the components of a mechanism, the steps of a process, or the relationship among parts of a system. Make diagrams only as complex as necessary. Diagrams can provide normal, cross-sectional, or exploded views of a mechanism. [Figure 17.6](#) provides a normal, external view of the Quantum Central Processing Unit (QCPU)

discussed in [Chapter 12](#). This view would be useful for describing the external physical attributes of the device.

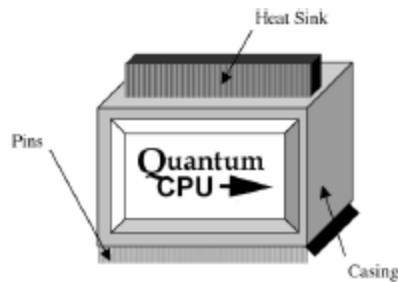


Figure 17.6

QCPU external view.

External View

Cross-Sectional View

[Figure 17.7](#) provides a 2-dimensional cross-section of the same device. This view would be useful for describing the internal parts of the QCPU and perhaps the process of its operation.

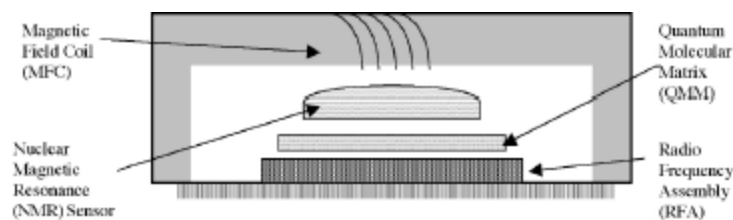


Figure 17.7

QCPU 2-D cross-section.

Exploded View

[Figure 17.8](#) provides a 3-D, exploded diagram of the same device. This view might be useful for describing the physical attributes of the internal structure, providing assembly instructions, or describing the process of its operation.

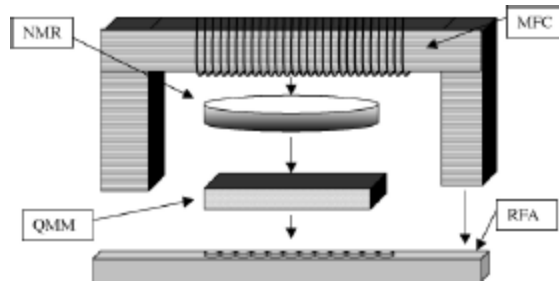


Figure 17.8

QCPU exploded 3-D view.

Graphs

Page 239

Graphs, which are also called charts or plots, are visual representations either of relationships among sets of numbers or of quantities and proportions of mathematical values. Graphs are great for presenting statistical information. Generally, the five types of graphs you will work with are **line charts**, **scatter charts**, **bar and column charts**, **Gantt charts**, and **pie charts**.

Line Charts

Line charts effectively show trends in data which have an intrinsic relationship, often time or distance. Normally, the vertical (y) axis is used to plot values of one or more dependent variables, and the horizontal (x) axis labels the independent variable. Each line plotted shows the changing value of a specific dependent variable. [Figure 17.9](#) provides a line chart showing the 5-year stock value trends for the fictitious Fink-Au-Pot-Products International Corporation (ticker symbol FINK).

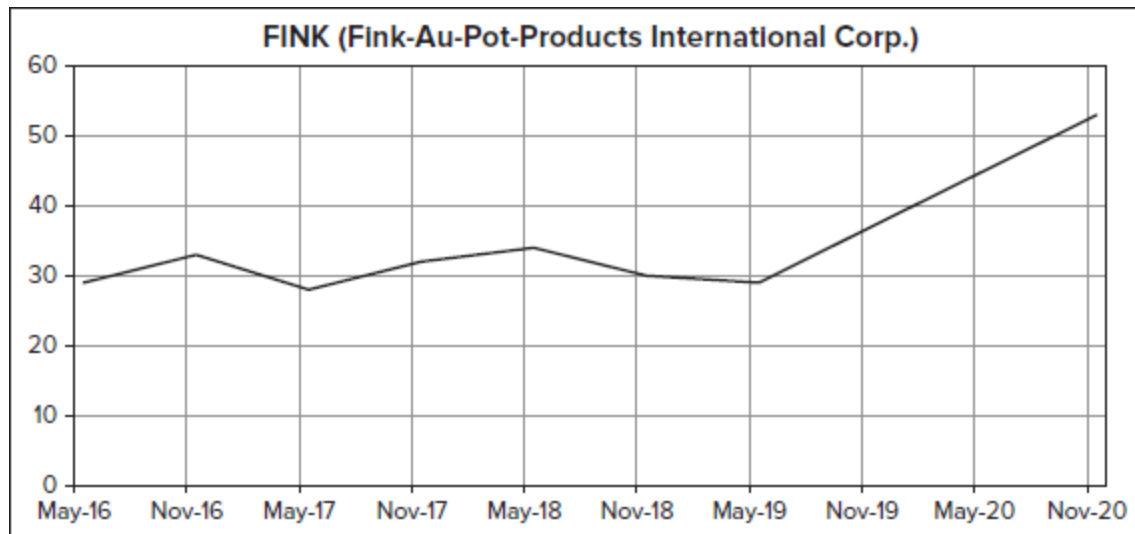


Figure 17.9

Line chart of critical point value trends adjusted for stock splits.

Source: "Powerhouse Companies," *Hyper-Investor Newsletter*, December 2020.

Scatter Charts

Scatter charts look like line charts in some ways but are typically used with two independent variables which do not have an intrinsic relationship. These charts are used to show correlation. [Figure 17.10](#) is an example of fictitious but reasonably realistic Havanese dog weights and heights. From the data points, you can see there are short overweight dogs and tall underweight dogs, along with all the ones who are proportional.

Adding Visual Interest to Line, Scatter, Bar, and Column Charts

In some situations, you might need to add visual interest to a chart to get your reader to look at it. A simple way to do that is to add a graphic or photograph to the chart. **Under no circumstances** should you ever use 3-dimensional charts.

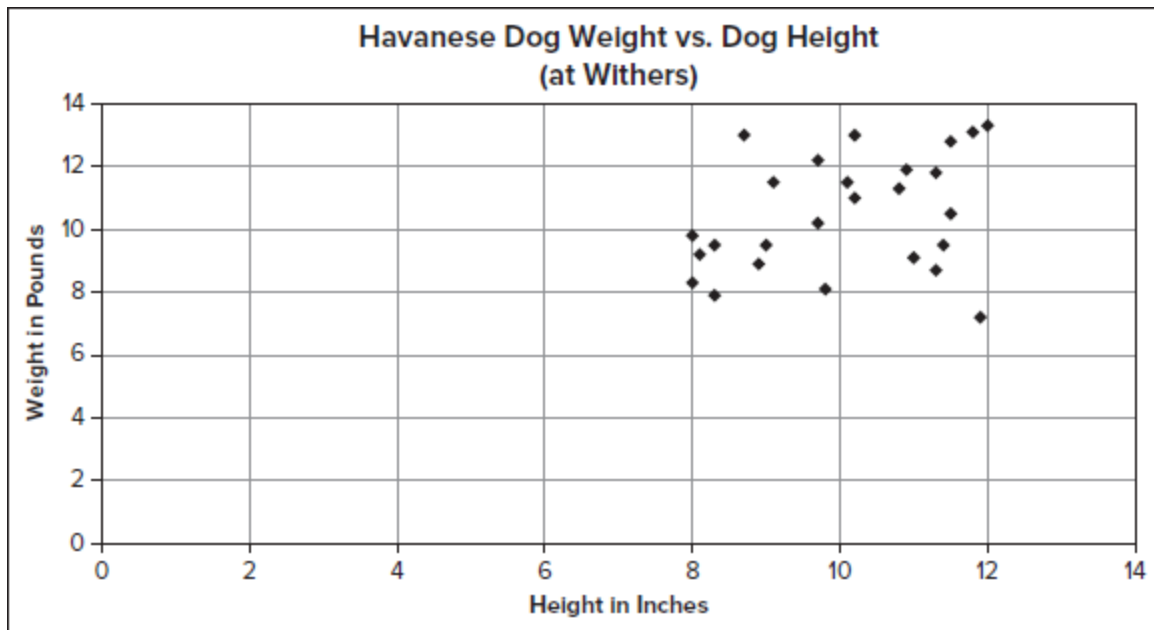


Figure 17.10

Scatter chart of Havanese dog weight vs. height as measured at withers.

Source: RandomDogData.com, accessed December 19, 2020.

Bar and Column Charts

Bar and column charts provide excellent tools for comparing discrete variables. Since each variable is a separate entity, a line chart would not be appropriate because no slope actually exists between data points. In other words, categorical data should **never** include a trendline. Normally, you use a bar chart ([Figure 17.11](#)) when the value labels are too long to conveniently fit on the horizontal axis. You use the column chart ([Figure 17.12](#)) when the value labels are short enough to fit well on the horizontal axis. Note that both figures provide discrete (and, unfortunately, fictitious) data regarding the authors' parentally preferred student behaviors.

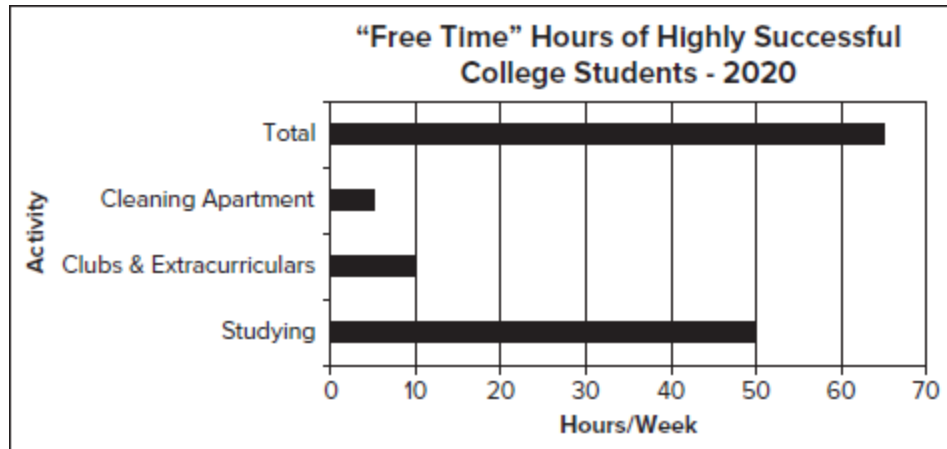
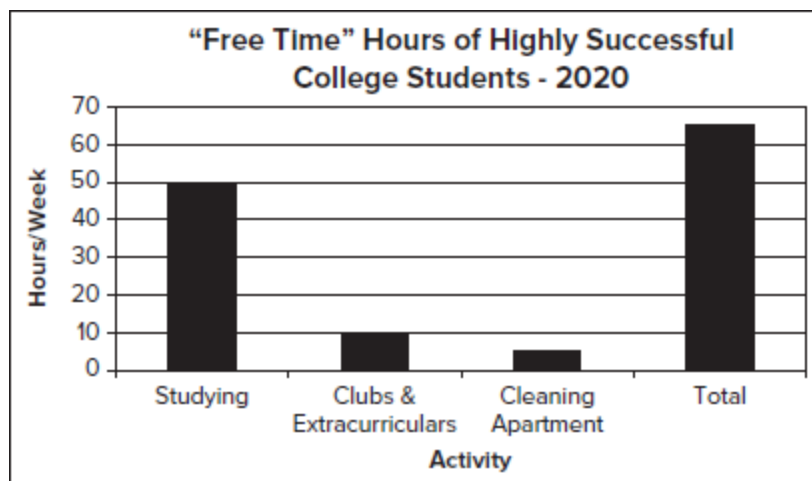


Figure 17.11

Bar chart of successful students' weekly time management.

Source: MomIsHappy.yay, accessed December 19, 2020.



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Figure 17.12

Column chart of successful students' weekly time management.

Source: MomIsHappy.yay, accessed December 19, 2020.

Note how the horizontal bar arrangement provides a larger space on the left for the category axis, and how the vertical column chart can be more intuitive than a horizontal chart since we normally think of larger amounts as moving up, not over.

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Gantt Charts

Gantt charts are specialized horizontal bar charts used as a management and production tool to track tasks and events over time ([Figure 17.13](#)). The graphical illustration they provide can be useful in coordinating, planning, or tracking specific tasks.

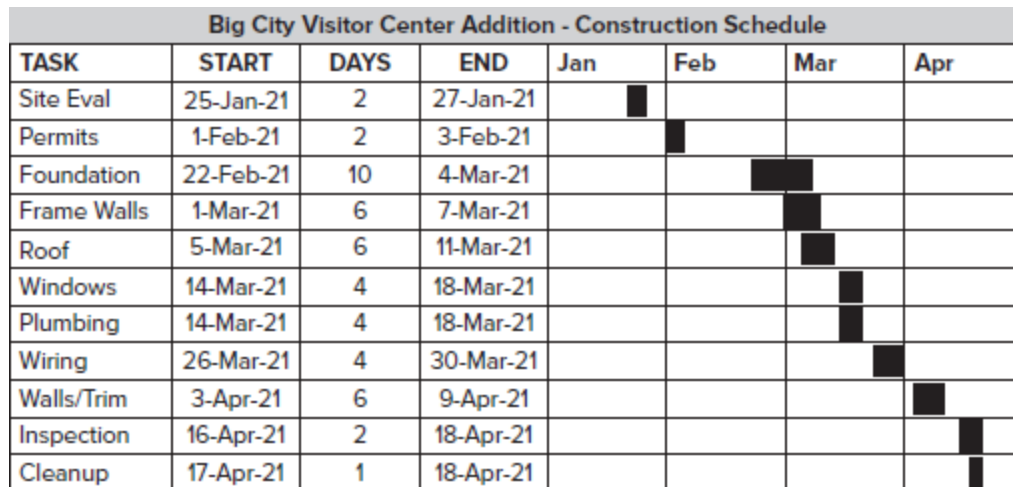


Figure 17.13

Gantt chart of construction schedule. Note that this chart is a hybrid; it integrates tabular data and a bar graph.

Pie Charts

There is extensive debate about the usefulness of pie charts. Typically, they are used when someone wants to create “eye-candy,” or communication that is more about catching an audience’s attention with color and glitz than effective messaging. We recommend that you avoid using them, as pie charts do not provide the same degree of visual precision as line, bar, column, or scatter charts. Under **no** circumstances should you ever use 3-dimensional pie charts. If you are unable to convince others to avoid pie charts, refer to [Figure 17.5](#) for an example of how best to format a 2-dimensional one.

Pictographs

Sometimes the most effective way to present technical information is to turn it into a pictograph. [Figure 17.14](#) is a pictographic chart of the authors’ fictitious students. This particular chart is analogous to the bar chart in

[Figure 17.11](#) and the column chart in [Figure 17.12](#). Instead of bars or columns, [Figure 17.14](#) uses alarm clocks as analogs for the values of each category. Obviously, this chart lacks the precision of either the bars or columns, but it is more interesting visually. In this chart, the clocks replace the distance analogs of a bar or column graph.

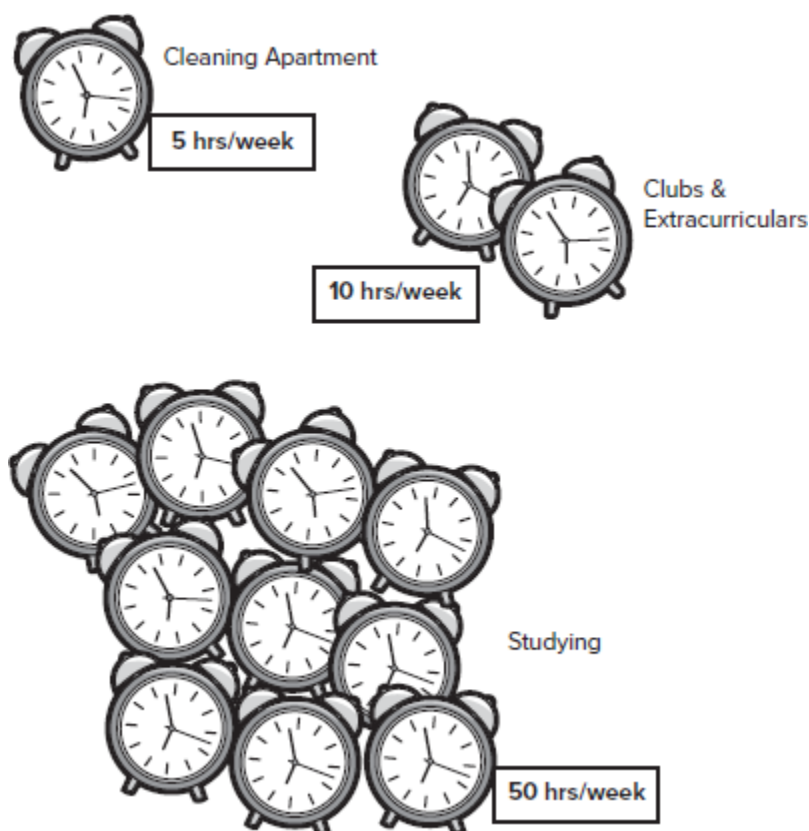


Figure 17.14

Pictograph of successful students' weekly time management.

Source: MomIsHappy.yay, accessed December 19, 2020.

In the chart below ([Figure 17.15](#)), a pictograph is used to portray what is basically a listing, but with much greater visual interest. In this chart, the train cars constitute items in a list.

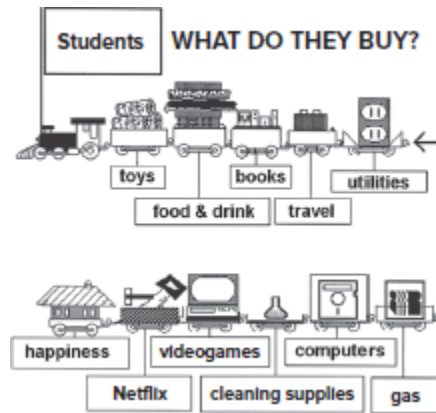


Figure 17.15

Pictograph of a purchase list.

Schematics

Page 243

Schematics visually represent a system's structure or the procedures involved in a process. For example, the flowchart in [Figure 17.16](#) schematizes (and oversimplifies) the writing process that one might use to produce a report.

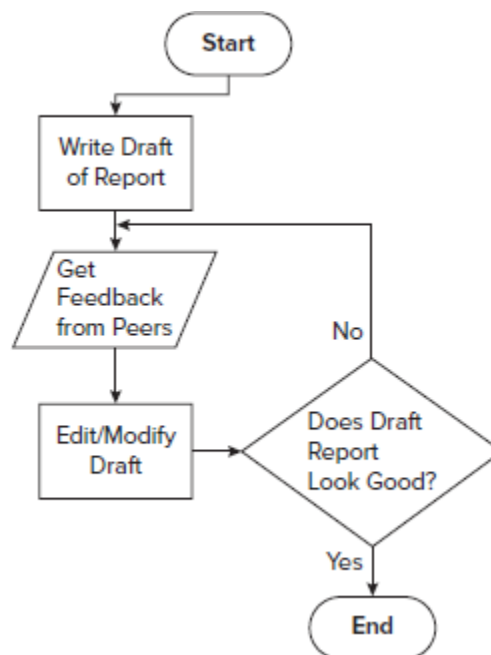


Figure 17.16

Schematic of report-writing process.

Schematics are also used frequently to provide circuit diagrams, such as the power amplifier circuit in [Figure 17.17](#), or like the schematic of a fuel cell complete with chemical equations in [Figure 17.18](#).

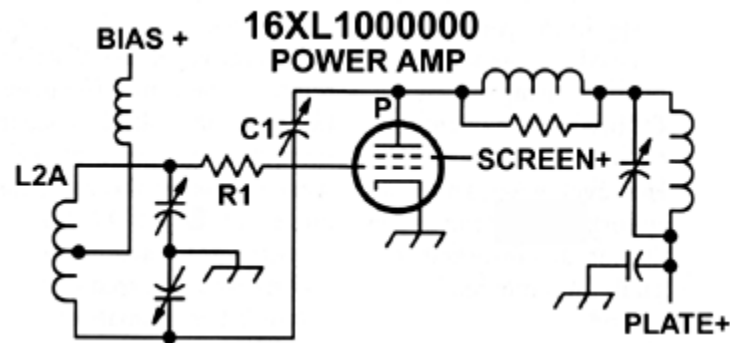


Figure 17.17
Power amplifier circuit.

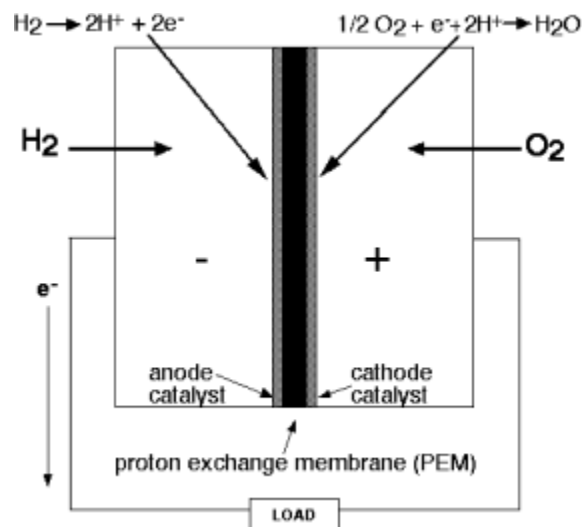


Figure 17.18
PEM fuel cell schematic.

Note how [Figure 17.18](#) schematizes the fuel cell assembly and integrates the chemical reactions that are occurring into the diagram.

Tables

Tables are orderly arrangements of data and information in columns and rows. Tables are the most precise way to display an array of data. They can also be used for a wide range of technical writing applications, from simple listings to complex troubleshooting spreadsheets. [Table 17.1](#) provides, in tabular form, the precise hourly data on which [Figures 17.11](#), [17.12](#), and [17.14](#) are based.

Table 17.1 Successful students' weekly time management.

Source	Hours/Week
Studying	50
Clubs & Extracurriculars	10
Cleaning Apartment	5

Source: MomIsHappy.yay, accessed December 19, 2020.

[Table 17.2](#) shows a troubleshooting chart for setting up wireless connectivity on a computer, while [Table 17.3](#) provides the same information presented in [Figure 17.15](#).

Table 17.2 Wi-Fi troubleshooting table.

Problem	Causes and Remedies
Speed slow	Test internet speed, change channel on router, reset router
Nothing can connect	Try Ethernet cable, reset router, replace router
Recent update broke Wi-Fi	Remove update and wait for patch

Table 17.3 Two-column tabular listing of students' purchases.

Students: What Do They Buy?	
Toys	Gas
Food	Computers
Drink	Cleaning Supplies
Books	Videogames
Travel	Netflix
Utilities	Happiness

Images

Photographs

Photographs are visual reproductions, typically encoded as digital data, but in some cases recorded on film emulsion, that accurately reproduce the appearance of objects or events. They can add interest, credibility, and extreme visual detail to a technical report. Photographs are high-bandwidth tools that can give your reader lots of credible, nonverbal information quickly.

For example, consider how photographs are used in the Page 245 following brief discussion of lava flows.

Lava is a type of igneous rock created when molten rock called **magma** reaches the surface through volcanic action. Certain volcanoes, known as **shield volcanoes**, produce large amounts of magma (2100°F). Magma pouring from the volcano often creates expansive flows of hardened lava. One of the most active shield volcanoes with extensive lava flows is Kilauea on the island of Hawaii.

Kilauea tends to produce two main types of lava: **Pahoehoe** and **‘a’a**. Pahoehoe lava is formed when a relatively shallow flow (1 to 2 feet thick) of magma cools and starts to harden. The crust that is formed in Pahoehoe lava can have a smooth, glassy surface; or it can have a ropelike, wrinkled texture caused when the hardening surface, which cools first, is dragged and pushed by the molten lava below. On the other hand, ‘a’a lava flows can be several yards thick and are characterized by rubble that has broken off from the flow’s leading edge, forming a cascade of debris (see [Figure 17.19](#)).

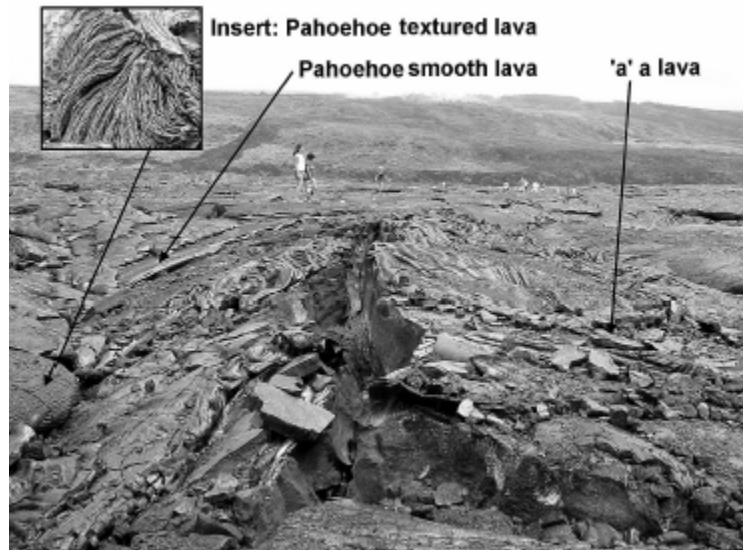


Figure 17.19

Kilauea lava flow.

Leo Finkelstein

The photograph adds a dimension to this discussion that allows the reader to better understand and appreciate the nature of lava flows. For one thing, looking at the photograph better enables the reader to visualize and appreciate the size and scope of the flow. Also, the photograph clarifies the appearance of the different types of lava, effectively complementing the text description.

Photographs can be used readily to enhance almost any technical presentation. Image processing software also provides an effective means of easily enhancing, editing, and modifying images.

We have two precautions for the use of photographs in technical documents:

- First, make sure each and every photograph serves a purpose relevant to the topic at hand. Do not just include a photograph because it is easy to do or because you think you need to pad the document (a favorite practice of some students).
- Second, when you are working with digital images, always ensure they have adequate pixel density for the purpose at hand. A density of 2.1 megapixels is adequate for normal use, while a density of at least 3.2 megapixels is preferred when you are enlarging images to fill a large

part of a typical 8.5" × 11" page. (The author shot the photo in [Figure 17.19](#) with a 2.1-megapixel digital camera while exploring the Hawaii Volcanoes National Park.)

Image Alteration

Page 246

With computer and phone hardware and software, it is simple to combine, edit, and modify high-resolution images down to the pixel level. This capability can be used not only to enhance photographs, but also to “modify” reality—in effect, to create false images. For example, one could replace specific people’s images in a crowd to imply diversity of race, ethnicity, culture, or gender; replace the signs behind someone’s image to change location; or even change someone’s shape to appear larger or smaller than they truly are.

Photographic editing and composition are powerful tools that can enhance the process of technical writing, but any changes or edits must be ethical and acknowledged. Any edited image is usually, to some extent, a false image; however, that does not mean such images cannot and should not be used. On the contrary, they can be employed in the technical communication process if they are used properly. The appearance of objects can be altered to clarify a point or even to show something that normally cannot be seen (see [Figure 17.20](#)).



Figure 17.20

Altered image of a lightning bug inside of a glass container. The light emitted by the insect is enhanced to make it easier to see.

Anita Patterson Peppers/Shutterstock

Acknowledging Alteration

Digital imaging and the power of alteration can raise serious ethical issues. To avoid ethical problems, when the alteration of an image has fundamentally changed the information it represents, always acknowledge in the caption or title of the image that the image has been altered. Minor adjustments such as contrast, brightness, or hue are not normally acknowledged because they usually do not change the information that the image represents. For more on ethics, see [Chapter 2](#).

Non-Photographic Images

The same rules and guidelines about making and acknowledging alterations apply to non-photographic images. For example, at the time we are writing this textbook, the State of Iowa's Public Health Department is running a billboard campaign to highlight the importance of vaccination. The

billboards are using cartoon images of viruses—more specifically, measles, flu, pertussis, HPV, and mumps—smiling and excited with the caption “Armed & Dangerous. Get Vaccinated!” It would be unethical to take that billboard and change the words to something like, “We Make Your Immune System Stronger” or “Don’t Vaccinate. We didn’t harm your grandparents; we won’t harm you.”

Process Limitations

As mentioned earlier, when you are developing visuals, it is important to keep in mind the limitations of the intended output or production process for the document. These limitations are particularly important when you are using photographic images that will be printed. Since the photos in [Figures 17.19](#) and [17.20](#) were intended for a single-ink production process, the original color information was discarded, and the photographs were converted to grayscale. This not only reduced the file sizes of the photographs, but also allowed a more accurate preview of how they would appear when printed. Remember, reproducing photographs requires that grayscale or color information be rendered properly in the final document, and this requirement alone often dictates special treatment and additional costs in the printing or duplication phase.

Fonts

Visuals should be designed with audience in mind; this includes making careful decisions about the fonts in both your visuals and in your document. Choose visuals and fonts that increase audience comprehension rather than indulge your creative whims.

Some fonts are easier to read than others. For example, **serif** fonts are easier to read than **sans serif** fonts (“sans” means “without”). Serif fonts have extra curves and flourishes that make discerning the difference between letters easier, and thus, make the text easier to read (see [Figure 17.21](#)). Use serif fonts for blocks of text that you want your audience to read and consider, but use sans serif fonts for headings that you want your audience to use as visual cues for sections of text (things that you want your audience to easily find when they visually scan documents). Serif fonts include Times New Roman and Cambria; sans serif fonts include Arial and Calibri. Fonts can create or undermine your credibility, depending on the

level of professionalism they connote. Comic Sans, for example, is perceived as being childish and should not be used professionally. Consider the following comparisons in [Table 17.4](#).



Figure 17.21
Difference between serif and sans serif fonts.

Table 17.4 Examples of font choices.

Font type	Example of font using the first paragraph of “Fonts”
Times New Roman	Visuals should be designed with audience in mind; this includes making careful decisions about the fonts in both your visuals and in your document. Choose visuals and fonts that increase audience comprehension rather than indulge your creative whims.
Arial	Visuals should be designed with audience in mind; this includes making careful decisions about the fonts in both your visuals and in your document. Choose visuals and fonts that increase audience comprehension rather than indulge your creative whims.
Comic Sans	Visuals should be designed with audience in mind; this includes making careful decisions about the fonts in both your visuals and in your document. Choose visuals and fonts that increase audience comprehension rather than indulge your creative whims.

Font size is measured using the term “point” (abbreviated as pt). There are two things to consider when choosing font size. First is that the font size should be large enough to be easily legible, but not so large that it makes a bad impression. For example, compare the differences in [Table 17.5](#).

Table 17.5 Comparison of font sizes.

Size	What it looks like	Impression it gives
8 pt	Visuals should be designed with audience in mind; this includes making careful decisions about the fonts in both your visuals and in your document. Choose visuals and fonts that increase audience comprehension rather than indulge your creative whims.	Writer does not want reader to be able to see what is written.
10 pt	Visuals should be designed with audience in mind; this includes making careful decisions about the fonts in both your visuals and in your document. Choose visuals and fonts that increase audience comprehension rather than indulge your creative whims.	Professional, unless the font type is too small at 10 pt.
12 pt	Visuals should be designed with audience in mind; this includes making careful decisions about the fonts in both your visuals and in your document. Choose visuals and fonts that increase audience comprehension rather than indulge your creative whims.	Professional, unless the font type is too large at 12 pt, in which case maybe the author does not have enough to say and is making the document longer by increasing font size.
14+	Visuals should be designed with audience in mind; this includes making careful decisions about the fonts in both your visuals and in your document. Choose visuals and fonts that increase audience comprehension rather than indulge your creative whims.	Author is manipulating font size to increase the length of the document.

Size is not universal across all fonts; that is, fonts are not all the same size at 10 pt and 12 pt. For Times New Roman, the most common and legible size is 12 pt, but for Cambria it can be as low as 10 pt. Just be aware of font size and choose accordingly.

In addition to choosing the appropriate font size for reading text, you should also choose the appropriate font size to reflect the hierarchy of headings. The title of the document should have the largest heading, the section headings the next largest, the sub-section headings the next largest, etc.

When you want to emphasize content, whether a word, phrase, or entire sentence, different methods are more effective than others. Capitalization reduces the visual differences between letters and underlining can connect the lower edges of letters together. The use of all capitals and/or underlining makes text difficult to read. Both are also relics of typewriter days, when letters were spaced far enough apart that all caps and underlining did not connect the individual letters.⁵

VISUALS SHOULD BE DESIGNED WITH AUDIENCE IN MIND; THIS INCLUDES MAKING CAREFUL DECISIONS ABOUT THE FONTS IN BOTH YOUR VISUALS AND IN YOUR DOCUMENT. CHOOSE VISUALS AND FONTS THAT INCREASE AUDIENCE COMPREHENSION RATHER THAN INDULGE YOUR CREATIVE WHIMS.

Visuals should be designed with audience in mind; this includes making careful decisions about the fonts in both your visuals and in your document. Choose visuals and fonts that increase audience comprehension rather than indulge your creative whims.

Visuals should be designed with audience in mind; this includes making careful decisions about the fonts in both your visuals and in your document. Choose visuals and fonts that increase audience comprehension rather than indulge your creative whims.

If you must emphasize content with formatting, consider using **bold** (and only bold!).

Conclusion

Visuals are a key element of modern technical writing. They aid immeasurably in describing complex topics precisely by providing a large amount of information.

Normally, we think of visuals as equations, formulas, diagrams, graphs, schematics, tables, and images, but font has also become a very important part of visual design. Equations and formulas in mathematics are symbolic expressions that represent a rule, statement, principle, or a relationship between two expressions. Formulas in chemistry are symbolic representations of atoms, molecules, and/or chemical reactions. Diagrams are drawings that show a mechanism or its components. Graphs, however, display and represent sets of numbers, values, quantities, and proportions. Schematics are visual representations of the structure of a mechanism or process, tables display the orderly arrangement of data or other information

in columns and rows, and images or photographs provide accurate visual reproductions. Fonts affect readability and require choices about type, size, and formatting.

Checklist for Visuals

- ☐ Have I identified my audience and if a visual will help them understand the message?
- ☐ Have I determined if there is context that must be considered or provided for the visual I want to use?
- ☐ Have I selected the best visual for the kind of information I am presenting?
- ☐ Have I accurately displayed the information?
- ☐ Have I numbered and titled each visual?
- ☐ Have I documented the source, if necessary, using a source line with the visual?
- ☐ Have I integrated the visual with the text discussion that references it?
- ☐ Have I referred to the visual prior to its placement in the report?
- ☐ Have I used labels in the visual that match the terms used in the text?
- ☐ If I altered a photograph, have I acknowledged that fact in the visual's caption or title?
- ☐ Have I considered the output publication process in my design of each visual?

- Have I considered readability and chosen fonts that facilitate audience understanding of my message?

Exercise

Look at the chart below ([Figure 17.22](#)) that models a typical half-life for 12.5 mg and 25 mg doses of the drug *Meclizine* taken at different intervals throughout the day. How well does it meet the checklist requirements? Also, assuming the line graph is technically correct and is designed for publication in this book, what is wrong with this chart and how could you improve it? Focus on the presentation and design deficiencies.

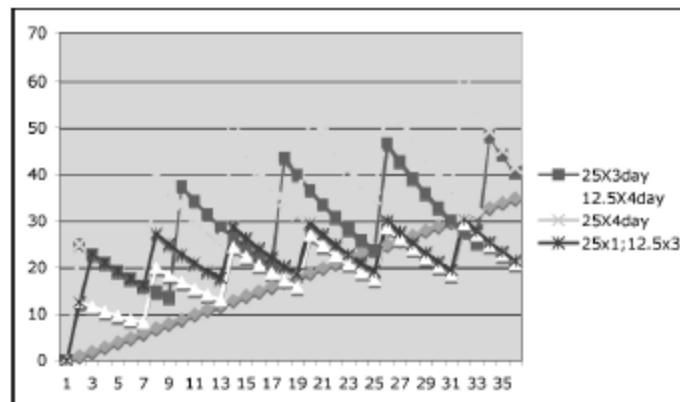


Figure 17.22

Meclizine half-life models.

-
1. National Eye Institute, "Color Blindness," July 3, 2019, <https://www.nei.nih.gov/learn-about-eye-health/eye-conditions-and-diseases/color-blindness>. Accessed December 17, 2020.
 2. For a primer on the tricks that people intentionally incorporate into their technical writing and data visualizations and how to spot them, see the seminal, 60+-years-old book, "How to Lie with Statistics" by Darrell Huff, W. W. Norton & Company, 1993.
 3. Tufte, Edward. *The Visual Display of Quantitative Information*, 2nd ed. Graphics Press, 2001.
 4. Article explaining the *Journal of Irreproducible Results*, https://en.wikipedia.org/wiki/Journal_of_Irreproducible_Results. Accessed December 16,

2020.

5. Another relic of old-fashioned typewriter days is including two spaces after a sentence, which is no longer necessary because word processing software automatically adjusts the spacing for us.

==

CHAPTER EIGHTEEN

Presentations



Question: what characteristics do you use to describe an engaging presenter? We know you have thought about it, or at least have been subjected to a range of presenter engagement in your lifetime. Perhaps energetic, enthusiastic, and fearless come to mind? How about intelligent, capable, and keen? No one enjoys “death by PowerPoint.” We all appreciate speakers who can keep our attention as they share information. It turns out that these adjectives are also some of the many used to describe the winningest dog breed of Westminster: the Wire Fox Terrier, or Foxie. In general, Foxies do not enjoy boredom, a sentiment with which most audiences would concur. Just as Foxies are not born with an ability to win dog shows (they must work at their showing skills!), most people are not born with outstanding public speaking skills. It is important to understand that presenting involves skills that **can** be learned and improved (see [Figure 18.1](#)).

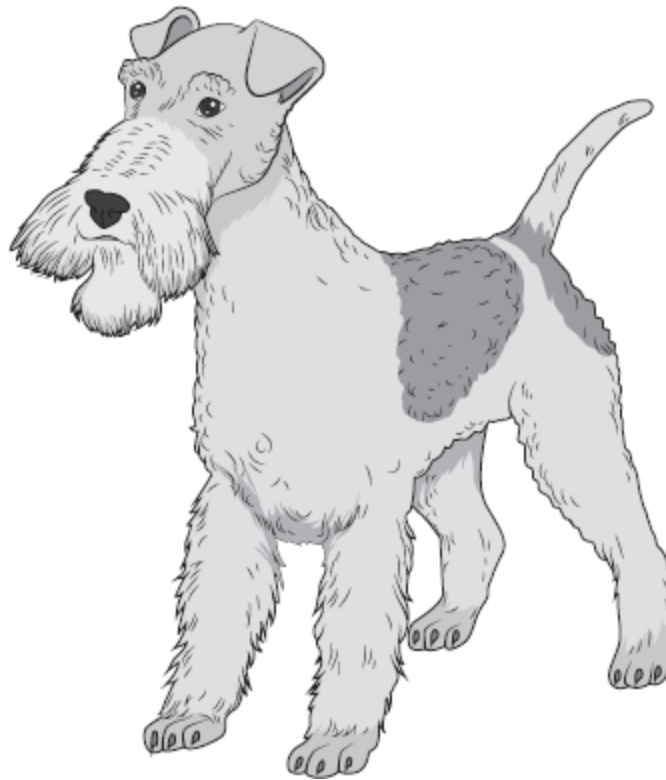


Figure 18.1

Developing excellent presentation skills takes practice.

The competitive world of science, technology, and business is built around technical presentations. There are “shows” every day, and you will be the star of some of them! Increasingly companies rely on their technical experts (you!) to present information to various audiences. Topics of such presentations can be quite varied, from plans and programs, to goods and services, to advanced theories and concepts. The audiences can range from your coworkers in a staff meeting to thousands of people in an international video conference. This chapter can only begin to touch on a topic that has whole courses, degrees, and professions dedicated to it, but effective presentations is a topic critically important to your success.

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What Are Presentations?

Presentations are interpersonal performances in which concise technical information is provided to an attending audience. These performances,

whether live or posted online as recorded video, are augmented by visuals and other media. Making effective technical presentations is not easy. Doing so requires a creative imagination, a bit of courage, technical expertise in the subject, and the ability to communicate accurately and effectively.

There are **many** kinds of presentations. While most do not have specific names, there are some that do. You might have heard of an **elevator speech** (or pitch), an **update report** (also called a briefing), or a **keynote address** (the most famous example might be a TED Talk¹). Generally speaking, these three presentation types span the formality spectrum. Each of these types has their own audience and purpose, and of course, their own situational context for every specific occasion to consider as well (see [Table 18.1](#)).

Table 18.1 Typical presentation considerations and examples.

TYPICAL CONSIDERATIONS		ELEVATOR SPEECH	UPDATE REPORT	KEYNOTE ADDRESS
Formality		Informal	Either	Formal
Who presents		You	You; a team including you	You
Audience	Audience size	1–3 people	5–20 people	20–2000 people
	Audience type	Various—customers, operators, managers, suppliers, instructors	Team members, peers within a department or discipline, clients, operators	Often people who share knowledge/appreciation of your expertise
Purpose	Potential purpose examples	Explain to an operator why you are collecting data; convince a recruiter to give you an interview; share with the CEO why the company should invest in new test equipment	Present to classmates about a project; update your engineering department on project progress; help a client understand why product changes are necessary; explain to shop floor workers why accreditation is happening	Inform technical professionals about trends in the industry at a conference; deliver division or company quarterly update; present at a professional meeting
Situational Context For Formulation	Medium	In-person, synchronous	Usually in-person, synchronous, but can also be online synchronous or video asynchronous	In-person synchronous, online synchronous, or online asynchronous
	Delivery	Impromptu	Extemporaneous	Preferably extemporaneous, but depending on criticality of information details, could be manuscript
	Visual aids	Not usually	Often slides, video, etc.	Depends
	Typical length	1–3 minutes	5–20 minutes	20–40 minutes
	Sitting or standing?	Either	Either	Standing
	Technology	None	Range from none to high tech using slides, video examples, etc. and if meeting is online, using video conference software	Slides

Examples of situations where you might give a presentation Page 253 include the following:

- For a course (~30 audience members, presenter standing, using slides)
- In a graduate degree defense for a faculty committee (5 audience members, presenter likely standing if in person/sitting if in video conference, using slides)
- In a 1-on-1 conversation with a shop floor employee explaining about process changes related to product redesign (1 audience member, presenter standing, no slides)

- In a small group roundtable with peers updating them on research findings (5 audience members, presenter sitting, possibly using slides or screen sharing, either in-person or online video conference)
- At a weekly department meeting (10–60+ audience members, presenter sitting, using slides)
- At a plant-wide or division quarterly update meeting (~200 audience members, presenter standing, using slides)
- For a current customer (5 audience members, presenter sitting, online using slides and screen sharing)
- For potential customers (unlimited audience members, presenter sitting/standing, asynchronous video posted online)

When you are asked by your supervisor to make a technical presentation—and sooner or later you will be—do not despair. There are ways that you can channel anxiety so that it will help you give a good presentation. Before you focus on that, though, the following will help:

- Establish who your audience is. Who are they? What do they know? What is their stake in your presentation? What are their fears, concerns, and needs?
- Identify the presentation's purpose and required message. Determine what you need to happen after you are done with your presentation. What do you want your audience to do?
- Identify context. What is the situation? What factors outside of your control could affect your message?
- Determine the content. What information will your audience require to take necessary action? What information could help alleviate adverse situational context issues?
- Identify the formulation requirements. Is the presentation live or recorded, in-person or video conference, formal, time-constrained, software specified, etc.?
- Organize ideas clearly and effectively for **your audience's understanding**. Remember that communication is about how the audience understands and perceives your message.

- Tailor the presentation and its message to the audience and purpose at hand.
- Rehearse the presentation **multiple** times.

The big secret (which is no secret at all!) to effective technical presentations is to consider audience appropriateness for five key factors:

- Substantive ideas with precise, accurate information
- Clear, coherent organization
- Terminology and concepts
- Effective supporting media (see the General Guidelines section later in the chapter)
- Professional performance

Substantive Ideas

The first consideration when you give a technical presentation is to have something worth presenting. In technical presentations, the subject matter and purpose are usually well defined. You might present a proposal on behalf of your company, the status of your project, the results of a laboratory study, or information from a research project. In every case, the purpose and topic will dictate what substance should be included.

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Clear, Coherent Organization

Once you identify the message, organize the material so you can present it effectively. Live oral presentations differ from both recorded video and written reports in that live, spoken words are transient. If those watching your recorded video presentation or reading your writing have trouble following what you are saying, they can review it, think more about it, maybe ask someone else, and perhaps, in time, figure it out.

Those listening to a live oral presentation, however, do not have that luxury. What happens if the members of your audience get lost or cannot follow your line of thought? What happens is mostly bad.

- They might stop to think about it, in which case they will not be listening to what you are saying next.

- They might ignore what you said and try to keep up, in which case they will have missed your earlier point.
- They might just get frustrated and stop listening altogether, in which case they are no longer part of your audience.

Sometimes audience members can interrupt your presentation to ask questions, but often that is unlikely. Depending on the audience and purpose of the presentation, asking a question requires a tacit admission by the audience member, in front of their supervisors and peers, that they do not understand what everyone else **seems** to understand. Often audience members remain quiet in such situations, resulting in the exact opposite of what you want: to accomplish shared meaning of your message.

That is why, especially for live and recorded oral presentations, it is important to organize what you must say clearly and in a way that is obvious to your audience. Ideas—even difficult theoretical concepts—are much easier to follow and understand if they are organized logically and coherently. It is also critically important to note that the technical writing in presentations is not the same as that in written documents. For example, if you use a slide to present bulleted information, you will want to use 30 or fewer words and no punctuation. The key to good technical writing in a presentation is that it **supports you** as the deliverer of information and does not demand center stage.

Terminology and Concepts

Use words that are appropriate for your audience and type of presentation. Informal presentations can use informal language, whereas formal ones should be more “proper.” Never use words that your audience will not understand, or if you must use them, define them at your audience’s knowledge level. This is often harder than it sounds, depending on your anticipated audience. If you will be presenting to a peer group of engineers, you have some insight into the terminology with which they are familiar, but if you will be presenting to the general public, best practice is to assume familiarity at an eighth grade level of education. While readers can stop and look up something that they do not understand, if your listeners do not accurately comprehend your words as you say them, you will lose their attention. Of course, they may also misunderstand what you are saying. This situation can be worse than when the audience does not understand,

because now your listeners may believe they understand when, in fact, they do not.

Professional Performance

You have identified what to say, you have organized the material into a coherent and logical structure, and you have selected the proper language. Exactly how you deliver your presentation to the audience depends on the rhetorical situation. This is important. If you are speaking in an informal meeting, you might stand and talk from your position at the table, or you might even stay seated. If you are making a formal presentation, you will probably do so from the focal point of the room, like a lectern or head table. In some cases, you might even stand on a podium. If you are giving an update, you will probably have slides—which means confirming that the necessary hardware and/or software are available and compatible with your technology choices. For example, if you build your presentation slides on a Mac using Keynote, will they work seamlessly on a Windows computer with an older version of PowerPoint? Do not wait until the middle of your presentation with a potential client to determine this.

You might find yourself speaking impromptu without any Page 255 technology to a small group, or you might use a sound system with thousands of people in the audience. Often you will have a time limit. You might advance your own slides, or you might remotely signal someone else to change your slides, if, for example, the battery dies in your remote mouse. And, increasingly, you might speak in front of a video camera to many people in geographically remote locations.

Obviously, you must adapt your presentation to the situation at hand. However, some general principles apply to any presentation. First, make sure you look professional and act professionally. It is always better to err on the side of being too conservative; for example, if you are unsure if the situation requires business formal or business casual attire, choose business formal. If you are using slides, make sure they are professional in content and appearance. Some listeners may interpret inappropriate personal appearance or substandard slides as a lack of interest or technical capability on your part. Once again, mechanics errors play an oversized role in your audience's perception of you. You might be the smartest person in the room when it comes to flux capacitors used in time travel, but if you have

spelling or grammar mistakes on your presentation slides, you will quickly lose your audience's confidence.

Know how much time you have and use it appropriately. If you have a limit, do not go over it; in fact, save a minute or two for questions at the end of your time slot. Managers feel disrespected and sometimes get downright cranky when you go over your time limit—you are not doing yourself any favors. It is also important to be aware of your audience's attention span. Most adults lose focus after twenty minutes; if your presentation can be shorter than that, good! If it must be longer, try to incorporate changes in delivery that get your audience involved or require them to change how they are listening to you.

If you are giving your presentation via video conference, there are some additional things to consider. If possible, make sure that your background is neutral (e.g., avoid personal expressions of politics), and if possible, uncluttered. As a student, it is common to do your assignments from your dorm or apartment; consider hanging a sheet behind you to block your unmade bed and laundry strewn across the chair and close your closet doors. You should also do a test with the lighting in the room. It is best to have natural light in front of you, with no lighting behind you. Make sure that your Wi-Fi or internet connection is stable and fast enough to support your video conference platform along with any other software that you intend to use during the presentation. Familiarize yourself with the video conference software to be used: while Zoom, Google Meet, Microsoft Teams, and WebEx have similar features, each has its own nuances, and of course, requires a different download or application. Again, do not wait until the presentation is supposed to be happening to try to figure out how to share your screen.

If you are recording a presentation to share with others through either file dissemination or by posting on a website, consider the following. Your look, tone, material, and capabilities will be captured and available for posterity; it is the digital equivalent of putting yourself on a billboard along the highway in a never-ending presentation. What impression do you want to give others which will be accessible to anyone for the rest of time? You might think recording a video is private and will only be shared in your organization. The reality is that once created, digital media exists forever and because of a few malicious bad actors in the world, it can potentially be

hacked and shared. Always assume that videos of you will be public; let that inform how you comport yourself.

One final note regarding delivery of your presentation: it is normal to be nervous. To some extent, nervousness is a positive thing. When channeled, it can give you an edge that will make your presentation livelier. However, nervousness, when not controlled, can distract your audience and degrade your presentation. It can cause you to stumble over your words, lose track of time, move without purpose, and speak too rapidly. So, what do you do for nervousness?

First, know your material backward and forward. Be familiar with what you plan to say, as well as with the theory and details behind what you are saying. The odds are that in many cases, you will be the expert in the room on the topic you are presenting, so take comfort in the knowledge that you know more about your topic than anyone else.

Second, rehearse your presentation—on your feet (if you will be standing), in front of a camera (if you will be on video), and out loud—to the point where you are comfortable giving it. When you practice, the more sensory feedback you have, the easier it is to improve. For example, if you tend to “dance” while speaking (either shifting your weight back and forth or actually stepping back and forth without purpose), practice presenting while standing on a plastic bag from a store (bubble wrap works, too, but it’s more expensive!). This way when you move, your feet will “talk” to your brain, informing you that they are nervous and unsure of what they should be doing. Similarly, if you tend to use filler words when you present, practice in front of a mirror or record yourself with your phone. When you see yourself say “uh,” you will stop allowing yourself to do this. It will be uncomfortable at first, but you will learn to pause rather than generate unnecessary sounds. If you are using slides, change them as though you were actually giving the presentation. If you are presenting with a team, practice where each of you will be throughout the presentation so that you can avoid awkward crossovers and handoffs. Practice might not make perfect, but it helps build your confidence and capability.

Third, if possible, visit the physical location where you will Page 256 make the presentation. Make sure you know how to raise or lower the lectern; how to connect your laptop, flash drive, or to the cloud; and if needed, how to use the audio system. Knowing both your material and your surroundings will help control your nervousness.

Fourth, a backup plan will also help reduce nerves. It is a good idea to print and have a paper copy of your presentation which you can reference in the case of technology issues. Nothing inspires more confidence in an audience than the speaker who gracefully overcomes unplanned issues; this will only happen if you anticipate and plan for all the things that might go wrong.

Speaking Situations

In the technical world, you will generally see three distinctly different types of speaking situations. Here they are, along with a few hints for dealing effectively with them.

Impromptu

Impromptu means spontaneous. Picture this situation: You are a systems administrator working for a large company. You have been invited to attend the morning staff meeting. One of the vice presidents mentions hearing on the morning news something about a hacker breaking into a competitor's web server and substituting a clown's picture for that of the CEO. Everyone laughs—except your company's CEO. With a deadly serious look on their face, they turn toward you and ask you to describe the security measures you are taking to counter such threats. Everyone turns, looks at you, and waits. You are on!

You have found yourself in an impromptu situation, where you must talk intelligently on a complex topic with virtually no preparation time. Although such speaking would be highly unlikely in a formal situation, it does occur frequently in such informal settings as staff meetings. Clearly, this situation is risky. The wrong choice of words, topics, and arguments can be hazardous to your career. And if you say something obnoxious or unintelligible, people will not remember that you did so in an impromptu mode—only that you said something obnoxious or unintelligible. So here are a few suggestions for handling impromptu situations.

First, if you think there is a chance you will be put on the spot, think through what you will say in advance. Clearly, as a systems administrator,

you should have known about the hacker; and once you were invited to the staff meeting, you should have realized that the topic might come up.

Second, when you are unexpectedly put on the spot, do what forensic coaches teach their competitive speakers to do: buy some time to think about it. One way to do that is to divide the topic in some generic way. For example, everything has a past, present, and future. So, in this case, divide computer security into the past, present, and future—and talk, by way of introduction, about how it used to be in the days before firms hired cybersecurity engineers. While buying time, you should be able to gather your thoughts for what you are going to say next. Of course, this assumes you know what you are talking about. If you are not sure, it is best not to make up facts attempting to bluff your way through the presentation. If you get caught, you will lose your credibility, which is difficult to get back.

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Extemporaneous

Extemporaneous means planned but conversational. Extemporaneous speaking is the preferred mode for a technical presentation because the presentation is well prepared but not precisely scripted. We also call this delivery style “off the cuff,” again meaning you know what you’re talking about, and you have a planned presentation, but you follow an outline using your own unscripted words to explain the material.

Extemporaneous speaking can be effective in technical situations, but only if you know the information and have practiced the presentation. You can use note cards for facts and figures and outlines for the presentation, but the best notes are the slides that you show to your audience. Do **not** read from full pages of text. It is easy to get lost in a full page of text, and when reading it, you will often come across as stilted and insincere. Do **not** read full text from slides, either. Besides being boring (remember, “death by PowerPoint”), the audience does not need you if they can just read the slides.

Manuscript

Manuscript presentations are totally prepared in advance. When you give one of these, all you do is read the script, with gestures and inflection added at appropriate places in the manuscript for emphasis. Avoid these kinds of presentations in most business settings. They come across as insincere and

stuffy, and when lengthy, they often generate boredom and despair among those stuck in the audience.

However, manuscripts do have a place in technical presentations, especially where high precision in detail and word selection is essential. For example, technical presentations that are going to be translated into several languages need careful word selection, especially where international business and commerce are involved. Also, if you are making a legal statement on behalf of the company, it is usually best to read, verbatim, what the legal staff has prepared.

One more point: Never try to recite a presentation from memory. If you are distracted or have any kind of memory lapse, you will find yourself “hung out to dry” in front of an audience with nowhere to go and nothing to say.

Speaking Purposes

Knowing the purpose for a technical document is critical to how you write it. In the same way, knowing the purpose for a technical presentation is critical to what information you include and how you present it. Generally speaking, technical presentations are **informative**, **demonstrative**, or **persuasive**.

Informative

In an **informative presentation**, your primary goal is to give the audience facts and other information. Informative presentations often take the form of a background briefing, where no decisions are required, and no particular response is expected from the audience. A briefing on the characteristics of a new proton exchange membrane for fuel cells would be an informative presentation. As you might imagine, these presentations are relatively nonthreatening events that normally are considered low risk.

Demonstrative

The primary goal of a **demonstrative presentation** is to show the audience how to do something or how something works. Teaching students in a computer class how to access and use a simulation on the web would be

such a presentation. These kinds of presentations often require audience interaction. They also tend to depend on having tools, equipment, and materials available during the presentation. Consequently, while relatively nonthreatening to the audience, they can be higher in risk because of your dependence on the equipment working properly when needed. Test the equipment thoroughly **before** the presentation, understand what you are doing, and have a backup plan should something go wrong. Also, make sure you take all safety precautions, if warranted, so that you are not endangering anyone by your demonstration.

Persuasive

A **persuasive presentation** tries to convince the audience to make a particular decision or take some specific action. Often these presentations take the form of decision briefings, where you ask your supervisor to fund your project or approve an organizational change. Of all the technical presentations, these are potentially the highest risk because scarce resources, such as dollars and people, are often involved. Your presentation may be part of a zero-sum game; if you win, someone in the audience (one of your coworkers) loses. The best advice is to know your topic, have supporting facts and figures readily available, and keep your cool under fire.

Technical Updates—Combining Purposes

Technical updates are focused oral presentations that often combine purposes and use visual aids normally referred to as slides. These slides can be run from a laptop, a flash drive, a phone, or the cloud. They provide an outline of the presentation and, like technical documents, include words, illustrations, photographs, video, charts, and tables (see [Chapter 17](#) for a refresher on good visual design). They add visual interest and support transitions in the presentation.

General Guidelines for Effective Supporting Media

Use slides to punctuate the presentation with short phrases and visuals, but do not provide a manuscript on the screen for the audience to read. Here are a few tips for producing presentation slides:

- Take full advantage of software such as Microsoft PowerPoint or web-based applications like Google Slides for business and technical presentations. Such applications provide many sophisticated capabilities, yet they are usually simple to learn. Producing professional presentations, using standard templates and color schemes, is easy and inexpensive with the options commonly available. As we mentioned earlier, you can also use these programs to print backup paper copies (often called **hard copies**), in case the computer system goes down when it is your turn to speak.
- Make sure the slides to which you will be speaking will be readable in the room and on the screen. Normally, you will want to use at least an 18-point font, but you may need larger font if the room is larger than normal (if you can, stand in the back of the room to check the legibility and readability of your text well before you present). Avoid script and fancy fonts because they can be difficult to read.² Standard templates that come with software packages such as PowerPoint are not created equal; some are very cluttered or have difficult font defaults, so evaluate color schemes and fonts when you choose a template.
- The general rule of thumb for slide text is to keep the number of words below 30. Never read the content of a slide word-for-word to your audience unless you are quoting material for effect. Use short phrases (called **bullets**) on your slides, not complete sentences. Or sometimes better yet, use a single main idea as the title of the slide, and only include a visual to support this idea in the body of the slide.
- Pick your colors carefully. Lower-contrast combinations (such as light blue on darker blue, or white on yellow) may look fine on your video display and on quality projection systems, but they could be indiscernible with different hardware/software systems or different ambient lighting. Additionally, avoid light-colored fonts on dark backgrounds (inverted text) if you are unsure of the projection system and room. Under good conditions they can look dramatic; however, you will find that dark fonts on light backgrounds are

readable in marginal situations when inverted combinations are not. Keep in mind that stronger colors such as red can overpower some members of your audience, while weaker colors such as yellow may fade out on them.

Like technical documents, technical updates are straightforward and easy to organize. Typically, a technical update contains the following slides:

1. Title slide
2. Overview slide
3. Discussion slides
4. Summary slide
5. Reflection slide (optional)
6. Questions/contact information slide (optional)

Programs such as PowerPoint provide layout masters for the different types of slides used in a presentation. They also provide numerous color schemes and template designs, all of which can be customized with your own color preferences, artwork, and images. Pick a standard template and modify it, or develop your own specific color scheme and design. Settle on something you like, then use that look consistently throughout the entire presentation. Consistency helps to create a theme and ties your presentation together visually from beginning to end. Some organizations have standardized templates, which they require employees to use, so that logos and other high-level branding decisions are consistently maintained.

Title Slide

The **title slide** leads off your presentation by telling your audience the topic and any other pertinent information. If you are updating your peers in a departmental meeting, you probably will not be as formal about including information like your own name, position, and affiliation. However, if you are presenting to an external audience or to an internal audience comprised of people with whom you do not typically interact, you will want to include such identifying information, perhaps even including your email address and telephone number. [Figure 18.2](#) shows a sample title slide done in PowerPoint using a modified standardized template. The slide in this figure

features the logo and tagline of Dudley's Dog Treats. Notice that the name of the presenter, Sagan Rettop (the fictitious niece of the fictitious Lena McFinkel), has been included on this slide, along with her title and position.



Figure 18.2

Title slide.

Caia Image/Glow Images

Overview Slide

The **overview slide** lists the main topics to be discussed in your briefing. It is also called an agenda. It describes to your audience how your presentation is organized and what information will be included in it. Be sure to use only bullets on this slide, not complete sentences. [Figure 18.3](#) provides a sample overview slide. In this case, the bullets have been further reduced to three visuals, which are repeated on future slides as **signposts**. Signposts are used to help the audience follow along within the presentation and are helpful for brain-mapping the information being shared. In using such techniques, the likelihood of the audience understanding your message increases.



Figure 18.3

Overview slide.

Discussion Slide

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Discussion slides constitute the body of your presentation. Normally, you will have one or more discussion slides for each topic listed on your overview slide. We start by showing an example of what **not** to do in a slide in [Figure 18.4](#): never cut and paste the text from a report into a slide and call it done. This is bad for so many reasons; you do not want to read to your audience, they don't want you to read to them, and your points are buried in all of this text. Without reducing the information to key points, you have made yourself as the presenter irrelevant. Just email your report to your audience—they can read it without you doing it for them. As soon as this slide is shown, best case you have lost them, and worst case you have irritated them.

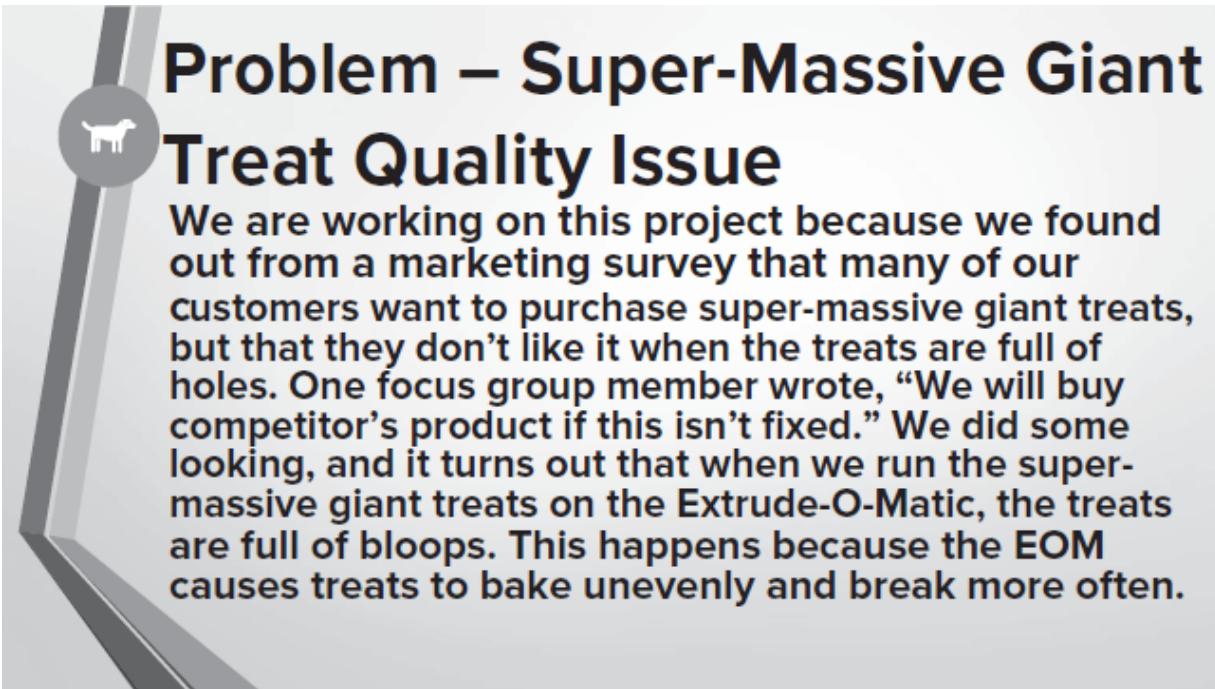


Figure 18.4

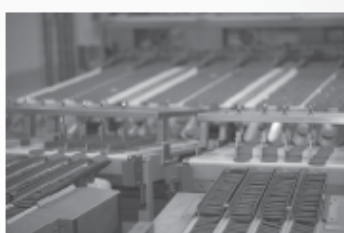
Discussion slide example of what **not** to do.

[Figure 18.5](#) shows two different styles of slides with key points. [Figure 18.5\(a\)](#) provides samples of two discussion slides that use bullets rather than complete sentences, while [Figure 18.5\(b\)](#) shows that same content on slides with a more direct format of sharing information with the audience. The slides in [Figure 18.5\(b\)](#) take advantage of the human brain's ability (or lack thereof) to multitask. Instead of asking the audience to read bullets silently as you read them out loud and embellish as you go along, you capture the audience's attention with the main idea in the title, and then embellish orally. You are guaranteed to be the focus of the presentation in [Figure 18.5\(b\)](#).

Problem – Super-Massive Giant Treat Quality Issue

Impetus • Customers want this size treat, but not treats full of holes • Focus group: “We will buy competitor’s product”	Problem • When we run SMG treats, they are full of holes (bloops) • Causes treats to bake unevenly and break more often
---	--

Methods

• Team: Jian, Ahmed, Kara, Willem • Process: A3 • Root Cause Analysis: 5 Whys • Solution identification: Weighted criteria matrix	 • Process problem identified • Occurs on Extrude-O-Matic (EOM) machine
---	--

(a)

Figure 18.5

The slides in [Figure 18.5\(a\)](#) use bullets to summarize the speaker’s information. The audience might pay less attention to the speaker because they are reading all of the words.

Jess2717/Shutterstock

 **Problem: Customers don't like that 80% of SMG Treats have blops**



 **Methods – Team used A3, 5 Whys, & WCM to evaluate EOM and find solution**



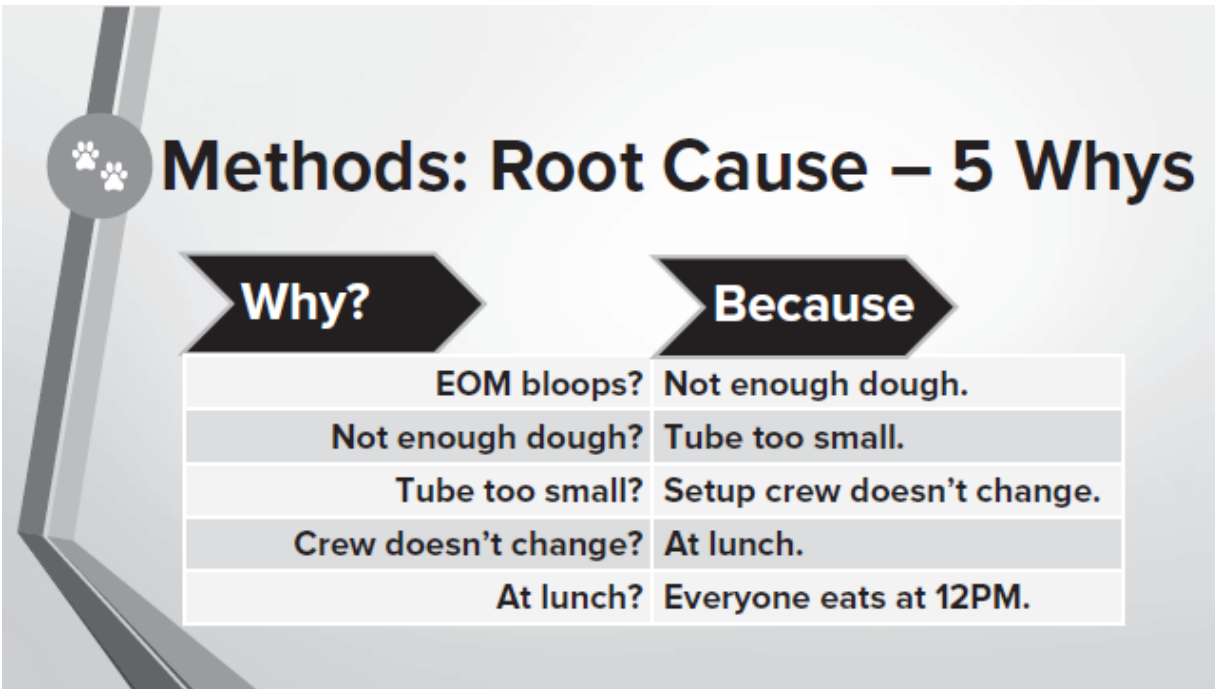
(b)

Figure 18.5

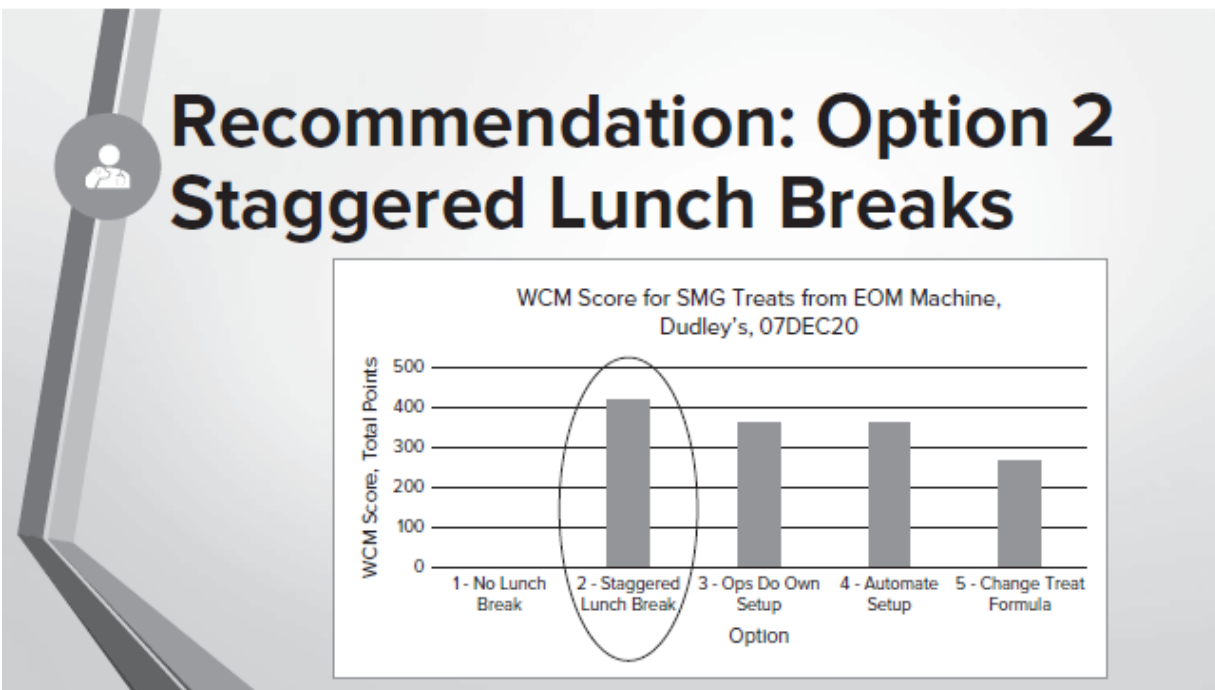
The slides in [Figure 18.5\(b\)](#) use minimal text, allowing the audience to focus more brainpower on the presenter.

Jess2717/Shutterstock

[Figure 18.6](#) provides two sample discussion slides that use Page 264 visuals to tell the story, including a root cause analysis tool in (a) and a bar graph which presents data in (b).



(a)



(b)

Figure 18.6

Discussion slide examples using graphics; (a) visual tool and (b) graph.

Summary Slide

The **summary slide** provides your audience a brief summation of the important points of your presentation. In most cases, you should be able to use your overview slide to inform your summary slide. [Figure 18.7](#) provides an example of a summary slide. Note that this summation should include actual key points of information and should not just be a restatement of the categories of information provided. In other words, do not say, “I told you about the problem, our methods, and our recommendation,” and then move on to questions. This misses the point of a summary. Audiences are more likely to remember your message if they hear the same information multiple times. By restating your key points, you increase the odds that they will remember your **takeaways**.

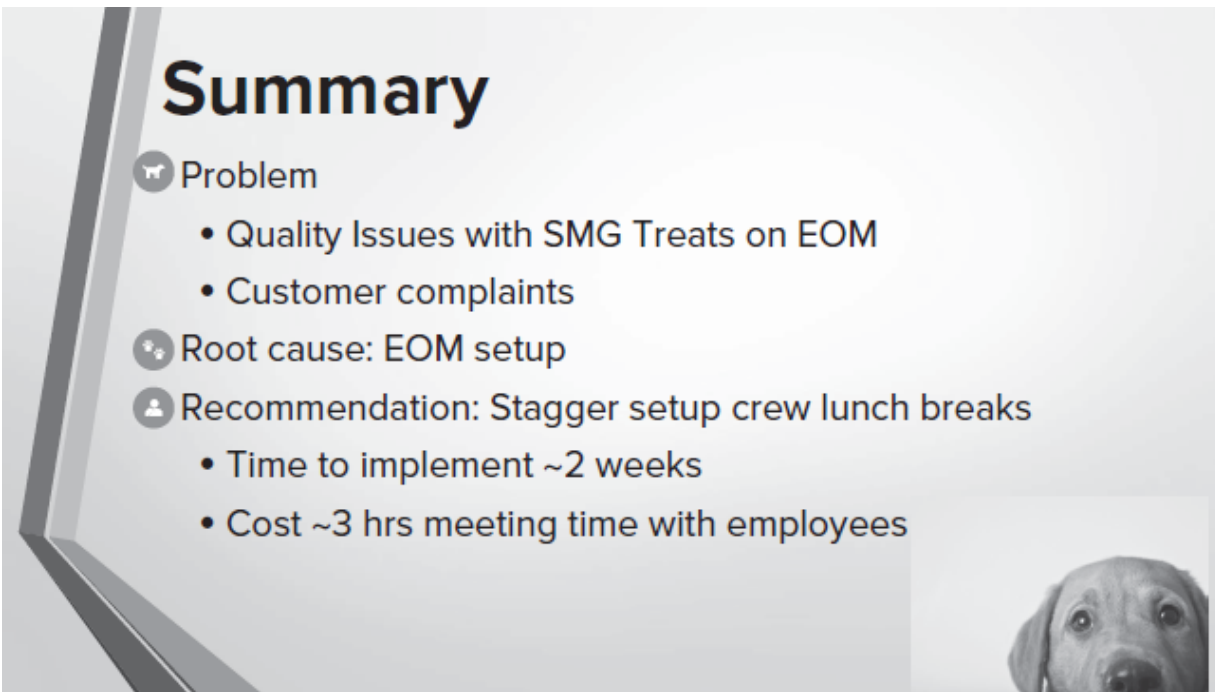


Figure 18.7

Summary slide.

Caia Image/Glow Images

Reflections Slide

Some organizations emphasize reflection as part of their operating philosophy and request that you assess how well a project is going. This is a

separate conversation from the project update but is included in the update presentation. In such cases, you might include a **reflections slide** like the one shown in [Figure 18.8](#).

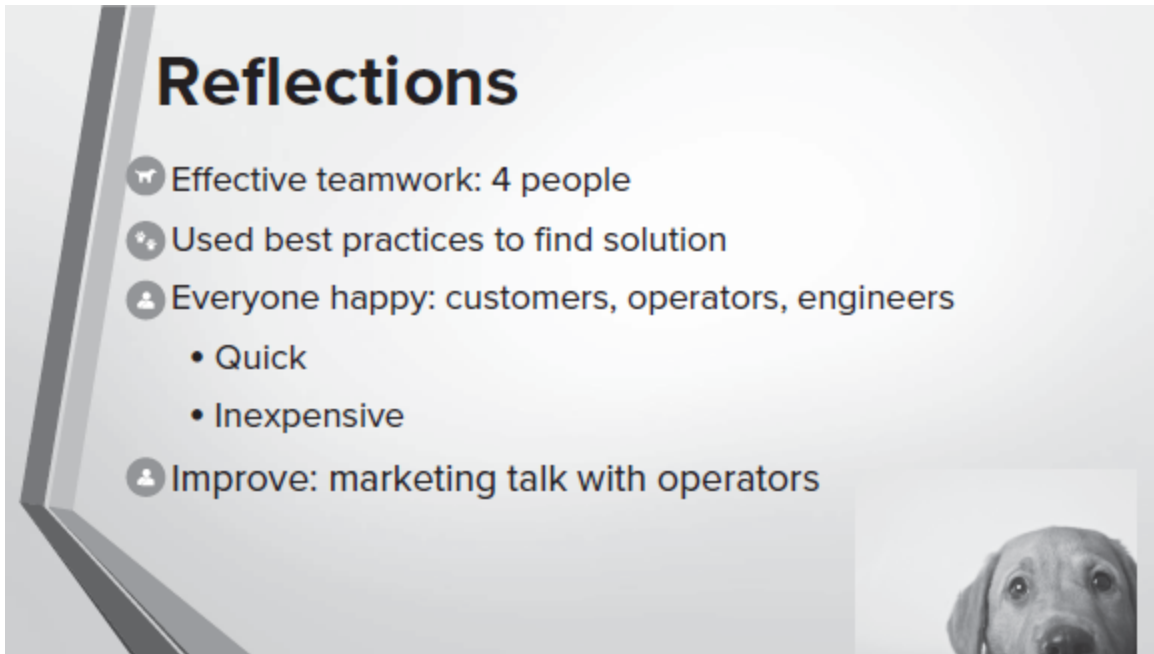


Figure 18.8

Reflections slide.

Caia Image/Glow Images

Finally, depending on your audience, you might decide to end Page 266 with a slide titled “Questions” or “Contact information.” Either of these is a clear indicator to your audience that your planned presentation is completed, which is appreciated by audiences much more than you saying, “So, yeah . . . that’s all.” The example slide shown in [Figure 18.9](#) combines the two functions of asking for questions and providing contact information. This is appropriate for audiences that need or appreciate easy access to your contact information.

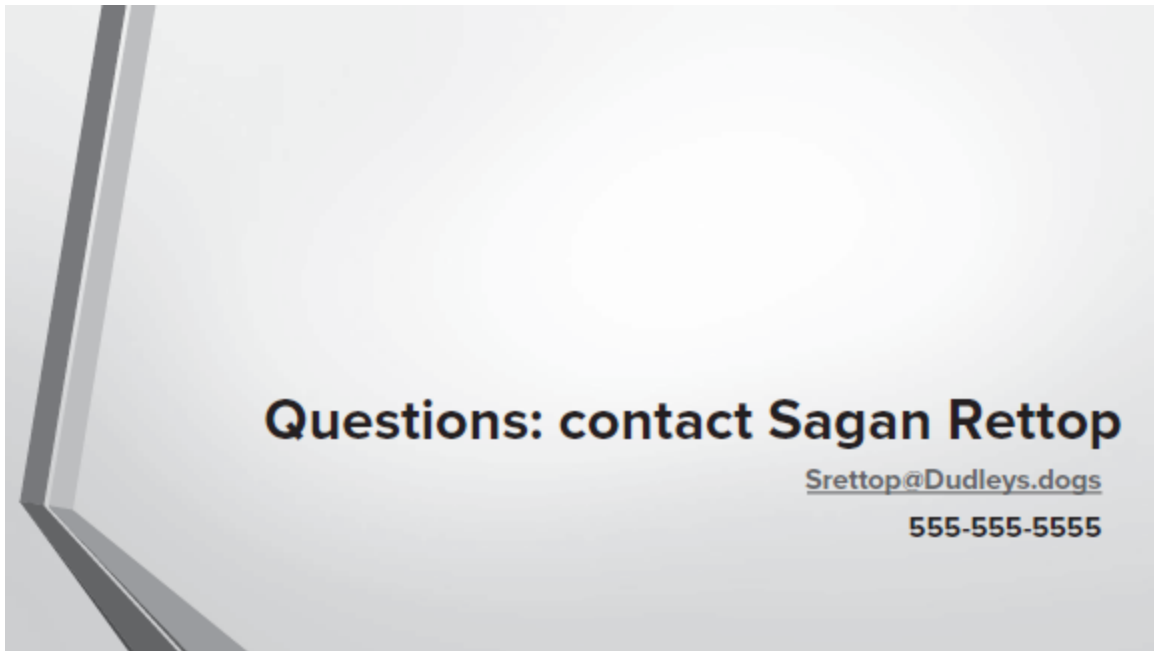


Figure 18.9

Questions/Contact information slide.

Controlling Complexity

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A guiding rule for producing technical reports is to use only the information that is necessary to get the job done. This guideline is even more important in technical presentations. Avoid adding anything to your presentation that does not have a specific purpose or serve a necessary function. However, that said, the structure of a presentation is quite different from any written genre. A presentation must include the key information of a written document, but purposefully place that information within the context of an introduction (which should include a hook!), body, and conclusion. A hook is how you get your audience's attention at the beginning of your presentation, and can be accomplished using many different techniques. Asking rhetorical questions, polling the audience, telling a story, and starting with an impressive solution ("We will save five million dollars if we do this.") are all ways to capture your audience's attention.

Visuals and Complexity

Imagine this: at the Westminster Dog Show, a well-known dog handler runs out into the ring—with an empty leash (see [Figure 18.10](#)). The audience gasps! The judge looks quizzically. The dog handler says, "I know you

can't see my dog, but . . .” Of course this would be a ridiculous entry. Unfortunately, it is not much different from similar words uttered every day by people giving technical presentations: “I know you can't see this but . . .”



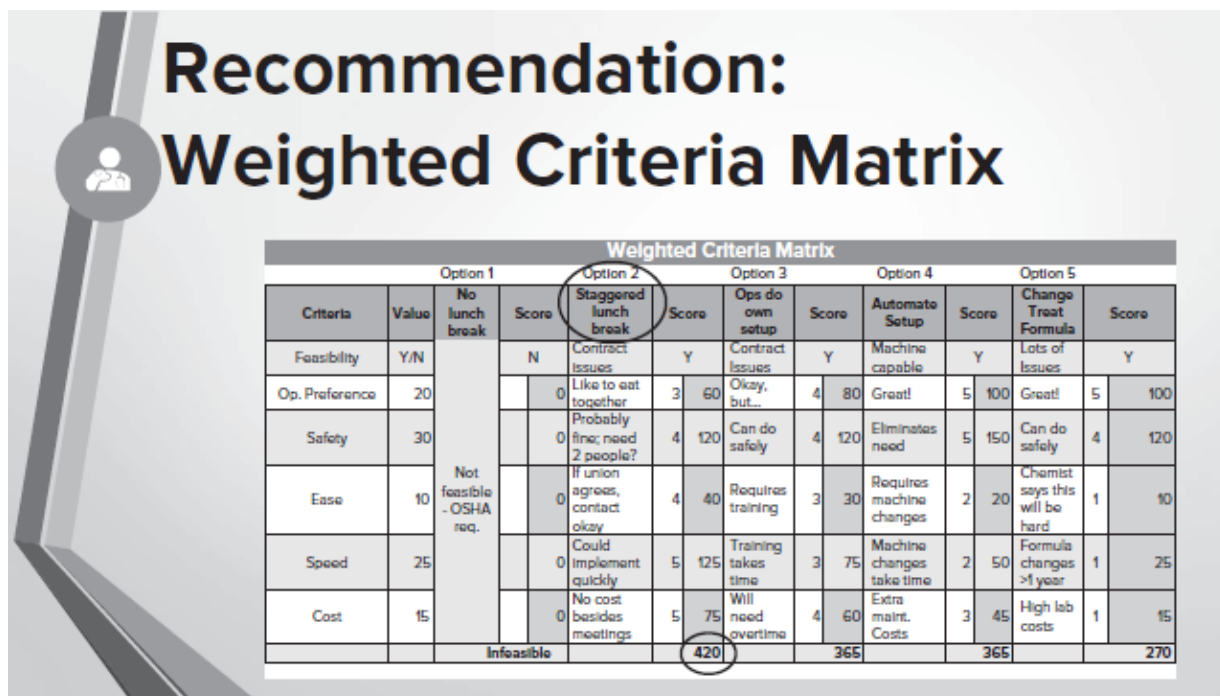
Figure 18.10

Visuals should be clear to the audience and complement the text.

Your audience can absorb only so much information on the Page 268 screen at one time. For example, a service technician can effectively use a large wiring diagram of a complex control system when that diagram is on their desk, and they have the time and need to trace through it. However, that same technician would find the same diagram useless, and even irritating, if you projected it onto a screen in a presentation. The detail would be difficult, if not impossible, to see or comprehend on the screen.

The same thing is true for complex tables or graphs of data on one slide. What if someone put up Figure 18.11 on the screen and then talked about each score and total for 30 minutes? The full table is so visually complex, and the font sizes of the entries are so small, that not even an eagle sitting in

the room could discern it, much less a human being. And the human being and eagle would have about the same level of interest and understanding. A good rule of thumb is that if you must drop the font size below 18 points to fit the information on the slide, then you are probably putting too much information on the slide. In defense of using the slide in [Figure 18.11](#), the one case for justifying its use is this: do not cover everything in the table, but simply show the table to help your audience have confidence in your methods, and highlight or magnify one piece of information from the table.



Recommendation:
Weighted Criteria Matrix

Weighted Criteria Matrix											
		Option 1	Option 2	Option 3	Option 4	Option 5					
Criteria	Value	No lunch break	Staggered lunch break	Ops do own setup	Automate Setup	Change Treat Formula	Score	Score	Score	Score	Score
Feasibility	Y/N	N	Contract Issues	Y	Contract Issues	Y	Machine capable	Y	Lots of Issues	Y	
Op. Preference	20	0	Like to eat together	3	60	4	80	Great!	5	100	Great!
Safety	30	0	Probably fine; need 2 people?	4	120	4	120	Eliminates need	5	150	Can do safely
Ease	10	0	If union agrees, contact okay	4	40	3	30	Requires machine changes	2	20	Chemist says this will be hard
Speed	25	0	Could implement quickly	5	125	3	75	Machine changes take time	2	50	Formula changes >1 year
Cost	15	0	No cost besides meetings	5	75	4	60	Extra maint. Costs	3	45	High lab costs
		infeasible		420		365		365		270	

Figure 18.11

This slide is overly complex if you are planning on covering every detail, but it might be appropriate if you are concentrating on one aspect of the tool or process.

The best approach is to decide whether all that information is really necessary for the purpose of your presentation. If not, get rid of what is not needed. For example, in [Figure 18.11](#), does your audience care about all of the alternatives that were considered? By showing your tool, will you be inviting a rehashing of decisions that were already made by the responsible team? Remember, more slides with less information are usually better than fewer slides with more. That said, a good rule of thumb is to use no more than one slide every one to two minutes of a presentation.

Another way of adding complexity to presentations is by incorporating special effects. In presentation software packages such as PowerPoint, it is a simple matter to add animation, video, and sound to your slides. You can have text flying in and out, diagrams dissolving into one another, animated arrows directing the audience's attention, and a whole range of other visual effects. You can punctuate every visual effect with a pop, bang, or squawk. You can stir the emotions with big band music and soothe them with the sounds of the ocean. You **can**, but should you?

The question here is not so much what you can do, but why you would want to use these effects. Special effects can be useful in small quantities—but be careful! Most of these effects get old quickly, even for the most patient audience. For example, using flying text or a humorous sound effect to highlight a point is fine, but using such effects throughout the entire presentation will become irritating and distracting. Too much of anything is usually bad.

Finally, avoid self-timed presentations, where you program the computer to change your slides for you at predetermined intervals. This is truly a recipe for disaster because you will never stay in sync with the computer, and the experience will not be the highlight of your career. Your audience will also be distracted, either waiting to see if you make it through the current slide before it changes, or watching you stand around with nothing to say as you wait for the next slide to appear.

While giving technical presentations can be intimidating at first, with knowledge and practice, you can improve. Set a stretch goal to be as successful as the Wire Fox Terrier, who has received more “Best in Show” awards than any other breed at the long-running Westminster Dog Show.³

Checklist for Presentations

General Checklist

- ☐ Do I understand who my audience is for the presentation?
- ☐ Do I understand the occasion and purpose of the presentation?

- Do I know the physical context in which I will present?
- Am I presenting information that is substantive and relevant?
- Have I organized my information effectively for my audience to understand it?
- Am I using language and media that are appropriate for my audience?
- Do I have all of the context that I need?

Checklist for Technical Presentations

Have I:

- Included a title slide with my name and organization?
- Included an overview slide listing the main topics of my presentation?
- Included at least one discussion slide for each topic on my overview?
- Included a summary slide (and reflection or questions/contact information slides if necessary)?
- Ensured that all my slides are readable when projected?
- Chosen an effective design scheme, and am I using it consistently?
- Avoided unnecessary complexity in all of my slides?
- Evaluated the briefing room and equipment beforehand (if possible)?

Notes

1. TED, <https://www.ted.com/talks>. Accessed December 8, 2020.
2. To learn more about fonts, consider reading more about the types (serif and sans serif) and which ones are most effective for the type of communication you're planning. Typography is a discipline and science of its own. For example, read a series of articles on typography (<https://www.fonts.com/content/learning/fontology/level-1>). Accessed December 10, 2020.
3. "'King' the wire fox terrier claims Best in Show title at 143rd Westminster Dog Show," <https://www.usatoday.com/story/sports/2019/02/12/westminster-kennel-club-dog-show-wire-fox-terrier-wins-best-show/2853664002/>. Accessed December 8, 2020.

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CHAPTER NINETEEN

Business Communications



Business Communications. Why should engineers and scientists want to involve themselves in communication that is non-scientific and non-engineering? After all, engineers and scientists are into technology, not business. Right? No, wrong. Business communications are not just for business majors, especially if you like to earn money and desire a good, rewarding career. To be successful, you must be able to communicate what you know to people who want, and sometimes who do **not** want, to hear what you know. This chapter addresses the primary forms and formats used on the business side of technology—email, memos, letters, etc.—that are expected in industry and will help you be a successful professional. While this chapter offers guidance about different business communication mediums, along with outlines and examples for both oral and written formats, its focus is on written communications after you have the job. Personal career-related communications (resumes, cover/application letters, and interviews) are covered in [Chapter 20](#).

Communication between scientists, engineers, and others in business is complicated, even more so than the complicated communication between dogs. Dogs who know and like each other need only a few seconds to express a positive greeting before they dive into the business at hand of enthusiastic play. But dogs who do not know each other often need time to understand the situation (see [Figure 19.1](#)). Who are the players? Are they giving off friendly or confrontational vibes? Do they like “paws-y” play? Where is everyone in the personality hierarchy (e.g., who is the primary Alpha and who is willing to be submissive)? What are the surrounding dynamics (e.g., just two dogs in a known territory or a pack of unknown

dogs at a new dog park?)) The list of considerations goes on, but you understand the point.

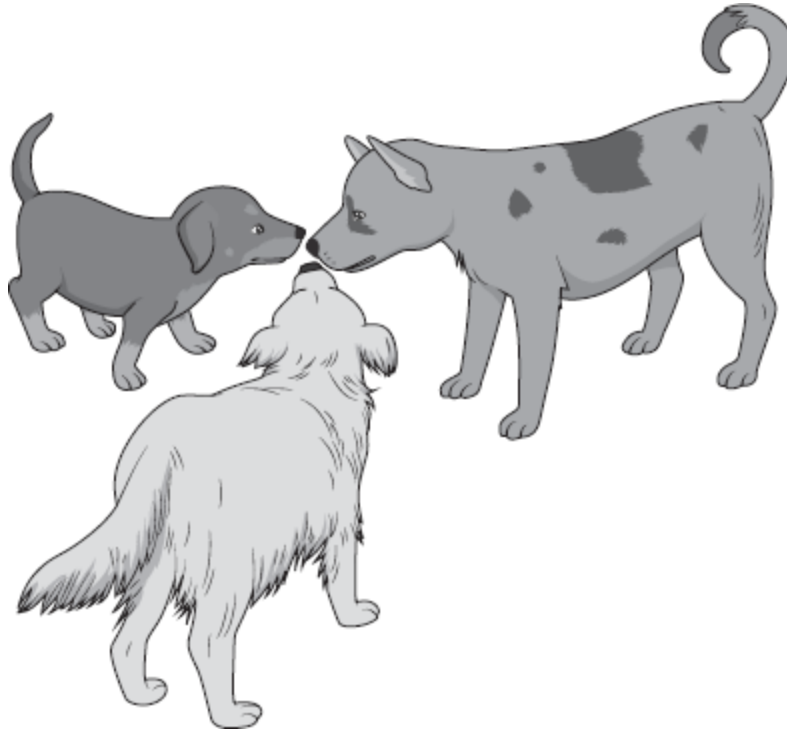


Figure 19.1

Just as all types of dogs must communicate effectively, so must all types of professionals. Doing so is complicated.

For every business communication to be successful, you must Page 272 consider (as always) audience, purpose, and context. Who will read your communication? Are they decision-makers, advisors, or implementors? Are you trying to build or maintain goodwill, persuade them to do something, or simply inform them? Do they need your information to do their job or do they need to know as a courtesy? What will they do with your communication? Have they asked you to communicate in a specific way, perhaps using a certain format, software, or deadline? Synthesizing the expectations and requirements in business communication is no simple task, and the outcome can range from benign to wildly successful to crash-and-burn. Understanding audience, purpose, and context will help you be successful.

Email

Most work-related written communication can be done via email; it is fast, inexpensive, and convenient. But just because it is quick does not mean you should use it without consideration. And while email communication is common, that does not mean it is consistently well done. Poorly constructed, obtuse, incoherent, or incorrect email messages can make you look really unprepared, can add to clogged inboxes, and can be costly, irritating, and even career-shortening. Effective email communication respects the reader's time, is succinct and on-point, and is sent to a receptive audience.

Note about Email

While email is superb for providing rapid communication worldwide, it can be easily compromised, filtered, spoofed, missed, or unintentionally deleted. Often, combining email with an attached letter or memo document can provide the rapid response required, while retaining the benefits of a separate letter or memo document.

Email overload is a serious concern, so respect your audience's time by following several best practices. Limit emails to only those that are necessary; this includes both emails sent directly to a person as well as copying (CC'ing) that person. Think carefully about what your communication needs are and if an email is actually the best method of communication. For example, often bad news is best delivered face-to-face (and then perhaps followed with an email). Limit emails to one topic and write specific subject lines so the recipient knows the topic of the email before they open it. The combination of one topic and specific subject lines makes it easier to find emails in mailbox searches. Keep emails short as best as you can. One screen long is ideal, but if you must write a longer email, provide an overview at the top of your email and include headings to allow your reader to easily scan the content. If you provide an attachment, make sure to reference it and preview its contents in the body of your email. If you need a response to your email, be sure to explicitly state your request for a response and give the date by when you need the response.

Most importantly, double check who will receive your email. Did you accidentally reply all when you meant to only respond to the sender? Did you choose the correct names or did autofill do you a disservice?

Creating a Record

Just like notes, emails are often part of the evidence that must be produced in legal disputes. See [Chapter 3](#) Note-taking for more discussion.

While email communication might seem informal like texting, it is not. Email communication is a formal method of workplace communication, a permanent record, and subpoenaable. Best practices for email communication include the following:

- **Provide specific subject lines.** State the topic of the message.
- **Use appropriate salutations** (Dear, Hello, Greetings, etc.). If you use a person's name in the salutation, use their professional title and not their marital status. For example, Dr. Smith—not Mrs. Smith/Miss Smith. If you use titles + last name, use the same formula for all recipients and not only those of a specific gender. If you are unsure of a person's title, err on the side of assuming the most respect (for faculty, assume Dr., but you can probably determine this from a quick search of your school's website). Professor is also a good choice when you are unsure. In industry, use the title Ms. for women until you are told otherwise.
- **Use appropriate capitalization.**
- **Write in complete sentences.**
- **Avoid using acronyms from texting.**
- **Make use of paragraphs.** Do not put your entire message in a data dump of a behemoth paragraph. Organize ideas in small chunk paragraphs with spaces between them.
- **Do not assume your email will be read.** If your message is that urgent or important, choose a different method of communication.

- **Refrain from ranting, complaining, venting, gossiping, or being otherwise non-professional** in any way. Your email is forever and can be easily forwarded to unintended audiences.
- **Do not use emojis.**

Emojis in Email ☺

This is a qualifying statement. Emojis are generally frowned upon in professional email, but as with all language, its use is evolving. It is best not to include them in emails that are being sent or could be/will be forwarded to anyone with whom you have a strictly business relationship. There are times, however, when you might be sending an email to a project team member, for example, with whom you have been working closely for many months. The two of you might be emailing about a problem-solving situation, and you might express a difference of opinion, but include a smiley face ☺ to indicate tone—that while you have a difference of opinion, you are not expressing it in a negative way. This is reasonable.

Some people will tell you to never use emojis in email. We tell you to be judicious about it and only include them when you have established a personal working relationship with the recipient and understand each other's intent.

Always assume two things about every email you write: (1) that it will be saved for posterity and (2) that it will be forwarded to someone else (possibly multiple people!).

In addition to the guidelines listed above, always write your emails so that your recipient will want to respond to you. Supervisors will usually be the kind of senders to whom recipients will want to respond, even if those supervisors are not well-liked or their emails do not follow the guidelines, merely because they are supervisors. However, employees and students must write their emails in ways that make the recipient want to be responsive. For example, read the emails in Examples 19.1 and 19.2. To which sender would you prefer to respond?

Example 19.1 Email Actually Received by One of the Authors

Email

From: student@anyuniversity.edu

Date: January 30, 2021

To: professor@anyuniversity.edu

Subject: a favor

Hey professor i am supposed to graduate in may an need your class.
when can i come to you're office so you can sign my add slip? Mike

Example 19.2 Fictitious Email That Would Have Made Its Recipient Happy to Help

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Email

From: student@anyuniversity.edu

Date: January 30, 2021

To: professor@anyuniversity.edu

Subject: ENGR 499

Professor Rao:

My name is Michael Attenborg and I am a senior in Industrial and Manufacturing Systems Engineering. I recently transferred and as I was completing my degree audit form, my advisor, Devon, noticed that I need to take your class to graduate.

I realize your class is currently full, but might you please consider adding me? I am a hard-working student with a 3.3 GPA (4.0 in my IE courses). I would be happy to talk with you if you wish.

Thank you for considering it,

Mike

Senior, Industrial Engineering, Any University

Email: mikeat@anyuniversity.edu

Cell: 123-456-7890

Outline 19.1 Email

Heading

- *From* Your (sender's) email address.
- *To* Recipient's email address.
- *CC* Email addresses of those who need to see your email or know that you have sent it, but who do not need to respond to your email.
- *BCC* Email addresses of those you want to see your email or know that you have sent it, but who you want to keep hidden from the recipient.
- *Subject* Identification of the email's topic.
- *Attachment* Any files attached to the email.

Body

- *Salutation* Greet and identify the recipient(s).
- *Purpose* State the purpose of the message.
- *Message* Give the message. Organize in paragraphs.
- *Close* Provide a conclusion, offer to answer questions, and provide contact information.
- *Closing* Provide a closing phrase.
- *Sign* Include your name.
- *Signature* Provide your signature information (title, organization, phone number, etc.).

Memoranda

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Memoranda (memos) are similar to, but more formal than emails, and similar to, but less formal than letters. Before emails and electronic communication became the norm, memos were often used within an organization for interoffice communications. Now they are currently used for formal communication within an organization. They can cover the same topics as emails, but tend to be used to document certain events or agreements, in which case they are attached to emails. Such memos have

many names, including a **memorandum of agreement**, **memorandum of understanding**, **memorandum for the record**, and **memorandum of inquiry**.

Memorandum or Memoranda?

Memorandum is singular. **Memoranda** is plural. For example:

We only need to send one memorandum on this topic.
We will need to send several memoranda on this topic.

Generally speaking, all memos use some variant of the format provided in [Outline 19.2](#), but many organizations have their own unique formats. As always, use the prescribed format if one exists for your organization—or if you are in a course, use the format prescribed by your instructor. If you have not been given a specific format, know that word processing software usually has templates.

Outline 19.2 Memoranda

Heading

- ***To*** Recipient's name and title.
- ***From*** Sender's name and organization/office
(Note: The sender's "signature" for a memo is their initials after their name on this line.).
- ***Date*** Date that the memo will be sent.
- ***RE*** Subject or purpose of the memo.

Body

- ***First paragraph*** Start with a summary about what will follow.
- ***Subsequent paragraphs*** Provide supporting materials and detailed explanations.
Provide a summary of your message. If

- **Summary paragraph**

you need a response, add a final line that invites a response. Note that memos do not require a salutation or close, but they may have either or both inserted as a matter of organizational style.

Example 19.3 Memo of Agreement

A memo of agreement outlines an agreement between two or more parties and documents their responsibilities and expectations on a large-scale project or toward reaching achievement of an outcome or objective. Consider, for example, an agreement that could be outlined for students and an instructor when starting an online course, a new learning medium for many, that outlines each party's responsibilities and expectations for the semester in order to reach the learning objectives for the course.

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Example 19.3 Memo

Interoffice Memorandum

TO: Brad Hunt, Student
FROM: Darshana Elin Rao, Instructor^{DER}
SUBJECT: Memo of Agreement for Online Course
DATE: January 25, 2021

The purpose of this document is to establish responsibilities for student and instructor for our online course this coming spring.

The Student

The student agrees to take responsibility for their own learning by assuming the following responsibilities:

- Familiarize themselves with the course structure and layout in course's learning management system.
- Confirm technical requirements so course materials can be accessed without problems.
- Identify and establish the communication channels of the course.
- Access course daily (Monday through Friday) and stay current.
- Attend weekly Q&A sessions and come prepared with videos watched, readings read, and assignments completed.
- Review course materials and announcements frequently, asking questions when necessary.
- Communicate professionally with the instructor. Do not expect a response to any correspondence on the weekend, holidays, breaks, or evenings.
- Submit all work following the provided guidelines and on time to avoid late penalties.
- Follow all course policies.

The Instructor

The instructor agrees to provide a safe and welcoming learning environment and help students learn by accepting the following responsibilities:

- Provide accurate course policies and schedule.
- Establish weekly work expectations with overview announcement every Monday by 8am.
- Communicate frequently via announcements.
- Check email and discussion board during workdays on normal academic calendar days. Best efforts will be made to respond to emails within 24 hours.
- Provide grades and feedback for submissions within a reasonable time frame. Best efforts will be made to have grades posted within ten days from the due date; however, the time may be longer for complex or long assignments. If an assignment is submitted early, there is no guarantee that it will be graded before the due date.

- Be available for individual conferences during weekly office hours and individually scheduled times.
- Provide accommodations as requested through the Student Accessibility Services Office.

Return the signed document to Dr. Rao (drao@anyuniversity.edu). Electronic signatures are accepted. A final copy will be provided for your records.

	Signature	Date
Student:	Brad Hunt	January 26, 2021
Instructor:	<i>Darshana Rao</i>	<i>January 27, 2021</i>

Letters

Whereas memos are used for formal communication within an organization, letters are used for formal communication external to an organization. Formats for letters can vary from one organization to another, and you should always use whatever is appropriate and expected.

In most cases, formal letters should be in the format of a general business letter as provided in [Outline 19.3](#). Normally, two possible layouts can be used: the **block letter** (Example 19.4) or the **modified block letter** (Example 19.5). In a block letter, everything is left-justified. In a modified block letter, everything is left-justified except for the date, the close, and the signature block, which are aligned with the center of the page. If your organization has a style manual, follow it. Note also that if you use company letterhead that has the address, you do not need to include your address at the top of the letter (Examples 19.4 and 19.5). If you do not use official letterhead, then include your address at the top of your letter (Example 19.6). That being said, most business letters follow [Outline 19.3](#).

Outline 19.3 General Business Letter

When producing business letters, always use standard fonts (e.g., Times New Roman, Cambria, etc.) of either 12- or 10-point type.

Sender's Address

- *Name* Your name.
- *Organization* Your organization's name.
- *Street Address* Street address (or post office box).
- *City, State, Zip* City, state, zip code.

Date

- *Date* Date the letter was signed by the sender.

Recipient's Address

- *Name* Recipient's name.
- *Organization* Recipient's organization's name (if applicable).
- *Street Address* Street address (or post office box).
- *City, State, Zip* City, state, zip code.

Body

- *Salutation* Greet and identify the recipient(s).
- *First paragraph* Start with a summary about what will follow.
- *Subsequent paragraphs* Provide supporting materials and detailed explanations.
- *Summary paragraph* Provide a summary of your message. Include a summary for informative letters, or your final pitch for persuasive letters. If you need a response, add a final line that invites a

	response. If applicable, offer to answer questions and provide contact information.
• Closing	Provide a closing phrase—e.g., “Best regards” or “Sincerely.”
• Sign	Sign your name.
• Signature	Provide your signature information (title, organization, phone number, etc.).

Example 19.4 Block Letter Format

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April 11, 2021

Maxine S. Beaudine, Mayor
1 City Hall Place
Big City, OH 45430

Dear Mayor Beaudine:

In response to your city’s RFP for a new wastewater treatment facility, Acme Enterprises is pleased to submit the attached proposal.

Acme Enterprises is a recognized leader in wastewater treatment, with a breadth of expertise and an arsenal of construction equipment that is perfect for this task. We understand the urgency of replacing the destroyed plant, especially given the unfavorable regulatory actions that

are pending. We at Acme Enterprises can and will meet all the requirements of this important and critical project.

The professionals at Acme Enterprises stand ready to quickly and effectively reestablish a wastewater treatment facility for your city, and we look forward to your reply. Please do not hesitate to contact me if you should have any questions.

Sincerely,

Jayden Jones

Jayden Jones, President
Acme Enterprises, Inc.

Encl: Proposal

Example 19.5 Modified Block Letter Format

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Maxine S. Beaudine, Mayor
1 City Hall Place
Big City, OH 45430

Dear Mayor Beaudine:

In response to your city's RFP for a new wastewater treatment facility, Acme Enterprises is pleased to submit the attached proposal.

Acme Enterprises is a recognized leader in wastewater treatment, with a breadth of expertise and an arsenal of construction equipment that is perfect for this task. We understand the urgency of replacing the destroyed plant, especially given the unfavorable regulatory actions that are pending. We at Acme Enterprises can and will meet all the requirements of this important and critical project.

The professionals at Acme Enterprises stand ready to quickly and effectively reestablish a wastewater treatment facility for your city, and we look forward to your reply. Please do not hesitate to contact me if you should have any questions.

Sincerely,

Jayden Jones

Jayden Jones,
President

Acme Enterprises,
Inc.

Encl: Proposal

Example 19.6 Block Letter Format Without Official Letterhead

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Jayden Jones, President
Acme Enterprises, Inc.
3020 Effluent Parkway
Opportunity, OH 43210

April 11, 2021

Maxine S. Beaudine, Mayor
1 City Hall Place
Big City, OH 45430

Dear Mayor Beaudine:

In response to your city's RFP for a new wastewater treatment facility, Acme Enterprises is pleased to submit the attached proposal.

Acme Enterprises is a recognized leader in wastewater treatment, with a breadth of expertise and an arsenal of construction equipment that is perfect for this task. We understand the urgency of replacing the destroyed plant, especially given the unfavorable regulatory actions that are pending. We at Acme Enterprises can and will meet all the requirements of this important and critical project.

The professionals at Acme Enterprises stand ready to quickly and effectively reestablish a wastewater treatment facility for your city, and we look forward to your reply. Please do not hesitate to contact me if you should have any questions.

Sincerely,

Jayden Jones

Jayden Jones, President
Acme Enterprises, Inc.

Encl: Proposal

The following sections discuss some of the most common Page 281
functions of business communications.

Transmittal Letters

Transmittal letters, sometimes called **cover letters**, formally transfer a document to its proper recipient.

These letters (1) specify the title of the document, (2) specify the requirement that the document fulfills, and (3) provide contact information for the sender to the recipient. Some cover letters, like those used to forward resumes in a job application, may include more extensive rhetorical strategies and information (see [Chapter 20](#)).

What is Rhetoric?

There are different definitions for the term rhetoric, depending on how it is used. It is generally understood to mean the language of persuasion. The ancient Greeks considered the combination of rhetoric, mechanics (yep, grammar!), and logic critical to education.

Examples 19.4 and 19.5 are all formal transmittal letters—they just have different letter formats. The following student letter (Example 19.7) provides an additional example—in this case, a student transmittal letter to a professor for a research report written for a biomedical engineering graduate course. Notice that this letter is short and to the point.

Example 19.7 Transmittal Letter for a Student Project

Elizabeth R. McFinkel
34421 Labrador Hall
Iowa State University

March 3, 2021

Charlie Jones
5442 Black Engineering
Iowa State University

Dear Professor Jones:

I am submitting the enclosed report, “Proposal for Standardizing Worldwide Data Collection of Food Emergency Needs for the United Nations Food and Agriculture Organization,” in fulfillment of the proposal requirement for IE 1400.

If you have any questions regarding this submission, please feel free to contact me at Elizabeth.Mcfinkel@UniversityEmail.edu.

Sincerely,

Elizabeth R. McFinkel

Elizabeth R. McFinkel
Student

Job Application Letter

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Job application letters, also known as cover letters, are designed to transmit a resume to a prospective employer. These cover letters are a critical part of any application package because they do more than simply transfer the resume. Properly crafted, job application cover letters establish the sender’s credibility, create goodwill, highlight key information, and invite a favorable response. Job application cover letters should do the following:

- Demonstrate that you are knowledgeable about the company and really want to work for it.
- Include examples of key skills and experiences that make you a desirable candidate.
- Demonstrate personal traits that make you a desirable candidate.
- Provide a conclusion that creates goodwill and invites a favorable response.

For a more extensive discussion of job application (cover) letters, including a fully developed example, see [Chapter 20](#).

Invitation Letters

Invitation letters simply invite someone to do something, usually for the sender's organization. These letters can take many forms, employ a variety of rhetorical strategies, and be used for many different business requirements. For example, a letter soliciting proposals for an upcoming project, often in conjunction with a Request for Proposal document (see [Chapter 8](#)), might take the form of a precisely worded letter that is consistent with accepted practice and various legal imperatives. On the other hand, a letter inviting an expert to be a guest speaker at an upcoming convention might require a less formal approach that focuses on the recipient's reputation, expertise, and the occasion for the speech. The form and content of invitation letters vary widely, and providing specific models for such letters is beyond the scope of this book.

Letters for the Record

Letters for the record define or document certain discussions, preliminary agreements, or events, often as a result of a meeting. Unlike minutes of a meeting, these letters are focused on particular aspects of the discussion or specific events that have occurred and may be confidential (see [Example 19.8](#)). Note: Often these business communications take the form of a **memorandum for the record**, especially when the context is internal to a single organization.

Example 19.8 Letter for the Record



Jason R. Barney, Recording Assistant
Acme Enterprises, Inc.
66 Roadrunner Plaza, Suite 3
Coyote, OH 45430

March 31, 2021

Jayden Jones, President
Acme Enterprises
3020 Effluent Parkway
Opportunity, OH 43210

In response to the referenced RFP #Y0643356 for a new wastewater treatment facility, Acme Enterprises convened its RFP Response Strategy Team at 1400 hours on 31 March 2021 to determine the company's approach for the Big City Wastewater Treatment Project Proposal. The meeting was chaired by President Jayden Jones.

The Team determined that Acme Enterprises has the resources to accomplish this project and would commit to this effort with the highest resource priority should a proposal be submitted and accepted. The Team also determined that the most effective rhetorical approach should, among other things, exploit the court-implemented remedies now in place for Big City, and the rapid, effective response Acme Enterprises could provide to satisfy these remedies.

Respectfully,

Jason R. Barney

Jason R. Barney, Recording Assistant
Acme Enterprises, Inc.

Letters of inquiry are requests for information from one organization to another. They often focus on the availability, suitability, and cost of certain resources that the sender's company needs or may need. For example, a manufacturer may send a letter of inquiry to a parts supplier regarding the supplier's ability to meet an upcoming surge requirement, or perhaps a letter to an overseas company to determine its potential efficacy as an outsourcing resource. These letters can also be very specific and may request detailed cost and availability information for, say, a parts list.

Letters of Inquiry use the standard letter formats provided in Examples 19.4 and 19.5.

Response Letters

Response letters simply respond in a formal way to inquiries, events, or other exigencies. These letters use the standard letter formats in Examples 19.4 and 19.5. Note: The response is often determined by the method of inquiry and frequently becomes a “response in kind”—e.g., an email inquiry would probably receive an email response.

Instant Messaging

You will be working with diverse professionals who have different comfort levels with communication mediums. Because of both technological and societal shifts toward online communication in recent years, much of the oral conversation that used to happen in bull-pen offices and by phone has changed. Where an employee might have walked down the hallway to ask someone a question before, they now send a text message. A project team working together where everyone is physically spread across floors, departments, buildings, and even around the world, might choose to use an instant messaging (IM) platform for much of their communication. Business communication often takes place on IM platforms like Slack or Discord. (Be cognizant of the comfort levels of team members—for IM to be an effective medium, you might need to teach people how to use it.) These platforms are less formal than email, but create easy-to-follow, on-going conversations which are often more structured than an email chain because of their ability to follow threads and channels.

A Message about IM

There are many different platforms for instant messaging (IM). Besides Slack and Discord, you might be familiar with Rocket.chat, GroupMe, Skype, Jabber, and/or Keybase, to name just a few. Each has its own features and best case uses. Before using any IM platform for business communication, you should verify your organization's specific information security guidelines. Specific platforms might be allowed, required, or disallowed. It is important to understand and follow your organization's expectations.

Instant messaging is typically less formal than most of this chapter, but there are still expected, socially acceptable behaviors. For example, writing in all capital letters (all caps) is considered yelling on any platform. An instant message is also **not** an email; the appropriate length of an IM is between one word and one short paragraph (anything longer should go in an email instead). Just like with team writing, if your team plans to use an online communication platform, it is best to establish ground rules at the onset to ensure efficient and effective use.

Some Communications Require a Personal Touch

Not all communication can be delivered via email or in print. In fact, there are many times that face-to-face meetings or phone calls are more effective than written communications. Knowing when **not** to write is also important to successful communications.

Face-to-Face Meetings

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Face-to-face communication, whether in person or virtual, is a good choice when the situation is complicated and clarity requires verbal and non-verbal cues, when you need to discuss something and need active back-and-forth with a colleague or expert (e.g., clarification, negotiation), when discretion is crucial (e.g., no paper trail), or when you need something immediately (e.g., getting a signature). Face-to-face communication is also a good choice when you want to cultivate a good working relationship.

When you do engage in face-to-face communication, be respectful of others' time and space, be prepared for the meeting, and be cognizant of signals to leave. If you are popping into your colleague's office, make sure it is a convenient time for them: "Excuse me, but do you have a few minutes?" If they say "no, not right now," accept this and come back later. Avoid pushing into their personal space-bubble (which differs by person). Do not hover over them or their desk. Do not move their possessions around to make space for yourself or any documents. If they give you permission to move things, it is acceptable, but do so in a manner that is subtle and respects their property. If you are seeking a face-to-face conversation because you need to clarify, negotiate, or discuss something complicated, bring any relevant documents with you. And finally, be on the lookout for your colleague's "it's time to leave now" signs. They could keep looking at the door or clock, repeatedly shift position, or stand up and start inching toward the door. They could even say, "Sorry, but I have a meeting coming up." Just like when a dog at the dog park grumbles, the other dogs know to stay clear; these are examples of your cue to thank them for their time and leave.

Phone Calls

Phone calls are a good choice when you do not require the non-verbal cues of a face-to-face meeting, but do need the verbal cues of tone of voice. While some organizations have calendar software which supports scheduling meetings, if you do not have this process available and must schedule a meeting, a phone call is a good choice. You can generally compare availability in a few minutes' time and secure the day/time in your calendar faster than back-and-forth emails where, in the meantime, the day/time has been scheduled for something else. Phone calls are also good if you need something immediately, like information for a time-sensitive document you are writing, and when it would benefit your working relationship to check in with your colleague. While phone calls can be used to discretely discuss topics (e.g., no paper trail), note that the date, day, time, and length of phone calls are recorded.

Much like with face-to-face communications, when you do need to make a phone call, be respectful of your colleague's time, polite in tone, and prepared to leave a message. Unless otherwise agreed upon because of deadline criticality, keep phone calls to within business hours, and make

sure to ask your colleague if they have time—and accept their answer if they do not. Do your best to call from locations without a lot of background noise and speak clearly and more slowly than you would in person. It is difficult to appreciate how much we rely on facial expressions until we do not have them. If your colleague does not answer their phone, do not just hang up—leave a message. Identify yourself, state the reason you are calling, and leave your number. For example,

Hello. My name is Samira Rao and I am calling because I need to schedule a meeting to discuss your proposal. Could you please call me back at 555-555-5555 as I will be in and out of the office today? Thank you.

While most phones do record caller numbers, technology can fail and sometimes people must be called back at a different phone from the one they used to place their call.

Thank You Notes

In business communication, sometimes the best thing you can do is send a thank you note. Very few people take the time to handwrite or even email a thank you note to someone who has helped out in a pinch (e.g., went out of their way to track down some information for you) or taken time out of their day to explain some hidden situational context you needed for a report. Why should you thank them? There are at least two reasons: (1) Because they put their own work aside to help (they might even have to stay after hours to get their own stuff done), and (2) Because the few minutes of effort it takes to write a thank you note will set you apart in the best ways possible. It will demonstrate courteous professionalism, a trait that is difficult to teach but so necessary for good business relationships.

Thank you notes can be emailed, printed and mailed, or Page 286 handwritten and mailed. There is no right way; truly, so few thank you notes are sent these days that any format will be appreciated. A thank you note does not need to be long; it just needs to be sincere and once again, reinforce to the recipient that you are the kind of person they want to continue working with in the future.

Conclusion

To be successful, you must be able to communicate what you know to people who want, and sometimes who do **not** want, to hear what you know. Effective business communication is constructed for a specific audience, purpose, and context. It gives the reader the information they need to do their work or to be able to help you do yours. Effective business communication fulfills its purpose, whether that is to inform, persuade, or build a working relationship. It also considers the context as best it can. When using written methods of communication, make sure to follow expected guidelines and formats, like using official organization memo templates or letterhead. In addition, make sure to follow best practices for each medium, including organization (beware the behemoth data dump of a long, one-paragraph message!), tone, and expression. Finally, be cognizant that not all communication should be delivered via email or in print. In fact, there are many times that face-to-face meetings or phone calls are more effective than written communications. Knowing when **not** to write is also important to successful communications.

Checklists for Business Communications

General

- ☐ Have you identified the recipient of your communication?
- ☐ What is your purpose in communicating?
- ☐ What, if anything, will you need the recipient to do after they have received your communication?
- ☐ What does the recipient need from you?
- ☐ Has the recipient asked you to communicate in a specific way, perhaps using a certain format or software, or meeting a specific deadline?
- ☐ Have you thoroughly analyzed the context for this communication?

- ❑ How much information, and how detailed, does your recipient need the message to be?
- ❑ Have you selected the appropriate medium for this communication?
- ❑ Is your writing grammatically and substantively correct?
- ❑ Have you properly formatted the communication—that is, fulfilled expectations for that type of communication?
- ❑ Have you crafted your message so that your recipient will want to respond to you?

Email

- ❑ Have you included anything in your email message that you would not want published on the front page of the *New York Times* tomorrow?
- ❑ If replying to specific addresses in a list, have you selected the proper reply mode for your email client?
- ❑ Have you verified that the proper people will receive the email?
- ❑ Have you considered whether you need to send this message now?
- ❑ Is your message a negative one? If so, reconsider whether email is the best medium.
- ❑ Have you proofread your email?
- ❑ Are you concerned that the message could be misinterpreted? If so, delete the email and schedule a face-to-face meeting or pick up the phone and call.

Letters and Memoranda

- Have you signed or initialed the letter or memo, as appropriate?
- Have you taken any security steps required to protect the information in the letter or memo?

Instant Messaging

- Have you established an acceptable instant messaging (IM) platform ahead of time?
- Have you confirmed that your IM choice is acceptable for communication within your organization?
- Have you followed generally accepted communication behaviors on your platform of choice (e.g., avoided all capital letters, used appropriate message length, etc.)?

Face-to-Face

- Have you made sure it is a convenient time for your colleague?
- Have you avoided invading their personal space?
- If you are meeting face-to-face to gain clarity, negotiate, or discuss a complicated topic, have you brought the necessary documents with you?
- Have you noticed and paid attention to any potential “it’s time to leave now” signs?

Phone Calls

- Are you prepared to leave a message that identifies yourself, states the reason you are calling, and leave your number?
- Have made sure it is a good time to call?
- Are you calling from a location relatively free of background noise?

- Are you speaking clearly?

Thank You Note

- Would your colleague appreciate a thank you note? If yes, then send one.

==

CHAPTER TWENTY

Communications with Future Employers (aka, Getting a Job)



The first step to getting any job is getting an interview. Getting an interview can happen in many ways, including submitting an excellently constructed **resume** to an online job posting, going to a job fair and having a great conversation with a recruiter, or **networking** at a meeting. The common link in these activities is communication. Usually, the communication tools used for job-seeking combine both written and oral skills. Often people in technical fields start job seeking by writing a resume. Providing a resume, which is usually a one-page summary of “your story,” is like knocking on a potential employer’s door. Sometimes you literally hand your resume to a recruiter in person, sometimes you email it to human resources, and sometimes you submit it to an online application process. In all cases, your audience, which could include humans, computer algorithms, or a combination of the two, quickly assesses whether to open the proverbial door. For the hiring company, the purpose of your resume is to determine if you might be a good fit for them. If they decide that you are worth further conversation, they invite you to interview—in other words, they open the door.

After you get the interview, it is time to demonstrate your worth. During the interview, the hiring organization will further assess if they have a position which is a good fit for you. It goes without saying—but we will say it anyway—not every company **is** a good fit for you. Do not let this dissuade you or make you feel bad about your prospects. When you or the company realize that the combination of you working for them is not a good fit, this is actually good news. It is much better to realize this as early in the

process as possible, before either of you invest significant time and/or money into the relationship. Consequently, seeking a job takes time. It is not for the weary. It requires tenacity. You must be determined, methodical, and resolute when looking for a job. For some people, looking for a job is a job all by itself.

In other words, you must be a Bulldog: stubborn but calm, persistent but people-oriented. Bulldogs are the heavy lifters of the dog world in terms of strength, courage, and tenacity. There is a reason why the Bulldog is the most popular college mascot¹; Bulldogs never give up (see [Figure 20.1](#)). For the past century, many athletes have been credited with different versions of “you can’t score if you don’t shoot.” The same is true when it comes to getting a job. The more you “work the room,” apply online, and show up to opportunities like career fairs and company presentations, the more likely you are to get interviews. The less active you are in seeking a job, the less likely you will be to land one.

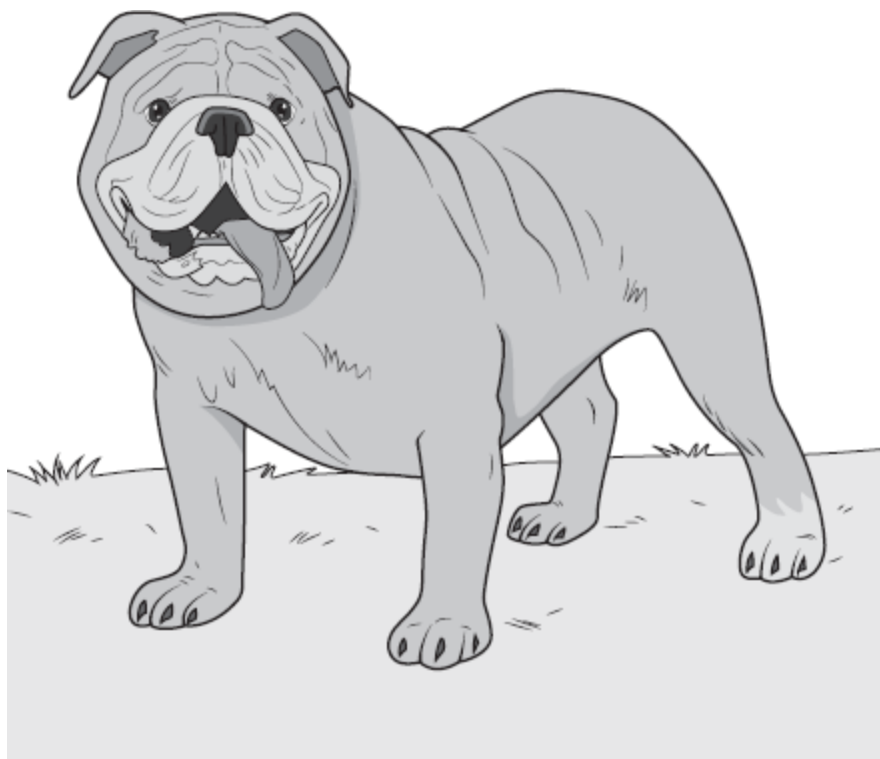


Figure 20.1

Bulldogs are friendly but tenacious. When searching for a job, you must be tactfully tenacious. It's not easy, but don't give up. You only need one job.

Job-Seeking Communication

Written

- Resume
- Cover Letter
- Thank You

Oral

- Job Fair
- Networking
- Interview

The world has changed dramatically since the start of the 21st century, including how we communicate when it comes to recruiting for and seeking jobs, so this chapter contains a mix of both written and oral communication tools now commonly used. While many of the tools remain the same in name, how they are constructed and executed has changed. Resumes, cover letters, and thank you notes (along with the emails sometimes used to send them) are still critical written tools. Job fair conversations, networking conversations, and interviews are critical oral communication tools. If you use all of them—tactfully, appropriately, and relentlessly—you will be successful. Be the Bulldog!

There are many online and print resources for resumes, cover Page 290 letters, business communication, interviewing, and job seeking.² The trick is knowing what information can be effectively generalized for all majors, professions, and companies, vs. what information has local, regional, and/or profession-specific preferences. To increase your chances of being seen and understood by your audience, it is important to understand what is generalizable and what are specific preferences. Like other chapters in this book, the topic of job-seeking communication is tremendously broad. There are entire books and even entire industries dedicated to the topics we describe here; we can get you started thinking about the right things, but if you need more detailed information, consider consulting your

college/university career services resources,^{3,4} discipline-specific resources like professional society websites,^{5,6} and even company-specific resources like suggested/required resume formulations and job requirements.^{7,8}

Written Communication

What Is a Resume?

A **resume** (pronounced reh'-zoo-may, from the French word *résumé*, which means summary) is a **summary** of **you**. The point of a resume is to convince a recruiter to offer you an interview. Let us start by addressing that the actual content you can and should include on a resume should be completely honest (see Chapter 2 Ethics). Before you can write about what you have done and who you are, you must actually be that person. Therefore, the best written resumes begin with well-balanced people doing the writing. Consider a three-legged stool—what happens when one leg is shorter than the others? It falls over, of course. You do not want to fall over; you want to balance academics, work experience, and activities/leadership (see [Figure 20.2](#)). By doing so, you show how well-rounded you are as a person. Potential employers can start to form very quick, positive opinions from how well-rounded you are.

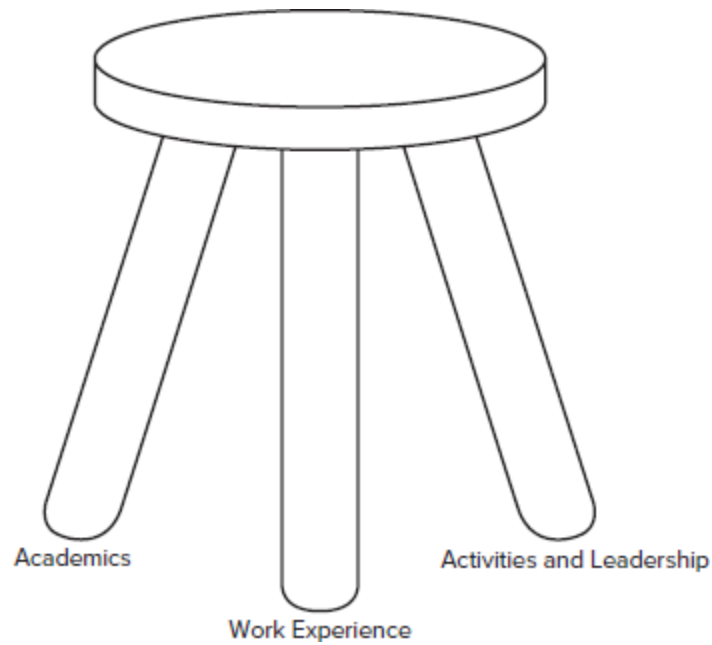


Figure 20.2

Much like a balanced three-legged stool, a good balance of grades, experience, and extracurricular activities on your resume will facilitate companies wanting to interview you.

Whoever you are and whatever balance you have, you must document that balance with words. A well-constructed resume allows a recruiter to quickly and clearly match your skills and experiences to their organization's needs. In doing so, you reassure them about the possible value you can bring to their organization. The words on your resume should resonate with your audience, making them want to talk with you. When a recruiter sees information they like on your resume, they will extend an invitation to you to interview.

Thus, good resumes “open the door” to further conversation. They accomplish this by setting you apart from others. They should contain audience-specific vocabulary, often including specific kinds of information that relate to the position desired. For example, if you are an industrial engineer, your resume should include the kinds of words that industrial engineers recognize and use in everyday work situations, like tasks, tools, and measurements. If you are a biologist, your resume should include the kind of vocabulary specific to the job for which you are applying, whether that be in education, medicine, or Peruvian Amazon herpetofauna.

It is not just word choices, however, that can make or break a resume. Format, font size, balance, numbers, readability, and many other things make a difference. The overriding design principle should reference Edward Tufte's data-ink ratio⁹: maximum message, minimum ink. Just as an abstract "sells" a journal article (see [Chapter 14](#) Abstracts and Summaries), your resume sells you. If you were trying to sell your car and get the most money possible for it, you would probably detail it so that it looked as good as it possibly could. When you write a resume, it is the same concept; details matter. Resumes must be flawlessly edited. Inherently, this means that you should not wait until the night before the job fair to write or even update your resume. Plan ahead so that you can ensure a flawless representation of yourself when you need it.

Another critical detail is the length of your resume. Until you are a mid-career professional, your resume should be **one page** long, no more and no less. Recruiters do not want to read more than one page (face it, they often will not even read one whole page). Submitting more than one page says to the world, "I was so uninformed about resumes that I got this wrong" or "I am so narcissistic that I believe I'm better than others and deserve more than one page." Neither of these creates a good first impression. Likewise, if your resume is less than one page, it says, "I was too apathetic to do more with my life," or "I was too apathetic to take the time to tell you about me." Again, not the first impression you are hoping to create. The one-page requirement is very challenging. Some first- and second-year college students find it difficult to fill a page with value-added information, while third- and fourth-year college students find it difficult to pare down their collegiate careers to a single page. We will help you with both in the following sections.

Suppose a firm is looking for an entry-level mechanical engineer to design embedded systems for mobile cranes. You have the qualifications to do that job, but what kinds of information should you highlight to increase an employer's certainty about hiring you? To accomplish this, your resume should have the kinds of information specified in [Outline 20.1](#).

Writing a Resume

Writing a resume is demanding work—it takes multiple iterations. Fortunately, you already know a lot about the subject; but determining your strengths and developing the document are still challenging. It is difficult to

be objective when you are the subject. Some people find it hard to “brag” about their accomplishments; it is not bragging to factually state what you have done. Take your time writing it, and then ask people you trust to provide you with honest, constructive feedback. Dr. Frank Peters, Industrial and Manufacturing Systems Engineering Associate Professor, recommends that you do a resume exchange with a friend and pay each other \$5 for every error found. You **want** your friends to find any errors; if they do, it will be the best \$5–20 you spend on looking for a job, because recruiters will toss your resume for a **single** mistake. Peters says, “If you are unwilling to risk the need to Venmo your friend some money or buy them some beers if they find mistakes, then you are not ready to distribute your resume to potential employers.”¹⁰

The outline in 20.1 is for a typical, early-industry (during or after college) resume. There are other types of resumes that people use appropriately:

- Mid-career industry: two pages
- Federal grant proposal bio: two pages
- Academic Curriculum Vitae: unlimited—can exceed 20 pages for someone who has been working in academia for 20+ years

Outline 20.1 provides a simple, straightforward approach for organizing a resume. When it comes to resume layout, think of it like an old-fashioned newspaper where placement on the page has value. Articles in a newspaper “above the fold” are the ones assigned a higher priority. The same is true on your resume. Information that you absolutely want recruiters to read should be in the top half of the page—this includes your name, contact information, objective, education, and most relevant work experience. The information in the bottom half of your resume might (or might not) get read. It can include less relevant or older work experience, publications, activities, leadership, awards, and skills. We note once again that every discipline will have preferences about the content, organization, and format

for a resume, so be sure to investigate any resources available to you through your professional societies, college career services, etc.

Outline 20.1 Resume

Name	List your name, email address, address, and telephone number.
Objective	Describe the kind of position you want and when.
Education	Document your formal education. • Degrees • Certifications • Honors • GPA
Experience	Include experience in order of relevance to the job objective.
Computer Skills	Document your computer literacy. • Languages (in descending order of fluency) • Operating systems (in descending order of expertise) • Applications (in descending order of expertise) • Platforms (in descending order of expertise)
Leadership/Activities	List your leadership positions and your activities.
Awards	List awards related to your profession.

Critical Information

Depending on the job for which you are applying, this could be industry and/or academic work experience. If you have not had work experience yet, volunteer experience can be described. This is “above the fold” information.

Flexible Information

These sections are more flexible and degree/position specific. A computer engineer will want to list all of their operating systems, applications, languages, etc. A mechanical or industrial engineer might want to list their fluency in other spoken/written languages. A chemist might want to list their most recent journal articles. Someone who has been recognized with a significant award might call this section Awards and include that information. This is “below the fold” information.

Name

As [Outline 20.1](#) shows, the header of your resume should include your name, email address, and telephone number. In our first example, we use Sunny Misumoto, a fictitious second-year student (based on a real engineering student). Make sure the email address you include is professional and one that you check regularly.

Sunny Misumoto
313 Dogwood
Ames, Iowa 50013
555-555-5555
smisumoto@anyuniversity.edu

You can include your postal address, but there is debate about how useful this is anymore. The likelihood of a company sending you a physical letter from your resume is low. If you do include your address, unless you are looking for a job near your hometown, use your school address. The (possibly?) subconscious thought recruiters have when they see a hometown address is that a student has not yet “cut the apron strings”—in other words, matured enough to live away from home and be fully independent.

Objective

Next, Sunny can decide if she wants to include a job objective. Based on numerous conversations with employers, about half of resumes have one and half do not. The preference for this should be determined by profession and/or industry. The specific feedback we hear about objectives is that **if** you have one, it should add value by informing recruiters what kind of job you want (internship, co-op, or full-time) and when you want it (starting spring semester, after May 15, etc.). You can use the objective to include preferences about the type of job that you would like, but keep in mind that this can also limit you. If you really only want a job doing quality engineering, it's a good idea to state that in the objective, but if you are not set on a certain type of job responsibilities, avoid listing preferences. If you are applying for a specific job from an advertisement, you can relate your preferences to the posted position for which you are applying. In Sunny's case, as a second-year student, she is not particular about the type of work that she would do—only that it starts in the summer. She is willing to take a summer internship or a longer co-op position.

Objective—to find a summer or co-op position where I can use my problem-solving and teaming skills to make a difference to the bottom line for my employer

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Note that if you have text which “consumes” part of a line, try to either reduce it and take fewer lines or further elaborate and use the line to your advantage. Every line of a resume is “real estate” that should not be wasted.

What does this objective tell a prospective employer about Sunny? First, it explicitly states the type of job she is available to do. She is not ready for full-time employment but will accept a summer or co-op position. This is helpful to recruiters so that they know into which pile to sort this resume.

Second, Sunny emphasizes that she has problem-solving skills. She does not pass judgment on how good her skills are; it is not advisable for you to tell recruiters how amazing and wonderful you are, but instead to provide **evidence** of your accomplishments in your Experience section and let them make that decision. However, Sunny is letting recruiters know that she understands their priorities—they need problem-solvers.

Third, she wants to be part of a team. She understands the nature of employment: that companies exist primarily to make money. She also understands that she will be working in a team environment, which is important because technical problems are rarely solved by one person. This kind of statement is supported by **demonstrating** teamwork in either work experience or activities later in the resume.

Education

Frequently, technical positions have specific degree requirements. This requirement may be a function of company policy, or it may be mandated by a client with whom the company is doing business. Start with your university or college and its location first, followed by your degree(s) and program(s). If you have not yet received your degree, include an “expected by” date. If you have multiple degrees, include them from highest to lowest and/or chronologically going backward in time.

Iowa State University, Ames, Iowa

Expected May 2023

Double Major: Mechanical Engineering, B.S. & Mathematics, B.A.

Honors Program

GPA: 3.6/4.0

No zip code needed. No one is going to mail your school based on this information.

Note that with a double major, both degrees should be listed. Minors should also be listed.

Include multiple majors, minors, certificates, and participation in honors programs. If you have studied abroad, this is the ideal place to include that information as well.

Students often ask about including GPA on a resume. The short answer is yes, you should include it. If your GPA (on a 4.0 scale) is a 3.0 or higher, you will meet the threshold of almost all companies who have one. If you have between a 2.0 and 3.0, there might be some companies who automatically sort you out of contention, but there are many who will still want to talk with you based on a holistic assessment of the information on

your resume. If your GPA is below 3.0, do not give up—be the Bulldog! We have heard recruiters say numerous times that they would rather hire a well-balanced individual than a 4.0 student with no work experience or activities. Omitting your GPA from your resume automatically invites a recruiter to assume that you are not confident. We include two tips for GPAs in the 2.0–3.0 range:

1. **Include your GPA “as is” and “roll” with it.** For example, at a career fair, when a recruiter says, “Tell me about your 2.0 GPA,” assuming you have justifiable reasons why, explain them. For example, if you had a very hard transition from high school to college with terrible grades your first year, but you are on track to do significantly better or have had good grades since then, explain that. If you are working 20+ hours per week to put yourself through school and must balance coursework with survival, tell them that. If you had one really bad semester because your mother was going through cancer treatments, share that. We’re happy to note that most recruiters are reasonable people; not only do they understand and appreciate circumstances such as these, they consider it a positive when a student overcomes a challenge or hardship.
2. **Include a subset of your GPA on your resume.** If you are Page 295 uncomfortable listing a lower cumulative GPA, but you had an outstanding third year, or you have a much higher GPA in your core/major courses, clearly state the subset and include it on your resume. For example, you might list “Core GPA: 3.3” or “Third Year GPA: 3.0.” This is not dishonest. It opens the door to a conversation about your strengths and weaknesses and indicates that you are capable.

Experience

Success breeds success: the people most likely to get a job have had a job. That makes getting the first job both challenging and critical. But it also means that if you have had **any** honest job, it is worth including on your resume. If you worked in a fast-food restaurant, as a farm hand, or delivering newspapers, it tells potential employers that you are motivated and understand something about the working world. Virtually any kind of successful prior employment can be valuable on a resume, but successful

employment in a job that relates to the objective can provide a tremendous advantage. This is especially true if you can show that you exceeded the expectations of the job. Volunteer experiences, especially those that show commitment, results, and skills, also demonstrate positive qualities to a potential employer.

Note: If you are applying to graduate school, higher GPAs are often required; you will want to understand those requirements early on in your undergraduate career so that you can work toward obtaining them. That said, every university, college, program, and major are different, so do not assume that just because you do not have a 3.2 or 3.5 or some other random threshold that you will not be accepted to graduate school. Ask!

You do not need to list all your experience on a resume. After your first year of college, you no longer need to include high school “evidence,” including work experience, unless it directly relates to the job you are trying to get. While it is acceptable to include work experience that pre-dates your college experience, it is best to elaborate on recent and/or relevant job experiences with more detail than you provide for jobs from your high school years.

Sunny’s resume lists her technical experience that relates to the job objective. However, her resume does not include her high school jobs as a page at the local library or fieldhand work detasseling corn. While those experiences were honorable, adding them to her resume will make it too crowded with information that is less relevant to the job she wants.

Work Experience

Jun 2020–present Iowa State University, Ames, IA, *Undergraduate Research Assistant*, Industrial Engineering—Dr. Leo Ellen

- Assisting PhD candidate with development of a novel digital surface roughness standard for castings
- Developing C++/C# GUI in Visual Studio for automated surface roughness calculation based on point clouds; expected to increase usability and enable adoption of the surface roughness standard

Jul–Aug 2020 *COVID-19 Planning Team Student Employee*

- Created fillable PDF seating charts used for contact tracing for ~150 university classrooms
- Documented standard process for creating and using forms; distributed by ISU Provost to all faculty
- Responded promptly to change requests based on faculty input and facility changes, and communicated changes with faculty via email

Research Experience (unpaid)

Jan–Apr 2020 First-Year Honors Mentor Program—Mathematics

- Learned linear algebra and differential equations concepts under Dr. Ravi Kotval
- Used MATLAB to solve 3 complex math problems, including optimization and controllability

Jun–Jul 2017 Secondary Student Training Program—University of Iowa—35 hours/week

- Conducted research for Dr. Gretchen Frank at the National Advanced Driving Simulator
- Modeled reaction time data using MATLAB and created code for future studies

Three key pieces of content should define your experience Page 296
“evidence”: active verbs, numbers, and impact. Verbs should be as descriptive as possible. For example, instead of “**improved** workstation safety,” use “**increased** workstation safety.” Be specific and precise with numbers. Instead of “wrote work instructions,” use “wrote **18** sets of work instructions.” Include impact as well by explaining how your effort made a difference: did it save \$5000/year or reduce changeover time by 20%? Include evidence of the value of your effort whenever possible.

Notice that Sunny’s experience includes employers, locations, job titles, and dates. It also describes her key responsibilities (using active verbs), uses numbers when possible, and includes impact. Because Sunny is a second-year student, she has not yet had an industry experience, but she includes both her paid and unpaid jobs. Note that one of her unpaid jobs happened during high school; while she has removed all other high school information from her resume, she made a conscious decision to include that research experience because she felt it was relevant to her interests going forward.

Computer Skills, Activities and Leadership, Awards, Volunteering, Skills, Etc.

The final sections of the resume should include more tailored, personal categories that help tell your story. You want to include information here that enhances your value to the company. For example, computers are essential tools in almost every workplace, and particularly in the technical and scientific workplace. Different disciplines have different expectations about listing computer skills on a resume. Those in MIS, computer programming, or computer, cybersecurity, or software engineering might be more likely to include all computer skills. Other disciplines might only include skills with which they are proficient, and which set them apart from others. Instead of listing computer skills separately, Sunny opted to include evidence of her computer skills (C++/C#, Visual Studio, and MATLAB) in her work experience. However, in the third-/fourth-year resume example in the Exercise at the end of this chapter, you will see how Jamal, a computer engineer, lists his computer experience separately.

Remember that technical qualifications are easily verifiable. For example, if you say on your resume that you are fluent in C++ or R, all an employer need do to verify your claim is ask you to implement a routine in C++ or R. That kind of thing happens in many interviews, either formally or informally. If it happens to you and you cannot do what your resume says, not only have you been unethical, but you have also failed the interview and forfeited the job opportunity.

If the jobs that you are seeking require that you be able to get a U.S. government clearance or that you must be a citizen, consider including this information in these optional sections. By choosing to include activities and leadership, you demonstrate for potential employers that you have the kind of personal balance they want. If you have held an office in an organization, you have experienced leadership responsibilities. If you have volunteered with a humanitarian organization, you demonstrate empathy. If you include intramurals, you show teamwork and a sense of competitiveness. If you include your fluency in a second language, you demonstrate having both a global perspective as well as a marketable skill. Consider including numbers and impact in these sections as well. The personal sections of Sunny's resume might include this type of information:

Activities, Leadership, and Awards

Dorm Floor President (where, responsibilities, dates)

Iowa State Putnam Prep / Problem-Solving Club – 2019 Score: 10/120 (dates)

First-Year Honors Program Leader – 13 students in class (dates)

Engineering Career Fair Ambassador (dates)

St. Jude Up 'Til Dawn (dates)

ISU Intramurals (basketball, soccer, volleyball, broomball, hockey) (dates)

Skills: basic German

See [Example 20.2](#) for a complete, formatted example of Sunny's second-year resume based on the information discussed in this section, as well as an example of a third-/fourth-year computer engineering student's resume in the exercise (Jamal). In Jamal's resume, note the following things:

- It is borderline too dense, but since he is applying for full-time jobs in his field, he felt it important to include the details that he did.
- Jamal's profession expects that job applicants list all of their computer skills, as well as provide examples of projects and personal learning.

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Ten Tips for Creating a Good Resume

1. **Be sure your resume is perfect both grammatically and stylistically.** Pay excruciating attention to detail: There should be absolutely no errors. Employers look for communication skills, and, fair or not, they will consider any grammar or style error to be evidence that you do not write well. They will also assume that if you are not willing to care about details when it comes to this single most important piece of paper, for which you have time to perfect, then you will not care about details on the job. Take advantage of grammar- and spell-checking software, but do not rely on it totally. You need human eyes looking at this; once again, consider enlisting trustworthy friends or family to give you constructive and thorough feedback.
2. **Tailor your resume for the position.** If you are attending a job fair, you can create multiple versions of your resume which emphasize different aspects of your experience. For example, if you are a computer engineer who is interested in both software development and

network administration, two different resumes, which can be handed to recruiters depending on the job they have available, makes good sense. If you do this, be careful in your delivery so it is not obvious that you have multiple resume versions (act like a card shark and deal off the bottom of the deck). If you are submitting your resume to an online application or emailing it in response to a specific job, then most definitely make your resume fit the job by presenting the most relevant and supportive data first.

3. **Always be truthful in your resume, but not self-effacing.** In other words, do not say, “I have a serious heart condition, but so far I have been stable with medication.” Factually state your accomplishments. Accentuate the positive about yourself, but do not go off the deep end. Do not write, “I am absolutely the most brilliant engineer, and I represent the quintessence of intellectual power and personal achievement.” While you might think that is a stretch, it’s common to see resumes that say, “Exceptional communicator” and the like. You shouldn’t make judgments about how great you are; instead, provide evidence of your accomplishments and let your audience decide. For example, if you presented a poster at the State Capitol building on how to operate a flight simulator, list that. Your audience will understand immediately that you were chosen for this role for a reason.
4. **Limit your resume to one page for industry positions until you are mid-career.** You can keep resume length in check by omitting minor details that the employer already knows or will not care about. For example, “references available on request” can be assumed. Also, non-job-related data, such as the names or occupations of your family, or that you like to swim, dance, and work for world peace, are facts that most employers are not interested in at the resume level of the interaction. For a mid-career industry or government grant proposal, a two-page resume is common. For a mid-career academic position, the curriculum vita will be multiple pages long (sometimes 20+). All the same holds true except that you have more room to list your experience and publications.

A two-page “Bio” (biography) is often requested for government grant proposals. The general layout of this document is similar to what we

recommend for resumes, but the sections are titled differently:

- a) Education (called “Professional Preparation”)
- b) Work Experience (called “Appointments”)
- c) Papers you have co/authored (called “Products”)
- d) Activities/leadership, personal accomplishments, awards, etc. (called “Synergistic Activities”)

You can find details and templates on the National Science Foundation (NSF) website:

<https://www.nsf.gov/bfa/dias/policy/biosketch.jsp>.

5. Remember that everything you include adds to the story of you.

Some people will not like everything about you; for example, if you include membership in clubs or groups that have political or religious preferences, you are taking a chance. You do not have to hide who you are, but know that by your choices, you affect how some people will think of you, and this can affect their decision whether or not to give you an interview. We are not recommending that you include or not include any specific kind of information—only that you consider the possible ramifications of your choices.

6. Do not use fancy paper or exotic folds, and do not include a photograph of yourself.

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Employers of engineers and scientists are usually very pragmatic and substance-oriented. They just want the information they want as efficiently as possible.

7. Avoid company-specific acronyms or jargon. For example, “improved the XLR1 production line” has no meaning outside of the company where you had your internship. Industry acronyms that are widely used are acceptable, but if you are only using the acronym one time, consider writing it out instead.

Just like with PowerPoint slides, just because Microsoft resume templates exist, does not mean you should use them, at least “as is.” In science and technology professions, your resume is not a good time to display your creative side. Stick to the basic format and organization.

8. **Never refer to yourself in the third person.** It comes across as stilted and sounds weird.
9. **Never put a date on a resume.** If you do, your resume will seem outdated almost immediately.
10. **Do not refer to salary requirements.** Salary is a topic better suited to the job interview or position negotiations. Sometimes you will be asked to fill out a formal job application form before the actual interview begins and address salary there. If you are pressed to include something on salary requirements, then just say that your salary is negotiable, or express your salary requirements in a range. If you are unsure of what the current market rate is, your college career services department might have data you can access.

Bonus: Use a distinguishing file name for your resume pdf, such as “Misumoto Resume.pdf.” You do not want to use generic file names that make it impossible to find: “Resume,” “Good Resume,” or “Sept 20 2020 version 13.11 res” are all unsearchable by a recruiter looking for **you**. Regardless of the software you use to create your resume, convert it to a PDF and verify that the formatting is still correct **before** you email it.

Cover Letters

Whenever you are emailing about a position, send your resume with a cover letter. However, simply including a note that says, “Here is my resume; give me a job” will not be productive. In fact, this wastes one of the most powerful tools you have—the opportunity to increase a recruiter’s confidence about investing interview time with you. A good cover letter is every bit as persuasive as a good resume. It supports your resume by including new and detailed information which cannot be included on your resume. A bad cover letter winds up in the trash, along with the resume attached to it. As with your resume, your cover letter can be run through computer software with the intent of identifying key words; consider this as you write it.

An effective cover letter does four things which contribute to establishing goodwill, informing, and persuading your audience:

- It demonstrates that you are knowledgeable about the company and really want to work for it.
- It includes examples of key skills and experiences that make you a desirable candidate.
- It demonstrates personal traits that make you a desirable candidate.
- It provides a conclusion that creates goodwill and invites a favorable response.

Develop a cover letter the same way Sunny develops hers in this chapter's example, with each paragraph having a particular function. First, start the letter with the name, address, and contact information used on the resume:

Sunny Misumoto

313 Dogwood
Ames, Iowa 50013
555-555-5555
smisumoto@anyuniversity.edu

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Next, whenever possible, address the letter to a person, not an organization—and not “To Whom It May Concern.” Sometimes you will not have a choice, but if you can find the name of the person running the office who supervises the job for which you are applying, use that name. Sunny has decided to apply to the firm that is seeking a mechanical engineering intern to design embedded systems for mobile cranes. Her letter is addressed to the senior engineer who is supervising mobile crane embedded systems:

Dr. Tracy Ann Burger
Senior Engineer, Embedded Systems
Cranes R Us
595 Miner Road
Wutherford Heights, Ohio 44132

In the body of the letter, the first paragraph should demonstrate that you know about the company and really want to work for it. Sunny has done some homework and shows that she is not just “shotgunning” this letter everywhere:

While doing career research to locate companies that might have embedded systems engineering opportunities, I read about the advanced work Cranes R Us has been doing in partnership with cybersecurity systems. In fact, the more I read about your company and its activities, the more convinced I became that I would like to be part of the Cranes R Us team. To that end, I have attached my resume with the hope that you will give me the opportunity to discuss further how I might fit in and make a measurable contribution at Cranes R Us during a summer internship.

Notice how this first paragraph both demonstrates Sunny's knowledge of the prospective employer and, as a bonus, shows that she wants to be a team player. She has taken the time to learn about the company's work with cybersecurity systems, and she clearly wants to explore how she might fit in with the Cranes R Us team.

The next paragraph briefly summarizes Sunny's skills and experience that make her a desirable candidate. Notice that she does not go overboard here because the details are available in the resume. Instead, she provides an example that further details one of the bullet points on her resume. This letter's goal is to get the employer to look at the resume, so providing different information here from what is on her resume allows her to increase the amount of detail she can provide overall.

I am currently a second-year student in Mechanical Engineering and Mathematics at Iowa State University. My degree program, which I will complete with University Honors in 2023, comprehensively blends the strong mathematics which form the fundamentals of cryptography and mechanical engineering concepts. During the past three years, I have had three different research positions, all of which have given me experience writing code for complicated systems, including optimizing and controlling robots. I learned through these previous experiences how much I enjoy the interface between systems and cybersecurity.

The third paragraph reviews Sunny's personal traits that make her a particularly desirable candidate:

One of the things that I have enjoyed the most in my courses at Iowa State was the group project we had in my first-year engineering course. I was one of a team of four students; we designed and built a prototype robot for climbing walls. We were able to quickly assess each team member's strengths and then divide and conquer on the four major components of our design. After our final presentations, our team won first place overall. The award was based on a combination of the success of our robot and how we presented it. While I look forward to continuously improving my written and oral skills, I have a good start on understanding and using effective technical communication skills.

A point worth noting: This letter is well written, and that alone supports Sunny's claim that she has a good start on effective communication skills. If she came across as inarticulate, that claim would be compromised, as would

the entire resume package. It is also critical that this letter be error-free. Dr. Mike Helwig, retired U.S. Navy, sums this concept up better than anyone: “No grammar mistakes. Period.”¹¹ Finally, the last paragraph provides a conclusion that creates goodwill and invites a favorable response:

Thank you for taking the time to review my resume; I realize how many letters and resumes you must receive. I know I can be a valuable resource for your organization, and I would welcome the opportunity to interview at your convenience.

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Notice that Sunny didn’t say, “I want a job—schedule me immediately for an interview!” You would be surprised how many people actually write something this brash in their cover letters and then wonder why they never get a response. See [Example 20.1](#) for the finished cover letter.

Thank You Note

Once again, we point out that the best thing you can do in this process is make it easy for recruiters to see how you would add value to their organization. One way to do this is to shine when it comes to professional behavior. Very few people take the time to handwrite or even email a thank you note to potential employers after an interview. Why should you thank them? There are at least two reasons: (1) Because it really is a lot of work for employers to try to determine who to hire. They invest significant time into you, and everyone appreciates being appreciated, and (2) Because the 30-minute effort of writing a thank you note will set you apart in the best ways possible. It will demonstrate courteous professionalism, a trait that is difficult to teach but so necessary for good business relationships.

Thank you notes can be emailed, printed and mailed, or handwritten and mailed. There is no right way; truly, so few thank you notes are sent these days that any format will be appreciated. A thank you note does not need to be long; it just needs to be sincere and once again, reinforce to the recruiter that you are the kind of person they want to hire. An example thank you note that Sunny might write and email to Cranes R Us is in [Example 20.3](#).

Finding Jobs (and Jobs That Find You)

It turns out that you are **always** interviewing, every single day. As a student, how you comport yourself in a course, including your interactions with both your instructors and peers, is under constant observation. Next year, when an instructor needs an undergraduate teaching assistant or research assistant,

they will remember. Likewise, next year when a peer who is an excellent student needs a team member, they will remember. In both cases, you have already “interviewed,” and your instructor or peers will know whether they want to work with you. The same is true after you graduate. When you work for a company, people are paying attention, and whether “the next” opportunity will avail itself to you depends largely on how you perform and behave for the months and years before that. Often the job opportunity you find available to you does not require your seeking it; it comes to you.

That said, most internships and first full-time employment opportunities **do** require some homework. There are many ways to find jobs. The order in which you use the techniques described in this chapter could vary in every circumstance, but it is always good practice to have a current resume ready to share at a moment’s notice. Different methods include

- Attending a job/career fair
- Submitting your resume to a company’s website application
- Sending an email with a resume attached to an individual at a company
- Networking at conferences, professional meetings, civic meetings, and through alumni groups
- Searching and applying to online job sites

At a career fair, you will most likely hand a recruiter your resume. After a networking conversation at a meeting or after a presentation, it is quite possible that you will be asked to follow up with a resume or at a minimum, contact information. Your excellent written and oral communications will be required, woven together as you find opportunities.

Oral Communication

Job Fair Conversation

If you are fortunate to attend a college or university that sponsors annual or semi-annual job fairs (also called career fairs), looking for a job cannot be made much more convenient. Sometimes cities or chambers of commerce will host job fairs as well. Prepare your resume well in advance, follow the

required rules for registration, do your homework on what companies will attend, and wear your interview attire. If you are unable to determine the companies in attendance ahead of time, consider doing a quick sweep when you arrive, choose whom you would like to approach, and do some quick research using your phone before you go talk with them (do not do this while you are standing in front of them—be sure to step to the side). One of the most frequent complaints from recruiters is that a student does not know what their company does.

You have approximately 10 seconds to make a first impression. You do this with your non-verbals as much as your verbal skills. How you look (clothes are professional, ironed, clean; hair is clean and groomed; jewelry is conservative or absent; you smile and look people in the eyes), how you smell, how you sound, and how you feel (if, after the pandemic is over and we go back to handshakes, you want a firm but not too-firm handshake, three pumps) all matter. When it is your turn to talk with a recruiter at a job fair, introduce yourself calmly but enthusiastically. Be prepared with your “elevator speech” (see [Chapter 18](#) Presentations). You have approximately 2–3 minutes to convince the recruiter that they want to talk with you longer. Have questions ready to ask—you want to appear interested and not like a deer in the headlights.

The purpose of a job fair conversation is to get a first interview. Sometimes these happen immediately onsite. Sometimes a recruiter will ask if you can meet in the following days. If you do not get a positive feeling (either they tell you they are not interested or they do not indicate any resolution of the conversation) but you are still interested, you can ask them what you could do going forward to make yourself a viable candidate for their organization.

In every case, when it is clear that the conversation is over, thank them and graciously exit their area. Recruiters are often alumni from your school and will return in the same role repeatedly over time. They remember students from job fair to job fair, and those first impressions make a difference. It is never too early to start making connections with recruiters, even if you are a first-semester student. Go and talk with them; begin to establish your network.

Networking

It turns out that every day and every interaction is actually an interview. One great way to get a job is to get your foot in the door by knowing someone who is already part of the company. This approach is frequently called **networking**. When it is time to go out and get a job, or to move from one job to another, networking with the right people in the right places can be extremely effective (see [Figure 20.3](#)).

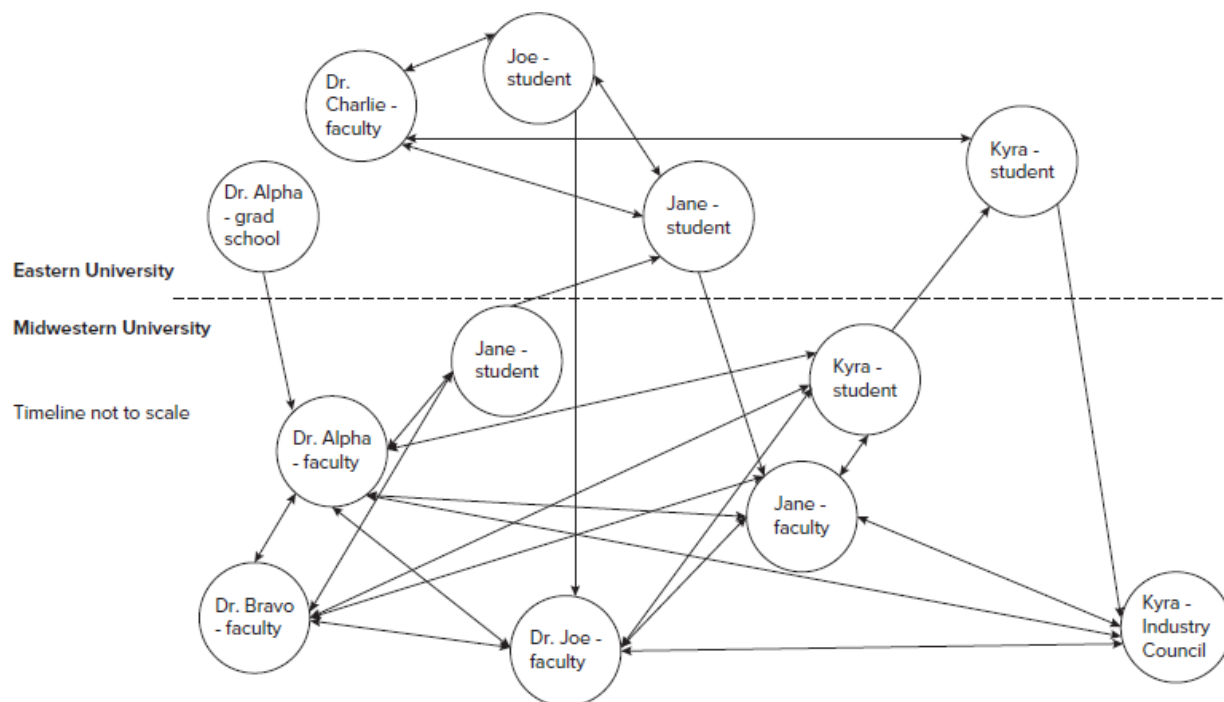


Figure 20.3

Developing your personal network can begin early; college is a great place to make connections. You can help others and benefit from your connections throughout your professional career.

Fun fact: an English Proverb is a saying that is generally well known and accepted as common sense. While there is some debate about “which” common sense is represented by the following, there is no arguing that the concept has outsized impact on how things get done in the current job environment:

“It’s not what you know. It’s who you know.” ~English proverb

Some insist that this proverb is about nepotism (hiring your own family and friends), but we believe it reflects the importance of **networking**. While it should **not** be true that knowing someone **gets** you the job, it is quite acceptable that knowing someone gets you the interview. Once you have

the interview, it is up to you to demonstrate to a potential employer why they should hire you.¹²

We demonstrate networking by telling a 33-years'-long true Page 302
story (names have been changed to protect the innocent!) and referring to the network that visualizes it (see [Figure 20.3](#)). A long time ago, Jane went to Midwestern University to study engineering and had Dr. Alpha and Dr. Bravo as faculty. She was a reasonably good student; she asked Dr. Alpha for advice about where to go graduate school. He suggested several, to which she applied and was accepted. She went to Eastern University for graduate school, which happened to be where Dr. Alpha had gone to graduate school. While there, she worked for Dr. Charlie and with his PhD student, Joe. Joe graduated with his PhD and interviewed at Midwestern University; before his interview trip, Jane shared with him all she knew about Midwestern University faculty, including the communication styles of those he was going meet. Dr. Joe had an excellent interview and was hired. Three years later, Midwestern University needed an instructor. Dr. Joe suggested Jane to Dr. Bravo because he knew her work, and Dr. Bravo remembered Jane as a student (and also because she had thanked him with a houseplant after he had written a letter of recommendation for her). She interviewed and was offered the job. During Jane's first year as a faculty member, Kyra was a student. Kyra asked Jane where she should go to graduate school. Jane suggested several schools. Kyra applied to Eastern University and both Jane and Dr. Joe wrote recommendation letters which led to her accepting and attending, where she worked for Dr. Charlie. Later, Kyra joined the industry council at Midwestern University and provided industry projects to current students. Everyone stayed connected.

This is just a small subset of the network developed by these six people. The important point to this story is that networking affected the jobs and lives of everyone in it. It is also noteworthy that over 33 years later, these people still value each other's opinions and seek them out when it comes to making decisions. Yes, it is messy. Yes, it is complicated. Yes, it is how it works. Of course, networking was not the only reason Jane, Joe, and Kyra got the jobs they did. Every single individual in this network is quite good at what they do. None of it would have happened if the individuals involved were incapable, unmotivated, or not personable. But they were (and are!), and because they "have the goods," networking has worked well for them.

Knowing people is not enough to get a job, but it is often enough to get an interview. In the words of Nike, how do you, “Just do it.”? Dr. Timothy Ewest summarizes the steps to approaching someone at a meeting or event quite nicely¹³:

1. Take the initiative
2. Thank them
3. Connect
4. Be specific and to the point
5. Ask a question
6. Thank them again

Interviewing

The purpose of a cover letter is to get an employer to read your resume. The purpose of a resume is to get an employer to interview you. The purpose of an interview is to get an employer to offer you a job. If you have the technical skills and personal characteristics an employer needs, and if you did a good job with your cover letter and resume, you will probably be invited to interview for the position. It is important to realize that interviews are stressful and risky for both the interviewer and interviewee. Both parties have much to gain or lose depending on how things go. Interviews are decision points: the employer must decide whether to make you an offer, and you must decide whether to accept it.

Interviews come in all shapes and sizes, and they might have different purposes. Some are screening interviews designed to narrow the field to two or three top candidates; these are often conducted by phone or videoconference. They are usually followed by in-person hiring interviews. The final decision to hire is normally made based on the results of this interview. If you make the cut in the screening interview, you will be invited to the hiring interview.

For both phone and video interviews, plan to be ready at least 30 minutes before the interview. “Ready” means being dressed in interview attire. Yes—even for a phone interview—you want to be dressed for success because it turns out that you will behave differently depending on how you are dressed. Wear sweatpants and you will slouch. This is detectable in your voice and your responses. So dress up. You should also have paper and pen

ready—take notes the old-fashioned way because you can do this with one hand while you hold a phone and it will not be as noisy or distracting to others as typing is. Do not use speaker phone; if you use ear buds or headphones, make absolutely sure (ahead of time!) that the microphone provides crystal clear audio for those listening to you. Write down names as your interviewers introduce themselves so that you can reference them later in the call.

Know this: You are being assessed from your very first interaction, including on the phone with an administrative assistant, at the guardhouse when you arrive and park your car, and as you walk through the hallway and interact with the custodian. Hopefully, how you behave is no different in these situations from when you speak with company managers, but either way, be aware that every current employee who interacts with you could be part of the hiring decision.

If your interview is via video, be sure to download the platform or application (e.g., Zoom, Microsoft Teams, Google Meet, WebEx, etc.), preferably to your computer, a day before the interview. Test it. Do you have enough bandwidth to run it without lags or interruptions? If not, consider connecting to the internet with a cable instead of using WiFi, or go to a quiet, private, public space that has excellent bandwidth, like a conference room that you can reserve. Make sure that your camera looks at your face—not at your stomach, up at your chin, or at the top of your head. The lighting should be in front of you—not behind you. Your background should be neutral and uncluttered—you want your interviewers to focus on you and not the gadgets or hallway traffic behind you. If you are interviewing from your dorm room, hang a sheet to hide the living space behind you.

Interviews can be as short as 15 to 30 minutes or they can last Page 304 several days. Some require only that you answer questions, whereas others require you to make a formal presentation about yourself and your work. They can also include personality and problem-solving tests. Interviews can be formal, where you sit down in a room full of important-looking people and respond to one or more interviewers' questions. Or they can be very

informal, such as a discussion at lunch or dinner. Dr. Mike Helwig offers that you be prepared to, “Tell me about yourself,” and asks, “Can you?”¹⁴

Two things should occur in any interview: The employer should determine whether you can effectively fill the position, and you should evaluate the employer to determine whether you want to work for that company. Sometimes when you report for an interview, you will be asked to complete job application information. Be sure to have complete contact information for each of your references with you, along with your employment history and extra copies of your resume.

Dr. Helwig elaborates: “Do you really think that the difference between a 3.2 and 3.4 GPA is so significant that the person with the 3.4 GPA will automatically be considered a more desirable prospect? Or that the person with three internships is so much stronger than the person with two internships? The answer is “No.” So if there are a lot of quality students with fairly similar resumes, what is the determining factor in whether or not you get a job offer? You guessed it—the interview. And that’s why it’s so important.”

Here are some tips for successful interviewing:

1. Fully research the prospective employer before your interview. You should have done some of this research when you wrote your resume and cover letter or before you approached them at a job fair. Now, however, you need more detailed information about the company, the kinds of work it does, and its track record of success. Also, try to assess the quality of the people working there, how stable their employment is, what sort of salary and benefits package the employer offers, the cost of living in that area, and so on. For many employers, much of this information is readily available on the internet or in a library. In fact, if the company is publicly held (i.e., its stock is publicly traded), its finances and business outlook are a matter of public record and can be found doing a quick search on the internet.

If the company is privately held, you can often find the information you need online; however, if not, you can request brochures and an annual report from the company before you interview. In any case, you

should also make discreet inquiries with those you trust who have experience with the company or who know people who work there. Dr. David Sly, President and CEO of Proplanner, a small business in Ames, Iowa, says the following: “As an employer, students who don’t know my company, or my industry are really a waste of my time. I don’t want their resume. Students need to show a real desire to work for me, and this needs to be an informed desire based on the fact that my job represents their life goal of WHERE they want to work, WHAT they want to do, and WHO (company size, market position, etc.) they want to work for. As an employer, any student just ‘looking for a job’ hasn’t earned the right to be employed as a working professional.”¹⁵

2. Be prepared to answer not only technical questions but also personal ones. For example, if you are interviewing for a civil engineering position involving the design and construction of highways, an interviewer might ask you about cyclical loading of concrete and asphalt, or fatigue and stress problems associated with aging bridges. But what if the interviewer asks you questions like these (not necessarily in this order)?

- “What is your greatest strength and greatest weakness?”
- “What are your long-term career goals?”
- “Where do you see yourself in five years?”
- “Why do you want to work for this company?”
- “How well do you work with others?”
- “What do you know about our company and this position?”

How would you respond to these questions? Think about that—these are probably the kinds of questions you will be asked first. Many new engineering or science graduates have no problem with the technical questions, but when asked the personal ones, they get nervous and incoherent or say something unwise.

3. Be prepared to discuss your salary requirements. You can find a variety of salary surveys on the internet (e.g., www.salary.com). Often college career services departments keep data on recent intern and graduate starting salaries. Identify the going rate for your particular skill level in the region where your employment will occur and decide, in advance, what your bottom line will be. On the job application, you

may be asked for your salary requirements. To keep from revealing your bottom line too early, just indicate on the form that your salary is negotiable or express your requirements in a range.

4. Listen carefully in the interview. Take your time and respond to what is being asked. If you need to think about the question for a moment, say so. Do not blurt out a careless answer in an effort to respond quickly to a difficult question. Also, if you are not sure what is being asked, request clarification.
5. Look professional and act professionally at all times. That does not mean you have to rent formal attire for the interview, but you should dress properly. For many companies, that might require a suit or similar interview attire, but find out in advance and do what is expected. Also, act professionally in your demeanor and in the substance of your discussions. Be on time for your interview, do not make mindless comments, and do not speak disparagingly about others, particularly former supervisors and coworkers. Always be positive.
6. Keep your phone off and put away. Be in the moment. Listen to others and afford them your full attention.
7. Take paper and a pen with you so that you can take notes, but ask before you do. A padfolio looks very professional for this and also serves as a way to keep your resume copies pristine.
8. Smile, be friendly, and be courteous. Do this sincerely. Enjoy the things you will learn.
9. Always be prepared to ask a good question. When an interviewer asks you if you have any questions, it is best to ask a thoughtful question that is related to the conversation you have just had. At the end of the day, when all of the interviewers come together to discuss you, they will compare notes about if and what questions you asked. If you did not ask any, that is a red flag. If you asked the same question of every employee, that is also a red flag.
10. Respect every person that you meet. Often the receptionist who greeted you upon your arrival for an on-site interview will have a say in the hiring process about how you treated them “while the interviewers weren’t looking.” This is a clear window into your interpersonal skills.

11. Be prepared to tell a story about EVERY bullet point on your resume.

Putting It All Together

Here is the complete resume package for Sunny, including the cover letter (Example 20.1) and the formatted resume (Example 20.2). Letter and resume formatting can vary, but in terms of the letter, it is generally best to use a traditional block or modified block style (see [Chapter 19](#)). The resume should be well organized with the most supportive information first, grammatically correct, and professional in every respect. Sunny's thank you note after her interview is also included (Example 20.3).

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Example 20.1 Cover Letter

Sunny Misumoto
313 Dogwood
Ames, Iowa 50013
555-555-5555
smisumoto@anyuniversity.edu

Dr. Tracy Ann Burger
Senior Engineer, Embedded Systems
Cranes R Us
595 Miner Road
Wutherford Heights, Ohio 44132

December 13, 2020

Dear Dr. Burger:

While doing career research to locate companies that might have embedded systems engineering opportunities, I read about the advanced work Cranes R Us has been doing in partnership with cybersecurity systems. In fact, the more I read about your company and its activities,

the more convinced I became that I would like to be part of the Cranes R Us team. To that end, I have attached my resume with the hope that you will give me the opportunity to discuss further how I might fit in and make a measurable contribution at Cranes R Us during a summer internship.

I am currently a second-year student in Mechanical Engineering and Mathematics at Iowa State University. My degree program, which I will complete with University Honors in 2023, comprehensively blends the strong mathematics which form the fundamentals of cryptography and mechanical engineering concepts. During the past three years, I have had three different research positions, all of which have given me experience writing code for complicated systems, including optimizing and controlling robots. I learned through these previous experiences how much I enjoy the interface between systems and cybersecurity.

One of the things that I have enjoyed the most in my courses at Iowa State was the group project we had in my first-year engineering course. I was one of a team of four students; we designed and built a prototype robot for climbing walls. We were able to quickly assess each team member's strengths and then divide and conquer on the four major components of our design. After our final presentations, our team won first place overall. The award was based on a combination of the success of our robot and how we presented it. While I look forward to continuously improving my written and oral skills, I have a good start on understanding and using effective technical communication skills.

Thank you for taking the time to review my resume; I realize how many letters and resumes you must receive. I believe I can be a valuable resource for your organization, and I would welcome the opportunity to interview at your convenience.

Sincerely,

Sunny Misumoto

Sunny Misumoto

Enclosure: Resume

Sunny Misumoto 313 Dogwood, Ames, IA 50013 555-555-5555 smisumoto@anyuniversity.edu			
Objective – to find a summer or co-op position where I can use my problem-solving and teaming skills to make a difference to the bottom line for my employer			This resume is meant for an internship or co-op, but could be revised for a full-time position.
Education Iowa State University, Ames, IA Expected May 2023 Double Major: Mechanical Engineering, B.S. & Mathematics, B.A. GPA: 3.6/4.0 Honors Program			Discipline-specific information like being in the Honors Program is included in the Education section.
Work Experience Iowa State University, Ames, IA <i>Undergraduate Research Assistant</i>, Industrial Engineering - Dr. Leo Ellen Jun 2020 - present - Assisting PhD candidate with development of a novel digital surface roughness standard for castings - Developing C++/C# GUI in Visual Studio for automated surface roughness calculation based on point clouds; expected to increase usability and enable adoption of the surface roughness standard <i>COVID-19 Planning Team Student Employee</i> Jul - Aug 2020 - Created fillable PDF seating charts used for contact tracing for ~150 university classrooms - Documented standard process for creating and using forms; distributed by ISU Provost to all faculty - Responded promptly to change requests based on faculty input and facility changes, and communicated changes with faculty via email			Using different formatting techniques makes this resume easy to read and understand (larger font section titles, boldface, italics)
Research Experience (unpaid) First-Year Honors Mentor Program - Mathematics Jan - Apr 2020 - Learned linear algebra and differential equations concepts under Dr. Ravi Kotval - Used MATLAB to solve 3 complex math problems, including optimization and controllability Secondary Student Training Program - University of Iowa - 35 hours/week Jun - Jul 2017 - Conducted research for Dr. Gretchen Frank at the National Advanced Driving Simulator - Modeled reaction time data using MATLAB and created code for future studies			Sunny has included 2-3 bullets of evidence for each of her work experience positions. Her critical information shows she is well-balanced.
Activities, Leadership, and Awards Dorm Floor President Jul 2020 - present - Dogwood, Iowa State University - Lead meetings, organize events, communicate information to ~50 undergraduates Iowa State Putnam Prep / Problem-Solving Club - 2019 Score: 10/120 Sep 2019 - present First-Year Honors Program Leader - 13 students in class Aug 2020 - present Engineering Career Fair Ambassador Spring 2020 ISU St. Jude's Up 'Til Dawn - member Jan 2020 - present ISU Intramurals (basketball, soccer, volleyball, broomball, hockey) 2019 - 2020 Skills - German (basic)			Sunny has excellent grades, very good work experience for a second-year student, and leadership/activities. She appears well-rounded.
			Sunny's header is visually balanced. She has dates right-justified so that they are easy to find.

Example 20.3 Thank You Note After an Interview

Dear Dr. Burger,

I hope this email finds you well. Thank you for the extremely interesting conversation and tour of your facility last Friday. My initial research of Cranes R Us made me believe it would be a good fit for my interests in embedded systems and cybersecurity. After talking with your team, I am sure of it. I have started researching the answer to Mr. Hansen's question about encrypting asymmetric keys because it is so fascinating, and I hope to use it as part of my cryptography course project.

If you have any more questions or need more information from me, I would be happy to provide it. I can be reached at this email or the phone number in my signature below. I hope that Cranes R Us continues to prosper, and that you have a great week! Thank you again.

Best wishes,
Sunny Misumoto
555-555-5555

Resume/Job Application Checklists

Cover Letter Checklist: Have I

- ☐ Demonstrated that I have researched the prospective employer and have a reasonable knowledge of the company and the position?
- ☐ Indicated that I really want to work for this company?
- ☐ Summarized my skills clearly and succinctly using personal examples?

- ☐ Summarized my personal traits that enhance my qualifications for this position using personal examples?
- ☐ Demonstrated my desire to discuss further how I might contribute to achieving corporate goals?
- ☐ Indicated my willingness to interview for the position?
- ☐ Included support for and/or details not on my resume without simply restating my resume?

Resume Checklist: Have I

- ☐ Put my name, email address, and phone number at the top of my resume?
- ☐ Optional: Provided a clearly stated job objective that is not generic?
- ☐ Listed my higher education, including degrees, study abroad, and honors?
- ☐ Listed my experience, listing the most relevant to the job objective first?
- ☐ Summarized my computer skills, activities and leadership, volunteering, or other personal information that helps to tell my story?

Interview Checklist: Have I

- ☐ Researched the company, and do I understand the position?
- ☐ Prepared myself to answer technical questions about what the company does?
- ☐ Prepared myself to answer personal questions about my career goals and how working for this company in this position fits in with those goals?

- Consulted salary surveys, and do I know my minimum salary requirements?
- Dressed properly, and do I have an appropriate appearance for the interview?

Exercise

The following resume is for a third- or fourth-year student (based on a real engineering student). Parts of it are done very well. Some things could be improved. What is done well and what could be improved?

555-555-5555

Jamal Gonzalez

jg@hackgonzalez.com

OBJECTIVE

To apply my knowledge and interest in cyber and physical security during an internship or co-op

EDUCATION

Iowa State University, Ames, IA

Expected 2020

B.S. in Computer Engineering, Minor in Information Assurance

GPA: 3.31/4.00

DEF CON Hacking Conference Attendee

'15, '16, '17, '18, '19

Black Hat Security Conference Attendee

'19

WORK EXPERIENCE

CLOUD SECURITY INTERN - Tractors, Inc., Ames, IA

APR '19 - Present

- Audited Amazon Web Services resources & security policies to ensure security and reliability of cloud systems
- Developed approximately 10 Python scripts to automate audit processes and enforce security guidelines
- Researched/developed containerized active defense systems for network security using Docker & ADHD Linux

TEACHING ASSISTANT (Cyber Security Fundamentals) - ISU, Ames, IA

AUG '19 - Present

- Led 70 students through weekly hands-on networking and security labs, explaining core concepts
- Manage up to 700 virtualized student lab machines across 70 virtual networks with VMware vCenter
- Assist students with troubleshooting problems and guide them to discovering solutions independently

DEVELOPER - Iowa State University Biology Department, Ames, IA

FEB '18 - DEC '18

- Designed and implemented features into BioLearn, an educational tool used by biology faculty in Bio 212 at ISU, used by hundreds of students per semester
- Wrote, documented, tested, & maintained C# code including algorithms for gameplay & animations

PERSONAL PROJECTS

Social Engineering Capture the Flag at DEF CON 26

JUN '18 - AUG '18

- Performed online open-source intelligence gathering against target (major US airline) and employees
- Wrote 5 pretexts using principles outlined by Mitnick and Hadnagy to establish rapport and justify contact
- Collected 10 flags and other information that could be used for malicious actions

CodeGreen - Python-based screen scraper and personal scheduling tool for hacking conference

SKILLS

Operating Systems - Linux Shell (Arch, Fedora Workstation, Ubuntu Server), Windows, Mac OSX & FreeBSD

Networking - Network setup, management, and auditing, using tools including Wireshark and Nmap

Programming Languages - Python, Java, C#, C, Verilog

Security and Virtualization Tools - Metasploit, VirtualBox, vCenter

Physical Security - Experience picking locks, knowledge of manual lock bypassing (shimming, under-door tools, etc.) and electronic lock weaknesses (thermal/motion sensors, badge cloning)

Social Engineering - Pretext development, extensive reading of fundamental literature

Open-Source Intelligence Gathering - Experience performing intensive online reconnaissance campaigns

ACTIVITIES AND LEADERSHIP

- | | | | |
|--|---------------|---------------------------------------|-----------|
| • ISU Cyber Defense Competition | '18, '19, '20 | • ISU Fall Hackathon, | '18 |
| • ISU Info. Assurance Student Group | '17 - '20 | • ISU Freshman Leaders in Engineering | |
| • SecDSM security organization meeting attendee | '19 | • President | '18 - '19 |
| • ISU Engineering Student Council Member | '18 - '19 | • Executive Board | '17 - '18 |
| • Volunteer with Food at First free meal service | '17, '18 | • Fundraising volunteer | '17 - '19 |
| • Tang Soo Do Karate, 2nd degree black belt | | | |

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5. "Careers: ACS Career Navigator." *American Chemical Society*.
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6. "Career Center." *Institute of Industrial & Systems Engineers*.
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<https://employmentboost.com/job-search/john-deere-careers-land-job/>. Accessed December 12, 2020.
8. "Job Search by Location." *Boeing*. <https://jobs.boeing.com/>. Accessed December 12, 2020.
9. Tufte, Edward. *The Visual Display of Quantitative Information*. 2nd ed. Graphics Press, 2001. https://www.edwardtufte.com/tufte/books_vdqi. Accessed December 12, 2020.
10. Dr. Frank Peters, Industrial and Manufacturing Systems Engineering, Iowa State University, Resume Roast, February 5, 2020.
11. Dr. Mike Helwig, Retired U.S. Navy, Associate Teaching Professor, Industrial and Manufacturing Systems Engineering, Iowa State University.
12. Ewest, Timothy. "True or False, 'It is not what you know, but who you know'?" *LinkedIn*. June 18, 2016. <https://www.linkedin.com/pulse/true-false-what-you-know-who-timothy-ewest>. Accessed December 11, 2020.
13. Ewest, Timothy. "True or False, 'It is not what you know, but who you know'?" *LinkedIn*. 18 June, 2016. <https://www.linkedin.com/pulse/true-false-what-you-know-who-timothy-ewest>. Accessed December 11, 2020.
14. Dr. Mike Helwig, Retired U.S. Navy, Associate Teaching Professor, Industrial and Manufacturing Systems Engineering, Iowa State University.
15. Dr. David Sly, PE, PhD, MBA, President and CEO of Proplanner.

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CHAPTER TWENTY-ONE

Team Writing



You find yourself in an engineering design course where you are grouped with three other students, none of whom you know. Your group is tasked to design a highly efficient, inexpensive, environmentally friendly snow and ice removal system to keep the engineering dean's parking place open during the winter months. Your grade in the course will be based primarily on the quality of your group's solution as documented in a technical report. The report must be produced by your group and delivered on the last day of class. Welcome to team writing!

In [Chapters 5](#) and [6](#), we described the mechanisms and processes associated with a Samoyed sled dog team. In this chapter, we examine the interactions of the **team members and their roles** rather than describing a final written product (see [Figure 21.1](#)). Each and every dog, along with the musher (human team member) who leads them, is critical to achieving a successful outcome. If any one dog decides to pull in a different direction, or perhaps decides to stop and rest while the others want to run, the entire team is slowed or stopped. If the musher ignores what the team needs in terms of resources (food, water, rest, shelter, and attention), takes a wrong turn, or fails to assist the team on hills, the team loses effectiveness and can even fail.



Figure 21.1

All of the members of a writing team must pull in the same direction, much like sled dogs, to achieve a successful outcome.

Team writing, often referred to as **group writing**, is the process Page 312 in which two or more authors work together to produce a document or set of documents to fulfill a requirement. Team writing is basically a “good news—bad news” thing. The good news is that you have several people with varied backgrounds, skills, and abilities from which to draw. The bad news is that you have several people with varied backgrounds, skills, and abilities from which to draw. The extent to which the news is good or bad depends in large measure on whether you are part of a *team* or a *group*, because the term group writing is really misleading.

If a collection of people wants to do well, they must be part of a team, not a group. A group implies only the existence of an aggregation of people. A team, however, is an aggregation of people organized and managed to achieve some purpose. This distinction might seem academic to some, but real differences exist between groups and teams—and understanding and dealing with these differences are essential if any group, including your group, is to prosper in the team writing environment. In both problem-solving and writing, companies and organizations want and need teams to capitalize on members having different perspectives; that is why teams exist

in the first place—because no individual has the capacity to experience and understand every aspect of the business, organization, or project.

The collective experiences, skills, and beliefs of the individual members can be a powerful force when properly managed, or working together can be characterized by hurt feelings, wasted resources, and worst of all, inferior work. Turning the collective diversity and talent of a group of writers into a successful team of writers is not easy. It requires the best effort of all involved, the right blend of skills, a great deal of common sense, and effective management and leadership.

Student versus Professional Team Writing

Team writing might represent a real adjustment for anyone whose writing experience has been limited primarily to individual work. As a student, you have probably gained most of your writing experience while working alone in courses. Some coursework, such as a team lab report or the group engineering design scenario mentioned earlier, does offer team writing experience—and getting that experience is great! However, you must take advantage of the opportunities provided. Student teams are established in many different ways. Sometimes you are randomly grouped with others through an unbiased process like counting off by the number of groups being formed. Sometimes you are grouped together by seat proximity in the classroom (and if you are sitting by friends, this outcome could be a happy accident or be the source of regret), sometimes the instructor will consciously use intake data to form teams by grouping students with similarities (e.g., all studying the same major, like chemical engineering) or differences (e.g., not all studying the same major, like chemical engineering, business, and communications). Other times you will be allowed to form your own teams. For capstone courses or other courses with significant projects where you get to create your own team, you might plan your roster months in advance.

There are many competing ideas about the best way to form teams in academic settings, and many faculty members say, “This is how it happens in industry.” The truth is, how teams are formed in industry is a mix. Sometimes team members are chosen based on expertise, but who actually

“fills the seats” is somewhat random; you might be asked to be part of a team simply because your engineering department needs a representative and you currently have the lightest load of responsibilities from among your peers because one of your projects just finished. Sometimes you will be assigned a project and told to build your team. If you know all the people you want and need, you must convince both them and their managers to allow them to participate, as time is a valuable resource.

Frequently, your supervisors and managers will decide that a team is needed and predetermine who they would like to be on the team. They do this based on job function, previous performance, and sometimes the mix of personalities. They will ask if you want to participate (keep in mind, being asked is often really being informed) and the team will commence. Because of the variety of team formation in industry, it is best to learn how to make your teams, regardless of how they are formed, quickly productive; college courses are a great place to practice these skills.

Once a team is identified, there is some work to do before the Page 313 writing can start. Note that there are entire textbooks, courses, and professions related to understanding team dynamics. One now very well-known concept was introduced by Bruce Tuckman in 1965 while he was at the Naval Medical Research Institute. In a paper called “Developmental Sequence in Small Groups,”¹ Dr. Tuckman labeled the stages of team behavior as **forming**, **storming**, **norming**, and **performing**. While the scope of this book cannot expand to include optimizing team dynamics, we recommend that every team address this concept to facilitate quicker productivity. Every team should start with understanding its team members. Personal backgrounds are important because perspective is important, and diversity of perspective can be both a strength (if you take advantage of it) and a challenge (if you let differences interfere with, instead of informing, your writing). Share preferences (Who likes to do research? Who likes to create visuals? Who likes to edit? Etc.), as well as strengths and weaknesses (Who is good at research? Who is good at creating visuals? Who is good at editing?). In addition, some instructors, institutions, and organizations use personality trait tests to help diversify personality types and approaches when building teams.

While you might not be the best or most comfortable writer, you still want and need practice. Everyone can get practice, and the team can still produce an excellent final product. It is important that **all** team members

understand they are individually responsible for generating written content, as well as responsible for the final team product. Every team member must contribute to the process. However, in your “forming” conversations as part of a technical course project, you might identify a team member who is a good writer and is comfortable taking the role of **lead writer**. When one person takes the role of lead writer, they can support other team members by synthesizing individual sections, writing transitions, establishing flow, creating a single voice, editing, and proofreading. This role is critical to creating a quality document, but they should never be the sole person responsible for creating the team’s document. The team should avoid having that person become the “team’s writer.”

The classroom experience frequently simulates or even works with the real world of business and industry, where the complexity of work often requires that technical writing be done by many members of a team. The scope and magnitude of the material, along with the required expertise in multiple disciplines, frequently prevent any one individual from writing the entire document. In the case of formal proposals, for example, proposal managers might lead several writing teams simultaneously. These managers often oversee various experts in applicable disciplines who work together to produce an effective proposal. In this case, a sub-team of technical experts might write the technical section, while functional or management experts might write the management section, and budget experts might write the cost section. Rigid roles are not always the case, though. Often these team members must work together simultaneously to develop the critical parts of a proposal. Depending on the document’s required formulation, these groups might also have to work together to write report sections which combine this information. In the end, whatever the formulation might be, all sections of the proposal are pulled together to form one coherent document—hopefully a winning one.

The Process of Team Writing

The team writing process consists of several steps that can vary significantly from one job to another depending on the requirements of the particular situation. As we described earlier, team formation happens in many ways, and establishing the team leader also varies. Some teams are

formed before the team leader is named; other teams elect their own team leader after being formed; and still other teams have the team leader designated by higher management, who then selects the members of the team. It is even possible, and sometimes preferred, that team members take turns with leadership roles on the team.

The nature of team writing has changed drastically with modern technology, and there is research that supports the idea of fluidity and flexibility in all aspects of teamwork, including roles. For an interesting article about team considerations, see “Teams in a New Era: Some Considerations and Implications,” by Benishek and Lazzara, made available by the National Institute of Health.² Expectations for roles within the team should be part of initial conversations. Some teams select writing portions for each specialty from within their group; others have professional technical writers assigned to them; and sometimes a single publication or marketing office in a company provides professional layout, design, and copyediting. Clearly, many variations might exist for any given situation. That said, [Outline 21.1](#) provides a generic model for the team writing process. It serves as a general guideline for team writing and includes three steps: determining requirements, taking preliminary actions, and producing the document. The document might be hardcopy (printed on paper), an electronic file or files, or both. Regardless of the medium, the process is the same.

Outline 21.1 Team Writing Process

Requirements

- Determine what must be done to solve the problem.
- Define the audience and purpose of the communication.
- Ascertain what kind of document will be needed and how it will be disseminated.
- Specify document themes and organization.
- Identify relevant context related to the problem.

Preliminary Actions

- Set team ground rules including constraints and expectations.
- Designate a leader and provide adequate authority and responsibility.
- Determine what specific resources (skills, data, and material) will be needed.
- Assign/recruit teams and/or individuals to provide these resources.
- Decide on the collaboration platform and process (e.g., word processing software like Microsoft Word shared via email or cloud storage, cloud applications like Google documents, etc.).

Document Production

- Define specific writing responsibilities and schedules.
- Task team members to produce draft inputs for their area(s).
- Edit and revise inputs as needed for the overall effectiveness of the document.
- Conduct a final review of all aspects of the document and deliver as required.

Here is a brief discussion of this three-stage process.

Requirements

Solving the project problem at hand is not the same as writing about it, but the two often inform each other. Normally, you would begin the team writing process with two basic assumptions. First, a need or opportunity exists that requires team action; and second, the output of the team action will be some type of solution detailed in a formal document. The first phase of this process, like all technical writing, involves defining both the audience and the purpose of the document, whether that is fulfilling a course requirement or winning a large business contract. In other words, you first must identify and understand the problem you are trying to solve before you jump into the process of solving it (see [Chapter 13](#) A3 Reports for a brief description of root cause analysis). Context is also very important

—what are the details that if understood and considered will make your document successful?

Next, you must determine what kind of document is necessary. If you are trying to sell your company's goods and services, or convince a professor to approve your team's design topic, a proposal is in order. If you belong to a team of experts charged with identifying which alternative is best to solve an existing requirement in the company, a recommendation report is the right choice. If you are developing knowledge about a new, evolving technology that your company is interested in pursuing, a state-of-the-art research report might be just the thing. Sometimes the required formulation is specified; other times, it is up to you to recognize what kind of document best fills the need.

In any case, once the type of document has been selected, the Page 315 next step is to decide on the controlling ideas (themes) to be emphasized and the sections (organization) of the document. Often, the style will be dictated by the audience, provided either formally by a specified style guide, or informally by a previously successful project of this type. Copying the style from a previous document sometimes works, especially when a style guide is not available; but be careful not to clone the document itself. That rarely works well and often brings with it a host of legal and ethical issues (see [Chapter 2](#) Ethical Considerations), not to mention the possibility that you might be copying a poorly written document. Finally, be sure to focus on themes—those recurring positive images of your solution, company, or team must be highlighted.

A Word about Themes (Tell the Story)

Themes, which are controlling ideas that recur throughout a document, play a powerful role in business writing, especially in proposals. Proposals, as discussed in [Chapter 8](#), are persuasive documents designed to sell goods and services, and as such, they succeed or fail depending on how well they convince the reader. Other technical documents, while not geared specifically to convince someone of something, still have important persuasive aspects. For example, a research report is designed primarily to provide information. However, depending on how well it is written and how effectively it appeals to the reader, the report can also enhance or degrade the image or reputation of the individual or company that produced it. Another way to think about themes is to consider “the story.” What story

are you trying to tell? People (read that as decision-makers and advisors) respond to good storytelling. Of course, in a technical document you must be factual and logical, but you want to tell your story in a way that compels your audience to care; using themes effectively can help accomplish this.

Highlighting themes in technical writing is all about creating persuasive images. In other words, you want to predispose your reader to be persuaded by their emotions, or convinced by your argument and credibility. The technical proposal discussed previously might be built around a well-defined problem, an effective solution, a precise statement of work, and an impressive list of resources—but it can still benefit from crafting a story based on high-quality work, solid performance, effective risk management, and creative problem-solving. Incorporating evidence of these concepts into your proposal will represent your organization well and encourage a positive, public image.

Preliminary Actions

The first preliminary action in the team writing process is to establish ground rules. Ground rules are agreements among team members about process, including constraints and expectations. They can be fluid; if **the team** decides they need adjustment partway through a project, then they should be changed. If your team decides it will identify a leader, make sure that person has the authority and responsibility needed to get the job done. Like team dynamics, a discussion of the philosophy of leadership is also beyond the scope of this book, but it is important to note that there are many different types of leadership, including **democratic (more consensus-building)** and **autocratic (more directive-oriented)** (see [Figure 21.2](#)). While different teams and team members will respond better to different leadership styles, in some cases, team writing leaders, once appointed or otherwise designated, benefit from having authority to make decisions that are not always built around consensus.

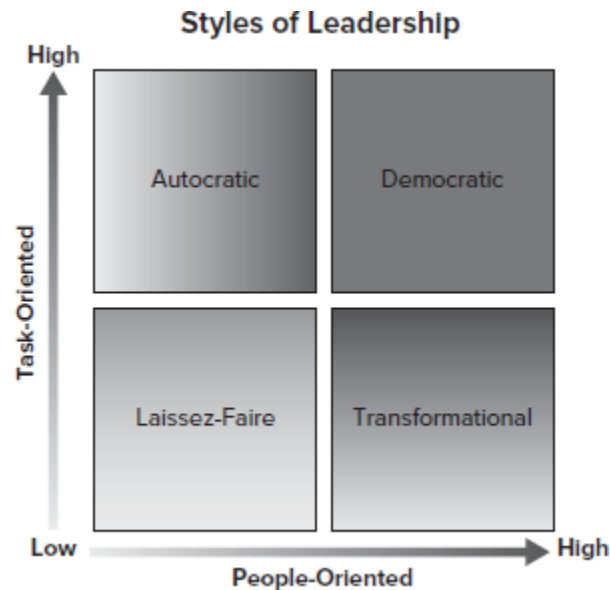


Figure 21.2
Leadership styles.

Leadership Styles

Many people have defined different leadership styles; some commonly accepted types are shown in [Figure 21.2](#). There is no “correct” leadership style. The effectiveness of a leader’s style depends on many things, including the specific project purpose, individual team members, and project context (scarcity of resources, deadlines, criticality of success to organization viability, etc.). In one situation, an authoritarian leader might be exactly what is needed to achieve a successful outcome, while in another, that same leadership style might send team members scurrying away to the corners of cyberspace. It is helpful to understand your own leadership style so that you can be flexible, when needed, to improve the likelihood of project success.

Another preliminary action is determining what resources are Page 316
needed. This includes human resources (the creativity and skills necessary to successfully complete the project) and material resources (the facilities, data, and equipment needed to properly research, write, and produce the report). This also includes specifying the platform upon which you will collaborate and the process for this collaboration. Very rarely will you pass printed paper back and forth; most collaborative writing is done using a combination of software, applications, and/or cloud storage.

Determine who has access to what, and if needed, procure the necessary access for all team members. For example, in writing this textbook, we determined ahead of time that we would use PDF and Microsoft Word documents. We chose at the onset to use cloud storage (like DropBox) for exchanging drafts; this was critical for version control. Also related to version control was our pre-determined file naming scheme, which included the chapter title and name, our initials, the save date, and sometimes the save time. We agreed to use email and texting to communicate status updates and resolve minor questions. We also agreed to use (perish the thought!) actual phone conversations to resolve more involved problem-solving. Process decisions should be discussed before work starts so that everyone understands and agrees to team expectations; this will increase the team's ability to become quickly productive.

Finally, the last preliminary action is to assign teams or individuals to specific areas of responsibility. In a student group, you might be responsible for developing and writing the technical approach to solving the problem, while another student might tackle researching and locating the required resources.

Document Production

The last major part of the team writing process is the actual writing. This is the point where the real challenge often begins because the team writing process is neither simple nor linear. In fact, many who should know better view the task of writing a report as consisting of three easy steps: getting information, organizing notes, and assembling the paper—as if the process were analogous to populating a circuit board and soldering the components in place. If only it were that simple! Team writing can be a complex, demanding process complicated by almost anything going wrong—from the unexpected illness of a member, to a software crash at the eleventh hour, to petty disputes over who gets to do what.

The first step in getting any team to write well is to define clearly what everyone must do. Each member must be aware of their responsibilities. Each member should know exactly what they are expected to do and when it must be done. Additionally, each person should understand how they fit in with the total effort. No one on a writing team works in isolation. Everything is coordinated and interconnected. Just like with the sled dog

team, failure on the part of one person or one sub-team ultimately affects everyone else and the success of the project.

In professional proposal writing, if the cost group fails to [Page 317](#) produce the cost section on time, it really does not matter how good the proposed solution is or how innovative the management philosophy might be. The proposal will fail, and everyone will lose. As a student, it is not hard to imagine the anxiety of approaching the last day of the last class before graduation, and having someone on your design team fail to produce a critical part of the required design report.

The bottom line is this: members of writing teams must work hard at working together to get the job done. The team leader's role is to determine, often through consensus in a student group, the best way to organize and develop the required elements of the report. The team's role is to produce those elements. The leader's role in this regard is to identify common ground; create a structure that encourages cooperation and collaboration; clearly identify goals, themes, and expectations; and ensure quality, on-time performance.

The leader is also responsible for editing and revising team inputs. They must ensure that the assembled inputs respond fully and adequately to the original tasking, and tell the project story. In addition, the leader must ensure that the document is coherent, reads as if written by a single voice, has the right tone, and is consistent in its style and structure.

The last step involves the final review of all aspects of the document and delivery of the final product. The final review is sometimes affectionately known as a **gut check**. The goal is to get one or more trusted, knowledgeable people (often called a **red team**—their function is to identify weaknesses) outside of the writing team to look over the document for silly mistakes. These kinds of errors are easy to make, especially if one is close to the writing process—which everyone on the writing team is. Sometimes we all see what we want to see, with glaring errors going totally undetected; and sometimes we overlook illogical or inconsistent statements that are right in front of our eyes. That is why, in a professional setting where the stakes are high and the environment is unforgiving, writing teams often employ the services of seasoned copy editors and consulting associates to help in this regard. As a student team, there are things you can do to facilitate proofreading; when team writing, members should proofread each other's work. See [Chapter 15](#) for more discussion on proofreading.

When you are ready to deliver the document, verify that you are using the audience's preferred/required method. If the requirement is to attach a PDF file to a transmittal email, do not share a cloud-based link. If the requirement is to upload a file to a server, do not send an email. There is possibly no worse moment in a project than missing the final step after all the time and effort that you put into writing the document.

Proofreading Tips from [Chapter 15](#)

- Reduce the width of the text window in which you are typing so that the paragraphs also change width. This causes words to move to new locations and allows your brain to find missing words and identify style issues.
- Cut and paste text from one software application to another (e.g., from email to a text editor). This has the same effect.
- Print what you have written on paper and read it.
- Read paragraphs in reverse order.
- Read your text slowly and out loud exactly as written. Pause a complete second for punctuation in a sentence (commas, colons, semi-colons) and pause two seconds for punctuation at the end of a sentence (periods, question marks, exclamation). Only stop for punctuation.
- Wait two hours (or two days) before proofreading.

Professional Team Writing

As an example of one of the most complex and demanding team writing activities in the professional business environment, we will focus on formal proposals involving the U.S. government (also see [Chapter 8](#)). You might think, “Wait, I need the best possible team to be successful, like a team of Samoyeds or Huskies to pull a sled, but my proposal team is a bunch of Poodles. What do we do?” With a good understanding of the team writing process, it can be done; check out a video from *The Tonight Show* on YouTube when John Suter, musher, is interviewed by Johnny Carson about

finishing the Iditarod sled race with poodles^{3,4} (see [Figure 21.3](#)). We will associate this highly structured process with the major elements of the model provided in [Outline 21.1](#) and schematized in [Figure 21.4](#). A typical proposal team organizational chart is provided in [Figure 21.5](#).



Figure 21.3

Team writing can be done effectively by any group that trains well and is motivated to be successful.

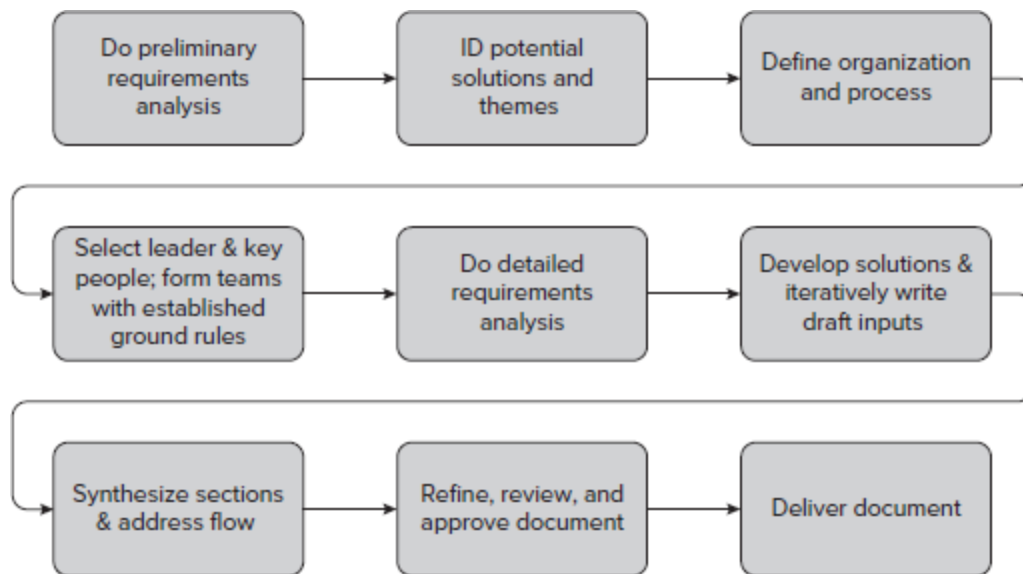


Figure 21.4

Professional team writing process.

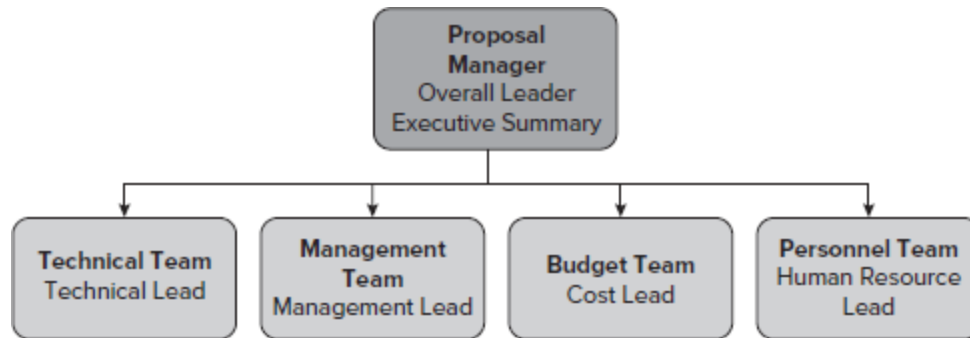


Figure 21.5

Typical proposal team constituents and associated contributions.

Requirements

This team writing process usually begins when someone called a **capture manager** receives advance information about an upcoming project from either a Request for Information (RFI), in which the government basically asks private industry for ideas on solving a problem; or a draft Request for Proposal (RFP), in which the government gives a “heads up” to industry that it will be inviting industry to submit proposals to solve some problem. The capture manager analyzes the information (including whether the necessary funds are available), determines whether the project represents a good opportunity for their company, and, if so, defines the problem and roughs out how the company might respond with a proposed solution. The capture manager also defines the major themes for the proposal. These themes are recurring images that will be used deliberately throughout the proposal to highlight the strengths of the company. Specific themes in this case might include evidence of solid past performance, great technical talent, and superb facilities.

The capture manager effectively puts the team writing process Page 319 into motion within their company. They prepare the company to receive the actual RFP when it is released and bring together both the human and the physical resources necessary to write the proposal. Often the capture manager will then become the **proposal manager**, the designated leader of the proposal effort. The proposal manager will be “on the hook” (i.e., responsible for) for the conduct and success of the entire project, and will serve as the document leader, the document coordinator, the editor of last resort, and the person with overall responsibility for the success of the

project. When the proposal manager takes over, the role of the capture manager is finished.

Preliminary Actions

The proposal manager now assumes the primary leadership of the project. They determine how the project will proceed, what exactly will be done, what the schedule will be, and who exactly will do the work. They determine the need for various functional or technical teams, might rough out expected approaches to be taken by each team, and then specifically task these teams. For example, the proposal manager would task the technical team to produce a refined, competitive technical solution to the problem and to draft the technical section for the proposal. In a similar manner, they might task the management team to develop the organizational structure and management approach to support the technical solution, and to draft the management section of the proposal. The same would be true of the accountants and financial managers producing the cost section, and so on.

The proposal manager then selects the various functional people who will serve in whatever specialty teams are needed, and might even appoint writers within the teams to draft their respective inputs to the proposal. The proposal manager might also recruit the services of technical writers and copy editors in a publications office, if one exists in the company. Either the proposal manager or the experts in the publications office would then define the style guide and procedures for actually producing the proposal.

Document Production

The proposal manager ensures that the RFP has been completely and thoroughly assessed and that every detail warranting a response has been identified and defined. Failure to respond to even a single element could cost the company the contract. A compliance matrix for each RFP requirement is built at this point. Ultimately, this matrix will be submitted to show evaluators where the RFP requirements are addressed in the proposal. Specific individuals or teams are then tasked to respond. The technical team, led by a technical team manager, develops the approach the company will propose to solve the problems defined by the RFP. Starting with the approach initially roughed out by the capture manager in the

requirements determination phase, the technical team develops the proposed solution.

Of course, this solution cannot be developed in a vacuum. The technical approach must be practical in terms of competitive cost and schedule, it must be consistent with the required management structure, and it must be doable in terms of the skills and resources available to the company. A great deal of coordination must occur between the technical team and other teams. The proposal manager provides the primary conduit for this coordination, although individual team managers might also work closely with one another.

During the document production phase, the proposal manager Page 320 and individual team managers continue to provide adequate oversight and guidance and often must make tough decisions on the spot. The emphasis must be on common sense and efficiency—one team cannot be “spinning its wheels” while another team is making progress. Additionally, when individual teams complete their draft sections, these drafts must be reviewed quickly and thoroughly, and then edited to make them consistent with the goals and themes of the overall proposal, as well as with the content of other teams’ draft sections. The proposal manager makes or coordinates changes at this point. The writing style and tone must be made consistent for all sections. Needless to say, the job of a proposal manager can be very complex and demanding, and proposal managers earn their pay!

Finally, the proposal manager reviews the entire document, including technical, management, cost, and human resource sections, along with the compliance matrix—and in the process, adds an executive summary that they often write. At this point, the entire project might receive a final review by a red team and approval by senior management. Additionally, the publications office might do final style and layout reviews. The document is then delivered according to requirements in the RFP. Note that some government RFPs have legal requirements, such as required clearances, secure computer interfaces, etc. The proposal manager must understand every detail to ensure success.

Student Team Writing

Student team writing is like professional team writing except that the experience base, skill levels, financial incentives, and authoritative structure are not present to the same degree in an academic setting as in a professional business environment. Additionally, student motivations are different. Obviously, grades are on the line, but grade pressure varies from professor to professor and from student to student. Some professors evaluate the entire group's performance, while others tend to hold each student more responsible for their contribution. Some students are strongly motivated to perform at high academic levels and would be upset with anything less than an A, while others might just want to get through the course and would be happy with any passing grade.

It is difficult to predict the dynamics and motives of any group of students involved in academic team writing. However, generally speaking, the process described in [Outline 21.1](#) and schematized in [Figure 21.6](#) is valid, and the principles described in this chapter do apply. Whether students are writing a proposal for a scientific study or producing a project report at the end of an engineering design sequence, writing team members must review the requirements of the project, take the necessary preliminary actions, and then produce the report.

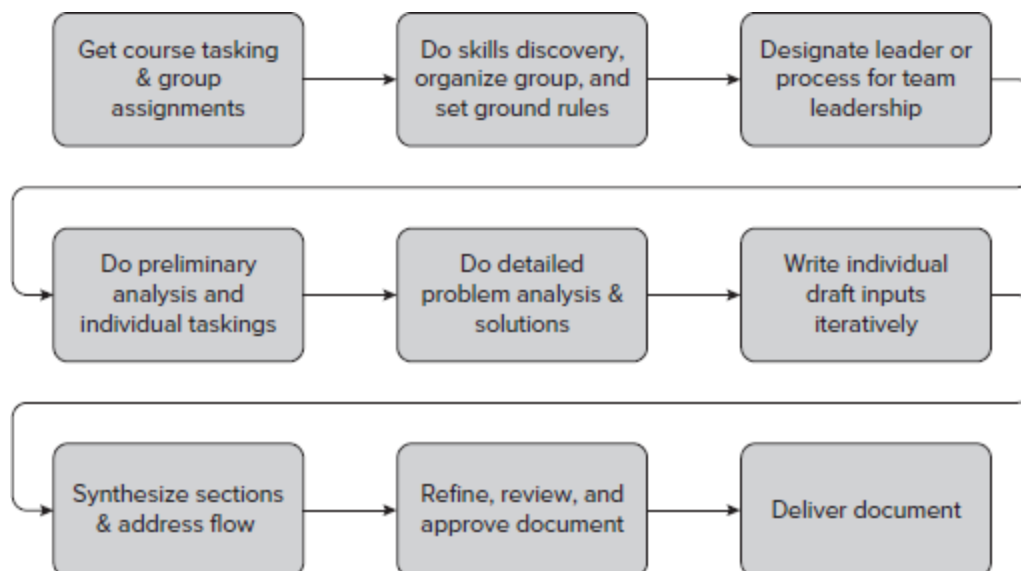


Figure 21.6

Student team writing process.

Preliminary Analysis

Do not forget: as with any technical writing, you must understand audience, purpose, and context.

Requirements

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In an academic setting, there is no need for a capture manager or requirements expert. The teacher often defines the problem and specifies the document formulation. In some design sequences, however, students are assigned to groups and charged with identifying problems that they will then propose to solve. This might include working with a company who specifies a problem area for the team. However, the teacher usually decides whether the problems proposed are worthy of being a course project and, specifically, what documents each group must produce to fulfill the course requirements. In all cases, you must determine who your audience is, how they will use your written document(s), and the context associated with your project.

Preliminary Actions

The preliminary actions phase is critical to the project's success. The first step for any student group is to get to know each other. You cannot form a team without some level of understanding of each member's personality and perspective. If a leader is not assigned by the instructor, your team must designate a leader or define how leadership roles will be shared. Oftentimes a natural leader evolves and is, unsurprisingly, the student considered by the team members to be the most qualified in the group—that is, the one who supposedly knows the most and who often has the most to lose. In other cases, the selected leader might be the one person who did not attend the initial meeting and was not present to decline the position. However the selection is made, the leader should be a highly qualified member of the group who is willing and able to exercise team-granted authority and serve as the focal point for the project. If not, the group must establish a new leader—it is just that simple! The leader then facilitates the group becoming a team by organizing and tasking the members. A good starting task is to have the team members jointly establish ground rules.

Similar to the professional business environment, the type of leadership that works best with fellow students varies, but a good starting place is usually more democratic than autocratic. The leader must work first to

establish consensus on the matters at hand. For example, if the teacher is leaving it up to the students to identify eligible problems, then team members should agree on what kinds of problems should be considered and what resources would be required to solve these problems. As part of this process, the leader should also guide their fellow team members through an informal skills discovery process to determine what everyone “brings to the table” in terms of knowledge and experience.

The goal of this process is (1) to identify and define potential projects for the group to solve; (2) to determine to what extent each potential project is doable, and then select the best one; and (3) to accomplish the tasks associated with solving that problem by determining who will be responsible for doing what. In other words, the team member who is the best computer programmer would do any required coding, the most experienced “metal bender” in the group would build any required device or apparatus, and the most accomplished writer would take the lead on writing the report.

Document Production

To produce a report, each member typically drafts the input for their part of the project. The lead writer designated by the team takes these inputs and synthesizes them into a cohesive report. Assembling a report that flows well, has a single voice (sounds like only one individual wrote it), and is consistently formatted is much more involved than simply setting up an outline in a shared file on the cloud. Synthesizing is a demanding, iterative process that requires the team leader to work with each team member as they craft a cohesive, consistent part of the story. The lead writer organizes the material, not only by content, but with transitional phrasing. They fill in gaps, smooth out uneven prose, establish a consistent tone, identify missing information, ensure consistent formatting, and make other changes, as necessary. To be successful, the lead writer and other team members must work very closely together throughout the process. It often takes a lot of time, and certainly requires many iterations. For these reasons, it is not surprising that in many cases the team leader becomes the report writer, but this is a disservice to the team. Do not be the team member who does not pull their weight in this process (sled dog pun intended!).

Definition: Iteration

To iterate is to repeat a process multiple times until the desired outcome is achieved. Iterative writing means writing multiple drafts (or versions). There are very few excellent finished drafts written on the first try.

Finally, once the project report has been fully drafted, each member reviews the entire document—not just their own section. Every team member is responsible for ensuring the consistency of the story and serves as an internal red team for the lead writer at this point in the process. Often it is much easier for someone not closely involved with writing a particular section to spot errors or inconsistencies. Also, by this point in the process, the writer and/or team leader might be too close to each of the sections to notice problems. In fact, if possible, the group should enlist the aid of an outside reader who can review the entire report for factual or stylistic errors. The last step is for each member to review and approve the document for submission.

Conclusion

Team writing is a demanding process whereby several people, often with specialized skills and knowledge, are brought together to produce a single document. The collective efforts of the writing team (or teams) have the potential of providing great synergistic advantage, but the team can also break down and, in some cases, totally fall apart if it is not managed well. Writing teams must have a clearly defined problem and clear management guidance with which to work. Before team writing begins, the audience must be understood, the specific requirements of the document must be well defined, the type of document must be determined, and the style and themes of the document must be specified.

Writing teams must also have a designated leader with overall responsibility of the project and with the authority and ability to make tough decisions. In professional team writing, this leader decides on the required skills and resources and assigns teams and individuals to

accomplish the necessary tasks. In student team writing, the interaction is often less formal, and the team leader works through consensus and negotiation. In any case, once the responsibilities for the project have been defined, the document can be produced. Specific writing tasks and schedules are laid out, and designated teams and members produce the required draft text, often iteratively. These drafts are edited and revised as necessary, and the final review and delivery of the document is completed.

Team Writing Checklist

- ☐ Am I committed to being a member of a team and not just a member of a group?
- ☐ Do I accept that team writing in college is helpful for preparing to work in teams as a professional?
- ☐ Do I accept that having a lead writer on my team does not mean that they are responsible for all the writing and that I do not have to do any?
- ☐ Do I know what requirements must be established before team building and work can begin?
- ☐ Do I know what preliminary actions must be taken before team writing can begin?
- ☐ Do I know what steps are required to produce my team's document?

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